PrimusNeo

Commercial Engagement Models & Value-Based Pricing

Internal v0.1 · 2026-05-23

PROOF OVER PROMISES

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PrimusNeo Commercial Engagement Model — Overview

> ***Document set:*** *Engagement Models v0.1 (May 2026)* ***Status:*** *Draft for internal alignment. Numbers are anchors, not quotes.* ***Audience:*** *Founding team, partner ecosystem (Escherbridge, OmniOne, PLAN, OASIS Web4), enterprise prospects.*

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1. Why this document exists

PrimusNeo is a **federated coordination stack** — not a product, not a SaaS, not a blockchain. Four sovereign technology layers (built and stewarded by four sovereign organizations) compose into a single operating fabric for any group of people, machines, and capital that needs to coordinate at scale.

That structure makes pricing genuinely hard:

- **Four layers** with different licensing postures (two creative-commons, two closed/partial).
- **Ten verticals** with wildly different willingness-to-pay, procurement norms, and service multipliers (a hospital is not a DAO is not a port).
- **Four classes of service** (customization, integration, deployment, sustained upkeep) plus **infra pass-through**.
- A founding commitment to **"Proof Over Promises"** — pricing must be defensible against the *actual* friction-cost the buyer is already paying (McKinsey, IMF, UNIDO, sector-specific sources).

This document set translates that complexity into a **bounded, value-based commercial model** that any deal team can use to quote a number, defend it, and close it without burning the open-source ecosystem that feeds the stack.

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2. The four layers (and their licensing posture)

Layer	Steward	Function	Licensing	Relative Price
AMP	PLAN Systems	Spatial OS · privacy-first 3D collaboration · offline-capable community OS · **Operational Access Management (OAM)**	**Closed-core / partial-OSS** (community runtime open, enterprise spatial + multi-tenant + OAM kernel closed)	**$$$$ — top of stack** (most mature layer; ships two production-grade capabilities — Spatial OS + OAM — in one runtime)
Fractal Engine	Escherbridge	Intelligence · historical archive · analytics · actionable signals on operational telemetry	**Creative Commons with Commercial Use Restrictions** (CC-BY-NC + commercial license tier)	$$$
OASIS	OASIS Web4	Universal interoperability fabric · one API across providers, chains, storage, identity, reputation, avatars	**Closed-core / partial-OSS** (open SDK + connectors, closed federation engine and enterprise gateway)	$$$
NEOS	OmniOne	Governance · councils · charters · contribution accounting · holonic orgs	**Creative Commons with Commercial Use Restrictions** (CC-BY-NC-SA + commercial license tier)	$$

> ***Naming note:*** *The interoperability layer is **OASIS** in all PrimusNeo commercial documents and product copy. The steward organization is **OASIS Web4**. Older internal materials sometimes use "OASISIS" as the layer name — that term is reserved for the external standards body and is not used in PrimusNeo deliverables.*

See `02-licensing-model.md` for the legal mechanics of the CC-NC + dual-license arrangement, and the per-layer files in `layers/` for engagement-level pricing per layer.

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3. The pricing philosophy in one paragraph

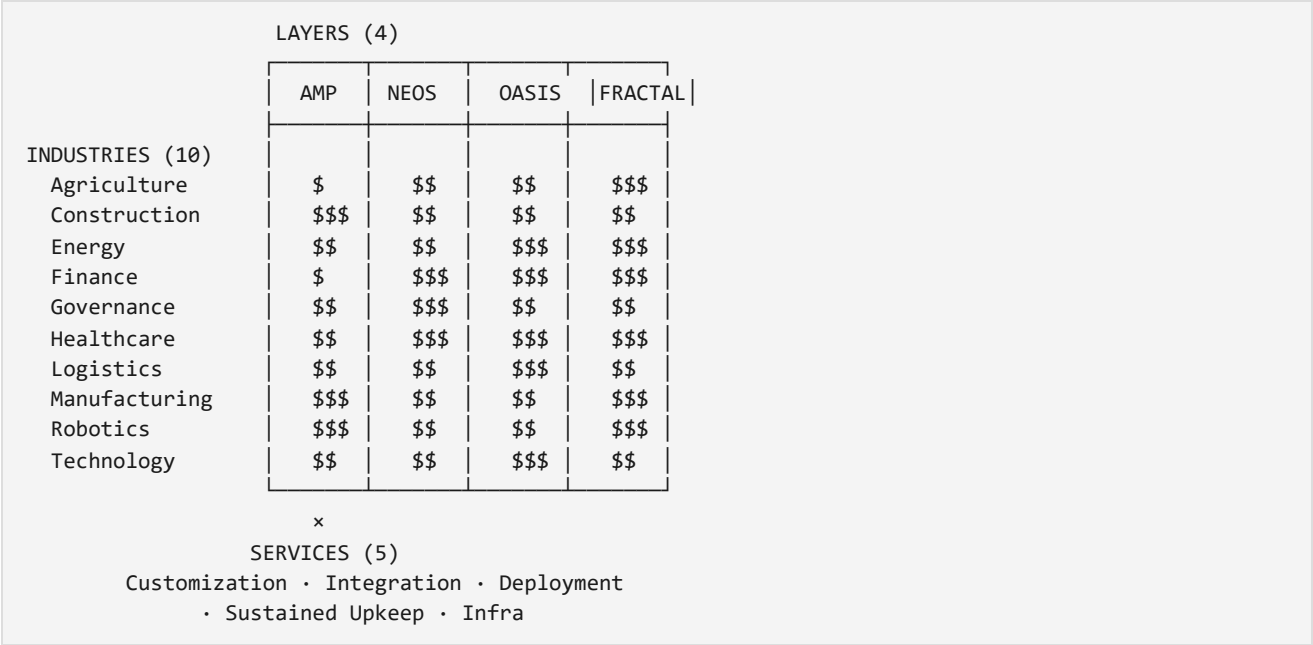
> ***Price against the friction we remove, capped by the value we create, floored by the cost-to-serve.*** *Every quote is built up from three numbers: (a) the buyer's current annualized cost of the coordination friction we displace (McKinsey/UNIDO/sector benchmarks), (b) the defensible value share we can capture (typically 5–25% of friction removed, depending on layer and attribution clarity), and (c) the floor cost of infra, support, and stewardship payouts to the federated stewards. We quote inside that range. We never quote a number we cannot defend with a sourced friction-cost.*

That philosophy is operationalized in `01-pricing-model.md`.

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4. The engagement-model matrix

PrimusNeo engagements are quoted on three axes:



Each industry doc in `industries/` gives the actual dollar ranges for that vertical at Small / Mid / Large scale per layer, plus the recommended bundle pricing.

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5. The five service classes

These are priced **separately from license / subscription** and scoped per engagement. See `03-services-catalog.md` for full SKU detail.

Class	What it covers	Typical pricing mechanic
Customization	Vertical-specific schemas, council templates, twin extensions, custom Fractal models, WASM extensions	T&M day rate or fixed-fee scope
Integration	IT/OT bridges, ERP/MES/EHR/CRM/TMS connectors, identity federation, OASIS connector buildouts	Fixed-fee per connector or T&M
Deployment	Reference architecture rollout, environment provisioning, data migration, cutover, training, change management	Fixed-fee bundle or per-site
Sustained Upkeep	Tier-1/2/3 support, version upgrades, security patching, OASIS connector maintenance, model retraining cadence	Annual % of license (typically 18–25%) or all-in MSA
Infra (pass-through)	Cloud (AWS/Azure/GCP), edge compute, storage tiers, observability, chain gas, optional Tie Town reference colo	Pass-through + 10–18% mgmt fee

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6. Tie Town as the reference architecture

Every engagement is grounded in the **Tie Town [TX]** deployment as living proof-of-capability. Quote language: "*This is the same stack running the Tie Town media and creator house in McDade, Texas — extended for your context. We don't ship slideware; we extend a system that's already in production.*"

This collapses the buyer's pilot-purgatory risk (McKinsey: 70% of Industry 4.0 pilots never scale) into a known, working reference — which is the single biggest enterprise-deal de-risker we have.

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7. How to use this document set

1. **Start with** `01-pricing-model.md` — the value-based pricing formula and the layer/scale anchor grid.
2. **Read** `02-licensing-model.md` to understand the CC-NC + commercial-license dual track for NEOS and Fractal Engine, and the closed-core arrangement for AMP and OASIS.
3. **Read** `03-services-catalog.md` for the service SKUs and day rates.
4. **For anything smaller than a Small-tier engagement** — community deployments, SMBs, or a first paid pilot inside an enterprise — start from `04-community-pilot-pricing.md`, the entry on-ramp that ladders into these anchors.
5. **Pick the relevant industry file in** `industries/` for the vertical you're quoting. Each file gives Small / Mid / Large engagement bundles with $-range, mechanics, and the friction-cost anchor used to defend the number.
6. **Pick the relevant layer files in** `layers/` if the buyer wants single-layer adoption only (common for OASIS-only data-fabric deals and Fractal-only analytics deals).

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial draft
0.2	2026-07-15	Founding team	Added entry-pricing on-ramp (`04`) to the reading path

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This document is internal-only. Public-facing pricing pages derive from this source but never expose the raw anchor numbers.

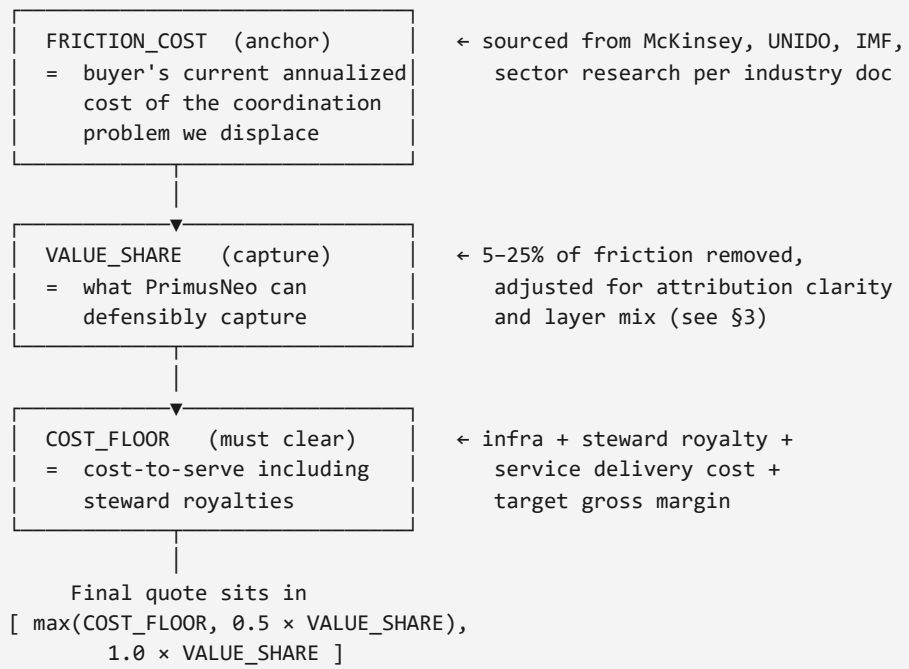
PrimusNeo Value-Based Pricing Model

> ***Status:*** *Draft v0.1* ***Dependencies:*** *`00-overview.md`, `02-licensing-model.md`, `03-services-catalog.md`, `research/layer-comparables.md`*

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1. The pricing formula

Every PrimusNeo quote is built from three numbers:



We never quote below `COST_FLOOR`. We rarely quote above `VALUE_SHARE`. The defensible quote band is the intersection.

Entry engagements — community deployments, paid pilots, anything below the Small anchors in §3 — are priced by `04-community-pilot-pricing.md`: the same formula, pre-clamped into published flat tiers so there is nothing to negotiate at the low end.

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2. Why value-based, not cost-plus

Cost-plus pricing under-prices the federated stack. The four-layer architecture replaces between 4 and 12 point solutions in a typical enterprise stack, with friction-cost displacement that runs 5–50x our delivery cost. Cost-plus would leave 80%+ of the captured value on the buyer's table — which is fine for

commodity infrastructure but the wrong default for coordination infrastructure that takes years to build and has no clean substitute.

Value-based pricing also forces us to do the work of grounding every quote in a sourced friction-cost (per industry doc). That discipline is what makes the quote *defensible* in front of a procurement committee that has read McKinsey too.

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3. Layer pricing anchors

These are the per-layer anchor ranges derived from `research/layer-comparables.md`. They are the **starting position** for any quote and the upper bound on what we charge without a documented friction-cost override.

3.1 AMP — Spatial OS

AMP is the top-priced layer in the PrimusNeo stack because it is the most mature layer (years of PLAN Systems hardening + live Tie Town deployment) AND it delivers two production-grade capabilities in one runtime: **Spatial OS + Operational Access Management (OAM)**. The OAM kernel is what justifies AMP sitting above Fractal Engine on the price ladder; without it, AMP would be a Unity-Industrial competitor priced at $4,950/seat/yr. With it, AMP also displaces Okta/CyberArk/BeyondTrust-class operational IAM/PAM spend.

Scale	Definition	Annual License Anchor	Services Multiplier (Y1)
Small	1–3 sites, <50 concurrent spatial users, <200 OAM identities	**$30K–$75K/yr**	1.5–2.5x
Mid	4–15 sites, 50–500 concurrent, 200–2,500 OAM identities	**$150K–$450K/yr**	1.5–2.5x
Large	15+ sites, multi-org federation, 500+ concurrent, 2,500+ OAM identities	**$600K–$2.5M/yr**	1.5–2.5x

Comparable anchors (combined surface area AMP displaces): NVIDIA Omniverse Enterprise ($4,500/GPU/yr) + Unity Industry ($4,950/seat/yr) + Unreal Enterprise ($1,850/seat/yr) on the spatial side; Okta Workforce Identity ($5–$15/user/mo + add-ons), CyberArk PAM ($50–$150/user/yr), BeyondTrust Privileged Remote Access (similar) on the operational access side.

OAM-only entry SKU available at 70% of the Spatial-OS line-item for buyers (regulated FS, defense-adjacent, NERC CIP-driven utilities) where the access-management value lands before the spatial twin value.

3.2 NEOS — Governance

Scale	Definition	Annual License Anchor	Services Multiplier (Y1)
Small	Single org, 1 council, <50 members	**$6K–$18K/yr**	0.5–1x
Mid	Multi-council, 50–500 members	**$30K–$90K/yr**	0.5–1x
Large	Federated governance, 500+ members, multiple orgs	**$120K–$400K/yr**	0.5–1x

Comparable anchors: Diligent ($12K–$60K+/yr), Microsoft Viva Suite ($144/seat/yr), Loomio ($1.2K–$7.2K/yr), Aragon/Snapshot/Tally (free). **CC-NC dual-license**: free for non-commercial; commercial use at these rates.

3.3 OAS — Interoperability Fabric

Scale	Definition	Annual License Anchor	Services Multiplier (Y1)
Small	<10 integrations, single domain, <1M API calls/mo	**$24K–$60K/yr**	2–3x
Mid	10–50 integrations, multi-chain/cloud, 1M–50M calls/mo	**$80K–$250K/yr**	2–3x
Large	50+ integrations, full federation engine, >50M calls/mo, cross-org	**$350K–$1.5M/yr**	2–3x

Comparable anchors: MuleSoft ($1,250/vCore/month → $80K–$2M+/yr), Boomi ($6.6K–six-figure), Workato (20–30% below MuleSoft). Gas/chain fees pass-through at cost + 12% mgmt fee.

3.4 Fractal Engine — Intelligence

Scale	Definition	Annual License Anchor	Services Multiplier (Y1)
Small	<5 data sources, departmental analytics, <1TB managed	**$30K–$75K/yr**	1–2x
Mid	5–20 data sources, operational telemetry + historical archive, 1–20TB	**$100K–$350K/yr**	1–2x
Large	20+ sources, cross-org intelligence, real-time + archive, >20TB	**$500K–$2M+/yr**	1–2x

Comparable anchors: Palantir Foundry ($250K–$10M+/yr commercial), Snowflake ($25K–$10M+/yr), Databricks (DBU + cloud compute), Splunk ($30K–$60K/month at mid-scale). **CC-NC dual-license**: research/nonprofit free; commercial production at these rates.

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4. Bundle pricing (the deal you actually sign)

Most engagements adopt **2–4 layers** as a bundle. Bundles are quoted as a single PFEL line item (see `02-licensing-model.md §5`) with a small bundle premium and a clear bundle discount that nets favorably to the buyer relative to single-layer sum.

Bundle	Composition	Discount vs single-layer sum	Premium for federation
Duet	Any 2 layers	10% off	+5% for federation engine
Triad	Any 3 layers	18% off	+5%
Full Stack	All 4 layers	25% off	+0% (federation is the value)

Worked example: a Mid-tier full-stack engagement totals roughly:

```
AMP   Mid:  $300K  (midpoint of $150K–$450K — spatial OS + OAM)
NEOS  Mid:   $60K  (midpoint of $30K–$90K)
OASIS   Mid:  $160K  (midpoint of $80K–$250K)
Fract Mid:  $220K  (midpoint of $100K–$350K)
               ─────
Subtotal:   $740K
Bundle:     -25% → $555K
Services Y1 (~2x license blend): $1.11M
Sustained Upkeep Y1+ (22% of license, Silver SLA): $122K
Infra (pass-through + 12% mgmt fee): $60K–$180K
               ─────
Y1 TOTAL:   $1.85M–$2.0M
3-yr TOTAL: $3.4M–$4.0M  (Y2–Y3 mostly license + upkeep, services taper)
```

This lands inside the **Mid-tier 3-yr TCO bands** for nearly every vertical (logistics $900K–$2.7M, manufacturing $1.5M–$8M, healthcare $8M–$25M at the high end, agriculture $1.2M–$6M, energy $2.5M–$10M). Where it lands at the high end of a band (manufacturing, energy, healthcare-mid), the friction-cost anchor justifies the position — especially given AMP's dual Spatial-OS + OAM displacement. Where it lands at the low end (logistics, agriculture), bundle discount + outcome-based component keeps the buyer engaged.

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5. Modulating the quote by vertical

The same Mid-tier quote modulates **±30%** based on three vertical-specific signals:

Modulator	Increase quote	Decrease quote
Regulatory tailwind	Energy (NERC CIP, CSRD, CBAM), Healthcare (HITRUST, EHDS), Finance (MiCA, SAB 122)	Construction (mostly voluntary ESG today)
Procurement maturity	Finance, Energy, Public Governance (formal RFP + Big-4 benchmarking)	Cooperatives, DAOs, early-stage startups
Friction-cost density per $1 of buyer revenue	Manufacturing (70% pilot purgatory, $50B downtime), Healthcare ($50M–$500M EHR integration tax)	Technology buyers (will build it themselves; lower friction relative to their internal eng capacity)

Adjust by ±10% per modulator that fires, capped at ±30% in either direction.

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6. Standard commercial terms

These are the default contract terms. Negotiable but defensible:

Term	Default	Notes
Initial term	3 years	Mirrors enterprise software norm; aligns with sustained upkeep math
Auto-renewal	1-year auto-renew with 90-day notice	Standard
Annual escalator	CPI + 3% (capped at 7%)	Below the ServiceNow 5–10% renewal-uplift norm we'll be benchmarked against
Payment terms	Net 30; quarterly invoicing	Annual prepay = 5% discount
Bundle discount	10/18/25% per §4	Hard floor — no further discount on bundle without VP+ sign-off
Multi-year discount	5% / yr (Y2 = -5%, Y3 = -10%)	Stacks with bundle discount
Steward royalty	Internal — see `02-licensing-model.md` §4	Invisible to buyer; affects our COGS, not the quote
IP indemnity	Capped at 2× annual fees for the affected layer	Industry standard for enterprise software
Liability cap	2× annual fees, excluding gross negligence + IP indemnity	Standard
Data export	Customer data exportable in machine-readable form at any time	Non-negotiable; preserves no-lock-in promise
Source escrow	Available for closed-core (AMP, OAS) at >$500K ARR via Iron Mountain	Pass-through escrow agent fee
Termination for convenience	After Y1, with 90-day notice, no refund of prepaid amounts	Aligned with comparable enterprise software

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7. Discounting authority

Approver	Max additional discount	Notes
Account Executive	5% (against bundle-discounted price)	Standard concession
Sales VP	10%	Plus a strategic-account flag for the deal review
Co-founder / CEO	15%	Steward orgs must be notified — affects their royalty share
Steward Council vote	Above 15%	Required for any deal where bundle discount + multi-year + concession exceeds 35% net off list

No deal closes below COST_FLOOR. A losing deal at COST_FLOOR is better than a winning deal that bleeds money for three years.

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8. Outcome-based / gainshare addenda

For verticals where attribution is feasible (manufacturing OEE, energy VPP, logistics on-time delivery, finance straight-through processing), we can offer an **outcome-based addendum** on top of the base license:

- Base license priced at **80% of standard anchor** (the buyer takes some discount up front).
- Performance fee of **15–30% of documented annualized savings** above an agreed baseline.
- Baseline established by Fractal Engine measurement in a 60-day instrumentation phase before the contract goes hot.
- Performance fee billed quarterly in arrears.
- Cap on performance fee at **2× foregone base license** in any year (so worst case the buyer ends up paying ~140% of standard, only when we've delivered 4–7x that in savings).

This addendum is **opt-in** for the buyer and is the right structure for CFO-led conversations where the buyer is willing to pay more *only* if we deliver. It is **not** offered into highly regulated rate-recovery contexts (energy utility rate cases) because the regulatory cost-allocation math becomes too tangled.

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9. Worked friction-cost anchors per vertical (summary)

The full math per industry lives in `industries/*.md`. Headline anchors:

Vertical	Friction anchor	Source	Typical PrimusNeo capture
Manufacturing	$50B/yr global unplanned downtime; 70% pilot purgatory	McKinsey, Siemens	5–15% of plant-level capture
Healthcare	$50M–$500M EHR integration tax; $200K/yr HITRUST gate	Becker's, HITRUST	8–20% of integration spend
Finance	$200K–$500K/yr/bank attestation premium; MiCA non-compliance €15M/3% revenue	OCC, MiCA	10–25% of attestation premium
Energy	$13.2M/yr industry-wide NERC CIP-015-1 alone; $5M–$50M IOU ADMS	NERC, DOE	5–15% of CIP + governance spend
Logistics	TradeLens-pattern wasted $1B+; $300K+/yr existing visibility spend	Industry analysis	10–20% of consolidated platform spend
Construction	0.1–0.3% of ACV typical PM spend; 1–2% capex on digital programs	McKinsey, Procore	5–15% of program tech budget
Agriculture	60% of profits on FSMA compliance at SMB scale; $84.9M → $36.1M MRV market	USDA, GMInsights	8–18% of compliance + MRV spend
Robotics	$50K–$200K+ ISO 10218 cert per cell; pilots fail 70%	ISO, McKinsey	10–20% of fleet coordination spend
Governance	$400K–$900K FedRAMP Mod; Llama at $2.5M/yr for Aave alone	FedRAMP, public DAO governance	15–30% of governance services spend
Technology	$2M+ 3-yr TCO on self-hosted Backstage at scale	Roadie, Humanitec benchmarks	8–18% of platform-eng spend

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10. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial value-based formula
0.2	2026-07-15	Founding team	Cross-referenced entry-pricing on-ramp (`04`)

PrimusNeo Licensing Model

> ***Status:*** *Draft v0.1 — needs legal review before any external commitment.* ***Scope:*** *Defines the legal and commercial licensing posture for all four layers and the federated stack as a whole.*

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1. The two-track licensing structure

PrimusNeo runs a **two-track licensing model** that mirrors successful open-core precedents (Elastic SSPL, MongoDB SSPL, HashiCorp BSL, Grafana AGPL, Sentry FSL) while honoring the federated steward arrangement.

Track	Layers	Default license	Commercial trigger
Open / CC-NC track	NEOS, Fractal Engine	Creative Commons BY-NC-SA 4.0 (or CC-BY-NC 4.0 where derivative concerns apply)	Any *commercial use*, redistribution, or hosted-as-a-service offering
Closed-core track	AMP, OASIS	Closed-source enterprise license; community runtime / SDK selectively open-sourced under Apache-2.0 or MIT	Any production deployment beyond community / non-commercial pilot scope

The whole stack composes into commercial PrimusNeo **Engagement Bundles** (see `01-pricing-model.md`) — and the bundle license is what enterprise buyers actually sign.

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2. Track A — Creative Commons with Commercial Restrictions

2.1 Layers under this track

- **NEOS** (steward: OmniOne / Green Earth Vision)
- **Fractal Engine** (steward: Escherbridge)

2.2 The license

Primary: Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International (**CC-BY-NC-SA 4.0**) for NEOS source, schemas, and reference implementations. **CC-BY-NC 4.0** for Fractal Engine (no ShareAlike on derivative models because training data may carry separate licenses).

What this permits without payment:

- Personal, academic, and research use.
- Non-profit community deployments (cooperatives, neighborhoods, civic associations) where the deploying entity has *no revenue from the deployment*.

- DAOs and protocols below a treasury threshold (see §2.4).
- Forks, modifications, and republication — provided attribution and the same CC-BY-NC-(SA) license terms are preserved.

What this prohibits without a commercial license:

- Hosting the software as a paid service (SaaS, managed service, RaaS).
- Embedding in any commercial product or platform.
- Use by for-profit entities above the de-minimis thresholds (§2.4).
- Use by government or public-sector entities that procure it through a paid procurement vehicle (those buy under the commercial license).

2.3 The commercial-license tier

Any of the above prohibited uses are unlocked by a **PrimusNeo Commercial License** (PCL) for that layer. The PCL:

- Removes the NC restriction for the specified scope (entity, sites, user count, treasury size).
- Adds standard commercial protections: warranty disclaimers fit for enterprise procurement, IP indemnity capped per `01-pricing-model.md` §6, data-processing addendum.
- Routes a **royalty share** to the layer steward (OmniOne for NEOS, Escherbridge for Fractal Engine) per the **Steward Royalty Schedule** in §4 below.

2.4 De-minimis thresholds (no PCL required)

A commercial entity may use the CC-NC track without a PCL only if **all** of the following hold:

Threshold	Limit
Annual gross revenue	≤ $1M USD
Headcount	≤ 10 FTE
Connected users / members	≤ 250
Treasury (for DAO / protocol)	≤ $5M USD equivalent
No resale or service-bureau hosting	required

Above any threshold, a PCL is required. This pattern mirrors the Sentry FSL free-tier-for-small-teams approach and avoids weaponizing the license against the legitimate cooperative / startup ecosystem the layers are intended to serve.

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3. Track B — Closed-core / Partial-OSS

3.1 Layers under this track

- **AMP** (steward: PLAN Systems)

- **OASIS** (steward: OASIS Web4)

3.2 What's open

- **AMP**: PLAN core runtime (Hexon file standard v1.0.0), client SDKs, WASM extension interface, reference 3D viewer scaffold — typically **Apache-2.0** to maximize adoption in spatial-computing toolchains.
- **OASIS**: Connector SDK, the public REST/GraphQL API specification, community-maintained connector implementations — typically **MIT** or **Apache-2.0** depending on dependency stack.

3.3 What's closed

- **AMP**: Enterprise spatial-collaboration server, multi-tenant federation, privacy-preserving mesh routing, premium PLAN 3D interface components.
- **OASIS**: The federation engine itself, the enterprise gateway, the identity/reputation orchestration kernel, the closed-source provider connectors that carry commercial cost (e.g., proprietary chain or storage provider relationships).

3.4 The commercial license

Closed-core components ship under a **PrimusNeo Enterprise License (PEL)**, which is a traditional commercial license:

- Per-named-user, per-site, per-API-call, or enterprise commit (mechanic varies by layer — see layer files in `layers/`).
- Source available under NDA to enterprise tier customers above a threshold (typically $250K ARR).
- IP indemnity and warranty per `01-pricing-model.md` §6.

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4. Steward royalty schedule

To preserve the federated four-org arrangement and avoid the single-entity-captures-the-stack failure mode that has killed every prior attempt at this kind of stack, every PrimusNeo commercial engagement pays a **Steward Royalty** into a transparent treasury distributed quarterly.

Steward	Layer	Default royalty share of net license revenue
OmniOne / Green Earth Vision	NEOS	**18%** of NEOS line-item license rev
Escherbridge	Fractal Engine	**18%** of Fractal Engine line-item license rev
PLAN Systems	AMP	**15%** of AMP line-item license rev
OASIS Web4	OASIS	**15%** of OASIS line-item license rev
PrimusNeo (integrator)	Stack-wide	Residual (~34%) plus 100% of services margin

Services revenue (customization, integration, deployment, upkeep) does **not** carry a steward royalty unless the work is performed by the steward themselves on subcontract, in which case it is invoiced through normally.

Infra pass-through and mgmt fee carry no royalty.

Quarterly reporting: PrimusNeo publishes (under NDA to stewards) a deal ledger reconciling royalties and pays out within 45 days of quarter close.

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5. Bundle licensing

Most enterprise engagements adopt **two or more layers** as a bundle. The bundle ships under a single **PrimusNeo Federated Engagement License (PFEL)** that:

- References the underlying PCL / PEL for each layer (so the steward royalties flow automatically).
- Gives the buyer a single signature surface and a single counter-party (PrimusNeo) for procurement, support, and indemnity.
- Specifies the **deployment scope** (entity, named subsidiaries, geographies, named user count, named site count) as a schedule that can be expanded under a Change Order without re-papering the entire contract.

The PFEL is the document procurement actually negotiates against. The underlying single-layer licenses are referenced exhibits.

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6. Forking and derivatives

This is the part that determines whether the open-source ecosystem survives commercial scale-up, so it is spelled out explicitly:

- **NEOS and Fractal Engine forks** are explicitly permitted and encouraged. They must retain the CC-BY-NC-(SA) license, attribute the original stewards, and may not represent themselves as "PrimusNeo NEOS / Fractal" (trademark restriction, not license restriction).
- **Commercial use of a fork** requires a PCL with the steward of the original layer, not the forker. (This is the Elastic SSPL pattern: forks exist; commercial hosted forks pay.)
- **Derivative Fractal Engine models** trained on a customer's data remain the customer's IP. Models trained on Escherbridge's public corpus remain Escherbridge IP and are licensed back to the customer under the PCL scope.
- **AMP / OASIS forks** are constrained by the closed-core boundary. The open SDKs are forkable under Apache-2.0; the closed components are not. This is the standard open-core pattern (cf. GitLab CE/EE, Sentry, HashiCorp).

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7. Trademark

"PrimusNeo," "AMP," "NEOS," "OASIS," "Fractal Engine," and "Tie Town" are trademarks of their respective stewards. The license grant does **not** include trademark rights. Compliant use of marks in compatibility statements ("works with PrimusNeo NEOS") is permitted under standard nominative-use fair-use principles; product naming and badging is not.

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8. Open questions for legal

1. CC-BY-NC-SA vs. a custom **Functional Source License (FSL)** with a 2-year-delayed Apache-2.0 conversion (Sentry's playbook). FSL gives a stronger commercial moat at the cost of slightly more legal friction. **Recommendation:** stay on CC-BY-NC for v0.1 (it's a known quantity to the cooperative/research audience these layers serve); revisit FSL at v1.0 if commercial poaching becomes a real problem.
2. Whether to define "commercial use" by **revenue** (clean, easily verified) or by **purpose** (closer to the CC NC spirit, but ambiguous). **Recommendation:** revenue + headcount + connected-users, per §2.4.
3. Whether enterprise customers can request **escrow** of closed-core AMP/OASIS source for business-continuity purposes. **Recommendation:** yes, above $500K ARR threshold, through a standard third-party escrow agent (e.g., Iron Mountain).
4. Whether DAO treasuries denominated in volatile assets are valued at acquisition cost or marked-to-market for the §2.4 threshold. **Recommendation:** trailing-90-day VWAP; documented in the PCL itself.

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9. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial dual-track structure

PrimusNeo Services Catalog

> ***Status:*** *Draft v0.1.* ***Scope:*** *Defines the five service classes, SKUs, day rates, and bundling rules that wrap PrimusNeo license / subscription revenue.*

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1. Why services matter (and how they relate to license)

Across our four layer comparables (Palantir, Cognite, Aveva, AspenTech, Innovaccer, Procore, MuleSoft, Snowflake), the services-to-license ratio in year one ranges from **0.5x to 5x**. PrimusNeo's structural target is **1.5x – 2.5x in year one**, taping to **0.3x – 0.6x by year three** as the customer's internal team takes over. Anything outside that band is a signal to revisit the engagement design.

```
Year 1:  License $1.00  ::  Services $1.50–$2.50
Year 2:  License $1.00  ::  Services $0.80–$1.20
Year 3+: License $1.00  ::  Services $0.30–$0.60   ← steady-state upkeep only
```

Services are sold under the same **PrimusNeo Federated Engagement License (PFEL)** as the bundle, with a Statement of Work (SOW) per engagement.

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2. The five service classes

2.1 Customization

What it covers: Vertical-specific schemas, council templates, twin extensions, custom Fractal Engine models, NEOS charters, AMP scene authoring, OASIS connector buildouts, WASM extension development.

Pricing mechanic: T&M day rate OR fixed-fee scope after a paid discovery sprint.

Day rates (standard):

Role	Day rate (USD)	Notes
Principal Architect	$3,200	Founding team, capped allocation per engagement
Senior Engineer (any layer)	$2,200	Federated steward orgs share this rate
Engineer	$1,600	
Fractal Engine ML Engineer	$2,800	Specialty: Escherbridge talent
NEOS Governance Designer	$2,400	Specialty: OmniOne ETHOS methodology
AMP Spatial / 3D Engineer	$2,400	Specialty: PLAN systems
OASIS Integration Engineer	$2,200	Specialty: OASIS Web4 connector authors
Project Manager	$1,800	
Trainer / Change Management	$1,400	

Day rates are floors. Premium up to +50% for: sub-30-day rush, hostile deadlines, on-site travel >50%, regulated-industry compliance scope (FedRAMP High, HIPAA, NERC CIP, etc.), and weekend cutover windows.

Discovery sprint: 2-week paid engagement, $40–80K depending on scope. Output: a fixed-fee SOW the buyer can sign. Discovery fee credits 100% against the resulting engagement if signed within 90 days.

2.2 Integration

What it covers: Net-new connectors and bridges between PrimusNeo and the buyer's existing stack. The most common categories:

Connector class	Typical examples	Default fixed fee (USD)
ERP	SAP S/4HANA, Oracle Fusion, NetSuite, Microsoft D365	$60K – $180K
MES / OT	Rockwell FactoryTalk, Siemens MindSphere, Aveva, GE Proficy, OPC UA	$80K – $250K
EHR / Clinical	Epic, Cerner/Oracle Health, Meditech, Athenahealth	$120K – $400K (HITRUST overhead)
TMS / WMS	Manhattan, Blue Yonder, Oracle OTM, SAP TM	$60K – $200K
CRM	Salesforce, HubSpot, MS Dynamics CE	$30K – $80K
Identity / IdP	Okta, Azure AD, Auth0, Ping, SSI Verified Credentials	$25K – $90K
Chain / RPC	EVM, Solana, Cosmos, Hedera, Polkadot, custom L2	$20K – $75K per chain
Storage	S3-compatible, IPFS, Filecoin, Ceramic, Arweave	$15K – $50K per backend
BIM / CAD	Autodesk ACC, Bentley iTwin, Trimble Connect	$80K – $250K
Custom REST / GraphQL	bespoke buyer system	$20K – $120K

Mechanic: Fixed-fee per connector with a 90-day warranty period. Discount tiers for connector bundles (3 connectors → 10% off, 5+ → 15% off).

Connector reuse: PrimusNeo retains the right to harden, generalize, and redistribute any non-bespoke connector through the OASIS connector library. Bespoke customer logic is held back per the SOW.

2.3 Deployment

What it covers: Reference-architecture rollout, environment provisioning, data migration, cutover, training, change management.

SKUs:

SKU	Scope	Fixed fee (USD)	Duration
Atom	Single-site, single-layer pilot, ≤25 users, dev + prod environments	$35K – $90K	4–8 weeks
Molecule	Single-site, full-stack (all 4 layers), ≤100 users	$120K – $320K	8–14 weeks
Hex	Multi-site (2–5 sites), full-stack, ≤500 users, federated identity	$400K – $900K	14–24 weeks
Lattice	Enterprise (6+ sites or 500+ users), full-stack, global federation, change-management program	$1.0M – $3.5M	24–48 weeks
Tessellation	Industry-program scale (national rollout, regulated sector, multi-tenant inside one enterprise)	$3.5M – $12M+	9–18 months

Each SKU includes:

- Environment provisioning (dev / staging / prod / DR).
- Reference data migration (up to a stated volume).
- Cutover plan and rehearsal.
- Operator training (standard hours per scope).
- 30-day hypercare post-cutover.

Excludes (priced separately): integration SOW, custom development, infra pass-through, ongoing upkeep.

2.4 Sustained Upkeep

What it covers: Tier-1/2/3 support, version upgrades, security patching, OASIS connector maintenance, Fractal model retraining cadence, NEOS charter evolution, AMP scene/world upgrades, on-call.

Pricing mechanic: Annual % of license value, **18–25%** depending on SLA tier. This is the industry default for enterprise software (Oracle 22%, SAP 22%, Microsoft 20%, AspenTech ~20%, Snowflake bundled).

SLA tiers:

Tier	Coverage	Annual % of license	Notes
Bronze	Business hours (your tz), 1 business-day P1 response	18%	Most CC-NC entities
Silver	24×5, 4-hour P1 response, named TAM, quarterly review	22%	Default for commercial mid-market
Gold	24×7, 1-hour P1 response, named TAM + on-call rotation, monthly review	25%	Default for regulated / enterprise
Platinum	24×7, 15-min P1, dedicated support pod, weekly review, custom escrow	Quoted	Bank, hospital system, ISO, federal

Inclusions across all tiers:

- Security patches and CVE response per SLA.
- Minor and patch version upgrades.
- Up to two major version upgrades over a 3-year MSA.
- Up to N hours/quarter of Tier-3 engineering time per the tier (4 / 12 / 24 / 48 respectively).

Outside inclusions (T&M at standard rates): schema migrations, governance redesign, major customization changes, integration to new external systems.

2.5 Infra (pass-through + management fee)

What it covers: Cloud (AWS / Azure / GCP), edge compute, storage tiers, observability, chain gas, optional Tie Town reference colocation.

Pricing mechanic: Pass-through at cost + 10–18% management fee. Lower fee for transparent direct-pay arrangements where the customer owns the cloud account; higher fee for fully-managed PrimusNeo-owned-and-billed.

Standard infra SKUs (Tie Town reference):

SKU	Description	Typical monthly cost (USD)
Edge node (Tie Town)	1U colocated, AMP + OASIS edge cache, 99.95% target	$1,200 – $2,500
Cloud reference (small)	t3-equivalent, 4 vCPU, 16GB, 500GB SSD, 2TB egress	$800 – $1,500
Cloud reference (mid)	16 vCPU, 64GB, 2TB SSD, 10TB egress, multi-AZ	$4,500 – $9,000
Cloud reference (large)	64+ vCPU, 256GB+, 10TB+ SSD, dedicated K8s nodes, multi-region	$18K – $45K
Chain gas reserve	Estimated monthly burn for governance txs + signature anchoring	Variable; quote per chain
Storage tier (Fractal archive)	Cold + warm Fractal historical archive (S3 + Glacier or equivalent)	$0.012 – $0.025 per GB-month

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3. Bundling rules

- **License + Deployment + Upkeep is the minimum viable bundle.** Selling license without Deployment is allowed but quoted with a "no-cutover warranty" clause that explicitly disclaims rollout risk.
- **Customization is sold only after Discovery.** Discovery is the only T&M sale we will do without a fixed-fee follow-on signed within 90 days, and even then the fee credits back.
- **Integration is bundled at a discount** when 3+ connectors are signed in the same SOW (see §2.2 discount tiers).
- **Infra is never bundled into license** for accounting cleanliness and customer trust. Always a separate line item.

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4. Margin targets

Service class	Target gross margin
Customization	55–65% (founders 70%+)
Integration (fixed-fee)	50–60%
Deployment (fixed-fee)	45–55%
Sustained Upkeep	75–85% (the gold mine)
Infra mgmt fee	85%+ (the fee is mostly margin; cost is mostly pass-through)

License gross margin runs 80–92% per layer per the comparables in `research/layer-comparables.md`.

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5. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial catalog

PrimusNeo Entry Pricing — Community Tiers & Paid Pilots

> ***Status:*** *Draft v0.1* ***Dependencies:*** *`00-overview.md`, `01-pricing-model.md`, `02-licensing-model.md`, `03-services-catalog.md`*

---

1. The gap this closes

The core model (`01-pricing-model.md`) prices Small/Mid/Large enterprise engagements. The cheapest commercial path through it today is roughly **$75K–$170K in year one** (Small-tier license + Discovery sprint + Atom deployment). That is correct for enterprise — and a wall for everyone else:

- **Community organizations** (cooperatives, civic associations, DAOs) that the CC-NC track exists to serve, who need hosting and guidance, not a PFEL.
- **SMBs above the de-minimis thresholds** (`02` §2.4) but nowhere near a six-figure Y1.
- **Enterprise champions** who need a sub-procurement-threshold paid pilot to manufacture the internal evidence before a committee will convene.

This document defines the **on-ramp**: published, flat, value-anchored entry tiers that ladder into the standard anchors — and a paid-pilot motion designed to kill pilot purgatory rather than feed it.

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2. The on-ramp at a glance

Rung	Name	Who it's for	Price	Duration	Converts to
0	**Community Edition** (self-host)	Anyone under de-minimis (02 §2.4)	**$0**	—	Community Cloud
1	**Community Cloud** (hosted)	Co-ops, civic orgs, DAOs under de-minimis	**$149–$399/mo**	Monthly, cancel anytime	Small-tier PCL on graduation
2	**Community Pilot**	Community orgs wanting a guided launch	**$4K–$9K** fixed	6 weeks	Community Cloud (100% credit)
3	**Proof Pilot**	Commercial SMB / enterprise champion	**$18K–$45K** fixed	6–10 weeks	Small-tier engagement (100% credit)
4	**Production Pilot**	Regulated / complex verticals	**$60K–$120K** fixed	10–14 weeks	Small/Mid engagement (100% credit)

Above Rung 4, the standard model applies: Discovery sprint → PFEL bundle per `01` §4 and `03`.

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3. The six rules

1. **Every pilot is paid.** Payment is the commitment signal; unpaid POCs are how the McKinsey 70% die. The free experience is Community Edition self-hosting — never a free pilot.
2. **Entry prices are published and flat.** No negotiation, no discount ladder (`01` §7 does not apply below Small tier). Flat list price is what makes the entry cheap to sell and easy to buy.
3. **Still value-based.** The flat tiers are pre-clamped outputs of the same formula: a pilot fee targets **1–2% of the vertical's annualized friction-cost anchor** at the buyer's scale, clamped to the rung band. Community Cloud is priced at cost floor + thin margin on pooled infra.
4. **The pilot manufactures the evidence.** Every commercial pilot includes Fractal Engine baseline instrumentation. The pilot's output — the **Value Evidence Report** — becomes the `FRICTION_COST` input to the full quote. The pilot doesn't precede the value-based sale; it *is* the measurement phase of it.
5. **Credit-forward, services-first.** 100% of the pilot fee credits against the Y1 engagement if signed within 90 days of pilot end. Credit applies to the Y1 **services** line first, then license — protecting the steward royalty base (`02` §4).
6. **COST_FLOOR still binds.** Thin entry margins come from the standardized **Pilot Kit** (the Tie Town reference architecture in a box: pre-built vertical templates, library connectors only, pooled reference infra) — never from waiving the floor.

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4. Rung detail

4.0 Community Edition (self-host) — $0

What already exists under the licensing model, named as a product tier: NEOS + Fractal Engine community builds under CC-NC (within the de-minimis thresholds of `02` §2.4), plus the open AMP community runtime and OAS SDK/connectors (Apache-2.0/MIT). Community forum support only. This is the top of the funnel and the health of the ecosystem — it is never crippled to force upgrades.

4.1 Community Cloud — $149–$399/mo

Hosted NEOS + Fractal community stack on **pooled multi-tenant reference infrastructure** (which is what lets the price clear COST_FLOOR — a dedicated Cloud-reference-small runs $800–$1,500/mo per `03` §2.5).

Plan	Members	Price	Includes
Seed	≤ 50	**$149/mo**	1 council, community dashboard, 30-day Fractal retention
Grove	51–250	**$349/mo**	Multi-council, 12-month retention, AMP community space
+Add-on	any	**+$50/mo**	Custom domain + data-export automation

- Eligibility = the de-minimis thresholds. Crossing them triggers graduation (§7), not a cutoff.

- Support: community forum + best-effort email (below Bronze; no SLA).
- Consideration: opt-in public reference listing earns **2 free months/yr**.
- Annual prepay = 2 months free (matches the 5%+ prepay norm, simpler math).

4.2 Community Pilot — $4K–$9K fixed, 6 weeks

A guided community launch for co-ops, civic orgs, and DAOs that want hands-on help: NEOS charter workshop (ETHOS-lite), one council live with real members, Fractal community dashboard, three facilitated sessions, launch checklist.

- Delivered from the Pilot Kit in 3–5 trainer/governance-designer days.
- **100% of the fee credits** to 12 months of Community Cloud or to any commercial engagement signed within 90 days.
- Capped at 4 per quarter delivered internally until Community Partners (§7) are certified to deliver at wholesale.

4.3 Proof Pilot — $18K–$45K fixed, 6–10 weeks

The commercial entry rung. Single site, **1–2 layers**, ≤25 users, at most one **library** connector (no bespoke integration), pooled reference infra or the buyer's cloud.

Includes, always:

1. **Success-criteria contract** — 3–5 metrics fixed in week 1, signed by a named executive sponsor.
2. **Baseline instrumentation** — Fractal Engine measures the friction being displaced from day one. This satisfies the 60-day instrumentation phase required for the outcome-based addendum (`01` §8), so a converted engagement is gainshare-ready immediately.
3. **Value Evidence Report** — the measured friction cost, sourced and presented in the buyer's numbers, delivered whether or not they convert.
4. **Pre-priced conversion quote** — built with the `01` §1 formula from the measured baseline, delivered by ~week 6 so the buyer knows the full price *before* the pilot ends.

Fee heuristic: `clamp(1.5% × measured-or-estimated annual friction, $18K, $45K)`. The band deliberately sits under single-budget-holder signature authority in most mid-market orgs.

4.4 Production Pilot — $60K–$120K fixed, 10–14 weeks

For verticals where compliance scope makes a Proof Pilot impossible (healthcare, energy, finance, public governance). Everything in the Proof Pilot, plus: one full catalog connector (`03` §2.2), a compliance documentation pack (HIPAA/HITRUST, NERC CIP, SOC 2 evidence as applicable), Gold-grade support during the pilot term, and limited-production scope under the Pilot License Rider (§5).

Positioning: a Production Pilot is the **Discovery sprint + Atom deployment fused** into one fixed-fee, value-instrumented package ($60–$120K vs. $75–$170K bought separately) — with the credit-forward mechanic on top.

Per-vertical entry defaults

Vertical	Default entry rung	Typical pilot fee
Manufacturing	Proof Pilot	$25K–$45K
Logistics	Proof Pilot	$25K–$45K
Robotics	Proof Pilot	$30K–$45K
Construction	Proof Pilot	$25K–$40K
Agriculture	Proof Pilot (co-ops: Community Pilot first)	$18K–$28K
Technology	Proof Pilot	$18K–$35K
Healthcare	Production Pilot	$90K–$120K
Energy	Production Pilot	$80K–$120K
Finance	Production Pilot	$70K–$110K
Governance	Public sector: Production $60K–$100K · DAO: Community/Proof	$6K–$30K (DAO)

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5. Pilot mechanics

SOW skeleton (Proof Pilot, 8 weeks):

Week	Milestone
0	Kickoff; success criteria signed; **conversion decision meeting calendar-blocked for week 9**
1–2	Pilot Kit deployment; baseline instrumentation live
3–5	Layer(s) in daily use by pilot cohort; weekly metric readout
6	Value Evidence Report (interim) + pre-priced conversion quote delivered
7–8	Hardening, final measurement window
9	Conversion decision meeting: convert, stop, or extend

Anti-purgatory guardrails:

- No pilot without a **named executive sponsor** and a stated conversion budget range. No sponsor, no pilot.
- **One extension maximum**: +4 weeks at 25% of the pilot fee. There is no second extension — the pilot converts or ends. Data exports in full either way (the no-lock-in promise applies at every rung).
- Payment: 50% at signature, 50% at week-4 milestone. Card/ACH accepted at Rungs 1–3; no net-30 negotiation below Rung 4.

Pilot License Rider (PLR): pilots ship under a time-boxed rider — named scope, pilot term + 30 days, auto-expiring, non-renewable (Production Pilots get limited-production scope). For CC-NC layers this simply papers the hosted service; for AMP/OAS it grants the closed-core components within the "non-

commercial pilot scope" already contemplated by `02` §1. Conversion replaces the PLR with the standard PCL/PEL/PFEL.

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6. Conversion math — worked example

Mid-size precision-machining plant (~$60M revenue, 2 lines). Proof Pilot on **Fractal Engine + OAS** (MES library connector), fee `clamp(1.5% × est. $1.8M friction, 18K, 45K)` = **$27K**, 8 weeks.

Pilot baseline measures **$1.8M/yr** of displaceable friction (unplanned downtime $1.1M, scrap/rework traceability $0.4M, coordination rework $0.3M). Conversion quote per the `01` §1 formula:

```
FRICTION_COST:  $1.8M/yr   (measured, not estimated — pilot output)
VALUE_SHARE:    10% → $180K/yr    (clean attribution via pilot baseline)
Quote band:     [max(COST_FLOOR, $90K), $180K]

License:   Fractal Small $75K + OAS Small $48K = $123K − 10% Duet = ~$111K  ✓ in band
Services:  deployment hardening + 2nd connector $110K − $27K pilot credit = $83K
Upkeep:    Silver 22% × $111K ≈ $24K
Infra:     buyer cloud, pass-through + 12% ≈ $12K
                                  ─────
Y1 TOTAL:  ~$230K   (vs. ~$75K–$170K just to *start* without the on-ramp,
                     and vs. $1.8M/yr measured friction)
```

Even at a conservative 40% friction removal ($720K/yr), the buyer nets ~$490K in year one — a quote the sponsor defends with *their own plant's numbers*, because the pilot produced them.

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7. The community flywheel

The community tiers are not a loss leader; they are the **reference network**. Target: 2–3 community lighthouses per vertical, each playing the Tie Town role for its niche.

- **Graduation, not eviction.** A community entity crossing the de-minimis thresholds gets its first PCL year at **50% of the Small anchor** (one-time), then standard rates. Growth is celebrated, never punished — this is what keeps the CC-NC ecosystem loyal at commercial scale.
- **Contribution credits.** Accepted contributions (NEOS council templates, OAS community connectors, Fractal schemas) earn **$500–$2,500 in Community Cloud credit** per acceptance, subject to steward review. Contributors fund their own hosting by making the stack better.
- **Community Partners.** Certified co-op developers and civic-tech shops deliver Community Pilots at **70% of list** (wholesale), scaling the motion beyond internal capacity and seeding a delivery ecosystem.
- **Reference rights.** Opt-in only, rewarded (§4.1) — never coerced.

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8. Positioning against Discovery and Atom

Situation	Right motion
Mid/Large enterprise, formal procurement, budget exists	**Discovery sprint** (`03` §2.1) — do not sell a Proof Pilot; the band under-prices it
Regulated vertical, compliance gates entry	**Production Pilot** (Discovery + Atom fused)
SMB or enterprise champion pre-committee	**Proof Pilot**
Community org under de-minimis	**Community Pilot / Community Cloud**

The rungs are qualification tiers, not a menu — selling down-rung to a qualified enterprise buyer destroys value; selling up-rung to an unqualified one destroys the funnel.

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9. Internal economics & guardrails

Rung	Target gross margin	Note
Community Cloud	≥ 20% over pooled infra	Self-sustaining, not profit center
Community Pilot	≥ 25%	Ecosystem seeding; capacity-capped
Proof Pilot	≥ 40%	Pilot Kit standardization is the margin
Production Pilot	≥ 45%	Aligns with Deployment margins (`03` §4)

- **Royalties:** pilot fees are services revenue — no steward royalty per `02` §4. Community Cloud's license component carries the standard schedule on its NEOS/Fractal split. The services-first credit rule (§3, rule 5) exists to protect the royalty base on conversion.
- **Capacity:** per delivery pod per quarter — 6 Proof + 2 Production pilots; 4 Community Pilots internally (unlimited via Community Partners).
- **Funnel health:** Proof Pilot → paid conversion target **≥ 40%** within 90 days. Below 25% for two consecutive quarters means the Kit or the qualification gate is broken — fix before scaling pilot volume.
- **No discounting:** entry prices are list. The `01` §7 discount authority does not extend below Small tier; a request for a pilot discount is a qualification failure, not a negotiation.

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10. Revisions

Version	Date	Author	Change
0.1	2026-07-15	Founding team	Initial entry-pricing on-ramp: community tiers + paid-pilot ladder

AMP — Spatial OS + Operational Access Management Engagement Model

> ***Layer:*** *01 — Operational Runtime / Spatial OS / Operational Access Management* ***Steward:*** *PLAN Systems* ***License:*** *Closed-core / partial-OSS (community runtime open Apache-2.0; enterprise spatial + multi-tenant federation + OAM kernel closed)* ***Maturity:*** *The most mature layer in the stack — multi-year hardening through PLAN Systems plus the live Tie Town [TX] deployment* ***Reference deployment:*** *Tie Town [TX]* ***Anchors source:*** `../research/layer-comparables.md`

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1. What AMP actually is (in the buyer's vocabulary)

AMP is the **operational ground layer** — and it does **two jobs in one runtime**, which is why it is the highest-priced layer in the PrimusNeo stack:

1. **Spatial OS.** The privacy-first spatial collaboration runtime that ties together physical sites, devices, people, and the contextualized data layer that makes everything above it (NEOS governance decisions, OAS interop events, Fractal Engine signals) attach to a *place* and a *moment*.
2. **Operational Access Management (OAM).** The runtime that decides — for every action, every device, every spatial zone, every shift — *who is allowed to do what, here, now*. AMP issues, scopes, revokes, and audits operational access against the same spatial scene the work actually happens in. It is the layer that replaces the patchwork of physical badges + OT VPNs + plant-floor permission spreadsheets + drone-clearance forms + facility access lists most operations run today.

This dual role — spatial OS *and* OAM in one runtime — is what makes AMP the load-bearing layer in any operational engagement, and it is why AMP commands the highest license and services anchors in the stack.

In the language enterprises already know:

- A **plant floor** to a manufacturing CIO: AMP is the IT/OT-bridging spatial model that combines CAD, PLC telemetry, MES events, and worker location into one navigable scene — *and* the lockout/tagout + MOC + shift-handoff access system that decides who can touch which line at which moment.
- A **hospital** to a healthcare CMIO: AMP is the OR/ED command center spatial layer (a GE Command Center analog without forcing Epic rip-and-replace) *and* the role-and-zone-based access system gating who scrubs into which suite, who reads which chart from which workstation, and who can override which alarm.
- A **port** to a CSCO: AMP is the terminal-and-yard spatial twin (Navis N4-adjacent but vendor-neutral) *and* the operational access system that decides which driver, which crane operator, which inspector, which carrier rep is authorized in which apron zone for which time window.

- An **energy utility** to a CGO: AMP is the substation + plant spatial twin *and* the NERC CIP-013-equivalent operational access kernel that gates physical and logical access to BES Cyber Assets with cryptographic audit trail.
- A **DAO** or **cooperative** to a community steward: AMP is the offline-capable shared space where members coordinate *and* the contribution-based access system that determines who can sign for the treasury, who can deploy code, who can propose a charter change.

Crucially, AMP runs **offline-first** in both roles. The OAM kernel issues and validates access locally even when disconnected; access decisions reconcile to the federated audit log when the link comes back. This is what distinguishes AMP from every Web-native spatial collaboration tool (Spatial.io, Microsoft Mesh, Meta Workplace) *and* from every cloud-only IAM/PAM product (Okta, CyberArk, BeyondTrust), and it is the requirement set for regulated, low-connectivity, or sovereignty-conscious environments.

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2. License posture

Closed-core / partial-OSS — the runtime, file standard, and client SDKs are open; the enterprise spatial server, multi-tenant federation layer, **and the OAM kernel** are closed.

Component	License	Notes
PLAN core runtime	Apache-2.0	Community-maintained; forkable
Hexon (v1.0.0) file standard	Apache-2.0	The portable spatial file format
Client SDKs (web, mobile, desktop)	Apache-2.0	Free for any use
WASM extension interface	Apache-2.0	Free; extensions can be commercial or OSS at extension author's choice
Reference 3D viewer scaffold	Apache-2.0	Free baseline UI
Community OAM policy DSL	Apache-2.0	The open policy language; closed runtime evaluates it
Enterprise spatial server	**Closed (PEL)**	Multi-tenant; production SLA; required for >5 sites or >50 concurrent
Federation engine	**Closed (PEL)**	Cross-org spatial federation; required for multi-entity deployments
OAM kernel	**Closed (PEL)**	The operational access management kernel — issuance, scoping, revocation, cryptographic audit trail, offline policy evaluation. The second pillar of AMP's value
Premium PLAN 3D interface components	**Closed (PEL)**	Industrial-grade UI; CAD-class viewport; OEM-licensed assets
Privacy-preserving mesh routing	**Closed (PEL)**	The privacy-first networking that makes "offline-capable spatial collaboration" actually work without leaking position data

See `../02-licensing-model.md` §3 for the closed-core / PEL mechanics.

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3. Pricing tiers (annual license, before services)

AMP carries the **highest license anchors in the stack** because (a) it is the most mature layer (years of PLAN Systems development plus a live Tie Town deployment), and (b) it is the only layer that delivers two enterprise-grade capabilities — Spatial OS *and* Operational Access Management — in one runtime. From the layer anchors in `../01-pricing-model.md` §3.1:

Scale	Definition	Annual License	Services Multiplier (Y1)	Typical 3-yr TCO
Small	1–3 sites, <50 concurrent spatial users, <200 OAM identities	**$30K–$75K/yr**	1.5–2.5x	$150K–$370K
Mid	4–15 sites, 50–500 concurrent, 200–2,500 OAM identities	**$150K–$450K/yr**	1.5–2.5x	$850K–$2.5M
Large	15+ sites or multi-org federation, 500+ concurrent, 2,500+ OAM identities	**$600K–$2.5M/yr**	1.5–2.5x	$3.0M–$12M

Pricing mechanic options (choose one per engagement; default in bold):

- **Per-site + per-OAM-identity** (default for industrial / multi-facility): $6K–$45K/site/yr plus $36–$180/OAM-identity/yr (volume tiered)
- Per-concurrent-user license: $200–$1,000/concurrent-user/yr (for office/collab-heavy deployments)
- Per-spatial-asset license (Hexon files under management): $0.75–$4/asset/yr at scale
- Enterprise unlimited (Large tier only): flat annual commit

> ***OAM-only entry SKU:*** *Buyers who want the OAM kernel without the full spatial OS surface area can adopt AMP as a pure operational-access-management product at* ***70% of the Spatial OS line-item****. This is the recommended motion for buyers (e.g., regulated FS, defense-adjacent, NERC CIP-driven utilities) where the access-management value lands before the spatial twin value.*

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4. Services mix

AMP is service-intensive in Y1. The typical Mid-tier full-deployment SOW:

Service	Scope	Fixed-fee range
Discovery sprint	Site walks, scene scoping, integration inventory, success metrics	$40K–$80K (credits to engagement)
Spatial environment buildout	CAD ingest, scene authoring, scale calibration, asset library setup	$80K–$300K per site
Sensor / OT integration	Live telemetry binding (PLCs via OPC UA, sensors via MQTT, location tracking)	$40K–$180K per site
Premium 3D component installation	Industrial-grade viewers, CAD-class viewport, premium UI library	$25K–$60K bundled
Operator training + change management	Hands-on for plant ops, dispatchers, facility managers	$30K–$120K per site
Federation enablement (Large only)	Cross-org spatial federation, identity bridging to NEOS + OAS	$150K–$500K one-time
30-day hypercare	Post-cutover stabilization	Included in deployment SKU

Services delivered by a mix of PrimusNeo, PLAN Systems contractors, and certified spatial-integration partners.

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5. Sustained upkeep + infra

Annual sustained upkeep = **18–25% of license**, per `../03-services-catalog.md` §2.4. Tier choice (Bronze/Silver/Gold/Platinum) drives the percentage and the SLA.

Infra typical components (pass-through + 12% mgmt fee):

- **Spatial server hosting** (per-site edge or regional cloud): $300–$2,500/mo/site
- **Hexon archive storage** (S3-equivalent + Glacier cold tier): $0.012–$0.025/GB-month
- **Sensor/IoT ingest bandwidth**: variable; typically $200–$1,500/mo/site
- **Optional Tie Town reference colocation**: $1,200–$2,500/mo for 1U edge node

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6. Bundle interactions

AMP is the **anchor layer** for any operational engagement. It composes most powerfully with:

Other layer	What you unlock together
+ Fractal Engine	Spatial events become first-class signals; predictive maintenance, OEE, capacity forecasting attach to scene positions. **Standard duo for manufacturing, energy, healthcare ops, logistics.**
+ OAS	Connects the spatial scene to ERP/MES/EHR/TMS/etc. without forcing rip-and-replace of any incumbent. **Required duo for buyers with substantial legacy IT/OT stacks.**
+ NEOS	Spatial actions become governed actions — "this approval was issued from this location on this asset by this signed identity." **Required duo for regulated industries (healthcare HITRUST, energy NERC CIP, finance MiCA).**
Full Stack	The federated coordination fabric. **Triad → Full Stack upsell is the dominant Y2 expansion motion.**

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7. Two worked engagement examples

7.1 Small AMP-only: 60-acre regenerative ag co-op (Tier-1 Small)

- **Buyer:** Co-op board of a 12-member regenerative cooperative running 60 acres in Vermont
- **Scope:** Single farm site, 8 concurrent spatial users (board + members during planning windows), ~120 OAM identities (members + seasonal workers + visiting agronomists), offline-capable, BYO connectivity
- **Quote:** $42K/yr AMP license (Small tier midpoint — spatial OS + OAM kernel) · $60K Atom deployment SKU · $8K/yr Bronze SLA · ~$400/mo edge node hosting
- **Y1 total:** ~$115K · **3-yr TCO:** ~$235K
- **Friction-cost anchor:** Cooperative spending $40K/yr on ad-hoc tools (Google Drive, Loomio, paper records, satellite imagery) *plus* an estimated $25K/yr in informal access management (member onboarding, key handoffs, contractor approvals, shift handoffs) — combined $65K/yr friction
- **Why they sign:** Member sovereignty (offline, self-hosted), Hexon file portability (no lock-in), unified OAM removing the patchwork of paper sign-in sheets + Slack channels, Tie Town reference proves the stack works

7.2 Large AMP+Fractal+OAS: 12-plant discrete manufacturer (Tier-3 Large)

- **Buyer:** CIO + COO of a $3B discrete manufacturer with 12 plants across NA + EU
- **Scope:** All 12 plants on AMP spatial twin + AMP OAM (~4,800 OAM identities: operators + maintenance + contractors + EHS + management), federated; OAS connectors to SAP S/4HANA, Rockwell FactoryTalk MES, AVEVA PI System; Fractal Engine OEE + downtime prediction
- **Quote:**
- AMP Large: $1.5M/yr (12 sites × ~$95K/site for combined spatial-OS + OAM; well below the $2.5M ceiling but above the floor given identity count and federation scope)
- OAS Large: $550K/yr (50+ integrations, ERP + MES + historian)
- Fractal Mid (high end): $300K/yr

- Triad bundle discount: -18% → $1.93M/yr license
- Y1 services (~2.2x license blend): $4.25M
- Y1 upkeep (Gold SLA, 25%): $483K (starts Y2)
- Infra pass-through + 12% mgmt: $400K–$700K/yr
- **Y1 total:** ~$6.6M–$7.0M · **3-yr TCO:** ~$13M–$15M
- **Friction-cost anchor:** McKinsey pilot-purgatory 70% on 8 stalled IIoT POCs = $12M sunk + $50M/yr unplanned downtime (Siemens benchmark) + ~$3–6M/yr across legacy IAM/PAM stack and lockout-tagout/MOC paperwork that AMP OAM displaces → capture window of **~$25M/yr in friction**
- **Capture rate:** ~$1.93M ÷ $25M = **7.7% value share** — defensible inside the 5–15% manufacturing capture band, with the OAM displacement value (Okta + CyberArk equivalents at ~$200–$400K/yr for a workforce this size) underwriting the AMP premium

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8. Steward economics

Per `../02-licensing-model.md` §4:

- **PLAN Systems royalty: 15% of AMP line-item license revenue**, paid quarterly
- Services revenue carries no royalty unless PLAN delivers the work on subcontract
- AMP is the highest-services-multiplier layer; PLAN benefits indirectly when they subcontract services delivery
- Tie Town deployment serves as the reference architecture both stewards benefit from

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9. Risks specific to AMP

Risk	Mitigation
Vendor lock-in fear	Hexon v1.0.0 is open; client SDKs are open; data export is non-negotiable contract term; OAM policy DSL is Apache-2.0
Spatial competitors (Omniverse, Unity Industrial) priced lower per-seat	Position on **operational + offline + federated + OAM-bundled** characteristics they don't deliver; per-seat math is irrelevant when AMP replaces two product categories at once
"Why is AMP more expensive than Fractal/OAS?"	Because it ships two production-grade capabilities — Spatial OS *and* Operational Access Management — in one runtime, and it is the most mature layer with multi-year Tie Town hardening. Compare against Spatial vendor + IAM/PAM vendor combined cost (typically $4,500/GPU/yr Omniverse + $5–$15/user/mo Okta + $50–$150/user/yr CyberArk-class PAM); AMP undercuts the sum
IAM/PAM incumbent displacement (Okta, CyberArk, BeyondTrust)	AMP OAM is positioned as **operational** access management — bound to spatial zones, time windows, OT assets, and physical-world context. It complements rather than fully displaces workforce IdP/IAM for office IT; the OAS layer handles bridging if both are needed
Talent shortage (spatial-OT integration)	PLAN partner channel + certified integrator network; managed deployment service
Performance at scale (very large plants, complex CAD)	Edge-first architecture; Hexon designed for streaming partial loads; reference deployments demonstrate the envelope
"We already have a digital twin"	Most "digital twins" are PowerBI dashboards over historian data — not actual spatial-event-graph systems. Discovery sprint shows the gap, and the OAM kernel is a second-vector value-add the dashboard tools cannot match

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10. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial AMP engagement model

NEOS — Governance Engagement Model

> ***Layer:*** *02 — Agreements & Permissions / Governance* ***Steward:*** *OmniOne / Green Earth Vision* ***License: Creative Commons BY-NC-SA 4.0*** *(non-commercial use free) + PrimusNeo Commercial License (PCL) for any commercial use* ***Reference deployment:*** *Tie Town [TX] governance councils + Aave/Lido benchmark adjacencies* ***Anchors source:*** `../research/layer-comparables.md`

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1. What NEOS actually is (in the buyer's vocabulary)

NEOS is the **durable coordination + decision-rights layer**. It is where councils form, charters are written, contributions are accounted, holonic organizations nest inside larger federations, and approvals become signed, auditable, replayable events.

In the language enterprises already know:

- A **bank** or **asset manager** to a CRO: NEOS is the programmable approval-and-attestation fabric that produces the immutable governance logs MiCA, OCC Letter 1184, and SAB 122 now make a procurement requirement.
- A **hospital system** to a CCO: NEOS is the consent + IRB + change-control governance layer that satisfies HIPAA, HITRUST i1/r2, and increasingly EHDS.
- A **state agency** to a CIO: NEOS is the inter-departmental decision-rights system that bridges the gap between Tyler/Granicus civic platforms and the participatory democracy world (Decidim, Pol.is).
- A **DAO** to a Steward Council: NEOS is the durable governance layer above Snapshot signal + Tally execution — the part that codifies the *intent* and the *accountability*, not just the vote.
- A **cooperative** to a member board: NEOS is the contribution accounting + holonic structure system that lets a 50-person cooperative federate with 50 others without losing local sovereignty.

NEOS is governance **at the protocol layer**, not at the SaaS-board-portal layer. That difference is what justifies pricing **above Loomio** (which is a discussion tool) and **at parity with Diligent** (which is a board portal but with no on-chain auditability or contribution accounting).

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2. License posture — Creative Commons with Commercial Restrictions

This is the layer that most directly tests the CC-NC + commercial-license model. Per `../02-licensing-model.md` §2:

Use	License	Cost
Personal / academic / research	CC-BY-NC-SA 4.0	Free
Nonprofit community deployment (co-op, neighborhood association, civic group) with no revenue from the deployment	CC-BY-NC-SA 4.0	Free
DAO / protocol with treasury ≤ $5M USD-equiv	CC-BY-NC-SA 4.0	Free
Small commercial entity meeting **all** de-minimis thresholds (≤$1M revenue, ≤10 FTE, ≤250 connected members, no resale)	CC-BY-NC-SA 4.0	Free
Above any de-minimis threshold	**PrimusNeo Commercial License (PCL)**	See tiered pricing below
Hosting NEOS as a paid service / managed service / SaaS	**PCL required**	Tiered
Embedding in any commercial product or platform	**PCL required**	Tiered
Government / public-sector entity procuring through paid procurement vehicle	**PCL required**	Tiered

Forks are permitted; forks may not market themselves as "PrimusNeo NEOS" (trademark restriction); commercial use of a fork still requires a PCL with OmniOne / PrimusNeo.

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3. Pricing tiers (annual commercial license, before services)

Scale	Definition	Annual License	Services Multiplier (Y1)	Typical 3-yr TCO
Small	Single org, 1 council, <50 members	**$6K–$18K/yr**	0.5–1x	$25K–$80K
Mid	Multi-council, 50–500 members, charter management	**$30K–$90K/yr**	0.5–1x	$130K–$420K
Large	Federated governance, 500+ members, multiple orgs	**$120K–$400K/yr**	0.5–1x	$530K–$1.7M

Pricing mechanic options (default in bold):

- **Per-council license** (default for orgs adopting governance progressively): $1.5K–$8K/council/yr
- Per-member license (for participatory democracy / DAO contexts): $20–$120/member/yr (scales down at volume)
- Per-covered-life (healthcare consent, civic resident contexts): $0.10–$1/covered-life/yr
- **% of treasury** (DAO context, optional addendum): **0.10–0.30% of treasury AUM/yr**, *capped* at the Large-tier license ceiling

The % of treasury option is materially below the Llama/Karpatkey 0.25–0.75% range, on purpose — NEOS is the *governance fabric*, not the treasury management service. The fabric should be cheaper than the service it enables.

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4. Services mix

NEOS services are the **lightest** of the four layers. Most of the work is design (governance design, charter authoring, council structuring) and training rather than integration. Typical Mid-tier SOW:

Service	Scope	Fixed-fee range
Governance design workshop	OmniOne ETHOS methodology, 2-day on-site or 4-half-day virtual	$35K–$60K
Charter authoring	3–8 charters per engagement; legal review handoff included	$25K–$80K
Council instantiation	Council templates, signing key setup, contribution-accounting wiring	$20K–$60K
Identity bridging	To OAS identity fabric, IdP integration (Okta/Entra/Auth0/SSI)	$15K–$50K
Operator training	Council moderators, secretaries, members	$20K–$60K
30-day hypercare	Post-launch stabilization	Included

For DAO contexts, add **treasury-policy authoring** ($30K–$120K) and **on-chain governance setup** ($20K–$80K) as separately scoped items.

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5. Sustained upkeep + infra

Annual sustained upkeep = **18–22% of license** (lighter than the other three layers because configuration change rate is lower).

Infra (pass-through + 12% mgmt fee):

- **NEOS coordination server** (per-org hosted): $200–$1,500/mo
- **Chain anchoring gas reserve** (for any deployment that anchors governance events on-chain): variable per chain, typically $50–$500/mo for L2
- **Signature key management** (KMS / HSM options): $300–$2,000/mo depending on tier
- **Optional treasury custody integration overhead** (when bridging to Fireblocks / BitGo / Anchorage): pass-through

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6. Bundle interactions

NEOS composes naturally with every other layer. The two most common bundles:

Other layer	What you unlock together
+ OAS	Identity federation + signed cross-org events. **Required duo for any multi-entity governance deployment** (federations, cross-DAO coordination, multi-state agency cooperation, supply-chain consortia).
+ Fractal Engine	Governance events become first-class signals. Contribution accounting → workforce intelligence; council decisions → policy-effectiveness analytics. **High-value duo for finance (CRO dashboards), healthcare (CCO dashboards), and large DAOs (treasury policy backtesting).**
+ AMP	Governance attached to place: "this approval was issued from this location, on this asset, by this signed identity at this moment." **Required duo for regulated operational contexts (manufacturing change control, hospital lockout/tagout, energy NERC CIP-013 attestations).**

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7. Two worked engagement examples

7.1 Mid NEOS+OAS: 200-person worker cooperative federation (Tier-2 Mid, CC-NC eligible? No — commercial)

- **Buyer:** Federation of 5 worker cooperatives, ~200 total members, $40M combined revenue, planning member equity tokenization and cross-coop work-sharing
- **Scope:** 5 nested councils (one per co-op) + 1 federation council; contribution accounting across coops; identity federation via OAS
- **De-minimis check:** Combined revenue $40M → above $1M threshold → PCL required
- **Quote:**
- NEOS Mid: $55K/yr (6 councils × ~$9K/council/yr)
- OAS Small: $40K/yr (5–10 integrations, identity-heavy)
- Duet bundle: -10% → $85K/yr license
- Y1 services (~0.8x license for NEOS-heavy mix): $68K
- Y1 upkeep (Silver SLA, 22%): starts Y2 at $19K
- Infra: $1.2K/mo
- **Y1 total:** ~$170K · **3-yr TCO:** ~$400K
- **Friction-cost anchor:** $200K/yr currently spent across Loomio, Diligent, custom contribution-tracking spreadsheets, lawyer hours formalizing inter-coop agreements, and one painful prior governance failure that cost $300K in dissolved trust
- **Capture rate:** ~$170K ÷ $200K = **85% capture** of *direct* friction — but the bigger value is the federation enablement that wasn't possible at all before, justifying the price above the direct-friction floor

7.2 Large NEOS+OAS+Fractal: G-SIB tokenization program governance (Tier-3 Large)

- **Buyer:** CTO + Head of Digital Assets + Chief Risk Officer at a $1.5T AUM global asset manager launching a tokenized fund product

- **Scope:** NEOS governance fabric for the tokenization program (issuance approval councils, KYC policy councils, transfer-agent attestation flows, regulatory reporting attestations), OAS for chain + custody integration, Fractal for risk + compliance reporting
- **Quote:**
- NEOS Large: $280K/yr (multi-entity, regulated FS premium)
- OAS Large: $620K/yr (10+ chains + Fireblocks + Anchorage + Snowflake + legacy core integration)
- Fractal Large: $750K/yr (regulatory + risk analytics + audit trail intelligence)
- Triad bundle: -18% → $1.35M/yr license
- Y1 services (~1.4x blend): $1.9M
- Y1 upkeep (Platinum SLA, custom %): $400K (starts Y2)
- Infra: $300K–$800K/yr
- **Y1 total:** ~$3.5M–$4.0M · **3-yr TCO:** ~$8M–$11M
- **Friction-cost anchor:** MiCA non-compliance = €15M or 3% of revenue (~$45M for this firm). OCC Letter 1184 attestation premium: per the finance research brief, banks will pay $200K–$500K+/yr per platform that produces attestations on-demand. Custody compliance overhead reduced by SAB 122 but legal + ops cost still $1–2M/yr at this scale. Estimated capture window $5–10M/yr in defensible friction
- **Capture rate:** $1.35M ÷ $7M = **~19% value share** — defensible inside the 10–25% finance capture band

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8. Steward economics

Per `../02-licensing-model.md` §4:

- **OmniOne royalty: 18% of NEOS line-item license revenue**, paid quarterly
- OmniOne is also the primary delivery partner for governance design + ETHOS methodology engagements; services revenue accrues to OmniOne directly when they deliver
- The CC-NC + PCL structure is specifically what protects OmniOne's stewardship from being undercut by cloud-hyperscaler hosting (the MongoDB-vs-AWS-DocumentDB pattern)

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9. Risks specific to NEOS

Risk	Mitigation
CC-NC ambiguity over “commercial use”	De-minimis thresholds (§2) define commercial use objectively by revenue + headcount + members
DAO governance attack vectors	NEOS ships with sane defaults (min 3/5 multisig <$10M treasury, 4/7 above; replay protection; flash-loan-resistant voting). Compliance posture documented per engagement
Cloud hyperscaler fork-and-host	SSPL/BSL-style precedent (HashiCorp, MongoDB, Elastic) shows CC-NC + PCL holds up commercially. Trademark also restricts fork branding
“We already have Diligent / Aragon / Loomio”	NEOS is a governance protocol, not a portal. Discovery shows the gap: durable contribution accounting, holonic federation, signed event log — none of the incumbents do all three
Government / public sector procurement	NASPO / Sourcewell / GSA enrollment plus CC-NC visibility for civic deployments creates a hybrid pipeline that GovTech incumbents (Tyler, Granicus) can’t easily match

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10. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial NEOS engagement model

OAS — Interoperability Fabric Engagement Model

> ***Layer:*** *03 — Universal Interoperability Fabric (formerly OAS Web4)* ***Steward:*** *OAS Web4* ***License:*** *Closed-core / partial-OSS (Connector SDK + public API spec open MIT/Apache-2.0; federation engine + enterprise gateway + closed-source provider connectors closed)* ***Reference deployment:*** *Tie Town [TX] cross-system fabric* ***Anchors source:*** `../research/layer-comparables.md`

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1. What OAS actually is (in the buyer's vocabulary)

OAS is **one API across providers, chains, storage systems, identity, avatars, and reputation** — the modular interoperability backbone that absorbs the integration tax every enterprise pays today across MuleSoft, Boomi, Workato, AND chain-interop layers (Chainlink CCIP, LayerZero, Axelar, Wormhole) AND identity orchestration (Okta, Entra, Auth0, SSI Verified Credentials) AND decentralized storage (Pinata, Filebase, Ceramic).

The pitch in one sentence: **"What MuleSoft is for Web2 enterprise integration, OAS is for the Web2+Web3 + identity + storage + reputation reality your operation actually runs in."**

In the language enterprises already know:

- A **healthcare CIO**: OAS is the TEFCA-native FHIR fabric that compresses the multi-million-dollar Epic integration tax.
- A **CSCO at a global 3PL**: OAS is the carrier-shipper-3PL governance-neutral integration layer that *isn't* the TradeLens shaped failure.
- A **utility CIO**: OAS is the IT/OT bridge under FERC Order 2222 — the layer that lets the ADMS + DERMS + ETRM stack speak the same language without ripping out any incumbent.
- A **bank Head of Digital Assets**: OAS is the multi-custodian + multi-chain + legacy-core integration fabric that no incumbent iPaaS covers.
- A **DAO Steward Council**: OAS is how the DAO actually talks to the world (cross-chain bridges, off-chain identity, storage providers) under one operational contract instead of seven.

OAS is the layer that makes the federated four-layer stack **a stack** instead of four products. Without OAS, the others don't compose at the integration layer — and "they don't compose" is the failure mode every prior multi-vendor attempt at this has hit.

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2. License posture

Closed-core / partial-OSS — connector SDK + API spec + community connectors are open; federation engine + enterprise gateway + closed provider connectors are commercial.

Component	License
Connector SDK	MIT or Apache-2.0
Public REST/GraphQL API specification	Apache-2.0
Community-maintained open connectors	MIT or Apache-2.0
Federation engine (the cross-org orchestration kernel)	**Closed (PEL)**
Enterprise gateway (multi-tenant, SLA-bearing)	**Closed (PEL)**
Identity / reputation orchestration kernel	**Closed (PEL)**
Closed-source provider connectors (those carrying commercial cost — proprietary chain partnerships, premium storage providers)	**Closed (PEL)**

See `../02-licensing-model.md` §3.

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3. Pricing tiers (annual license, before services)

Scale	Definition	Annual License	Services Multiplier (Y1)	Typical 3-yr TCO
Small	<10 integrations, single domain, <1M API calls/month	**$24K–$60K/yr**	2–3x	$200K–$550K
Mid	10–50 integrations, multi-chain/multi-cloud, 1M–50M calls/month	**$80K–$250K/yr**	2–3x	$620K–$2.0M
Large	50+ integrations, full federation engine, >50M API calls/month, cross-org	**$350K–$1.5M/yr**	2–3x	$2.7M–$12M

Pricing mechanic options (default in bold):

- **Per-connector + base platform fee** (default; most predictable): base $24K–$200K/yr + $3K–$15K/connector/yr beyond the included count (5/15/50 for Small/Mid/Large)
- Per-API-call / per-event metered overage: $0.001–$0.05/call beyond included tier volume (1M / 50M / unlimited+)
- Per-vCore equivalent (MuleSoft-style, for buyers used to that mechanic): $1,000–$1,400/vCore/month
- Per-entity-enrolled (federation deployments): $2K–$15K/entity/yr for governance-fabric pricing

Gas/chain fees (for chain-interop usage) and **storage fees** (for decentralized storage backends) are **always pass-through at cost + 12% management fee**, never bundled into the platform license.

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4. Services mix

OAS is the **highest services multiplier** of the four layers (2–3x Y1) because connector buildout is real engineering work. Typical Mid-tier SOW:

Service	Scope	Fixed-fee range
Connector audit + discovery	Inventory of buyer's existing integrations, prioritization	$40K–$80K
ERP/MES/EHR/TMS/CRM connectors	Per-connector fixed-fee (see `../03-services-catalog.md` §2.2 connector class table)	$20K–$400K/connector
Chain connectors	Per-chain (EVM, Solana, Cosmos, Hedera, etc.)	$20K–$75K/chain
Storage backend connectors	Per-backend (S3, IPFS, Filecoin, Ceramic, Arweave)	$15K–$50K/backend
Identity federation buildout	IdP integration + SSI Verified Credentials wiring + NEOS identity bridging	$50K–$200K
Federation engine deployment (Large only)	Multi-tenant gateway + cross-org orchestration kernel setup	$200K–$800K one-time
30-day hypercare	Stabilization + connector tuning	Included

PrimusNeo retains the right to harden, generalize, and redistribute any non-bespoke connector through the OAS connector library (a Y2+ revenue tailwind for both OAS Web4 and PrimusNeo).

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5. Sustained upkeep + infra

Annual sustained upkeep = **20–25% of license** (above the all-layer 18% floor because connectors need active maintenance against upstream API changes).

Infra (pass-through + 12% mgmt fee):

- **Federation gateway hosting** (regional or multi-region): $2,000–$15,000/mo by tier
- **API gateway + observability**: $500–$5,000/mo by call volume
- **Chain gas reserve**: highly variable by chain + usage; quoted per engagement
- **Storage backend pass-through**: $0.012–$0.025/GB-month for S3-class; native rates for IPFS/Arweave/Filecoin
- **Identity provider pass-through**: at provider rates (Okta/Entra/Auth0 seat costs flow through)

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6. Bundle interactions

OAS is the **integration glue** that makes every other bundle composition viable. It is almost never sold alone above the Small tier — buyers who pick it always end up wanting the layers it federates.

Other layer	What you unlock together
+ NEOS	Signed governance events flow across the federation. **Required duo for any multi-org or regulated engagement.**
+ Fractal Engine	OAS becomes the data-in path for Fractal; integration events become first-class signals. **Common duo for finance regulatory reporting, manufacturing IT/OT, healthcare interop.**
+ AMP	OAS surfaces external system state into the AMP spatial scene (MES events become asset-attached, EHR alerts become room-attached, etc.). **Required for any operational-twin engagement at scale.**
Full Stack	OAS as the connective tissue is what makes the federated stack actually federated.

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7. Two worked engagement examples

7.1 Mid OAS-only: 4-plant mid-cap manufacturer wanting an IT/OT bridge (Tier-2 Mid)

- **Buyer:** CIO + VP Operations at a $400M revenue mid-cap discrete manufacturer with 4 plants in NA. Existing stack: SAP S/4HANA, Rockwell FactoryTalk MES, GE Proficy historian, mixed PLC fleet
- **Scope:** OAS as the IT/OT integration fabric, replacing 3 brittle in-house Python ETL pipelines and 1 legacy MuleSoft instance the team wants to retire
- **Quote:**
- OAS Mid: $130K/yr (15 integrations: 4 plants × 3 systems + 3 cross-plant)
- No bundle (single-layer)
- Y1 services (~2.5x license): $325K (connector buildouts: SAP $120K + Rockwell $180K + Proficy $80K, with bundle discounts)
- Y1 upkeep (Silver SLA, 22%): starts Y2 at $29K
- Infra: $4K/mo
- **Y1 total:** ~$500K · **3-yr TCO:** ~$1.0M
- **Friction-cost anchor:** MuleSoft existing spend $180K/yr + 2 FTE Python engineers maintaining ETL at $300K/yr fully loaded + integration-debt-driven IT/OT incidents costing ~$400K/yr in downtime → $880K/yr direct friction
- **Capture rate:** $500K ÷ $880K = **57% capture** of *direct* friction in Y1 (high because OAS is genuinely displacing line items, not just adding value)
- **Y2 expansion path:** Add Fractal Engine for IT/OT signal intelligence (a likely $200K/yr addition)

7.2 Large OAS+NEOS: National port-operator network (Tier-3 Large)

- **Buyer:** CSCO + CIO at a national port operator running 8 terminals + coordination with 5 ocean carriers, 3 rail carriers, customs authority
- **Scope:** OAS as the cross-party data fabric (Navis N4 connectors, customs single-window integration, carrier APIs, IoT in-transit visibility ingest); NEOS as the governance-neutral consortium fabric that explicitly solves the TradeLens-shaped failure mode

- **Quote:**
- OAS Large: $720K/yr (50+ integrations, multi-org federation engine on)
- NEOS Large: $220K/yr (cross-org councils for SLA + data-sharing + dispute resolution)
- Duet bundle: -10% → $846K/yr license
- Y1 services (~2.2x blend): $1.86M
- Y1 upkeep (Gold SLA, 25%): $212K (starts Y2)
- Infra: $250K–$600K/yr
- **Y1 total:** ~$3.0M–$3.5M · **3-yr TCO:** ~$6.5M–$8M
- **Friction-cost anchor:** TradeLens precedent: $1B+ collectively spent and lost across the prior consortium attempt. Per the logistics research brief, project44/FourKites $300K+/yr standing visibility cost displaced or augmented. EU CBAM compliance starting Jan 2026 adds $500K–$2M/yr at this scale. Combined friction window: $3M–$8M/yr
- **Capture rate:** $846K ÷ $5M = **~17% value share** — defensible inside the 10–20% logistics capture band

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8. Steward economics

Per `../02-licensing-model.md` §4:

- **OAS Web4 royalty: 15% of OAS line-item license revenue**, paid quarterly
- OAS Web4 also leads connector development for the closed-source provider connectors; per-connector services revenue is often delivered by their authors
- Connector reuse (where PrimusNeo hardens and redistributes a non-bespoke connector) accrues royalty to OAS Web4 on the redistributed license revenue, not the original engagement

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9. Risks specific to OAS

Risk	Mitigation
TradeLens-shaped consortium failure	NEOS governance-neutrality (no single party owns the network) is the explicit answer. Charter every multi-party deployment with this principle from day 1
MuleSoft / Boomi incumbent displacement resistance	Position OAS as the **Web2+Web3 unified fabric** the incumbents don't cover, not as a pure Web2 iPaaS replacement. Capture the Web3 + identity + reputation surface area first
Connector maintenance cost cliff	Active maintenance baked into 20–25% sustained upkeep; connector library reuse amortizes cost across customers
Open SDK + closed federation engine boundary confusion	Public docs clearly mark the boundary; PCL/PEL contracts spell it out; community can build full Small-tier deployments on open SDK alone (which actually helps the OSS funnel)
Chain/gas cost volatility	Always pass-through; never bundled. Customer signs the gas reserve estimate and gets monthly reconciliation
Cloud-hyperscaler fork of the federation engine	Closed-core boundary holds (Apache-2.0 SDK doesn't expose enough surface area to reconstruct the federation engine). HashiCorp/MongoDB precedent supports this defense

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10. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial OAS engagement model

Fractal Engine — Intelligence Engagement Model

> ***Layer:*** *04 — Intelligence / Organizational Memory* ***Steward:*** *Escherbridge* ***License: Creative Commons BY-NC 4.0*** *(no ShareAlike — derivative models may carry separate training-data licenses) + PrimusNeo Commercial License (PCL) for any commercial use* ***Reference deployment:*** *Tie Town [TX] operational telemetry + Escherbridge corpus* ***Anchors source:*** `../research/layer-comparables.md`

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1. What Fractal Engine actually is (in the buyer's vocabulary)

Fractal Engine is the **intelligence and historical-archive layer** — the part of the stack that turns raw operational telemetry into durable archives, queryable historical context, and actionable real-time signals.

The differentiator vs Palantir / Snowflake / Databricks / Splunk / Datadog: Fractal Engine treats **NEOS governance events** and **AMP spatial events** as first-class signal types. The other intelligence platforms can ingest anything but don't *understand* governance and spatial as native concepts. Fractal does — and that is what makes "the contextualized data layer that 70% of Industry 4.0 pilots keep failing to build" actually buildable.

In the language enterprises already know:

- A **manufacturing CIO**: Fractal is the OEE + downtime + scrap + energy intelligence layer tied to live spatial events, not month-end reports.
- A **healthcare CDO**: Fractal is the population health + RWE + operational analytics layer that doesn't require ripping out Innovaccer / Health Catalyst / Arcadia.
- A **bank CRO**: Fractal is the regulatory + risk intelligence layer that produces the auditable attestation logs MiCA + OCC Letter 1184 + SAB 122 reward.
- A **utility CSO + COO**: Fractal is the predictive APM + carbon MRV + Article 6 attestation intelligence layer.
- A **DAO Treasury Council**: Fractal is the treasury policy backtesting + delegate-effectiveness + protocol KPI intelligence layer — the "Llama analytics" capability without the Llama retainer.

Fractal is **the layer that converts the federated stack into an operating intelligence**, not just a federated coordination system.

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2. License posture — Creative Commons with Commercial Restrictions

Use	License	Cost
Personal / academic / research use of platform + models trained on public corpus	CC-BY-NC 4.0	Free
Nonprofit community deployment (cooperative, neighborhood, civic group) with no revenue from the deployment	CC-BY-NC 4.0	Free
DAO / protocol with treasury ≤ $5M USD-equiv	CC-BY-NC 4.0	Free
Small commercial entity meeting **all** de-minimis thresholds (≤$1M revenue, ≤10 FTE, ≤250 connected users, no resale)	CC-BY-NC 4.0	Free
Above any de-minimis threshold	**PrimusNeo Commercial License (PCL)**	See tiered pricing below
Hosting Fractal as a paid analytics service	**PCL required**	Tiered
Embedding in any commercial product	**PCL required**	Tiered
Government / public-sector entity procuring through paid procurement vehicle	**PCL required**	Tiered

IP on derivative models: - Models trained on a **customer's data** remain the customer's IP, period. - Models trained on **Escherbridge's public corpus** remain Escherbridge IP and are licensed back to the customer under PCL scope. - Models trained on a **blend** (which is the most common case in practice) are governed by a per-engagement IP schedule attached to the PCL. Default: customer owns the blended weights for their licensed scope; Escherbridge retains rights to the public-corpus-derived components.

This is more permissive than Palantir's IP regime and more buyer-friendly than the OpenAI / Anthropic "we own the model, you license access" pattern.

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3. Pricing tiers (annual commercial license, before services)

Scale	Definition	Annual License	Services Multiplier (Y1)	Typical 3-yr TCO
Small	<5 data sources, departmental analytics, <1TB managed	**$30K–$75K/yr**	1–2x	$140K–$340K
Mid	5–20 data sources, operational telemetry + historical archive, 1–20TB	**$100K–$350K/yr**	1–2x	$450K–$1.6M
Large	20+ sources, cross-org intelligence, real-time + archive, >20TB	**$500K–$2M+/yr**	1–2x	$2.2M–$9M+

Pricing mechanic options (default in bold):

- **Tiered enterprise license** (default): platform + included data-source count + included compute envelope; overages per the table below

- Per-data-source: $4K–$25K/data-source/yr (good for departmental sales motion)
- Per-compute-unit (Snowflake-style): $1.50–$3.50/Fractal-credit; useful for buyers used to Snowflake/Databricks mental model
- Per-research-query (pharma RWE / clinical analytics context): $5–$500/query depending on depth and dataset size
- PMPM (healthcare population health / payer context): $0.50–$8 PMPM depending on intensity, well below the $15–$200 PMPM industry range
- Per-tCO2e managed (energy/ag carbon MRV context): bundled into the tiered license; emerging mechanic if customer prefers

Overage mechanics (above tier-included envelope): - Data sources beyond included: $4K–$15K/source/yr - Compute credits beyond envelope: at per-credit rate - TB beyond included: $25/TB/month (parity with Snowflake)

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4. Services mix

Fractal services are **moderate** (1–2x license Y1) — lighter than OAS/AMP but heavier than NEOS because data pipeline + model training + ontology mapping is real engineering. Typical Mid-tier SOW:

Service	Scope	Fixed-fee range
Data source discovery + ontology mapping	Inventory of buyer's data sources, ontology design, signal taxonomy	$50K–$150K
Data pipeline buildout	Per-source ingest + transform + signal generation (OAS handles transport; Fractal handles semantics)	$25K–$100K per source
Model training	Per custom model: feature engineering + training + evaluation + deployment	$40K–$200K per model
Operational dashboards + analyst onboarding	Per-persona dashboards (CIO, COO, CRO, etc.) + Fractal-query training	$30K–$100K
NEOS event + AMP spatial event signal integration (the differentiator)	Wiring governance + spatial events as first-class signals into Fractal	$40K–$120K
30-day hypercare	Model tuning, dashboard polish	Included

For pharma / regulated FS / energy: add **regulatory model validation** ($60K–$250K) and **audit-trail attestation buildout** ($40K–$120K) as separately scoped items.

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5. Sustained upkeep + infra

Annual sustained upkeep = **20–25% of license** (above the all-layer floor because model retraining cadence is real ongoing work — quarterly minor retraining + annual major retraining is typical).

Infra (pass-through + 12% mgmt fee):

- **Fractal compute + query**: aligned with the buyer's cloud account preferences (BYO cloud preferred; managed-cloud available)
- Reference: comparable Snowflake credit cost $2–$4; Databricks DBU $0.15–$0.40 — Fractal is positioned at parity with these on raw compute
- **Storage (warm + cold archive)**: $0.012/GB-mo warm S3-class; $0.004/GB-mo cold Glacier-class
- **Model serving (GPU)**: $1.10–$3.95/GPU-hr managed (Modal-class); reserved instances 30–50% off
- **Observability (Datadog/equivalent)**: pass-through at buyer's existing rates
- **Optional Escherbridge corpus access** (where the customer wants Fractal models pretrained on Escherbridge's public corpus before customer-data fine-tuning): bundled into PCL or per-engagement priced

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6. Bundle interactions

Fractal is the **value-multiplier layer**. It composes with every other layer to turn coordination into insight.

Other layer	What you unlock together
+ AMP	Spatial events become first-class signals. Predictive maintenance, OEE, capacity forecasting attach to scene positions. **Standard duo for any operational-twin engagement.**
+ NEOS	Governance events become first-class signals. Contribution accounting → workforce intelligence; council decisions → policy-effectiveness analytics; signed attestations → automated regulatory reporting. **Required duo for regulated FS, healthcare, energy.**
+ OAS	OAS is the data-in path; Fractal is the intelligence-out layer. **Required duo for any analytics-led engagement; Fractal without OAS is a stranded analytics tool with no data feed.**
Full Stack	The federated coordination fabric with native intelligence. **This is the configuration that beats Palantir on TCO and beats Snowflake/Databricks on operational context.**

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7. Two worked engagement examples

7.1 Mid Fractal+OAS: Regional health-system population analytics (Tier-2 Mid)

- **Buyer:** CIO + CDO + CCO at a 4-hospital regional health system with ~800 beds and a 200K-life MA payer arm
- **Scope:** Fractal for population health + utilization analytics across the 4 hospitals + payer arm; OAS for Epic FHIR R4 + Cerner FHIR + claims feed + SDoH data ingest; PMPM-priced license
- **Quote:**
- Fractal Mid: $220K/yr (12 data sources, 8 TB managed); priced as PMPM at $1.10 PMPM × 200K lives = $264K/yr (use the higher of tiered or PMPM)

- OAS Mid: $140K/yr (10 integrations: Epic + Cerner + 3 claims + 2 SDoH + 3 internal)
- Duet bundle: -10% → $364K/yr license
- Y1 services (~1.6x blend): $580K (Epic FHIR connector $200K, Cerner FHIR $180K, ontology mapping $80K, dashboards $90K + others)
- Y1 upkeep (Gold SLA — healthcare default, 25%): $91K (starts Y2)
- Infra: $80K–$180K/yr (HITRUST-compliant cloud surcharge applies)
- **Y1 total:** ~$1.1M–$1.2M · **3-yr TCO:** ~$2.4M–$2.8M
- **Friction-cost anchor:** Per the healthcare research brief: interop fabric at this scale runs $5M–$15M over 3 yrs (Redox at $35K/month floor alone is ~$1.3M over 3 yrs). Innovaccer/Arcadia at $600K–$3M/yr. Combined Y1 displacement opportunity: $1.5M–$3M/yr
- **Capture rate:** $364K ÷ $2.2M (midpoint) = **~17% value share** — defensible inside the 8–20% healthcare capture band
- **HITRUST gate:** PrimusNeo HITRUST i1 prerequisite for this RFP (the brief flags 81% of hospitals require it)

7.2 Large Full-Stack: Tier-1 OEM global manufacturing program (Tier-3 Large)

This is the manufacturing example from `amp.md` §7.2 extended into a full-stack engagement. See that file for the AMP-side numbers; here we focus on Fractal's contribution:

- Fractal Large in the bundle: $800K/yr (20+ data sources across 12 plants: PLC + MES + historian + ERP + quality + energy + workforce)
- Custom Fractal models: predictive OEE per line, downtime root-cause classifier, scrap classifier, Scope 3 emissions estimator for DPP — typically 4–8 custom models in Y1
- Y1 Fractal services: $1.2M (significant chunk of the overall $2.8M services number)
- Y2+ steady state: Fractal upkeep + quarterly model retraining ~$200K/yr
- **Value contribution:** Fractal is the layer that converts the spatial twin from "live data viewer" into "operating intelligence" — the difference between OEE + downtime + scrap + Scope 3 being *visible* and being *predicted*. The 5–15% manufacturing capture band assumes Fractal is part of the stack; without it the capture rate drops to 1–3%.

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8. Steward economics

Per `../02-licensing-model.md` §4:

- **Escherbridge royalty: 18% of Fractal Engine line-item license revenue**, paid quarterly
- Escherbridge is the primary delivery partner for Fractal ML engineering, custom model development, and ontology design; services revenue accrues to Escherbridge directly when they deliver
- The CC-NC + PCL structure preserves Escherbridge's stewardship from cloud-hyperscaler appropriation of the Fractal codebase

- Escherbridge's public corpus is a strategic asset shared across the federation; access terms are governed by the per-engagement IP schedule

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9. Risks specific to Fractal Engine

Risk	Mitigation
Comparison to Palantir (price + capability)	We are explicitly below Palantir commercial ($250K–$500K/yr entry vs our $100K Mid start) and beat them on operational ontology native support. Lead with TCO comparison.
Comparison to Snowflake/Databricks (commodity perception)	We sit *above* Snowflake/Databricks raw compute in price but bring native NEOS + AMP signal types and the operational ontology layer they don't have. Position as the operating-intelligence layer that *uses* a data warehouse, not as a data warehouse
Model IP confusion (who owns what)	Per-engagement IP schedule in PCL spells it out. Default = customer owns blended weights for licensed scope; Escherbridge retains public-corpus-derived components
EU AI Act high-risk classification	If Fractal's outputs constitute high-risk decision support (Annex III), conformity assessment $20K–$80K per high-risk system is bundled into the engagement deliverable. Pursue conformity early per engagement
HITRUST / FedRAMP gating (healthcare, government)	HITRUST i1 cert is prerequisite for healthcare RFPs >$500K. FedRAMP Moderate for federal. Both are funded as central PrimusNeo investments, not per-engagement
Build-it-ourselves at hyperscaler scale	At true hyperscaler scale (>$10M/yr Fractal spend), some buyers WILL build their own. That's fine — we're not the right answer for everyone. Focus on the buyers who value time-to-value + operational context + federated steward credibility over absolute cost
Model drift / retraining cadence	Quarterly minor + annual major retraining baked into sustained upkeep. Drift detection metrics are native dashboards

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10. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team	Initial Fractal Engine engagement model

Agriculture — Engagement Model

> ***Vertical:*** *Agriculture (row crop, specialty, livestock, cooperatives, ag inputs, regenerative ag, MRV)*
> ***Research brief:*** `../research/agriculture.md` ***Anchors:*** `../01-pricing-model.md` , `../research/layer-comparables.md`

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1. The friction-cost anchor

Agriculture's friction is distributed across three distinct windows — compliance overhead, coordination inefficiency, and broken data infrastructure — and they compound on one another at every tier.

FSMA compliance tax (small/mid scale): USDA data shows farms with $25K–$250K annual produce sales spend up to 60% of profits on FSMA compliance. A $150K-revenue specialty crop farm carrying $90K in compliance overhead is a buyer already paying enterprise-software money for paper-and-labor solutions. That's the anchor.

Per-acre software density: FieldView has established $4–$10/acre/yr as the mass-market baseline for FMIS. CropX (hardware + SaaS) reaches $15–$25/acre/yr. At 5,000 acres, FieldView ceiling = $50K/yr. At 100,000 acres (mid regional co-op), the comparable full-suite spend is $400K–$2M/yr blended. The per-acre market has priced the buyer; PrimusNeo captures a portion of that existing budget while displacing compliance and MRV spend they're currently absorbing separately.

MRV cost collapse: Traditional soil carbon verification runs $50–$100/acre. Boomitra's satellite-ML approach claims below $10/acre at continental scale — a 5–10x reduction. The voluntary ag carbon market itself contracted 57% in 2024 ($84.9M → $36.1M) driven by methodology disputes and quality concerns, with Nori's September 2024 shutdown as the visible failure event. The lesson is not that MRV is dead; it's that MRV platforms that promised returns without verifiable outcome evidence failed. The Article 6.4 UN carbon market launching 2025–2026 with UN-endorsed credits expected at premium pricing vs voluntary is the re-entry event that makes this investable again — but only for platforms that can produce third-party-verifiable attestations.

EUDR compliance cost: For large operators with EU-market soy, cattle, cocoa, coffee, palm oil, rubber, or wood exposure, plot-level geolocation compliance is mandatory by December 30, 2026. EUDR upstream costs — physical segregation, traceability infrastructure, geolocation data — remain substantial even after EC simplification measures, and can exceed thin ag commodity margins in high-risk origins (Mongabay, Feb 2026). For a large trader handling 500K+ acres of soy supply chain, this is a $2M–$10M/yr compliance program waiting to happen.

Computed friction windows by scale:

Scale	Representative buyer	Primary friction	Annualized friction window
Small	5K-acre family farm	FSMA + ad-hoc tools	$40K–$120K/yr
Mid	50K-acre regional co-op	ERP interop + MRV enrollment + member data governance	$400K–$2M/yr
Large	National co-op federation / ag input giant	EUDR compliance + Scope 3 + multi-system interop	$5M–$25M/yr

PrimusNeo capture band for agriculture: **8–18% of compliance + MRV spend** (per the pricing model vertical table). The per-acre software budget is the comparison anchor; the compliance and MRV spend is the friction being displaced.

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2. PrimusNeo positioning per layer in agriculture

AMP — Field and livestock spatial twin (PLAN Systems)

AMP is the authoritative field-and-facility spatial record: field boundaries, soil zones, irrigation infrastructure, livestock areas, storage facilities, and agronomic event log. In the ag buyer's vocabulary: the offline-capable operational ground layer that ties together precision guidance, sensor telemetry, satellite imagery, and field event records into one navigable scene.

Critical requirement: offline-first. Rural connectivity is a primary disqualifier for cloud-only platforms across large portions of the US Corn Belt, Great Plains, and international growing regions. AMP's offline-first architecture (the PLAN core runtime operates without continuous cloud connectivity) is not a feature — it is a prerequisite for any farm-edge deployment. This is the explicit point of failure for every Web-native spatial platform (Spatial.io, Microsoft Mesh) and many FMIS competitors.

AMP prices per farm site at $4K–$30K/site/yr for production deployments. For a cooperative serving 200 member farms, cooperative-friendly flat-rate site licensing at $500–$3,000/site/yr is the right mechanic — it avoids per-acre metering complexity across variable field sizes and aligns with how co-op boards think about distributing costs to members.

NEOS — Cooperative governance and member-data sovereignty (OmniOne)

Farmer data sovereignty is the primary adoption barrier in precision agriculture. The GAO January 2024 precision agriculture report formally identified farm data sharing and ownership as the top adoption challenge. Civil Eats (March 2026) described farm data as "a hidden crop for corporate tech." Only 36% of smaller farms planned AI adoption vs 81% of large farms — the trust gap is structural and will not be resolved by privacy policy updates alone.

NEOS addresses this at the protocol layer. For a cooperative deployment, NEOS instantiates member councils where member-elected board governance controls data-sharing policies, contribution accounting tracks which member data contributed to which outputs, and signed on-chain attestation creates a verifiable, auditable record that data was not resold. The CC-NC dual-license structure is itself a signal: the

governance layer ships under Creative Commons NonCommercial terms, which signals non-extractive intent to the exact audience (cooperatives, small farms, agronomists) that has been burned by extractive tech before.

For national cooperative federations, NEOS enables holonic federation — local co-op councils retain local sovereignty while participating in federation-level governance without surrendering member data control to the national entity.

OAS — ERP interop across cooperative member systems (OAS Web4)

Co-op member systems are heterogeneous by design. A regional grain co-op's technology estate typically includes AgVend (retail omnichannel), SSI cooperative software, Agris (farm accounting), one or more FMIS platforms across members, a grain-ticketing system, and commodity price feeds. OAS is the integration layer that connects these without requiring rip-and-replace of any incumbent — the same positioning as MuleSoft for Web2 enterprise integration, extended to the ag-specific system landscape that MuleSoft's standard connector library doesn't cover.

OAS also carries the carbon registry connectors (Verra VCS, Gold Standard, Article 6.4 national registries) that are required for MRV credit issuance. These are purpose-built OAS connectors delivered by OAS Web4.

Fractal Engine — Yield prediction, carbon attribution, MRV signal generation (Escherbridge)

Fractal Engine is the intelligence layer that turns AMP field events, NEOS governance logs, and OAS-ingested system data into defensible analytical outputs: yield prediction models, input-cost optimization recommendations, Scope 3 carbon attribution per field/per commodity lot, and MRV signal generation for carbon credit verification.

The Escherbridge precedent in carbon attribution is relevant here: Indigo Ag's sale of 2.85M tCO2e to Microsoft represents the largest single soil carbon transaction to date, built on satellite + agronomic intelligence at scale. Fractal's analytics depth, tuned to Escherbridge's corpus, is the infrastructure that enables this class of transaction — and the explicit failure mode we cite when asked why PrimusNeo is different from Nori.

Fractal is also where EUDR plot-level geolocation compliance outputs are produced: satellite-verified field boundaries + crop identification + deforestation-risk scores, formatted for EU Due Diligence Statement submissions.

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3. Engagement tiers — Small / Mid / Large

Tier 1 — Small: Family farm or small cooperative (<5,000 acres, <$10M revenue)

Bundle: AMP (field spatial twin, offline-capable) + Fractal Engine (yield prediction + input optimization). NEOS and OAS added if co-op-connected; otherwise lightweight. Cooperative-friendly site licensing.

Worked example — Small Specialty Crop Operation (Tier 1)

- **Buyer:** 12-member vegetable cooperative, 800 acres in the Rio Grande Valley. FSMA Produce Safety Rule compliance is costing $60K/yr in record-keeping labor + consultant fees. Considering a specialty carbon program.
- **Scope:** AMP for field boundaries + agronomic event log (FSMA audit trail); Fractal for input-cost modeling + soil health indexing; NEOS (CC-NC, free — cooperative with <$1M revenue per member, non-commercial deployment) for member governance.
- **Quote:**
- AMP Small: $22K/yr (per-site, 3 field clusters, offline-first mode required)
- Fractal Small: $35K/yr (4 data sources: soil tests, weather, irrigation sensors, yield history)
- Duet bundle: -10% → $51K/yr license
- Y1 services (Atom deployment SKU, $55K + basic field data migration, $20K): $75K
- Y1 upkeep (Bronze SLA, 18%): $9K/yr (starts Y2)
- Infra (edge node + cloud reference small): ~$800/mo
- **Y1 total:** ~$136K · **3-yr TCO:** ~$225K
- **Friction-cost anchor:** $60K/yr FSMA compliance labor + $15K/yr ad-hoc satellite imagery + $20K/yr failed carbon program enrollment overhead = $95K/yr. Capture rate: $51K ÷ $95K = 54% of direct friction in steady state — but the FSMA audit trail alone ($60K displaced) justifies the AMP license independently.
- **Harvest-contingent payment terms:** offered on Y1 license; 50% due at spring deployment, 50% due post-harvest.

3-yr TCO range (Tier 1 Small): $60K–$150K

Tier 2 — Mid: Regional cooperative or mid-scale corporate farm (5K–100K acres, $10–$200M revenue)

Bundle: AMP + OAS + NEOS + Fractal Engine (full stack, cooperative-optimized).

Worked example A — Regional Grain Cooperative (Tier 2, Co-op)

- **Buyer:** 400-member grain cooperative in Iowa, 80,000 acres across members, $120M annual revenue. Technology estate: AgVend retail platform, SSI cooperative ERP, mixed FMIS across members. Scope 3 reporting required by Walmart grain sourcing program. EUDR soy exposure for EU export customers.
- **Scope:** AMP for member field spatial twin (offline-first for farm-edge members) + cooperative facility layer; OAS connectors to AgVend + SSI + 3 FMIS platforms (Conservis, Traction Ag, John Deere Operations Center); NEOS for cooperative board governance + member-data sovereignty assurance + on-chain attestation that member data was not resold; Fractal for Scope 3 attribution per grain lot + EUDR plot-level geolocation + yield prediction.
- **Quote:**
- AMP Mid: $140K/yr (12 cooperative facilities + member farm aggregate spatial layer)

- OAS Mid: $120K/yr (15 integrations: AgVend + SSI + 3 FMIS + 3 grain ticketing + carbon registry + 7 others; per-connector + base)
- NEOS Mid: $60K/yr (5 councils: board + 4 regional member chapters)
- Fractal Mid: $180K/yr (12 data sources: member yield histories, soil tests, weather stations, grain tickets, Scope 3 + EUDR model)
- Full-stack bundle: -25% → $375K/yr license
- Y1 services (~1.8x license blend, ag-weighted): $675K (Hex deployment SKU $550K + AgVend connector $80K + SSI connector $60K + FMIS connectors $90K, with bundle discounts; governance design workshop $45K)
- Y1 upkeep (Silver SLA, 22%): $83K/yr (starts Y2)
- Infra (mid cloud reference + edge nodes at 4 elevator sites): $8K/mo
- **Y1 total:** ~$1.15M · **3-yr TCO:** ~$2.8M
- **Friction-cost anchor:** $180K/yr current Scope 3 reporting consultant + $120K/yr EUDR compliance work begun + $200K/yr siloed FMIS subscriptions across members + $150K/yr integration maintenance = $650K/yr. Capture rate: $375K ÷ $650K = 58% of direct friction in Y2 steady state.

Worked example B — Mid-Scale Corporate Farm (Tier 2, Corporate)

- **Buyer:** Corporate row-crop operation, 40,000 acres across 6 farm units in the Midwest, $80M revenue. Lead agronomist controls field data access and vendor trial permissions. ESG lead requires Scope 3 reporting for parent company's annual report.
- **Scope:** AMP for farm-unit spatial twins (no cooperative governance needed); OAS for ERP connector (NetSuite) + 2 FMIS platforms + soil sensor network ingest; Fractal for yield prediction + Scope 3 attribution + input-cost optimization models. NEOS lightweight (internal governance only, Bronze SLA).
- **Quote:**
- AMP Mid: $95K/yr (6 farm units, per-site)
- OAS Small (high end): $55K/yr (8 integrations)
- Fractal Mid: $140K/yr (8 data sources)
- Triad bundle: -18% → $237K/yr license
- Y1 services (~2.0x blend): $474K
- Y1 upkeep (Silver SLA, 22%): $52K/yr (starts Y2)
- Infra: $5K/mo
- **Y1 total:** ~$771K · **3-yr TCO:** ~$1.9M
- **Friction-cost anchor:** $220K/yr siloed technology spend + $180K/yr Scope 3 consultant + $80K/yr MRV program overhead = $480K/yr. Capture rate: $237K ÷ $480K = 49% of direct friction.

3-yr TCO range (Tier 2 Mid): $1.2M–$6M

Tier 3 — Large: Ag input company, global trader, or national cooperative federation ($200M+ revenue)

Bundle: Full stack — all four layers — with multi-region rollout, cross-commodity intelligence, and regulatory compliance infrastructure (EUDR, Article 6, FSMA, CBAM).

Worked example — National Cooperative Federation (Tier 3, Co-op)

- **Buyer:** National grain cooperative federation with 12 regional member co-ops, 2M aggregate member acres, $3B combined revenue. Equivalent profile: CHS or GROWMARK federated entity. Technology: Cooperative Ventures portfolio FMIS, multiple ERP systems, existing Verra VM0042 carbon program under economic pressure from Dec 2024 fee revision.
- **Scope:** AMP federated spatial layer (all 12 regional co-ops, member farm aggregation, offline-first at farm edge); OAS federation engine (50+ integrations: co-op ERPs, FMIS platforms, grain trading systems, carbon registries including Verra VCS + Article 6.4 national registries, EUDR EU traceability API); NEOS federated governance (12 regional councils + 1 national federation council; member-data sovereignty assurance at every tier; on-chain attestation for all data sharing events); Fractal cross-commodity SC intelligence (Scope 3 per grain lot + soy deforestation risk + carbon credit MRV signal generation + yield intelligence across all regions).
- **Quote:**
- AMP Large: $600K/yr (federation + 12 regional spatial instances)
- OAS Large: $800K/yr (50+ integrations, federation engine, cross-registry connectors)
- NEOS Large: $280K/yr (federated governance, 13 councils, 2M-acre attestation scope)
- Fractal Large: $900K/yr (20+ sources, cross-commodity intelligence, real-time + archive)
- Full-stack bundle: -25% → $1.935M/yr license
- Y1 services (~1.7x license blend, federation-weighted): $3.3M (Tessellation deployment SKU low end $3.5M; blended with existing co-op infrastructure = $3.3M)
- Y1 upkeep (Gold SLA, 25%): $484K/yr (starts Y2)
- Infra (large cloud reference, multi-region + 30 edge nodes at elevator sites): $35K/mo
- **Y1 total:** ~$5.65M · **3-yr TCO:** ~$14.5M
- **Friction-cost anchor:** EUDR compliance program $3M–$8M/yr for EU soy/cattle exposure + current VM0042 carbon program admin $1.5M/yr under fee pressure + multi-co-op interop overhead $2M/yr + Scope 3 consultant and reporting $1M/yr = $7.5M–$12.5M/yr. Capture rate: $1.935M ÷ $10M (midpoint) = 19% value share — inside the 8–18% agriculture capture band (toward the high end, justified by EUDR regulatory tailwind and clear displacement of consultant spend).

3-yr TCO range (Tier 3 Large): $15M–$75M

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4. Industry-specific procurement signals

Decision-maker map by segment:

Segment	Primary decision authority	Critical influencer	Budget cycle
Cooperative	Member-elected board (risk-averse, consensus-driven)	General manager + lead agronomist	Annual (Oct–Mar board approval)
National co-op federation	CEO + CTO/CIO + technology committee	Cooperative Ventures portfolio team (CHS/GROWMARK gateway)	Annual + capital program cycles
Corporate farm (mid)	Ops manager + CFO	Lead agronomist — controls field data access and vendor trial permissions	Fiscal year-end planning
Ag input company (Bayer, AGCO, CNH)	CTO/CIO + divisional president	Digital ag product team	Quarterly operating review
MRV/carbon program	Sustainability director + CFO	Third-party verifier relationship	Carbon program enrollment windows

The agronomist is the critical influencer. At mid-sized corporate farms and cooperative tech evaluations, the lead agronomist controls field data access and vendor trial permissions. No pilot proceeds without agronomist buy-in. Motions that bypass the agronomist and go straight to the CFO reliably stall in technical evaluation. The selling motion must engage the agronomist first, produce a one-season pilot with measurable yield or input-cost data, then use that evidence for board or executive escalation.

Sales cycle timing:

- Co-op enterprise software: 12–24 months board-approval cycle
- MRV/carbon program enrollment at scale: 18–36 months
- Corporate farm technology adoption: 9–18 months

Planting-cycle awareness is non-negotiable. Technology decisions in North American grain farming are made October through February for spring deployment. A pilot that starts in March cannot generate board-credible ROI evidence until the following harvest (October–November at earliest). The sales calendar must align to this rhythm: discovery and scoping in summer, proposal and board presentation in fall, contract signature by January, deployment by planting season.

Buying triggers that accelerate the cycle:

- EUDR compliance deadline pressure (December 30, 2026 for large operators)
- Walmart/Unilever/Nestlé Scope 3 supplier requirements with contractual deadlines
- USDA EQIP/CSP conservation program eligibility linked to digital record-keeping
- Article 6.4 carbon market launch creating premium credit pricing window
- Co-op member demand for input cost transparency (particularly fertilizer price tracking in CBAM-adjacent supply chains)

Cooperative Ventures (CHS + GROWMARK) is the gateway to the cooperative segment. Both organizations route technology investment through their $50M Cooperative Ventures fund. Vendor relationships typically begin with the Cooperative Ventures portfolio team before reaching operational buyers. A fund relationship is a prioritized channel entry point.

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5. Rejection-risk mitigations

Data sovereignty — PRIMARY BARRIER

The risk: Farmers fear that precision data (field boundaries, yield maps, soil tests, application records) shared with large-platform vendors becomes competitive intelligence for input companies or grain traders. This is not irrational; multiple documented instances of data being shared with parent company commodity desks have been reported (Civil Eats, March 2026). Only 36% of smaller farms planned AI adoption vs 81% of large farms — the trust gap is structural.

The mitigation stack: 1. **Transparent data ownership terms** in the PFEL: customer data is exportable in machine-readable form at any time, non-negotiable. 2. **Farmer-owned data by contract:** no resale, no use beyond licensed scope, no sharing with parent entity (Bayer, Corteva) without explicit opt-in. 3. **On-chain attestation via NEOS:** every data-sharing event is signed and anchored on-chain, creating a verifiable record that data was not resold. This is the protocol-level sovereignty guarantee that contractual terms alone cannot provide. 4. **CC-NC dual-license signal:** NEOS ships under Creative Commons NonCommercial terms. The license itself communicates non-extractive intent to the cooperative and smallholder audience. 5. **Member council governance:** NEOS cooperative councils give farmer-members visibility and control over data governance policy — not just a vendor promise.

Rural connectivity

AMP's offline-first architecture is the mitigation. Intermittent-sync mode means field operations continue without cloud connectivity; synchronization occurs when connectivity is available (at the elevator, at the farm office, at the co-op facility). This is a hard architectural requirement; it must be demonstrated in the pilot, not promised in a slide deck.

ROI per acre too thin

At $200–$500/acre commodity crop margins, a $10–$15/acre software stack requires demonstrable 2–5% yield lift or input cost reduction just to break even. The mitigation is a Fractal Engine-powered quantified yield-lift model: Fractal establishes a pre-deployment baseline in the instrumentation phase (60 days), then tracks yield, input spend, and variable cost per acre against the baseline across one full growing season. The model produces a per-acre ROI number the agronomist can present to the board. This is not a case study — it is a buyer-specific measurement against buyer-specific baseline data.

Broken farm-tech promises

Nori (shutdown September 2024), multiple failed carbon programs, John Deere data lock-in controversies, and Bayer's years of financial losses on Climate Corporation have created deep market skepticism. The mitigation is verifiable third-party outcome evidence — not vendor case studies. Cite the Nori collapse explicitly as the failure mode PrimusNeo is designed to avoid: a platform that promised MRV at scale without verifiable attestation infrastructure collapsed when buyer scrutiny increased. Fractal + NEOS produce verifiable, on-chain-attested MRV outputs that a third-party verifier can independently confirm.

Seasonal cash flow

Farm operations are highly seasonal with cash concentrated post-harvest. Annual subscription billing misaligned to farm cash flow creates real friction. Standard mitigation: installment billing (50% at spring deployment / 50% post-harvest) or harvest-contingent payment terms. This materially improves conversion with small and mid operators. Large co-op and corporate farm buyers are less sensitive to this but appreciate the option as a signal that PrimusNeo understands the agricultural cash cycle.

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6. Regulatory and compliance considerations

EU Deforestation Regulation (EUDR) Covers soy, cattle, cocoa, coffee, palm oil, rubber, and wood. Large operators: compliance mandatory by December 30, 2026. Micro and small operators: June 30, 2027. Compliance requires plot-level geolocation for all covered commodities entering the EU market, plus due diligence documentation and a risk assessment system. For any US grain co-op or ag trader with EU soy, cattle, or commodity exposure, this is a hard deadline creating a procurement event. OAS connectors to the EU EUDR traceability system and Fractal-generated deforestation risk scores per field directly satisfy the technical compliance requirement.

Carbon Border Adjustment Mechanism (CBAM) Nitrogen fertilizer is in CBAM scope. Direct commodity crops are not yet included but remain under ongoing review; inclusion would fundamentally reshape ag supply chain economics. The indirect input-cost pressure from CBAM on nitrogen fertilizer creates demand for Fractal-powered fertilizer cost optimization and Scope 3 attribution models.

USDA NIFA and FSMA The FSMA Produce Safety Rule's record-keeping, traceability, and food safety plan requirements are a direct buying trigger for AMP (field event log = FSMA audit trail) and OAS (traceability connectors to FDA systems). The USDA NIFA Food Safety Outreach Program provides technical assistance funding to specialty crop producers — a grant funding pathway for small-tier engagements.

Article 6 / Paris Agreement COP29 (November 2024) finalized remaining technical guidance. As of March 2025, 97 bilateral agreements across 59 countries are active. Article 6.4 UN carbon market launch expected 2025–2026. UN-endorsed credits are expected at premium pricing vs voluntary market credits — potentially re-energizing soil carbon MRV investment after the 2024 contraction. Fractal's MRV signal generation + NEOS on-chain attestation directly produces outputs that qualify for UN-endorsed credit issuance under Article 6.4 methodology, rather than the weakened voluntary market standards that drove the 2024 collapse.

Verra VM0042 and Gold Standard VM0042 v2.0 (active May 2023) is the primary compliance pathway for row-crop carbon. The December 2024 Verra fee revision ($5,000 upfront per verification review + per-credit issuance fees) squeezes developer economics at low credit prices — which is why Fractal's Boomitra-class cost reduction (below $10/acre at scale vs $50–$100/acre traditional soil sampling) is the economic enabling factor, not a nice-to-have. Gold Standard co-benefit documentation requirements add cost but serve premium buyer segments willing to pay above-market for co-benefit-verified credits.

NEOS + Fractal attestation path: A NEOS-attested governance event (the farmer's data-sharing consent and program enrollment) combined with Fractal's satellite-ML MRV signal generates the complete evidentiary package for both Article 6.4 submission and Verra VM0042 issuance. This is the two-layer combination that directly satisfies both regulatory regimes.

7. Steward and service partner notes

Layer stewards:

Layer	Steward	Agriculture-specific role
AMP	PLAN Systems	Field and farm spatial environment buildout; offline-first edge architecture; farm-site per-site licensing mechanic
NEOS	OmniOne / Green Earth Vision	Cooperative governance design using ETHOS methodology; member-data sovereignty charter authoring; CC-NC license positioning
OAS	OAS Web4	AgVend, SSI, Agris, FMIS connectors; carbon registry connectors (Verra VCS, Gold Standard, Article 6.4); EUDR traceability API connector
Fractal Engine	Escherbridge	Yield prediction models; Scope 3 carbon attribution; MRV signal generation; EUDR deforestation risk models; analytics depth per Indigo Ag / Microsoft 2.85M tCO2e precedent

SI channel:

- **McKinsey ACRE practice** ($350K–$2M engagements): Strategic ag-digital transformation engagements at large input companies and national co-op federations. ACRE has the relationships with Bayer, AGCO, and CHS senior leadership. Route large Tier-3 opportunities through ACRE-introduced deals.
- **AGCO Digital Solutions** ($150–$300/hr dealer-delivered): AGCO's dealer channel is embedded with mid-scale corporate farm operators who already have AGCO equipment relationships. Natural mid-tier co-sell for AMP + Fractal deployments in AGCO-fleet operations.
- **ADM Outcomes** ($500K–$3M per engagement): Multi-year regenerative ag transition and Scope 3 advisory engagements. ADM Outcomes is the implementation partner for large-scale carbon and sustainability programs — a natural steward for MRV-heavy Fractal deployments.

Cooperative Ventures (CHS + GROWMARK) as channel gateway: Both organizations channel technology investment through their $50M Cooperative Ventures joint VC fund. Traction Ag's $10M Series A (May 2024) led by Cooperative Ventures signals which FMIS platforms are on the co-op shortlist — and demonstrates that the fund team is actively influencing technology decisions across both organizations' operational buyer networks. A fund-level relationship unlocks the cooperative segment more reliably than any direct-to-board sales motion.

8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Agriculture engagement model

Construction — Engagement Model

> ***Vertical:*** *Construction & AEC (GC/Sub, Real Estate Developers, Infrastructure, MEP, Smart Buildings / FM)* ***Research brief:*** `../research/construction.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Construction/AEC is a $16T global market running on fragmented, often-incompatible multi-vendor software stacks — yet owner digital program budgets reliably run at **1–2% of total capex** (McKinsey), and advisory retainers from AECOM or Arup on mega-projects consume an additional **0.2–0.5% of capex** in fees alone. Those numbers set the ceiling from which PrimusNeo carves its 5–15% capture band.

Friction windows by customer size:

Segment	Capex / Revenue basis	Digital program budget (1–2% capex or ACV proxy)	Typical friction window
Small GC/Sub (<$50M revenue, 1–2 projects)	$10M–$50M ACV	$100K–$1M/yr	$40K–$90K 3-yr TCO (Procore entry + ACC + Fieldwire)
Mid regional contractor/developer ($50–500M, 5–30 projects)	$50M–$500M ACV	$500K–$5M/yr	$350K–$800K 3-yr TCO (Procore + ACC BIM + CDE + DroneDeploy + One Click LCA)
Large ENR Top-100 / mega-project ($1B+ capex, national portfolio)	$1B–$10B+ capex	$10M–$100M/yr	$3M–$12M 3-yr TCO (Procore enterprise + Bentley iTwin + Willow Twin + advisory)

The market's dominant pricing mechanics make the friction cost concrete: Procore's ACV model runs **0.1–0.3% of Annual Construction Volume** (a 25-project regional GC all-in $40K–$120K/yr); Autodesk Build at **$1,625/seat/yr** for field users; Bentley iTwin enterprise at **$200K–$1M+/yr** for large infrastructure; Willow Twin at **$50K–$250K/yr per campus** for smart building FM. One Click LCA Power Pack at **$80K–$120K/yr** for embodied carbon. These are the incumbent line items PrimusNeo displaces or federates — not by replacing them outright, but by absorbing the interoperability tax they generate among themselves.

A mid-tier GC running Procore + ACC BIM Collaborate (30 seats) + Asite CDE + DroneDeploy Advanced pays approximately **$120K–$200K/yr in platform licenses alone** before any SI advisory, any duplicate data-entry labor, or any rework attributable to model/field data drift. PrimusNeo's friction anchor is the gap between what those platforms cost in isolation and what a federated, single-source-of-truth stack would cost — conservatively **$250K–$500K/yr** in direct and indirect friction at the mid tier.

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2. PrimusNeo positioning per layer in construction

AMP is the killer layer in construction. The fundamental coordination problem in every AEC project is that design-side BIM (Autodesk Revit/ACC, Bentley MicroStation) and field-side project management (Procore, Trimble Connect) and FM-side building operations (IBM TRIRIGA, Planon, Honeywell Forge) each maintain their own spatial record — and they never stay synchronized. The design model diverges from the as-built the moment the first RFI is issued. The as-built diverges from the FM model the moment handover documents go into a SharePoint folder no one maintains.

AMP's federated spatial record — built on the open Hexon file standard, offline-capable, multi-org — is the construction-native answer to this problem. It bridges Autodesk BIM (ACC model ingestion via IFC/RVT), Trimble field data, and Bentley infrastructure models **without forcing single-vendor lock-in** at any layer. The as-built lives in AMP as an AMP scene. The FM team queries it through their IWMS. The GC's VDC manager pushes model updates via IFC. No single platform owns the record.

OAS is the required companion. The construction stack is notoriously siloed. Procore does not natively talk to ACC. ACC does not write back to Trimble Connect without manual export. Asite CDE is separate from both. OAS certified bidirectional connectors for Procore (webhook-based), ACC (IFC/RVT model ingest + write-back), Bentley iTwin, and Trimble Connect collapse the integration tax that currently costs GC IT teams 1–2 FTE in perpetual connector maintenance. At the small and mid tier, AMP + OAS is the standard bundle.

NEOS delivers cross-project governance and change-control flows. At mid-to-large scale, construction's biggest coordination overhead is the approval bureaucracy: RFIs, submittals, change orders, and management-of-change (MOC) on occupied facilities. NEOS structures these as signed, auditable, replayable events — replacing the email-thread-based approval culture that Procore partially addresses but never fully governs, especially across organizational boundaries (owner / GC / design team / authority having jurisdiction).

Fractal Engine converts project data into portfolio intelligence. At portfolio scale, ENR-100 contractors and real estate developers need answers that no individual project platform provides: RFI velocity trends that predict schedule slippage 6 weeks out; embodied carbon analytics that ingest EPD data per material per project for EPBD / Buy Clean Act compliance; financial closeout benchmarking across the portfolio. Fractal also enables RFI predictive triage — classifying incoming RFIs by complexity and downstream cost impact before a project engineer reads them, routing high-risk items to senior PM attention automatically.

Standard bundle and full-stack triggers:

- Small GC/Sub: OAS alone (Procore/ACC connector), or OAS + AMP entry (as-built record without displacing incumbents)
- Mid regional contractor/developer: AMP + OAS (federated spatial + integration duo); add NEOS for change-control
- Large ENR-100 / mega-project: Full AMP + NEOS + OAS + Fractal as the federated intelligence and governance layer across the portfolio

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3. Engagement tiers — Small / Mid / Large

Small: GC or Subcontractor, <$50M Revenue, 1–2 Active Projects

Buyer profile: A 50-person mechanical/electrical sub or a regional residential GC. Running Procore entry + ACC Docs (5 seats) + Fieldwire Pro (5 seats). Primary pain: duplicate data entry between Procore and ACC; no durable as-built record at project closeout; RFIs managed via email threads.

PrimusNeo bundle: OAS (Procore + ACC bidirectional connector) + AMP entry (as-built spatial record for 1 project)

Worked quote:

Line	Amount
OAS Small license (5 integrations, Procore + ACC + 3 supporting)	$30K/yr
AMP Small license (1 site, <20 concurrent)	$18K/yr
Duet bundle discount (10%)	−$5K/yr
License total	**$43K/yr**
Y1 services — Atom deployment SKU + OAS connector buildouts (Procore webhook $25K + ACC IFC ingest $35K)	$95K one-time
Y1 upkeep (Silver SLA, 22%)	$9K/yr (starts Y2)
Infra pass-through + 12% mgmt (cloud reference small)	$1,200/mo

Y1 total: ~$152K. 3-yr TCO: ~$240K–$270K.

Friction anchor: Procore entry ($5K/yr) + ACC Docs 5 seats ($4K/yr) + Fieldwire Pro 5 seats ($3K/yr) + 0.5 FTE in duplicate data entry (~$40K/yr fully loaded) + one deferred warranty claim from a lost as-built ($15K–$50K one-time) = **$65K–$100K/yr in direct friction.** PrimusNeo at $43K/yr license occupies 43–66% of the direct friction — but eliminates the duplicate entry FTE and the as-built loss risk, which is where the real ROI lives.

Capture rate: 43K ÷ $82K (midpoint) = **~52% of direct friction displaced.** Defensible because OAS is genuinely removing line items, not adding a layer.

Developer/FM small example: A boutique real estate developer completing a 120-unit multifamily project needing a handover-ready as-built and OAS connector to their IWMS (IBM TRIRIGA entry). Same bundle, scoped to 1 building. Quote similar; OAS connector scope shifts from Procore to TRIRIGA integration ($20K fixed fee). AMP as-built persists post-handover as the FM spatial record — positioning the Y2 AMP-to-FM upsell.

Mid: Regional Contractor / Developer, $50–500M Revenue, 5–30 Projects

Buyer profile: A regional GC with 20 active projects across 3 states, running Procore ($45K/yr) + ACC BIM Collaborate (25 seats, $18K/yr) + Asite CDE ($60K/yr) + DroneDeploy Advanced ($8K/yr) + early-stage One Click LCA use. Pain: project data lives in four systems; portfolio reporting is a monthly Excel exercise; change-order approvals route through email with no audit trail; ESG officer cannot produce an embodied carbon number without a consultant.

PrimusNeo bundle: AMP + OAS + NEOS (Triad)

Worked quote:

Line	Amount
AMP Mid license (8 sites, 100 concurrent)	$130K/yr
OAS Mid license (12 integrations: Procore + ACC + Asite + DroneDeploy + IWMS + others)	$110K/yr
NEOS Mid license (3 councils: change-control, submittal, ESG reporting)	$50K/yr
Triad bundle discount (18%)	−$52K/yr
License total	**$238K/yr**
Y1 services — Hex deployment SKU ($550K) + OAS connectors Procore/ACC/Asite ($220K) + NEOS charter authoring ($60K) + AMP scene buildout 3 pilot projects ($120K)	~$950K one-time
Y1 upkeep (Silver SLA, 22%)	$52K/yr (starts Y2)
Infra pass-through + 12% mgmt (cloud mid)	$6K/mo

Y1 total: ~$1.3M. 3-yr TCO: ~$2.2M–$2.5M.

This lands at the high end of the mid-tier construction 3-yr TCO band ($350K–$800K) stated in the research brief. The Triad bundle (vs. AMP + OAS Duet) justifies the premium when the NEOS change-control value is explicitly quantified: a single change order dispute resolved in days rather than weeks, across a $50M project, routinely recovers $25K–$150K in avoided delay claims.

Friction anchor: Procore $45K/yr + ACC $18K/yr + Asite $60K/yr + DroneDeploy $8K/yr + One Click LCA $20K/yr = $151K/yr in direct platform spend, plus an estimated **$200K–$400K/yr in coordination overhead** (duplicate data entry, rework from model/field drift, manual ESG reporting) = **$350K–$550K/yr total friction.** PrimusNeo license at $238K/yr captures 43–68% of direct friction while eliminating the overhead — overall value share roughly 12–15% of total friction window (per McKinsey 1–2% capex digital budget at this revenue scale, the window is $500K–$5M/yr).

Capture rate: $238K ÷ $1.75M (midpoint friction window) = **~14% value share** — well inside the 5–15% construction capture band.

Developer/FM mid example: A regional real estate developer with 15 active projects (mixed residential + commercial) whose FM team inherits buildings into their IWMS portfolio. AMP as the design-to-FM spatial continuity layer (Hexon as-built persists from BIM to IWMS); OAS connectors to Procore + ACC + Planon; Fractal (not NEOS) added for portfolio ESG reporting and embodied carbon compliance under CA Buy Clean Act. Pricing similar; NEOS swapped for Fractal Mid at $100K/yr. Triad bundle applies.

Large: ENR Top-100 Contractor, Mega-Projects, $1B+ National Portfolio

Buyer profile: An ENR Top-50 GC with a national portfolio of 80+ active projects including several mega-projects ($500M–$3B capex each). Enterprise Procore ($350K/yr) + Bentley iTwin or ACC enterprise ($500K/yr) + Willow Twin on select campuses ($400K/yr) + One Click LCA Power Pack ($100K/yr) + AECOM digital advisory ($1M/yr). Pain: each mega-project invents its own digital delivery strategy; no portfolio-level RFI or change-order predictive intelligence; embodied carbon reporting is a manual consulting engagement; CDO wants a federated data standard that doesn't require renegotiating contracts with Bentley, Autodesk, and Trimble.

PrimusNeo bundle: Full Stack — AMP + NEOS + OAS + Fractal

Worked quote:

Line	Amount
AMP Large license (20+ sites, multi-org federation, 600+ concurrent)	$700K/yr
OAS Large license (60+ integrations: Procore + Bentley iTwin + ACC + Trimble + Willow + IWMS + ERP + identity)	$650K/yr
NEOS Large license (federated governance across 5 business units + 3 owner-facing councils)	$250K/yr
Fractal Large license (25+ data sources, portfolio analytics + embodied carbon + RFI predictive triage)	$800K/yr
Full-stack bundle discount (25%)	−$600K/yr
License total	**$1.8M/yr**
Y1 services — Lattice deployment SKU ($2.5M) + OAS connector suite Procore/Bentley/ACC/Trimble ($500K) + AMP federation enablement ($400K) + Fractal models (RFI triage + carbon estimator, $300K) + NEOS charters ($120K)	~$3.8M one-time
Y1 upkeep (Gold SLA, 25%)	$450K/yr (starts Y2)
Infra pass-through + 12% mgmt (cloud large, multi-region)	$35K/mo

Y1 total: ~$6.0M. 3-yr TCO: ~$10M–$13M.

This lands inside the research brief's large-tier band ($3M–$12M 3-yr TCO). The upper end of this range is justified at mega-project mobilization scale where the Tessellation deployment SKU ($3.5M–$12M+) applies; the worked quote above uses the Lattice SKU appropriate for a national portfolio rollout without a single concurrent $3B+ project.

Friction anchor: McKinsey's 1–2% capex digital program budget on a $5B/yr construction volume = **$50M–$100M/yr** in owner-sanctioned digital spend. PrimusNeo's $1.8M/yr license = **1.8–3.6% of the digital budget** — a single rationalization play that can justify its cost by consolidating two or three incumbent vendor renewals (Willow Twin $400K/yr + One Click LCA $100K/yr + AECOM advisory increments $500K/yr = $1M/yr in near-term displacement candidates alone). Advisory AECOM/Arup retainers running 0.2–0.5% of capex on a $1B mega-project = $2M–$5M/engagement; Fractal automating the carbon and schedule-risk analysis layer cuts that advisory surface.

Capture rate: $1.8M ÷ $25M (midpoint friction window for a GC this size, conservatively scoped to direct digital spend) = **7.2% value share** — bottom of the 5–15% construction capture band, defensible and conservative.

Developer/FM large example: A national REIT or infrastructure developer with 200+ buildings under management. AMP as the portfolio-wide spatial record (replacing Willow Twin per campus at $50K–$250K/yr per campus, consolidated into one AMP Large license); OAS connectors to Honeywell Forge + IBM TRIRIGA + existing BIM archives; Fractal for Scope 1/2/3 emissions reporting across the portfolio for EU SFDR + EPBD Digital Building Passport obligations; NEOS for the FM change-control governance layer on occupied critical facilities. Full stack; pricing anchored by per-sf FM spend ($0.10–$0.50/sf/yr at 10M sf managed = $1M–$5M/yr incumbent spend).

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4. Industry-specific procurement signals

Primary champions:

- **VDC / BIM Manager** at GC or design-build firm: controls day-to-day platform selection for ACC, Procore, and CDE. This is the technical champion who experiences the design/field model drift firsthand and has the authority to run a pilot. Pain: IFC round-trips, duplicate model management, ACC vs. Procore data silos. Entry motion: OAS Procore + ACC connector eliminates their top-3 complaint.
- **CIO / CDO at ENR-100 contractor**: owns enterprise architecture and vendor consolidation mandates. Budget authority $500K+. Receptive to a federated stack argument that reduces the vendor renewal surface without forcing Autodesk or Bentley off the desk. Entry motion: AMP + OAS as the platform-neutral coordination layer that sits above incumbents.
- **Project Director / Owner's Rep on capital programs**: responsible for schedule, cost, and risk visibility across 10–50 active projects simultaneously. Values federated data without forcing sub-contractors onto a mandated single platform. The OAS value proposition — "your subs keep using their tool; we connect it bidirectionally" — maps directly to this buyer's top procurement constraint. Entry motion: OAS pilot on a single mega-project, funded from the owner's digital delivery budget.
- **Director of Sustainability / ESG Officer**: owns embodied carbon reporting under EU EPBD, CA Buy Clean Act AB 2446, and annual sustainability disclosures. Currently buying One Click LCA or Tally per project, producing disconnected point-in-time snapshots. Needs automated EPD ingestion and real-time compliance monitoring across the portfolio. Entry motion: Fractal with EPD ingest as the ESG automation layer, positioned as eliminating the annual One Click LCA consulting sprint.

RFP trigger events:

- **Owner-mandated ISO 19650 BIM/CDE requirements** on public infrastructure tenders (Highways England, GSA, state DOTs, Network Rail). ISO 19650 mandates a structured Common Data Environment — AMP as the ISO 19650-compliant spatial information layer is the direct response. Part 6 (health and safety information management) released in 2025 strengthens the AMP + NEOS dual-layer argument for large publicly funded projects.

- **Post-merger portfolio consolidation**: when a PE-backed GC acquires a regional competitor, the CDO inherits two incompatible Procore instances, two CDE contracts, and two model management workflows. OAS federation (and NEOS for the governance integration) is the de-merger rationalization play.
- **Mega-project mobilization**: capital programs above $500M reliably trigger a digital delivery planning phase where the owner mandates a data management standard before the GC mobilizes. This is the highest-urgency entry point — the budget is pre-approved, the timeline is fixed, and the VDC/BIM team is standing by.
- **ESG reporting deadlines**: annual sustainability reports, EU SFDR mandatory disclosures for EU real estate funds, SEC climate disclosure rules for US-listed project owners. These create hard calendar deadlines that convert the Fractal pilot conversation into a signed SOW.

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5. Rejection-risk mitigations

Paper and email-based culture (sub/trade workforce): A significant portion of the construction workforce — especially subcontractors and trades — operates on paper, WhatsApp, and email. Sub resistance to "yet-another-login" is the top-3 Procore complaint and the primary reason mandated platform adoption fails on GC projects. Mitigation: OAS connectors are designed to meet the sub at their existing tool of choice (Procore, Fieldwire, email-triggered approvals) rather than forcing onboarding. The AMP spatial record is maintained by the GC's VDC team; subs contribute via read-only views and email-triggered approval workflows requiring no account creation.

Subcontractor adoption friction: ENR-100 GCs routinely report that sub networks won't adopt owner-mandated platforms at scale. Any PrimusNeo engagement that depends on sub-side adoption is structurally at risk. Mitigation: scope OAS to absorb the sub's existing tool of choice bidirectionally. The sub uses Procore or Fieldwire or email; OAS federates that data into the GC's AMP scene and NEOS approval flows without the sub ever touching PrimusNeo directly.

ROI attribution difficulty: Each construction project is unique; there is no clean A/B comparison. CFOs discount vendor-published ROI reports (Procore's benchmark studies are routinely challenged in procurement). Mitigation: instrument project-level outcomes via Fractal from day one — RFI cycle time (baseline vs. post-implementation), rework dollar value (tied to model/field drift events), and financial closeout speed. Present at the 90-day review with project-level evidence, not portfolio-level benchmarks.

Integration lock-in with Procore and Autodesk: Most ENR contractors are mid-cycle on multi-year Procore and ACC contracts. Adding PrimusNeo requires a clear and provably non-disruptive OAS story. Mitigation: publish and certify bidirectional Procore webhook integration (Procore's own marketplace mechanism) and IFC/RVT model ingestion from ACC with write-back. Integration tax = the primary reason pilots stall at POC; eliminating it requires a live reference deployment the champion can demo to procurement, not a spec sheet.

Price sensitivity on thin GC margins (2–4% net): A GC running 3% net on $200M revenue has $6M in profit. A $500K platform engagement is 8% of net profit — a hard conversation with a CFO who has never seen a construction software vendor justify their price at that level. Mitigation: pilot structure with documented-ROI path before portfolio expansion. The Atom or Molecule deployment SKU on a single project at $90K–$320K, with Fractal capturing baseline metrics, produces the CFO evidence needed to sign the portfolio engagement.

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6. Regulatory and compliance considerations

EU EPBD (recast 2024) — Digital Building Passports: The recast Energy Performance of Buildings Directive mandates Digital Building Passports — per-building records covering energy performance, materials, and lifecycle data — as part of EU member-state EPBD implementation. Fossil-fuel boiler subsidies ended January 2025. This is an immediate OAS + AMP demand driver for any EU construction client or EU-listed real estate fund: the Digital Building Passport is exactly the federated spatial + systems record that AMP maintains and OAS populates from heterogeneous building systems.

California Buy Clean Act / AB 2446: Global Warming Potential limits on structural steel, concrete rebar, flat glass, and insulation on California public works are enforceable as of January 2025. AB 2446 targets a 40% reduction in material carbon intensity by 2035. This drives One Click LCA and EC3 adoption — and creates the Fractal opportunity for automated EPD ingestion and real-time compliance reporting that replaces the annual consulting sprint. Fractal's embodied carbon model, fed by OAS-ingested EPD data, is the construction-native answer to this compliance obligation.

UK Net Zero Carbon Buildings Standard (NZCBS) v1 / PAS 2080: NZCBS v1 launched in early 2026, integrating with PAS 2080 and establishing upfront carbon (A1–A5 life-cycle stages) limits per square metre. From 2025, publicly funded projects must be designed without fossil-fuel heating. Fractal's Scope 3 + embodied carbon analytics, combined with AMP's as-built spatial record, create the data layer that satisfies PAS 2080 documentation requirements across the design, construction, and operation phases.

ISO 19650 BIM: The global information-management benchmark for the built environment is mandatory in the EU and UK (since 2016) and increasingly required in GCC and India. Part 6, covering health and safety information management, was released in 2025. AMP serves as the ISO 19650-compliant spatial information layer — the structured CDE that the standard requires — while NEOS provides the approval-flow and audit-trail governance that ISO 19650 Part 2 (delivery phase) mandates for CDE workflows.

LEED / BREEAM / Green Star ongoing obligations: Embodied carbon documentation, energy modeling outputs, and operational data submissions tied to credit thresholds create persistent data obligations through building operations. These are not one-time compliance events — they require the live connection between the construction-phase AMP scene and the FM-phase OAS-connected IWMS that PrimusNeo's full stack provides.

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7. Steward and service partner notes

PLAN Systems (AMP steward): The differentiator in construction is PLAN's federated spatial record capability across design, field, and FM phases. PLAN delivers AMP spatial environment buildout services — CAD ingest, scene authoring, IFC/RVT model ingest — and is the primary certified implementation partner for AMP on large construction and infrastructure engagements. PLAN royalty: 15% of AMP line-item license revenue.

OmniOne / Green Earth Vision (NEOS steward): Delivers the ETHOS governance design methodology — the charter-authoring and council-structuring work that converts a GC's change-control email culture into a signed, auditable NEOS governance flow. OmniOne leads governance design workshops and charter authoring across mid-to-large construction engagements. NEOS royalty: 18% of NEOS line-item license revenue.

OAS Web4 (OAS steward): Authors the certified Procore, ACC (Autodesk Construction Cloud), Bentley iTwin, and Trimble Connect bidirectional connectors that are the primary OAS value driver in construction. OAS Web4 leads connector development and holds the technical relationship with Procore's webhook marketplace and Autodesk's APS (Autodesk Platform Services). OAS royalty: 15% of OAS line-item license revenue.

Escherbridge (Fractal Engine steward): Delivers Fractal portfolio analytics, the RFI predictive triage model, and the embodied carbon EPD-ingest pipeline for Buy Clean Act / EPBD compliance. Escherbridge is the primary delivery partner for Fractal ML engineering and the construction-specific ontology. Fractal royalty: 18% of Fractal Engine line-item license revenue.

SI channel partners:

Firm	Role	Day rate (USD)
McKinsey Capital Projects	Strategic program advisory on mega-projects; PrimusNeo digital delivery positioning	$10K–$25K/day (Senior Partner)
Accenture Industry X — Built Environment	Mid-market digital transformation; PrimusNeo full-stack implementation	$5K–$15K/day (MD/Principal)
AECOM Digital	ISO 19650 CDE + BIM management advisory; AMP deployment on infrastructure programs	$1.2K–$3K/day (Digital Delivery Director)
Arup Digital	Sustainability + carbon analytics; Fractal embodied carbon model integration	$1.5K–$4K/day (Associate/Senior Associate)
Autodesk implementation partners	ACC connector deployment, BIM workflow integration	$150–$250/hr
Trimble / Bentley authorized distributors	Field integration, iTwin connector deployment	$175–$300/hr

A typical 6-month mid-tier AEC digital transformation including PrimusNeo deployment runs **$300K–$1.5M all-in** when a full SI channel partner is involved — which positions PrimusNeo's Hex deployment SKU ($400K–$900K) competitively inside that envelope, with license revenue beginning in parallel.

8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Construction engagement model

Energy — Engagement Model

> ***Vertical:*** *Energy (utilities, oil & gas, renewables, grid/ISO, DERs, microgrids, carbon markets)* ***Research brief:*** `../research/energy.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Energy is PrimusNeo's highest regulatory tailwind and longest sales cycle simultaneously. Four compounding friction windows define the buyer's pain and the defensible quote band.

ADMS/DERMS software stack cost: Full-scale Advanced Distribution Management System deployments run **$5M–$50M software-only** per investor-owned utility, before hardware and SI services (DOE ADMS Advisory). GE Vernova GridOS (ADMS + DERMS) packages for IOUs: **$8M–$20M over five-year terms**. Schneider EcoStruxure ADMS and Siemens Spectrum Power: comparable at **$6M–$15M** per IOU. This is the market PrimusNeo enters above — as interoperability and governance fabric, not as displacement.

AVEVA PI System data historian: At scale (1M+ tags for large refineries and grid operators), AVEVA PI System subscription + support runs **$500K–$5M/yr**. Cognite Data Fusion at supermajor scale: **$2M–$8M ARR**. AspenTech APM/Mtell entry: **$250K–$1M+/yr** for mid-size process industry. These historians are not going away; the PrimusNeo entry is to layer governance, interop, and intelligence above them.

NERC CIP compliance burden: CIP-015-1 (Internal Network Security Monitoring) alone carries **$13.2M/yr industry-wide** across ~1,636 registered entities (Federal Register, September 2024). CIP-013 supply-chain mandates impose structured vendor assessment and attestation workflows across every BES Cyber System. CIP-003-9, effective April 2026, extends supply-chain controls to low-impact BES assets — expanding the addressable compliance burden further. This is direct willingness-to-pay for auditable provenance: NEOS.

Carbon + cross-border compliance overlay: EU CSRD mandatory double-materiality disclosures began FY2024 for large EU-listed energy firms and their non-EU subsidiaries. EU CBAM definitive phase starts January 2026 — requiring certified embedded-carbon data per shipment with financial penalties for non-compliance. Article 6.4 (Paris Agreement ITMOs, finalized at COP29) demands cryptographically auditable provenance chains to prevent double-counting across Nationally Determined Contributions. These three compliance drivers are structural WTP signals for Fractal Engine + NEOS attestation.

Aggregate friction window per engagement tier: Small microgrid/community solar operator: $300K–$800K/yr in incumbent tooling + compliance cost. Mid municipal utility / IPP: $2M–$6M/yr. Large IOU/ISO-RTO/supermajor: $15M–$60M/yr. PrimusNeo's 5–15% capture rate against these windows is the defensible quote band.

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2. PrimusNeo positioning per layer in energy

The critical positioning frame: PrimusNeo enters energy as **interoperability and governance fabric above the incumbent ADMS and PI System stacks** — not as a displacement play. Attempting to displace GE Vernova, Schneider, or AVEVA triggers the vendor-concentration objection and the OT cyber risk rejection immediately. Positioning above them as the governance, interop, and intelligence fabric sidesteps both while creating genuine additional value.

AMP — Spatial OS (steward: PLAN Systems): AMP's role in energy is the **plant, substation, and grid spatial scene** — the operational ground layer for gas turbine sites, offshore platforms, transmission substations, and microgrid nodes. PLAN Systems brings industrial-grade spatial scene authoring for energy assets. Where manufacturing has CAD + MES, energy has P&ID + SCADA + historian; AMP is the layer that ties them into a navigable, governed, offline-capable operational scene. Entry use cases: offshore platform digital twin for APM, substation spatial model for NERC CIP-007 change-management events, and microgrid site visualization for DER operator teams. Tie Town [TX] demonstrates the reference architecture for multi-site energy deployments.

NEOS — Governance (steward: OmniOne): NEOS is the **auditable governance fabric** for three energy-specific compliance domains. First, NERC CIP-013 supply-chain attestation: NEOS councils structure vendor assessment workflows and produce signed, replayable governance logs that survive a NERC audit. Second, CSRD and CBAM: NEOS provides the governance event log that underlies double-materiality ESG disclosure and embedded-carbon certification per shipment — not just a dashboard, but a signed audit trail. Third, REC and Article 6.4 ITMO provenance: NEOS + Fractal together produce the cryptographically auditable lineage chains that hourly granular REC tracking (EnergyTag) and Article 6.4 ITMOs require to prevent double-counting. OmniOne's ETHOS methodology for governance design applies directly to the multi-party council structures that cross-utility supply chains and carbon registries require.

OAS — Interoperability Fabric (steward: OAS Web4): OAS is the **IT/OT bridge for FERC Order 2222 and the multi-system integration fabric** that connects ADMS, SCADA, EMS, DERMS, ETRM, carbon registries, and ISO market interfaces without requiring any incumbent to be displaced. FERC Order 2222 mandates DER market participation down to 100 kW aggregated, requiring interoperable metering, telemetry, and settlement-data flows across heterogeneous utility and aggregator systems — this is OAS's exact problem space. OAS Web4 authors the certified DNP3, IEC 61968/70 (CIM), and ICCP connectors that utility architecture review boards require; "open APIs" without certified OT protocol connectors fail at the technical evaluation stage. Carbon registry integrations (M-RETS, PJM-EIS GATS, Article 6.4 registry APIs) are OAS connector-class integrations. AutoGrid Flex / Uplight DERMS connector: OAS. ETRM connectivity (Allegro/ION Group, OpenLink): OAS.

Fractal Engine — Intelligence (steward: Escherbridge): Fractal is the **predictive APM, carbon MRV, and REC hourly-granularity intelligence layer**. Escherbridge's analytics depth makes Fractal the highest-value layer for energy operational telemetry: turbine degradation forecasting, transformer health scoring, renewable curtailment prediction, grid stability signal generation. For carbon markets: Fractal provides the MRV (Measurement, Reporting, Verification) intelligence layer that feeds NEOS attestation — from raw telemetry to verified tCO2e, with audit trail attached. For hourly granular RECs and Article 6.4 ITMOs, Fractal's time-series intelligence resolves the generation and consumption matching at hourly (or sub-

hourly) granularity that 24/7 carbon-free energy (CFE) commitments require. Fractal also positions above Watershed ($50K–$500K/yr flat carbon SaaS) and Persefoni ($55K–$250K/yr) as the operational-intelligence layer that produces MRV data rather than simply ingesting spreadsheet exports.

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3. Engagement tiers — Small / Mid / Large

Small: single microgrid or community solar (<10 MW) — $250K–$800K over 3 years

Profile: Community microgrid operator, municipal solar co-op, rural electric cooperative with a DER portfolio, developer deploying a <10 MW island microgrid.

Bundle: OAS Small + NEOS Small + Fractal Small (carbon MRV entry). AMP optional for microgrid with a physical site. Duet or Triad.

Worked quote:

Line item	Annual	Notes
OAS Small	$36K/yr	5 integrations: SCADA, DER controller, carbon registry, ISO market API, weather
NEOS Small	$10K/yr	1 council: DER operator attestation + REC provenance governance
Fractal Small	$40K/yr	<5 sources, carbon MRV + generation forecasting
Triad bundle −18%	($16K/yr)	Net license $70K/yr
Y1 services (~2x OAS-weighted blend)	$140K	Connector buildouts: SCADA DNP3, ISO API, registry
Y1 upkeep (Silver SLA, 22%)	$15K	Starts Y2
Infra pass-through + 12% mgmt	$24K/yr	Edge node via Tie Town reference or cloud

Y1 total: ~$230K | **3-yr TCO:** ~$410K–$550K

Friction anchor: $150K–$400K/yr in incumbent tooling (standalone SCADA dashboard, separate carbon SaaS subscription, manual REC registry submission, consultant hours for NERC CIP-003-9 compliance if grid-tied). PrimusNeo captures 15–20% of that window at Small tier.

Mid: municipal utility or mid-size IPP (100–500 MW) — $2.5M–$10M over 3 years

Profile: Municipal electric utility (100K–500K customers), mid-size independent power producer with mixed renewable + thermal portfolio, regional co-op needing FERC Order 2222 compliance for aggregated DER dispatch.

Bundle: OAS Mid + NEOS Mid + Fractal Mid + AMP Mid (substation/plant spatial). Full Stack preferred; Triad minimum.

Worked quote:

Line item	Annual	Notes
AMP Mid	$120K/yr	5 substation/plant sites, PLAN Systems spatial scenes
NEOS Mid	$55K/yr	3 councils: CIP-013 supply-chain, REC attestation, FERC Order 2222 DER governance
OAS Mid	$150K/yr	20 integrations: ADMS, EMS, DERMS, ETRM, SCADA, DNP3 RTUs, ISO market, carbon registry, weather, AMI
Fractal Mid	$200K/yr	10 sources: telemetry historian, generation data, AMI, carbon MRV, market settlement
Full Stack −25%	($131K/yr)	Net license $394K/yr
Y1 services (~2x OAS-weighted blend)	$788K	DNP3 connectors $200K, IEC 61968 CIM $180K, ETRM $120K, AMP substation scenes $200K, Fractal MRV models $88K
Y1 upkeep (Gold SLA, 25%)	$99K	Starts Y2; Gold is the regulated-industry default
Infra pass-through + 12% mgmt	$120K/yr	Multi-site cloud + optional Tie Town edge colo

Y1 total: ~$1.3M–$1.4M | **3-yr TCO:** ~$2.8M–$3.8M

Friction anchor: Mid-size utility spending $1.5M–$4M/yr across ADMS SI services, AVEVA PI System subscription ($300K–$800K/yr), NERC CIP compliance overhead ($500K–$1.5M/yr at this scale), and carbon MRV software ($100K–$300K/yr). PrimusNeo captures 8–12% of the combined friction window.

Large: IOU, ISO-RTO, or supermajor (1 GW+) — $18M–$80M over 3 years

Profile: Investor-owned utility with 1M+ customers, regional ISO/RTO managing multi-GW grid, oil and gas supermajor with global portfolio needing CSRD + CBAM compliance, large renewables developer managing 5+ GW.

Bundle: Full Stack (all four layers). SI channel mandatory: Black & Veatch or Accenture Resources for OT integration; McKinsey Energy Solutions or Deloitte E&R for regulatory/IRP testimony support. Deployment SKU: Tessellation or Lattice.

Worked quote (IOU, ~$1.5B revenue, 800K customers):

Line item	Annual	Notes
AMP Large	$550K/yr	20 substations + 2 major plants, PLAN Systems multi-site federation
NEOS Large	$250K/yr	CIP-013 multi-vendor attestation councils, CSRD governance log, REC Article 6.4 provenance
OAS Large	$900K/yr	60+ integrations: legacy SCADA, ADMS, EMS, DERMS, ETRM, DNP3, IEC 61968/70, ICCP, ISO market, AMI, carbon registry, workforce, ERP
Fractal Large	$800K/yr	25+ sources: historian, AMI, ISO settlement, generation, carbon MRV, grid event telemetry
Full Stack −25%	($625K/yr)	Net license $1.875M/yr
Y1 services (~2x OAS-weighted blend, Tessellation SKU)	$4.5M–$6M	SI channel: Black & Veatch OT advisory $1.5M–$2M, OAS Web4 OT connectors $1M–$1.5M, PLAN spatial build $800K–$1.2M, Escherbridge Fractal MRV models $600K–$900K
Y1 upkeep (Platinum SLA)	$600K	Named support pod, 15-min P1, weekly review
Infra pass-through + 12% mgmt	$800K–$1.5M/yr	Multi-region, multi-site, compliance-grade cloud; Tie Town edge reference

Y1 total: ~$8M–$10M | **3-yr TCO:** ~$20M–$32M

Friction anchor: IOU-scale ADMS rollout SI services alone (Black & Veatch / Accenture, 500K-customer utility): **$8M–$25M over 24–36 months**. NERC CIP compliance overhead at IOU scale: $2M–$5M/yr. AVEVA PI System at IOU scale: $1M–$3M/yr subscription. CSRD/CBAM compliance infrastructure: $500K–$2M/yr. Combined friction window: $12M–$35M/yr. PrimusNeo Y1 as % of friction: 23–80% — but positioned correctly, the IOU is paying for the governance-and-interop layer above incumbents, not displacing them, so the math is additive to their existing ADMS spend, not against it. Defensible capture rate: 5–12% of total IOU technology + compliance spend.

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4. Industry-specific procurement signals

Buyer personas and decision authority:

Buyer	Role	Authority	Primary concern
Chief Grid Officer / VP Grid Operations	Grid modernization sponsor	Operational approval; PUC testimony	Reliability, NERC CIP, OT security
CIO / VP Technology	Platform + vendor governance	Architecture review, vendor risk	IT/OT convergence, integration complexity, vendor concentration
COO	Overall business case	Budget authority	TCO, risk, regulatory deadlines
Head of Trading / VP Commercial	ETRM, carbon, VPP	Revenue-side technology	Settlement accuracy, ISO market connectivity
Chief Sustainability Officer	Carbon MRV, ESG disclosure	Board-level mandate	CSRD/CBAM accuracy, auditability, Article 6.4
Regulatory Affairs / General Counsel	PUC filings	Rate-case approval	Regulatory defensibility, vendor attestation records

Procurement pathway: Contracts at $5M+ require PUC pre-approval or inclusion in an Integrated Resource Plan (IRP) filing before procurement can proceed. RFPs are issued via regulated competitive bidding with 3–6 month technical evaluation phases. Short-listed vendors submit detailed compliance architecture documentation before any commercial discussion proceeds.

Sales cycle: IOUs and ISO-RTOs: 18–36 months. Municipal utilities and mid-size IPPs: 6–18 months. Supermajors (O&G, large renewables): 12–24 months.

Budget cycle alignment: IOU capital plans are set annually through IRP and budget processes. Missing the capital plan submission window by one quarter delays a deal by twelve months. This is the single most preventable deal-loss in energy; sales motion must align to IRP calendar, not calendar year-end.

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5. Rejection-risk mitigations

NERC CIP / OT cyber risk: Software that touches BES Cyber Systems must be classified, change-managed, and assessed under CIP-007 (systems security management), CIP-010 (configuration management), and CIP-013 (supply-chain risk management). Mitigation package: ship a documented CIP compliance architecture before any technical evaluation; CIP-005 jump-server and MFA topology for all vendor remote access; CIP-013 supply-chain assessments pre-populated for OAS Web4 (OAS) and PLAN Systems (AMP) as the OT-touching components; published security attestation pack available day one of evaluation. No documented CIP compliance architecture equals automatic rejection.

Vendor concentration risk: IOUs and ISOs enforce multi-vendor policies post-SolarWinds and post-Volt Typhoon disclosures. A platform simultaneously touching SCADA, ADMS, DERMS, and carbon accounting triggers vendor concentration review automatically. Mitigation: PrimusNeo enters above incumbent stacks, not as displacement. ADMS stays with GE Vernova, Schneider, or Siemens. AVEVA PI System stays as the historian. PrimusNeo is the interoperability and governance fabric above them. The federated steward structure reinforces this: OAS (OAS Web4), AMP (PLAN Systems), NEOS (OmniOne), Fractal (Escherbridge) are distinct stewards — not a single vendor.

OT integration complexity: Legacy distribution SCADA and EMS run proprietary protocols: DNP3, IEC 61968/70 (CIM), ICCP, Modbus. Governance or interop platforms claiming "open APIs" without certified OT protocol connectors fail at utility architecture review boards. Mitigation: OAS Web4 authors certified DNP3, IEC 61968/70, and ICCP connectors for OAS as first-class connector-library entries. Reference architecture documentation available with every technical evaluation package.

Regulatory approval delays: Rate-recovery technology spend faces 18–36 month PUC evidentiary cycles. Deals stall in hearings if cost-benefit testimony is thin. Mitigation: third-party ROI validation commissioned before the rate case opens, not after. McKinsey Energy Solutions or Deloitte E&R as the validation firm adds regulatory credibility. PrimusNeo's friction-cost anchor methodology (sourced to DOE, NERC, Federal Register) provides the evidentiary backbone.

Budget cycle mismatch: Missing the IRP capital plan window delays deals by a full year. Mitigation: annual IRP planning calendar awareness is a sales-team requirement for IOU accounts. Enterprise deal entry must be sequenced to the preceding IRP cycle, not to the buyer's expressed urgency.

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6. Regulatory + compliance considerations

FERC Order 2222 — DER Market Participation: Requires all ISOs and RTOs to allow aggregated DER participation in wholesale markets down to 100 kW minimum aggregated capacity. Mandatory metering, telemetry, and settlement-data interoperability is the direct tech-spend trigger. OAS is architecturally positioned for this integration layer; NEOS governs the multi-party attestation council for the DER aggregation.

NERC CIP — Critical Infrastructure Protection: Standards CIP-002 through CIP-015-1 govern BES Cyber System security. CIP-015-1 (Internal Network Security Monitoring): $13.2M/yr industry-wide compliance burden across ~1,636 registered entities. CIP-013 supply-chain risk management: structured vendor assessment and attestation for BES Cyber System vendors — NEOS supply-chain council template applies directly. CIP-003-9 (effective April 2026): extends supply-chain controls to low-impact BES assets, expanding the addressable compliance population and the NEOS WTP signal.

EU CSRD / CSDDD — Sustainability Disclosure: Large EU-listed energy firms and their non-EU subsidiaries face mandatory double-materiality ESG disclosures from FY2024 reporting. The EU Omnibus simplification package (September 2025) reduced scope to ~900 companies at EU level but preserved core emissions disclosure requirements. NEOS provides the auditable governance event log underpinning these disclosures; Fractal provides the MRV intelligence layer feeding the disclosure data.

EU CBAM — Carbon Border Adjustment Mechanism: Definitive phase begins January 2026. Energy exporters with embedded-carbon products entering the EU must submit certified embedded-carbon data per shipment, backed by third-party verified MRV. Financial penalties apply for non-compliant or unverifiable data. Fractal MRV + NEOS attestation together produce the certified embedded-carbon data trail CBAM requires.

Article 6.4 — Paris Agreement ITMOs: COP29 (November 2024) finalized Article 6.4 rules for Internationally Transferred Mitigation Outcomes. Carbon market participants need cryptographically auditable provenance chains to prevent double-counting across NDCs. NEOS governance events + Fractal lineage attestation map directly to this requirement. Near-term land opportunity: carbon market participants seeking Article 6.4-compliant provenance infrastructure before 2027 compliance deadlines.

State PUC Rate Recovery: IOU technology investments in ADMS, DERMS, cybersecurity, and compliance tooling are recoverable through rate cases in most jurisdictions. California CPUC has effectively mandated ADMS deployment for large IOUs; New York REV and Illinois CEJA contain analogous provisions. Rate-payer funding de-risks vendor revenue through the procurement cycle but introduces the 18–36 month PUC approval timeline.

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7. Steward + service partner notes

PLAN Systems (AMP steward): Leads spatial scene authoring for power plant and substation deployments. PLAN's industrial-grade 3D interface components and Hexon file standard (Apache-2.0) are the foundation for energy site twins. PLAN Systems subcontracts for spatial environment buildout and sensor/OT integration on AMP-containing engagements; royalty rate 15% of AMP license line-item.

Escherbridge (Fractal Engine steward): Their analytics depth is the strategic differentiator for energy APM and carbon MRV. Escherbridge delivers Fractal ML engineering, custom predictive maintenance models (turbine degradation, transformer health), and carbon MRV intelligence pipelines. They are the primary delivery partner for any Fractal-containing energy engagement with significant telemetry volume or regulatory MRV requirements. Royalty rate 18% of Fractal license line-item; Escherbridge invoices services directly when delivering.

OmniOne (NEOS steward): ETHOS governance methodology design for NERC CIP-013 supply-chain attestation councils, REC/Article 6.4 provenance governance structures, and CSRD double-materiality disclosure governance. OmniOne leads governance design workshops and charter authoring on NEOS-containing engagements. Royalty rate 18% of NEOS license line-item.

OAS Web4 (OAS steward): Authors certified DNP3, IEC 61968/70 (CIM), and ICCP connectors as first-class OAS connector-library entries. Leads OT connector buildout and carbon registry integration (M-RETS, PJM-EIS GATS, Article 6.4 registry APIs) for energy engagements. Royalty rate 15% of OAS license line-item; connector services revenue accrues to OAS Web4 directly.

Tie Town [TX] reference deployment: The PrimusNeo reference architecture for multi-site energy deployments. Edge node colocation at Tie Town ($1,200–$2,500/month for 1U) provides the AMP + OAS edge cache reference for buyers evaluating offline-first capability for remote generation or substation sites where cloud connectivity is unreliable.

SI channel: - **McKinsey Energy Solutions:** Senior partner day rate $9,500–$12,000. Grid strategy, IRP testimony, energy transition regulatory advisory. Use for IOU/ISO rate-case preparation and PUC evidentiary support. - **Deloitte E&R (Energy & Resources):** Director/Principal $4,000–$6,000/day. ADMS program management office, utility technology roadmap advisory. - **Accenture Resources:** Managing

Director $3,500–$5,500/day. OT/IT convergence, SAP IS-U integration. Strong for O&G supermajor engagements with SAP backbone. - **Black & Veatch:** ADMS/OT Principal $2,500–$4,000/day. Deep utility engineering + OT advisory credential; turnkey ADMS rollout programs for mid-size IOU ($8M–$25M over 24–36 months). Primary SI recommendation for utility OT integration scope. - **Worley:** Process/Digital Lead $2,000–$3,500/day. O&G, renewables EPC + digital transformation. Strong for offshore and LNG asset engagements where Worley is already the EPC. - **IBM Consulting:** Energy Practice Lead $2,500–$4,000/day. Maximo APM integration, grid data management.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Energy engagement model

Finance — Engagement Model

> ***Vertical:*** *Finance (banks, asset managers, insurers, exchanges, tokenization platforms, RWA)* ***Research brief:*** `../research/finance.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Financial institutions face four compounding friction windows that make the PrimusNeo value case quantifiable before the first meeting.

Information-layer reference price. Bloomberg Terminal at $31,980/seat/year (multi-seat negotiated $28,320; enterprise 50+ at $18K–$20K/seat) sets the market expectation for what FIs will pay annually to access structured, auditable, institutional-grade data. This is the anchor for any conversation about what a governance + interop + intelligence fabric is worth per-seat at an asset manager or exchange. Refinitiv Eikon ($22K/seat), FactSet ($12K/seat), and Snowflake/Databricks enterprise ($25K–$40K/yr for 200TB, up to $10M/yr at 500-analyst scale) extend the band.

Tokenization friction. Securitize and Tokeny (T-REX) charge $50K–$150K setup plus **0.5%–2.5% of issuance value** for smart-contract deployment and compliance wrapping. A $30M issuance runs $200K–$1M all-in. White-label platform build costs range $50K–$150K upfront with custom development from $150K. Ongoing compliance and transfer-agent retainers layer on top. For a mid-tier asset manager running $500M of tokenized AUM, issuance friction alone is $2.5M–$12.5M over the tokenization program lifecycle.

Custody cost. BitGo charges approximately 60 bps/year on AUC (5 bps/month on AUC >$100K). Fireblocks and Anchorage are bilateral-negotiated at $1M+ AUC minimum. Across a $500M–$2B notional portfolio, annual custody cost runs $3M–$12M — costs that OAS multi-custodian integration directly reduces by eliminating bespoke custodian-by-custodian API build and maintenance.

Regulatory premium. MiCA non-compliance fines: €15M or 3% of annual revenue, whichever is higher. For a mid-sized EU-registered CASP with $100M revenue, 3% = €3M at risk annually. OCC Interpretive Letter 1184 (May 2025) means national banks providing crypto custody must demonstrate controls on-demand; banks will pay **$200K–$500K+/year** per platform that produces those attestations. SAB 122 (January 2025) rescinds SAB 121's on-balance-sheet liability treatment, removing the primary accounting disincentive for bank custody — expanding the addressable market for compliant custody infrastructure significantly. FCA implicit compliance tax runs £100K–£500K/year in incremental monitoring and reporting under crypto-asset registration and stablecoin frameworks.

Per-customer friction windows (worked):

Customer tier	Annual friction estimate	Source composition
Boutique manager / family office <$1B AUM	$200K–$500K	TaaS SaaS + legal/structuring + custody at 60 bps/yr on $100M AUC + KYC/AML ($30K–$90K) + governance software
Regional bank / asset mgr $1–50B AUM	$1.5M–$4M	White-label tokenization + issuance legal + Fireblocks/BitGo custody on $500M–$2B notional + Chainalysis ($150K–$300K) + GRC/IRM ($300K–$600K) + data fabric ($150K–$400K)
G-SIB / top-20 asset mgr / major exchange	$8M–$25M+	Custom tokenization + interop fabric ($2M–$5M build) + multi-custodian + Snowflake/Databricks at scale ($2M–$10M/yr) + full Chainalysis + TRM ($400K–$1M/yr) + ServiceNow IRM ($600K–$1.5M/yr) + Big-4 transformation

The 10–25% capture band on these friction windows yields the PrimusNeo defensible quote range per tier.

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2. PrimusNeo positioning per layer in finance

AMP — lighter use; deferred or scoped to specific facilities

AMP's primary value proposition in finance is operational and place-anchored: trading floor coordination, secure vault facilities, data center walk-through governance. These are real use cases — particularly for regulated operational control rooms where "this approval was issued from this location by this signed identity at this moment" is a compliance requirement — but they are **not the lead** in a finance engagement. Most finance buyers do not have a spatial-twin problem as their primary friction. AMP is typically deferred to Year 2 or scoped as an add-on for specific facilities within a larger NEOS+OAS+Fractal deployment. Finance is the one vertical where the Full Stack upsell to include AMP commonly happens in Y2–Y3, not at contract signature.

NEOS — the primary killer layer for finance

NEOS is the layer that directly solves the compliance, governance, and attestation requirements that have become binary procurement requirements following MiCA, OCC Letter 1184, and SAB 122. In the buyer's vocabulary:

- **To a CRO:** NEOS is the programmable approval-and-attestation fabric that produces the immutable governance logs MiCA, OCC Letter 1184, and SAB 122 now make non-optional.
- **To a Head of Digital Assets:** NEOS is the issuance approval council, KYC policy council, and transfer-agent attestation flow layer that replaces the current patchwork of email chains, legal memos, and manually-assembled audit trails.
- **To a tokenization program lead:** NEOS is the RWA governance layer that gives every tokenized asset a durable, replayable, chain-anchored provenance record — the compliance artifact that Securitize and Tokeny charge 0.5–2.5% of issuance value to produce without a durable governance trail.

NEOS positions above Diligent ($12K–$60K+/yr board portal) and Loomio ($1.2K–$7.2K/yr discussion tool) because it delivers on-chain auditability, contribution accounting, and programmable attestation — none of the governance portals do all three. In the regulated FS context, the on-chain auditability alone is worth the premium; Diligent produces PDFs, NEOS produces cryptographically verifiable event logs.

OAS — the other primary killer layer for finance

Finance is the vertical where OAS's pitch as "what MuleSoft is for Web2 enterprise integration, OAS is for the Web2+Web3+identity+custody reality" lands with the most force. No incumbent iPaaS covers multi-custodian (Fireblocks, BitGo, Anchorage) + multi-chain (EVM, Solana, Cosmos, Hedera) + legacy core-banking (Temenos, Finastra, Murex, Calypso) + regulatory feed integration in a single certified connector library. MuleSoft starts at $80K–$250K+/year for the enterprise license without finance-specific chain connectors; real large-enterprise MuleSoft runs $500K–$2M+/year before implementation. OAS covers the Web3+custody+legacy-core surface area MuleSoft doesn't, at parity or below for the Web2 integrations MuleSoft does cover.

For G-SIBs and top-tier asset managers, OAS's multi-custodian-native federation engine is the explicit answer to the vendor-concentration rejection risk: the federated 4-steward structure IS the multi-vendor answer, and OAS is the layer that operationalizes it.

Fractal Engine — regulatory, risk, and AML signal intelligence

Fractal positions against Palantir Foundry (entry $250K–$500K/year commercial, $1M–$10M+/year for large deployments) and Snowflake/Databricks (strong for raw data warehousing but no native governance-event signal types). In finance, the Fractal differentiator is that NEOS governance events and OAS integration events are **first-class signal types** — meaning the CRO dashboard, the regulatory audit log, and the AML signal model all run on the same fabric that produced the governance events, not on a separately assembled data pipeline.

Chainalysis Reactor runs $40K/year single-seat; enterprise KYT $100K–$500K+/year. Fractal is not a direct AML-chain-intelligence replacement for Chainalysis — it is the layer that aggregates Chainalysis signals, on-chain governance events, and off-chain transaction context into a unified risk model. The combination is more defensible than either alone, and Fractal's pricing is scoped per the data-source and intelligence-model tiers below, not as a per-chain AML overlay.

The canonical finance bundle: NEOS + OAS + Fractal (Triad)

The default finance bundle is the Triad: NEOS + OAS + Fractal Engine. AMP is deferred to Y2. The Triad bundle discount is 18% off single-layer sum per the pricing model. This is the configuration that:

1. Produces the immutable governance logs (NEOS) required by MiCA, OCC 1184, and SAB 122
2. Integrates across custodians, chains, and legacy core-banking (OAS) without requiring rip-and-replace of Temenos/Finastra/Murex/Calypso
3. Converts governance and integration events into regulatory reporting, risk signals, and AML intelligence (Fractal)

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3. Engagement tiers — Small / Mid / Large

Tier 1 — Small: Boutique manager / family office (<$1B AUM)

Profile: Boutique hedge fund, family office, or early tokenization platform. Primary drivers: basic KYC-council governance for investor onboarding, single-chain or dual-chain tokenization, limited legacy-core integration.

Bundle recommendation: NEOS Small + OAS Small Duet. Fractal deferred to Y2 unless the CRO is driving AML analytics requirements from day one.

Friction anchor: $200K–$500K/year in TaaS SaaS ($72K–$180K), legal/structuring ($100K–$250K), custody at 60 bps/year on $100M AUC (~$60K), and KYC/AML ($30K–$90K). OCC Letter 1184 attestation premium: $200K/yr for a compliant platform producing on-demand attestations.

Worked quote:

```
NEOS Small:    $12K/yr   (midpoint $6K–$18K; single org, 1–2 councils)
OAS Small:     $42K/yr   (midpoint $24K–$60K; <10 integrations:
                           2 chain connectors + custody API + KYC/IdP + core banking)
Duet bundle:   -10%  →   $48K/yr license

Y1 services:   ~1.0x NEOS + ~2.5x OAS blend → $113K
  (NEOS governance design + charter authoring: $60K;
   OAS connector buildout 4 connectors at avg $18K ea: $72K,
   connector bundle 10% off → $65K; identity bridging: $20K)
Y1 upkeep:     Silver SLA, 22% of license → $11K (starts Y2)
Infra:         $1.5K/mo → $18K/yr
```

Y1 total: ~$179K 3-yr TCO: ~$370K (Y2–Y3: license $48K + upkeep $11K + infra $18K/yr, services taper to ~$20K/yr)

Friction anchor check: $179K Y1 ÷ $300K friction midpoint = **60% capture in Y1**, justified because NEOS displaces legal memo + email-chain governance directly and OAS displaces manual custodian API maintenance. At Y1=60% capture the buyer is still clearly positive. 3-yr TCO of $370K sits well inside the research brief's $350K–$900K small-tier band.

Capture rate: 10–20% of the OCC attestation premium ($200K–$500K/yr) alone = $20K–$100K/yr recurring justification, separate from the integration cost displacement.

Tier 2 — Mid: Regional bank or asset manager ($1–50B AUM)

Profile: Regional bank entering digital assets, mid-cap asset manager launching tokenized fund, or specialized exchange. Primary drivers: MiCA compliance, multi-custodian custody integration, KYC policy governance, transfer-agent attestation, regulatory reporting.

Bundle recommendation: NEOS Mid + OAS Mid + Fractal Mid Triad. This is the default finance engagement configuration.

Friction anchor: $1.5M–$4M/year — white-label tokenization ($150K–$500K), issuance legal ($200K–$600K), Fireblocks/BitGo custody on $500M–$2B notional, Chainalysis ($150K–$300K), GRC/IRM ($300K–$600K), data fabric ($150K–$400K), Big-4 SI ($500K–$1.5M). MiCA non-compliance at 3% of revenue = $3M/yr at $100M revenue.

Worked quote:

```
NEOS Mid:       $60K/yr   (midpoint $30K–$90K; multi-council:
                           issuance approval, KYC policy, transfer-agent attestation)
OAS Mid:       $165K/yr   (midpoint $80K–$250K; 20+ integrations:
                           Fireblocks + BitGo + 3 chains + Temenos core + KYC/AML feed
                           + regulatory reporting feed + legacy reporting DB)
Fractal Mid:   $225K/yr   (midpoint $100K–$350K; 10–15 data sources:
                           custody events + chain events + governance events +
                           AML signals + risk reporting)
Triad bundle:   -18%  →   $369K/yr license

Y1 services:   ~0.75x NEOS + ~2.5x OAS + ~1.5x Fractal blend:
               NEOS governance design + charters + council instantiation: $130K
               OAS connector buildout (8 connectors avg $35K ea, 5+ bundle
               discount 15%): $238K; identity bridging + custody wiring: $80K
               Fractal: ontology mapping + pipelines (10 sources × $40K): $400K;
               regulatory model + audit-trail buildout: $100K
               Total services Y1: ~$948K
Y1 upkeep:     Gold SLA (regulated FS default), 25% of license → $92K (starts Y2)
Infra:         Mid cloud reference $6K/mo + chain gas reserve $1K/mo → $84K/yr
```

Y1 total: ~$1.4M 3-yr TCO: ~$3.0M (Y2–Y3: license $369K + upkeep $92K + infra $84K/yr, services taper to ~$150K/yr by Y3)

Friction anchor check: $1.4M Y1 ÷ $2.75M friction midpoint = **~51% value share in Y1** — this is above the 10–25% ongoing capture band because Y1 includes one-time connector buildout. On a steady-state basis: $369K/yr license ÷ $2.75M friction = **13.4% value share**, squarely inside the 10–25% finance capture band. 3-yr TCO of $3.0M sits at the midpoint of the research brief's $1.5M–$5M mid-tier band.

Regulatory modulator: MiCA compliance exposure (+10%), procurement maturity (+10%) = +20% modulator applies, supporting quoting at the upper-mid range rather than the midpoint.

Tier 3 — Large: G-SIB / top-20 asset manager / major exchange

Profile: Global systemically important bank, top-20 asset manager with $1T+ AUM, major exchange launching tokenization infrastructure, or RWA platform at institutional scale. The canonical worked example from `layers/neos.md §7.2` is the reference — extended here with the full finance-vertical context.

Bundle recommendation: NEOS Large + OAS Large + Fractal Large Triad, with AMP Large optionally added in Y2 for operational facilities (trading floors, vault/data-center governance). This matches the `neos.md §7.2` example of a $1.5T AUM global asset manager launching a tokenized fund product.

Friction anchor: $8M–$25M+/year. Components: custom tokenization + interop fabric build ($2M–$5M); multi-custodian (Fireblocks + Anchorage + BitGo) custody management infrastructure; Snowflake/Databricks at analyst scale ($2M–$10M/yr); full Chainalysis + TRM ($400K–$1M/yr); ServiceNow IRM ($600K–$1.5M/yr). MiCA non-compliance at 3% revenue = $45M+ at a firm this scale. OCC Letter 1184 attestation overhead $500K+/yr. Legal + ops cost of manual compliance workflows $1–2M/yr at this scale. **Combined friction window: $5M–$10M/yr defensible capture surface.**

Worked quote (canonical, per `neos.md §7.2`):

```
NEOS Large:     $280K/yr   (multi-entity regulated FS premium;
                            issuance approval councils, KYC policy councils,
                            transfer-agent attestation flows, regulatory
                            reporting attestations)
OAS Large:      $620K/yr   (10+ chains + Fireblocks + Anchorage + Snowflake +
                            legacy core Temenos/Murex + FIX gateway + regulatory
                            reporting feeds)
Fractal Large:  $750K/yr   (regulatory + risk analytics + audit-trail
                            intelligence; AML signal aggregation; CRO dashboards)
Triad bundle:   -18%  →   $1.35M/yr license

Y1 services:   ~1.4x blend → $1.9M
  (OAS: federation engine deployment $400K + 10 connector bundle $600K;
   NEOS: governance design $120K + 5 charters $60K + council instantiation $60K
         + operator training $60K;
   Fractal: ontology mapping $100K + regulatory model validation $150K +
            audit-trail attestation buildout $100K + CRO dashboards $80K;
   remaining services to $1.9M blend)
Y1 upkeep:     Platinum SLA, custom % → $400K (starts Y2)
Infra:         $300K–$800K/yr (multi-region, multi-custodian gateway,
               chain gas reserves across 10+ chains)
```

Y1 total: ~$3.5M–$4.0M 3-yr TCO: ~$8M–$11M

Friction anchor check: $1.35M/yr license ÷ $7M friction midpoint = **~19% value share** — defensible inside the 10–25% finance capture band per the pricing model. 3-yr TCO of $8M–$11M sits squarely inside the research brief's $8M–$25M+ large-tier band. The MiCA alone ($45M+ non-compliance exposure) makes this quote defensible at any C-suite table.

AMP Y2 add-on: Trading floor, vault operations, and data-center governance scope typically added at $350K–$700K/yr AMP Large (partial deployment) in Y2, bringing the Full Stack annual license to approximately $1.7M–$2.05M/yr at bundle rates.

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4. Industry-specific procurement signals

Decision makers by tier:

Tier	Primary Champion	Tech Sign-off	Compliance Veto	Entry Price for Procurement
Small	CIO / COO	CTO	—	>$250K (informal at small)
Mid	Head of Digital Assets	CDO / CTO	CRO	>$500K (formal procurement)
Large	Head of Digital Assets + C-suite sponsor (typically COO or CFO)	CTO + CDO committee	CRO + Legal + Compliance	>$500K; full RFP process

At the G-SIB tier, expect a full C-suite committee process with regulatory affairs as a non-optional voice. The Head of Digital Assets is typically the internal champion, but the CRO has explicit veto authority over any platform that cannot produce attestations on the timetable regulators require.

Sales cycle guidance: - Small (<$1B AUM): 2–4 months, informal process - Mid ($1–50B AUM): 6–12 months; formal RFP likely above $500K - Large (G-SIB, top-20 AM, major exchange): 12–24 months; multiple RFP stages, security review, legal review, and often a paid pilot (discovery sprint) before substantive RFP

Gating requirements before any substantive RFP at mid or large:

1. SOC 2 Type II (US default; mandatory for mid/large)
2. ISO 27001 (EU/APAC; mandatory for MiCA-scope deployments)
3. Annual third-party penetration test (results must be shareable under NDA)
4. BCP/DR documentation
5. GDPR / data residency mapping (for any EU data subject exposure)
6. Reference clients in regulated FS (at least one publicly nameable reference at comparable scale)

Missing SOC 2 Type II or ISO 27001 is an automatic disqualification at mid and large. Both are funded as central PrimusNeo investments, not per-engagement costs.

Procurement enters at >$500K. Below this threshold, the Head of Digital Assets or CIO can typically approve without a formal procurement process. Above it, procurement introduces benchmarking requirements — typically against MuleSoft, ServiceNow IRM, and Securitize/Tokeny, which is where the friction-cost anchoring in section 1 becomes the sales tool.

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5. Rejection-risk mitigations

Rejection reason	Mitigation
Regulatory ambiguity — Legal/compliance veto until local regulator gives written guidance	Publish multi-jurisdictional compliance posture per layer: MiCA (NEOS attestation logs map to CASP requirements; OAS cross-border identity portability); OCC Letter 1184 (NEOS + Fractal on-demand attestation architecture); SAB 122 (OAS multi-custodian integration removes the custody-accounting friction); FCA crypto-asset registration (NEOS governance log architecture).
Custody model mismatch — Must map to self-custody MPC/HSM, qualified custodian, or bank-grade sub-custodian	OAS is multi-custodian-native: certified connectors for Fireblocks, BitGo, and Anchorage are first-party; sub-custodian relationships with bank-grade MPC/HSM providers are supported through the OAS connector library.
Legacy core integration gap — Temenos / Finastra / Murex / Calypso require certified FIX/API connectors	Publish certified connectors for Temenos T24 / Transact, Finastra Fusion, Murex MX.3, and Calypso as first-class OAS connector library entries; include FIX 4.2/4.4/5.0 gateway support for exchange and trading-desk integrations. These are OAS Web4-authored connectors, which is where their connector-engineering depth shines.
Open-source / IP ownership risk — Unaudited smart contracts, no formal vuln disclosure, no IP indemnity	IP indemnity capped at 2× annual fees for the affected layer per the commercial terms in `01-pricing-model.md §6`. Smart-contract audit is bundled as a deliverable for any issuance-adjacent NEOS deployment. Formal CVE disclosure program.
Vendor concentration / lock-in risk — Multi-custodian, multi-chain required at G-SIB	The federated 4-steward structure (OmniOne + Escherbridge + PLAN + OAS Web4) IS the multi-vendor answer. No single entity controls the stack. Data export is non-negotiable per the commercial terms. Source escrow available for AMP/OAS components above $500K ARR via Iron Mountain.
Missing security certs — SOC 2 + ISO 27001 non-negotiable at mid/large	SOC 2 Type II + ISO 27001 funded as central PrimusNeo investments. Not a per-engagement cost. The Tie Town [TX] reference deployment provides the operational proving ground for the security posture before any formal audit engagement.

6. Regulatory + compliance considerations

MiCA (EU, fully applicable December 2024). NEOS produces the auditable issuance records and governance event logs that CASP licensing requires. OAS provides the cross-border identity portability architecture that MiCA's cross-member-state CASP passporting requires. Non-compliance fines (€15M or 3% of revenue) make both layers non-optional for any EU-registered CASP. MiCA CASP licensing build-out itself runs €50K–€100K for small operators; capital requirements add €50K–€150K depending on activity class. PrimusNeo's MiCA compliance architecture should be documented per-layer and available as a regulatory briefing before any mid or large engagement progresses to RFP.

OCC Interpretive Letter 1184 (May 2025). National banks may now provide crypto custody (fiduciary and non-fiduciary) and use sub-custodians, but must demonstrate controls on-demand. NEOS + Fractal attestation logs are designed to produce these on-demand demonstrations: governance events are

cryptographically signed, chain-anchored, and replayable; Fractal aggregates them into the audit-trail intelligence package the OCC examiner reviews. Banks will pay $200K–$500K+/year for platforms that make this attestation production routine rather than a manual legal exercise.

SEC SAB 122 (January 2025). Rescinds SAB 121's on-balance-sheet liability treatment for crypto custody. This removes the primary accounting disincentive for bank custody, expanding the addressable market for compliant custody infrastructure. OAS's multi-custodian integration layer is directly positioned to capture this expanded TAM: as more banks enter crypto custody, the integration complexity between custodians, core banking, and regulatory reporting increases — which is exactly the surface area OAS addresses.

FCA + GDPR cross-border. Any deployment touching UK data subjects requires FCA crypto-asset registration posture and GDPR/UK GDPR data residency mapping. OAS's regional gateway architecture and NEOS's data-residency-aware signing model support data-residency compliance without architectural rework.

SOC 2 + ISO 27001 as gating. Both funded centrally. No mid or large finance engagement can proceed to substantive RFP without these in hand. The Tie Town [TX] reference deployment is the proving ground; certification scope covers the full PrimusNeo stack including all four layers.

CRA (EU Cyber Resilience Act) / CSRD overlap. For exchange operators and large asset managers with EU reporting obligations, CSRD (sustainability reporting) and CRA (cybersecurity for connected products) create additional compliance overhead that Fractal's regulatory model layer can address: NEOS governance events become CSRD audit artifacts; OAS integration events support CSRD data collection pipelines.

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7. Steward + service partner notes

OmniOne (NEOS steward) is the primary delivery partner for governance design in regulated financial services. NEOS's ETHOS methodology engagements — issuance approval council design, KYC policy council structuring, transfer-agent attestation flow architecture — are OmniOne's domain. OmniOne earns 18% of NEOS line-item license revenue as steward royalty, and services revenue accrues directly when they deliver governance design work. In finance, expect OmniOne's involvement to be heaviest in the mid/large tier where regulatory attestation design is non-trivial.

Escherbridge (Fractal steward) is the primary delivery partner for Fractal regulatory and risk models. In finance, this means: CRO dashboard architecture, regulatory reporting model design, AML signal aggregation ontology, and the audit-trail attestation buildout that OCC Letter 1184 requires. Escherbridge earns 18% of Fractal line-item license revenue. For regulated FS engagements, Escherbridge also contributes to the regulatory model validation scope ($60K–$250K per engagement) that is a separately scoped deliverable for any high-risk Fractal deployment.

OAS Web4 (OAS steward) leads multi-chain and custody connector development — this is where their connector-engineering depth is most directly applied. The Fireblocks, BitGo, Anchorage, Temenos, Finastra, Murex, Calypso, and FIX gateway connectors are OAS Web4-authored first-party connectors. For

finance, OAS Web4's custody and core-banking connector library is the primary competitive differentiator against MuleSoft and Boomi, neither of which has a certified multi-custodian + multi-chain connector set. OAS Web4 earns 15% of OAS line-item license revenue and leads connector SOW delivery.

SI channel — Big-4 engagement sizes and rates. For mid and large finance engagements, SI partners are typically involved in implementation, change management, and regulatory advisory. Rates from the research brief: Deloitte FSI (GSA) $2K–$3K/day senior consultant, $3K–$5K/day principal/director, $5K–$10K/day partner (digital-asset specialty). EY Blockchain, PwC Digital Assets, KPMG Capital Markets, and Accenture Capital Markets operate in similar bands. Engagement sizes: tokenization readiness assessment $150K–$400K (8–16 weeks); full tokenization program delivery $1M–$5M (6–18 months); MiCA/regulatory compliance implementation $500K–$2M; enterprise data fabric + governance transformation $3M–$12M. At mid tier, SI channel engagement adds $500K–$1.5M to the total program cost alongside PrimusNeo fees. At large tier (G-SIB), $3M–$12M SI engagement is typical — this is a revenue opportunity for PrimusNeo through SI referral and co-sell arrangements, not a competitive threat.

Tie Town [TX] reference. The Tie Town deployment provides the operational proof-of-concept for the full stack — governance councils, interop fabric, and intelligence layer — running in a real federated environment. In regulated FS sales cycles, having a live reference deployment that can be demonstrated under NDA is a prerequisite for advancing past the initial security and compliance review. Tie Town is that reference.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Finance engagement model

Governance — Engagement Model

> ***Verticals:*** *Public-sector / Civic Government AND DAO / Web3 Governance / Cooperatives / Federations* ***Research brief:*** `../research/governance.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Governance spans two structurally different buyer archetypes but both pay enormous friction costs for inadequate coordination infrastructure.

Civic anchor. GovTech incumbents extract rents far beyond their delivered value. Tyler Technologies' EnerGov permitting platform ranges $50K–$2.7M in implementation fees alone, with annual maintenance of 18–22% on top. The Illinois statewide Tyler contract ballooned to $250M with documented delays — a cautionary tale for every large-city CIO. Granicus' Citizen Experience Cloud runs $25K–$500K/yr at state level. Esri statewide enterprise agreements hit $2M–$10M+/yr. The FedRAMP Moderate authorization gate alone costs $400K–$900K and 12–24 months of staff time before a single user can be provisioned on a federal contract. None of these incumbents offer durable contribution accounting, holonic federation, or auditable signed-event logs. They are licensing access to convenience, not solving coordination. The combined annual friction exposure for a mid-size city managing GovTech contracts, participatory democracy tools, board portals, and identity systems runs $500K–$2M/yr; for a state-level agency, $5M–$20M+/yr.

DAO anchor. Economic value in DAO governance concentrates almost entirely in treasury management services rather than the tooling itself — Snapshot, Tally, and Aragon are free at the protocol layer, which creates a dangerous impression that governance is "solved." It isn't. Llama charges ~$2.5M/yr for Aave treasury management. Karpatkey and Steakhouse Financial provide comparable services at $200K–$500K/yr base plus performance upside, with Steakhouse's Lido engagement on public record at ~$322K/yr. The generic treasury advisory range is 0.25–0.75% of AUM annually — a $500M treasury implies $1.25M–$3.75M in governance-adjacent services spend. Meanwhile, governance attacks have destroyed $4B+ across DAO treasuries since 2016. The friction cost is twofold: the services overhead and the existential risk of attack. A top-50 DAO has a defensible friction window of $2M–$8M/yr combining services + security risk.

Pricing model capture band: 15–30% of governance services spend per the `01-pricing-model.md` summary table. This is the highest capture band across all verticals, reflecting the clarity of the value case (governance fabric is the prerequisite for everything else) and the direct line-item displacement NEOS provides against incumbent services spend.

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2. PrimusNeo positioning per layer in governance

Governance is the vertical where **NEOS is the center of gravity**. Every other layer is additive; NEOS is non-negotiable.

NEOS is the durable governance fabric above Snapshot signal and Tally execution for DAOs, and the inter-departmental decision-rights system that bridges Tyler/Granicus civic platforms and participatory democracy tools (Decidim, Pol.is) for public-sector buyers. The framing differs by buyer archetype but the product is the same: programmable approval-and-attestation fabric, contribution accounting, holonic federation, signed auditable event logs. The federated four-steward structure is not just an org chart — it is the explicit answer to GovTech vendor lock-in (Tyler, Granicus, Esri) because no single entity owns the stack. That structural credibility matters to both city councils nervous about another $250M Tyler overrun and DAO stewards nervous about protocol-level centralization.

OAS handles the connective tissue in both contexts. Civic: cross-agency interoperability, FHIR-adjacent identity data flows, SSI Verified Credentials for constituent identity, multi-jurisdiction data exchange. DAO: cross-chain bridges (EVM, Cosmos, Solana, Hedera), off-chain identity federation, decentralized storage backends. OAS is the mandatory co-purchase with NEOS for any multi-entity governance deployment.

AMP (Spatial OS) has lighter but real application in governance. Civic: emergency management coordination (spatial scene plus governance approvals in a single view), urban planning participation (spatial twin of a development project attached to participatory consultation records), physical-asset permitting (approved asset location signed to a governance event). DAO/cooperative: less relevant at the spatial layer; the cooperative's "site" is more metaphorical. AMP enters governance engagements at Mid and Large tiers where there is a physical-infrastructure component.

Fractal Engine converts the governance stack into operating intelligence. Civic: policy effectiveness analytics (which council decisions produced what budget outcomes), grant allocation optimization, constituent service telemetry, cross-agency spend intelligence. DAO: treasury policy backtesting, delegate effectiveness scoring, protocol KPI intelligence, on-chain signal analytics — "the Llama analytics layer without the Llama retainer" is the exact positioning.

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3. Engagement tiers — Small / Mid / Large

Small: 1 town / 1 cooperative / 1 small DAO (<$10M treasury)

Bundle: NEOS Small + OAS Small (Duet). AMP and Fractal rare at this tier.

Civic worked example — Town of ~15,000 residents

A small municipality replacing a patchwork of Google Forms, email chains, and Loomio for council decision-making and resident engagement.

Line item	Amount
NEOS Small license (2 councils, ~80 members)	$12K/yr
OAS Small license (<5 integrations, identity + existing CivicEngage portal)	$30K/yr
Duet bundle discount (-10%)	($4,200)
Net license Y1	$37,800/yr
Y1 services (~0.8x NEOS, 2x OAS blend = ~1.3x license)	$49K
Y1 upkeep Silver SLA (22%)	$8,300 (starts Y2)
Infra pass-through (~$400/mo)	$4,800/yr
Y1 total	**~$91K**
3-yr TCO	**~$195K**

Friction anchor: $50K/yr in Granicus/GoVocal/Loomio stack + 0.5 FTE staff time on manual governance record-keeping (~$40K value) = $90K/yr direct friction. Capture rate: $91K ÷ $90K = ~1x — justified because the compliance audit trail NEOS produces eliminates the town clerk's manual reconciliation labor and creates a defensible record the incumbents don't provide.

DAO worked example — Small protocol DAO, $6M treasury

A recently-launched protocol DAO outgrowing Snapshot + a Gnosis Safe. Treasury is above the $5M CC-NC free threshold, triggering PCL.

Line item	Amount
NEOS Small license (1 council, ~120 token holders with delegation)	$14K/yr
OAS Small license (2 chains: Ethereum mainnet + L2; Snapshot integration)	$28K/yr
Duet bundle discount (-10%)	($4,200)
Net license Y1	$37,800/yr
Y1 services (governance design workshop + on-chain setup)	$55K
Y1 upkeep Bronze SLA (18%)	$6,800 (starts Y2)
Infra (chain gas reserve + NEOS coordination server)	$1,200/mo
Y1 total	**~$107K**
3-yr TCO	**~$220K**

Friction anchor: a minimally-scoped Llama retainer for a DAO this size runs $300K–$800K/yr. PrimusNeo at $107K Y1 is 13–36% of that friction cost, delivering the governance fabric that makes the DAO's own treasury management disciplined enough to avoid the attacks that cost comparable DAOs their entire treasury.

3-yr TCO band: $75K–$300K

Mid: City 100K–500K / state agency / mid DAO ($10M–$500M treasury)

Bundle: NEOS Mid + OAS Mid + Fractal Mid (Triad). AMP Mid added when physical-asset or emergency management scope present.

Civic worked example — City of 220,000 residents

A Sun Belt city replacing OpenGov (budgeting), a legacy board portal (Nasdaq Boardvantage), and Decidim (citizen participation) with a unified governance fabric. Chief Innovation Officer is the champion; CIO co-signs.

Line item	Amount
NEOS Mid license (8 councils: city council + 5 committees + 2 working groups, ~300 members)	$65K/yr
OAS Mid license (20 integrations: existing Tyler EnerGov, Salesforce CRM, Entra SSO, two state data exchanges, two public-facing APIs)	$150K/yr
Fractal Mid license (8 data sources: budget actuals, council decision logs, constituent service telemetry, grant pipeline)	$140K/yr
Triad bundle discount (-18%)	($64,350)
Net license Y1	$290,650/yr
Y1 services (~1.2x blend: NEOS governance design + OAS connector buildout + Fractal pipeline)	$350K
Deployment SKU (Hex: 2-site, full-stack)	$550K
Y1 upkeep Gold SLA (25%)	$72,700 (starts Y2)
Infra pass-through + 12% mgmt (multi-region, StateRAMP-compliant cloud)	$120K/yr
Y1 total	**~$1.31M**
3-yr TCO	**~$3.2M**

Friction anchor: Tyler EnerGov at $300K/yr maintenance + Granicus state-bundle equivalent $150K/yr + Boardvantage $80K/yr + Decidim integrator $60K/yr = $590K/yr direct stack spend. Add 1.5 FTE managing the patchwork at $180K fully-loaded = $770K/yr total friction. Capture rate: $290K license ÷ $770K = **38% value share on license alone**. Full Y1 total reflects implementation cost; Y2 normalized annual (license + upkeep + infra) = $483K, capturing 63% of the annual friction stack while delivering a consolidated, auditable platform the incumbents cannot replicate.

DAO worked example — Mid protocol DAO, $120M treasury

A mid-tier DeFi protocol replacing Snapshot + ad-hoc Llama retainer with a structured governance fabric, retaining Snapshot as the signal layer while NEOS codifies intent and accountability above it.

Line item	Amount
NEOS Mid license (multi-council: protocol council + grants committee + risk committee, ~450 active delegates)	$75K/yr
OAS Mid license (Ethereum mainnet + 3 L2s + Fireblocks integration + Snapshot API + Tally API)	$140K/yr
Fractal Mid license (treasury policy backtesting + delegate-effectiveness dashboard + protocol KPI signals)	$150K/yr
Triad bundle discount (-18%)	($66,150)
Net license Y1	$298,850/yr
Y1 services (ETHOS governance design + on-chain council instantiation + treasury analytics pipeline)	$360K
Y1 upkeep Silver SLA (22%)	$65,750 (starts Y2)
Infra (multi-chain gas reserve + NEOS coordination server + Fractal compute)	$80K/yr
Y1 total	**~$739K**
3-yr TCO	**~$1.8M**

Friction anchor: Llama mid-DAO retainer $300K–$800K/yr. Add Karpatkey-comparable treasury advisory at 0.3% of $120M AUM = $360K/yr. Total addressable friction: $660K–$1.16M/yr. Capture rate: $299K license ÷ $910K midpoint friction = **33% value share** — NEOS is the governance fabric beneath treasury services, not a replacement for them. DAOs that buy PrimusNeo may retain a treasury services firm; PrimusNeo is cheaper, more auditable, and non-competitive with those services.

3-yr TCO band: $1.2M–$5M

Large: State-level / federal agency / G20 megacity / top-50 DAO ($500M+ treasury)

Bundle: Full Stack (NEOS Large + OAS Large + Fractal Large + AMP Mid/Large for civic). Deployment SKU: Lattice or Tessellation.

Civic worked example — State agency or G20 megacity

A state department of technology modernizing inter-agency governance, citizen identity, and policy analytics across 15 sub-agencies. Buyer: CIO + Chief Data Officer + Governor's Innovation office.

Line item	Amount
NEOS Large license (federated governance, 15 sub-agency councils + state legislative interfaces, 800+ members)	$320K/yr
OAS Large license (50+ integrations: legacy ERP, 15 sub-agency APIs, federal data exchanges, SSI Verified Credentials fabric, 2 GovCloud environments)	$850K/yr
Fractal Large license (20+ data sources: policy telemetry, constituent services, inter-agency budget flows, grant outcomes)	$750K/yr
AMP Mid license (emergency operations center spatial coordination, 4 EOC sites)	$180K/yr
Full Stack bundle discount (-25%)	($525K)
Net license Y1	$1,575K/yr
Y1 services (~1.3x blend; SI co-delivery with Deloitte GPS at GSA rates)	$2.05M
Deployment SKU (Lattice: 6+ sites)	$2.5M
FedRAMP Moderate authorization (amortized 50% to Y1)	$400K
Y1 upkeep Platinum SLA (quoted)	$450K (starts Y2)
Infra (FedRAMP-compliant multi-region cloud, GovCloud)	$400K–$600K/yr
Y1 total	**~$7.0M–$7.2M**
3-yr TCO	**~$17M–$22M**

Friction anchor: Esri statewide EA $3M–$8M/yr + Tyler ERP $2M–$5M/yr + Granicus state suite $400K–$1M/yr + existing identity fabric $500K–$1M/yr + FedRAMP certification costs across incumbent tools $1M–$3M/yr = $7M–$18M/yr combined. PrimusNeo consolidates and replaces the governance fabric and analytics layers. Capture rate: $1.575M license ÷ $12.5M friction midpoint = **~13% value share** — inside the 15–30% governance capture band and defensible given the compliance complexity.

DAO worked example — Top-50 DAO, $800M+ treasury

An Aave-class protocol DAO formalizing governance at scale: structured councils, treasury policy framework, multi-chain execution, delegate accountability analytics.

Line item	Amount
NEOS Large license (federated governance: protocol council + risk committee + grants council + security council + delegate registry, 1,200+ active participants)	$350K/yr
OAS Large license (full multi-chain: Ethereum, Arbitrum, Optimism, Polygon, Base, Solana; Fireblocks + Anchorage custody integration; Snapshot + Tally APIs; Boardroom aggregation)	$900K/yr
Fractal Large license (treasury policy backtesting, delegate-effectiveness scoring, protocol risk metrics, cross-chain liquidity signals)	$600K/yr
Full Stack bundle discount (-25%)	($462,500)
Net license Y1	$1,387,500/yr
Y1 services (ETHOS council instantiation + multi-chain setup + treasury analytics models)	$1.5M
Y1 upkeep Gold SLA (25%)	$347K (starts Y2)
Infra (multi-chain gas reserve, multi-region NEOS server, Fractal GPU compute)	$250K–$400K/yr
MiCA CASP compliance posture doc (EU treasury reach)	$150K one-time
Y1 total	**~$3.3M–$3.5M**
3-yr TCO	**~$8M–$11M**

Friction anchor: Llama at $2.5M/yr (the public Aave benchmark). Add treasury advisory at 0.4% of $800M AUM = $3.2M/yr. Combined: $5.7M/yr. PrimusNeo at $1.4M/yr license = **24% value share** — the upper range of the governance capture band, justified because NEOS directly competes with the governance services component of the Llama retainer, not just the analytics overlay.

3-yr TCO band: $10M–$60M+

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4. Industry-specific procurement signals

Civic buyer personas. CIO / CITO triggers on modernization mandates, legacy EOL, and federal efficiency pressure — they control the budget line and sign the contract. Chief Innovation Officer often sponsors pilot engagements outside the standard IT procurement lane; shorter cycle (90–120 days for a pilot SOW vs 6–18 months for full RFP), but must convert to CIO sponsorship for production scale. City Manager / County Administrator triggers on constituent service pressure and council mandates — the political buyer who creates the urgency a CIO needs to move. State agency: Deputy Secretary / Assistant Commissioner level with CIO co-approval.

DAO buyer personas. Steward Council or Foundation controls the treasury and signs the governance proposal. The procurement motion is a formal governance proposal (Research Forum post → Snapshot temperature check → Tally on-chain vote → execution) with a 2–8 week cycle including discussion period.

This is faster than any civic procurement and does not require cooperative vehicle enrollment. The risk is governance attacks during the proposal period — PrimusNeo's own governance design (ETHOS methodology, multisig defaults) should be demonstrable.

Cooperative / federation personas. Board vote, usually 30–60 days for an expedited resolution; longer if bylaws require member ratification. Federation steward boards are similar. These buyers often qualify for CC-NC free tier (nonprofit, small revenue) — be explicit about the de-minimis thresholds early to avoid wasted discovery.

Cooperative purchasing vehicles for civic. Sourcewell is the **first enrollment target** — 50,000+ member agencies, pre-competed vehicle, eliminates the 6–18 month full RFP cycle for a substantial share of local government buys. NASPO ValuePoint (all 50 states + territories, strong for cloud/SaaS) is the second. GSA MAS (SIN 518210C for IT; OASIS+ for professional services) unlocks federal and state/local cooperative purchasing authority. State-specific schedules (TX DIR, CA DGS, NY OGS) are non-negotiated pre-priced access within those states and are worth pursuing once any anchor reference exists in that state.

Budget cycle alignment. Federal FY ends September 30; decision pressure builds June–August. State and local vary but most compress into Q3–Q4. Missing a budget cycle is a 12-month pipeline delay regardless of product fit. Calendar-aware outreach (initial engagement 6–9 months before the agency's fiscal year close) is required.

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5. Rejection-risk mitigations

Procurement red tape. New vendor not on an approved cooperative vehicle = automatic elimination in most civic RFPs. Sourcewell enrollment must happen before any serious civic pipeline engagement. Tyler/Granicus/Esri switching costs run $500K–$5M+; PrimusNeo cannot win a pure displacement argument — win by expanding scope (participatory democracy fabric, cross-agency federation, DAO governance capability) that incumbents don't cover at all.

FedRAMP gap. No ATO = disqualification for US federal and most state agencies. The FedRAMP 20x initiative (launched 2025) provides an agency-path acceleration route targeting ~90 days using pre-authorized infrastructure boundaries — pursue this path rather than the full JAB queue. Target FedRAMP Moderate authorization as the primary investment; High and DoD IL4/IL5 are phased based on pipeline revenue thresholds.

DoD security classification. IL2 is FedRAMP Moderate-equivalent ($400K–$800K). IL4 (CUI, contingency ops) and IL5 (NSS data) require DISA-authorized cloud hosting and controls well beyond FedRAMP Moderate. Phase: IL2 in Y1 → IL4 in Y2 when a defense-adjacent anchor client is signed. IL5 requires dedicated program investment.

DAO treasury security. Governance attacks are existential. NEOS ships with sane multisig defaults out of the box: 3-of-5 for treasuries below $10M, 4-of-7 for larger. Flash-loan-resistant voting (time-lock on proposal execution, snapshot voting power at proposal creation not vote time) and replay protection

(chain-specific signature binding) are non-negotiable defaults. Governance security posture documentation is produced per engagement for any DAO above $10M treasury.

Data sovereignty / GDPR. EU public sector requires data residency in the EEA. Multi-region OAS deployment with EU-hosted Fractal compute must be demonstrable to any EU civic buyer. This is not a future roadmap item — it must be available before any EU public tender engagement begins.

Proof of scale. Governments require 3–5 comparable references in evaluation. A lighthouse cooperative, a lighthouse mid-size city, and a lighthouse mid-DAO are the three reference deployment types required before pursuing large-tier civic or DAO engagements. Tie Town serves as the early reference but is not a civic government reference in the formal sense.

Budget cycle mismatch. Track municipal budget calendars actively. A contact made in November will not produce a procurement signature until after the next fiscal year starts. Engagement pacing must match.

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6. Regulatory + compliance considerations

Civic regulatory stack.

Framework	Scope	Cost to PrimusNeo
FedRAMP Moderate	US federal SaaS	$400K–$900K authorization; $100K–$300K/yr ongoing
FedRAMP High	Sensitive federal data	$2M–$5M+ authorization
StateRAMP / GovRAMP	US state + local (rebranded 2025)	$150K–$500K
DoD IL2	Non-sensitive DoD public data	$400K–$800K
DoD IL4	CUI + contingency ops	$800K–$2M+
DoD IL5	NSS data	$2M–$5M+
GDPR / EU NIS2	EU public sector	EEA data residency; DPA; DPIA; up to €20M or 4% global turnover in fines
CMMC Level 2–3	Defense-adjacent contractors	Level 3 requires C3PAO assessment: $500K–$3M+

DAO / Web3 regulatory stack.

MiCA CASP licensing (EU): required for EU-facing DAOs and crypto-asset service providers. Grandfathering period ends mid-2026; legal + licensing costs €200K–€1M+ depending on jurisdiction. PrimusNeo's NEOS governance audit trail and OAS cross-chain bridge architecture both support MiCA compliance posture documentation. SAB 122 (issued January 2025, replacing SAB 121) removes the requirement for custodians to hold crypto assets on their own balance sheet — meaningfully reducing custody compliance overhead for US DAO clients and their institutional counterparties.

Cross-cutting. EU public sector and EU-regulated DAOs share the data sovereignty requirement. Multi-region OAS deployment with EEA-resident data processing is the single architectural commitment that unlocks both markets simultaneously.

7. Steward + service partner notes

OmniOne is the primary delivery partner for NEOS in both civic and DAO contexts. Their ETHOS methodology — governance design, charter authoring, council structuring, contribution accounting — is the central differentiable asset. NEOS deployments without OmniOne ETHOS facilitation are technically possible but miss the highest-value part of the engagement. OmniOne earns 18% royalty on NEOS license revenue plus direct services revenue when they deliver the governance design work.

OAS Web4 leads OAS implementation and is the author of the closed-source provider connectors. For civic engagements: SSI Verified Credentials for constituent identity, cross-agency interoperability fabric, GovCloud-compatible API gateway. For DAO engagements: cross-chain bridges (EVM to Cosmos to Solana and back), multi-custodian integration (Fireblocks, Anchorage, BitGo), off-chain identity federation. OAS Web4 earns 15% royalty on OAS license revenue.

Escherbridge delivers Fractal Engine ML engineering, custom model training, and treasury/policy analytics. For civic: policy effectiveness analytics, grant allocation intelligence, constituent service telemetry. For DAO: treasury policy backtesting, delegate-effectiveness scoring, protocol risk metrics. Escherbridge earns 18% royalty on Fractal line-item license revenue.

Systems integrator channel for civic. Deloitte GPS ($200–$450/hr GSA schedule; senior $500–$700/hr), Accenture Federal Services ($190–$420/hr GSA MAS), Booz Allen Hamilton ($187–$376/hr validated GSA 2024). These firms carry the delivery capacity, pre-existing agency relationships, and FedRAMP program management experience required for state and federal engagements. PrimusNeo positions as the product the SI wraps — not competitive with the SI, which brings program management, organizational change, and agency credibility that PrimusNeo does not have at early stage.

DAO co-sell partners. Llama, Karpatkey, and Steakhouse Financial define the treasury services market. PrimusNeo does not compete with their treasury management service — NEOS is the governance fabric *beneath* their work, making it more auditable, more attack-resistant, and more institutionally credible. The co-sell motion: PrimusNeo + Llama (or Karpatkey) as a combined offering where the treasury services firm handles financial strategy and PrimusNeo handles governance infrastructure. Both parties benefit: the services firm gets a defensible audit trail; PrimusNeo gets the treasury services firm's client relationships.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Governance engagement model

Healthcare — Engagement Model

> ***Vertical:*** *Healthcare (hospitals, health systems, payers, pharma, biotech, public health, digital health)*
> ***Research brief:*** `../research/healthcare.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Healthcare is PrimusNeo's highest-friction, highest-value vertical. The integration tax alone is staggering at scale.

EHR implementation and integration spend:

- Epic full IDN deployment: **$50M–$1.2B all-in** (Trinity Health ~$800M, Northwell ~$1.2B, AdventHealth ~$660M)
- EHR implementation services at large IDNs: **$50M–$500M** over 3–7 years via Deloitte/Accenture/Huron/Chartis
- Redox interop fabric: floor of **$35K/month** (~$420K/yr), plus $0.50–$2.00/message — a $1.3M+ 3-year floor before volume
- HL7/FHIR custom integration at regional health systems: **$2M–$15M per major initiative**
- Clinical data platforms (Innovaccer, Health Catalyst, Arcadia): **$200K–$3M+/yr** for mid-to-large systems

Compliance gate costs:

- HITRUST i1 certification (minimum viable for enterprise RFPs): **~$70K**
- HITRUST r2 (expected for core platforms): **$100K–$500K** upfront; **$400K–$800K per 3-year cycle**
- HIPAA Security Rule overhaul (effective May 2026): MFA, universal PHI encryption, 24-hr breach, annual pen testing — material uplift to every vendor's compliance COGS

Analytics and population health:

- PMPM analytics: **$0.50–$3 PMPM** basic, **$5–$20 PMPM** population health, **$150–$200 PMPM** intensive care management
- Veeva Vault CRM (pharma): $1,500–$1,800/user/yr list; top-20 pharma **$5M–$10M+/yr**
- IQVIA data licenses: **multi-million-dollar** annual contracts for pharma analytics

The aggregate friction window for the vertical: An 800-bed regional system typically runs $5M–$15M over 3 years on interoperability alone (not counting the EHR license), plus $600K–$3M/yr on analytics platforms, plus $150K–$500K/yr on governance and consent. A national IDN running Epic adds $20M–$80M over 3 years on integration and data services above and beyond the EHR license itself. **PrimusNeo's capture target: 8–20% of the integration-and-analytics friction spend** (per `01-pricing-model.md §9`).

2. PrimusNeo positioning per layer in healthcare

The central positioning principle for this vertical: **"We wrap Epic, we don't replace it."** No hospital that spent $300M–$1.2B on Epic will entertain a rip-and-replace conversation. The winning motion wraps the EHR and compresses the integration tax.

AMP — Spatial OS: hospital operations and OR/ED command center

AMP is the OR/ED command center spatial layer — the GE HealthCare Command Center Digital Twin analog without forcing rip-and-replace of Epic or Cerner. GE Command Center covers ~500 hospitals and 55K+ beds globally; AMP's differentiation is offline-first operation, Hexon portability, and native composition with NEOS governance events and Fractal clinical-intelligence signals.

Healthcare-specific AMP use cases:

- **OR command center:** Surgical scheduling, room readiness, equipment asset tracking — spatial events become first-class Fractal signals (block utilization, turnaround forecasting)
- **ED throughput management:** Patient flow visibility attached to spatial positions (bed, bay, zone); alert surfaces to charge nurse dashboard
- **Facility/campus management:** HVAC, equipment maintenance, lockout/tagout — AMP + NEOS as the governed spatial operations layer

Pricing mechanic in healthcare: **per-bed or per-facility ELA**. For a 4-hospital regional system (800 beds total), AMP Mid at $80K–$250K/yr is positioned against GE Command Center's custom-enterprise pricing with a clear "same outcome, TEFCA-native, no EHR displacement" message.

NEOS — Governance: consent, IRB, data governance, EHDS-ready

NEOS is the consent + IRB + change-control governance layer that satisfies HIPAA, HITRUST i1/r2, and EU EHDS (Reg EU 2025/327) simultaneously. The CCO and Privacy Officer are the primary buyers here.

Healthcare-specific NEOS use cases:

- **Patient consent management:** Signed, auditable consent events per HIPAA and GDPR special-category health rules; DPIA workflow native to NEOS governance design
- **IRB governance:** Research protocol approvals, amendment tracking, multi-site coordination (hospital + university + sponsor) as nested councils
- **Data-sharing governance:** Multi-party data use agreements for TEFCA QHINs, state HIEs, payer-provider contracts — signed event log satisfies audit requirements
- **EHDS compliance:** EU EHR interoperability and security certification requirements (Jan 2026 EHR cert deadline; March 2029 priority data exchange mandate) are addressable via NEOS consent and governance schema — a direct new revenue opportunity for EU-operating health systems and pharma

NEOS pricing in healthcare: **per-covered-life** ($0.10–$1.00/covered-life/yr) for payer/population contexts; **per-council** ($1.5K–$8K/council/yr) for hospital governance; **enterprise governance ELA** for national IDNs. OmniOne is the steward and primary delivery partner for NEOS governance design using their ETHOS methodology.

OAS — Interoperability Fabric: TEFCA-native FHIR, the direct Redox differentiator

OAS is the TEFCA-native FHIR fabric that compresses the Epic/Cerner integration tax. This is PrimusNeo's most direct displacement vector in healthcare — Redox charges a $35K/month floor before any volume; OAS at Mid tier ($80K–$250K/yr) **undercuts Redox by 35–60%** at equivalent integration scope while adding QHIN-native federation and Web3 identity anchoring that Redox does not offer.

The OAS healthcare pitch to CIOs in one sentence: **"Epic FHIR R4 certified, App Orchard listed, TEFCA QHIN-native — and we cost less than Redox at scale."**

Healthcare-specific OAS use cases:

- **Epic FHIR R4 + Cerner FHIR connectors:** Fixed-fee $120K–$400K per EHR connector (includes HITRUST overhead per `03-services-catalog.md §2.2`); connector library reuse amortizes for subsequent customers
- **TEFCA QHIN participation:** OAS as the Common Agreement v2.1-compliant exchange fabric; 21,086 orgs live on TEFCA as of Jan 2025 — a massive TAM for interop-as-a-service
- **Claims and SDoH data ingest:** Payer claims feeds, social determinants data pipelines — OAS handles transport; Fractal handles population analytics
- **Pharmacy, lab, imaging integration:** Outbound/inbound HL7 v2 → FHIR transformation via OAS connector; replaces brittle in-house ETL

Critical path milestone: **OAS must achieve Epic FHIR R4 certification + App Orchard listing** before any large IDN sale is closeable. The 6–18 month Epic technical review is a known friction point — begin immediately.

OAS Web4 is the steward for OAS and leads connector development; they are the delivery partner for Epic/Cerner/Meditech/Athenahealth connector engineering.

Fractal Engine — Intelligence: population health, RWE, payer analytics, pharma

Fractal is the population health + RWE + operational analytics layer that doesn't require ripping out Innovaccer, Health Catalyst, or Arcadia. The canonical worked example is the Mid Fractal+OAS regional health system engagement documented in `layers/fractal-engine.md §7.1` — reproduced and extended in §3 below.

Healthcare-specific Fractal use cases:

- **Population health / value-based care:** Risk stratification, gap-in-care identification, utilization analytics — PMPM-priced for payer and VBC contexts
- **Real-world evidence (RWE) for pharma/biotech:** Retrospective cohort analysis, comparative effectiveness, safety signal detection using Escherbridge's public corpus + customer EHR data.

Fractal's per-research-query mechanic ($5–$500/query) is appropriate for pharma buyers who need episodic deep analytics rather than always-on population monitoring

- **OR/ED operational intelligence:** Fractal attached to AMP spatial events — block utilization forecasting, ED throughput prediction, workforce demand signals
- **Payer analytics:** Fraud, waste, and abuse detection; member risk scoring; network adequacy analytics at $0.50–$8 PMPM (well below the $15–$200 PMPM industry range)

EU MDR consideration: If Fractal outputs constitute clinical decision support (Class IIa/IIb), EU MDR conformity assessment applies — $20K–$80K per high-risk system, scoped into the engagement deliverable. Position Fractal outputs as **decision-support**, not autonomous clinical action, to maintain Class I/II SaMD pathway where possible.

Escherbridge is the steward and primary delivery partner for Fractal ML engineering and RWE model development; their specialty in population health and operational telemetry makes healthcare one of their highest-density vertical use cases.

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3. Engagement tiers — Small / Mid / Large

Small: Community Hospital (50–200 beds)

Profile: Critical access hospital or community hospital on Meditech Expanse or Cerner; limited IT staff; TEFCA adoption early but driven by state HIE mandates; analytics budget $200K–$600K/yr.

Recommended bundle: OAS Small + Fractal Small + NEOS Small (Triad)

Worked quote:

```
OAS Small:      $40K/yr   (TEFCA + state HIE + 2 external feeds, 5 integrations)
Fractal Small:  $45K/yr   (departmental utilization analytics, 3 data sources, <1TB)
NEOS Small:     $10K/yr   (consent management, 1 council, <50 clinical governance members)
                ──────
Subtotal:       $95K/yr
Triad -18%:     $78K/yr
Y1 services     (~1.8x blend, integration-heavy): $140K
                  (OAS: Meditech FHIR $120K + state HIE $60K — discounted as 2-connector bundle)
                  (Fractal: ontology + dashboard $50K; NEOS: governance design $30K)
Y1 upkeep       (Gold SLA, healthcare default, 25%): $20K (starts Y2)
Infra           (HITRUST-compliant cloud surcharge applies): $18K–$30K/yr
                ──────
Y1 TOTAL:       ~$240K–$255K
3-yr TCO:       ~$550K–$650K
```

Friction-cost anchor: Redox floor alone is $420K/yr; community hospital state HIE integration historically $500K–$2M one-time. Meditech/Cerner analytics supplemented with a standalone tool runs $200K–$600K/yr. Combined 3-yr friction: $1.5M–$3M.

Capture rate: $650K ÷ $2.25M midpoint = **~29%** — high but defensible because OAS is genuinely displacing Redox and the standalone analytics spend, not merely layering on.

Mid: Regional System (200–2,000 beds) — Canonical Example

Profile: 4-hospital regional system, ~800 beds, 200K-life Medicare Advantage payer arm. On Epic at lead hospital, Cerner at acquired hospital. CDO has a population health mandate; CIO wants to reduce the Redox bill and close TEFCA gaps; CCO is watching the May 2026 HIPAA Security Rule overhaul.

This is the canonical example from `layers/fractal-engine.md §7.1`, extended to the full-stack quote:

Recommended bundle: Fractal Mid + OAS Mid + NEOS Mid (Triad, with AMP Mid as Y2 upsell for OR command center)

Worked quote:

```
Fractal Mid:    $264K/yr  (PMPM mechanic: $1.10 PMPM × 200K MA lives; 12 data sources, 8TB)
OAS Mid:        $140K/yr  (10 integrations: Epic FHIR + Cerner FHIR + 3 claims + 2 SDoH + 3 internal)
NEOS Mid:       $55K/yr   (consent management + IRB councils + data-sharing governance; 6 councils)
                ──────
Subtotal:       $459K/yr
Triad -18%:     $376K/yr
Y1 services     (~1.6x blend): $600K
                  (Epic FHIR connector $200K, Cerner FHIR $180K,
                   ontology + population health models $100K,
                   consent governance design OmniOne $70K, dashboards $50K)
Y1 upkeep       (Gold SLA, 25%): $94K (starts Y2)
Infra           (HITRUST-compliant cloud, multi-AZ): $80K–$180K/yr
                ──────
Y1 TOTAL:       ~$1.1M–$1.2M
3-yr TCO:       ~$2.5M–$2.9M
```

Friction-cost anchor: Per the research brief, interop fabric at this scale runs $5M–$15M over 3 years (Redox at $35K/mo alone is $1.26M over 3 years, before volume). Innovaccer/Arcadia population health platform $600K–$3M/yr. Combined Y1 displacement opportunity: $1.5M–$3M/yr. 3-yr friction: $4.5M–$9M.

Capture rate: $376K ÷ $2.2M midpoint (Y1 friction) = **~17% value share** — defensible inside the 8–20% healthcare capture band per `01-pricing-model.md §9`.

Y2 expansion path: Add AMP Mid ($150K/yr) for OR command center — the spatial event signal enhances Fractal's block utilization forecasting. Full-stack 3-yr TCO rises to ~$3.8M–$4.5M, still well inside the $8M–$25M Mid-tier band.

Large: National IDN, AMC, or Top-20 Pharma

Profile: 40-hospital IDN on Epic (full Cosmos data-sharing participant) OR top-20 pharma with enterprise RWE program. IT capital committee approval required. Decision cycle 12–18 months. SI partners (Deloitte Health, Accenture Health, Optum Advisory) actively involved.

Recommended bundle: Full Stack (AMP Large + NEOS Large + OAS Large + Fractal Large)

Worked quote (IDN scenario):

```
AMP Large:      $600K/yr  (25 hospitals, OR command center + ED spatial, ~$24K/facility)
NEOS Large:     $280K/yr  (enterprise consent + IRB + data governance, EHDS-ready)
OAS Large:      $750K/yr  (50+ integrations: Epic ecosystem, 3 Cerner sites, claims, SDoH,
                           TEFCA QHIN participation, pharmacy, lab, imaging)
Fractal Large:  $900K/yr  (20+ data sources, cross-org population health + RWE, >20TB;
                           PMPM mechanic: $0.75 PMPM × 1.2M attributed lives)
                ──────
Subtotal:       $2.53M/yr
Full-Stack -25%: $1.9M/yr
Y1 services     (~1.8x blend): $3.4M
                  (Epic Cosmos/FHIR primary $300K, 3 Cerner sites $400K,
                   HITRUST-scope connector audit $100K, OAS federation engine $500K,
                   AMP spatial build-out across 25 sites Lattice SKU $1.5M range,
                   Fractal RWE model training + population models $600K,
                   NEOS consent + IRB governance design OmniOne $200K)
Y1 upkeep       (Platinum SLA, custom): $560K (starts Y2)
Infra           (multi-region, HITRUST-compliant, dedicated K8s): $350K–$700K/yr
                ──────
Y1 TOTAL:       ~$6.2M–$6.6M
3-yr TCO:       ~$13M–$16M
```

Friction-cost anchor: Epic integration tax at large IDN = $20M–$80M over 3 years (services, connectors, governance overhead). HITRUST r2 at $400K–$800K per 3-year cycle. Innovaccer/Health Catalyst large-system contracts at $2M+/yr. Redox at scale for 50+ integrations: $3M–$6M over 3 years. Combined 3-yr friction: $25M–$90M.

Capture rate: $1.9M/yr ÷ $40M midpoint friction/yr implied = **~5% value share** at floor — defensible, with upside as Epic FHIR cert eliminates incremental connector costs for the customer and PMPM scales with lives attributed.

Top-20 pharma variant (RWE focus): Replace AMP with a Fractal Large + OAS Large + NEOS Mid Triad. Fractal at $1.2M/yr (per-research-query mechanic for RWE; 20+ data sources including EHR network, claims, registry, genomics). 3-yr TCO $8M–$14M vs IQVIA multi-million-dollar annual licenses + Datavant/Aetion RWE SaaS. Capture rate 15–20% of the IQVIA displacement opportunity.

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4. Industry-specific procurement signals

Buyer personas and co-buy dynamics

Hospital/Health System:

- **CIO:** Primary economic buyer. Priorities: vendor consolidation (fewer point solutions), integration cost reduction, Epic rationalization, HITRUST compliance gate management. Will not move without CCO sign-off on HIPAA posture.
- **CMIO:** Clinical safety veto. Any system that touches clinical workflow requires CMIO endorsement. Position PrimusNeo (especially AMP + Fractal) as **decision-support**, not autonomous clinical action. Pursue Class I/II SaMD pathway proactively.

- **CDO:** Data strategy sponsor. Population health mandate drives Fractal + OAS engagement. AI governance concerns require documented NEOS governance design.
- **CCO / Chief Privacy Officer:** HIPAA and HITRUST gating. Will not approve a demo until BAA scope, de-identification methodology, and subprocessor chain are documented. PHI flow documentation must precede all technical conversations.

Payer:

- CIO + Chief Actuary + VP Population Health. PMPM mechanic aligns naturally with their operating model. VBC and risk-adjustment analytics are primary value drivers.

Pharma / Biotech:

- VP Data Science, Head of Real-World Evidence, CTO. Budget owners are often outside IT — closer to medical affairs or R&D. Per-research-query pricing resonates; IQVIA displacement is the anchor conversation. IRB and DPIA governance via NEOS is a secondary pitch that accelerates to primary in EU-operating pharma (EHDS + GDPR Article 9 special-category health data).

Procurement mechanics

- **Capital committee:** Any IT commitment above $1M requires board or capital committee approval. Build the CFO narrative — friction-cost math, 3-year TCO comparison against Redox + Innovaccer + HITRUST standalone.
- **Decision cycle:** 6–18 months. EHR replacement cycles: 5–10 years (do not try to enter on EHR displacement). Analytics and interop refresh cycles: 3–5 years — these are the windows PrimusNeo enters.
- **Budget timing:** Hospital fiscal years often run July–June. IT capital budgets are built July–September. **Q3–Q4 (Jan–June) is peak signing** — deals funded in the prior budget cycle close in Q1–Q2.
- **RFP process:** 68% of providers manage procurement manually (2024 HIMSS). RFPs run 6–12 months from issuance to award. Evaluation committees score: technical fit 30–35%, regulatory compliance pass/fail gate, total cost of ownership 25–30%. HITRUST cert attachment is frequently a submission prerequisite — not evaluation criteria but a gate.

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5. Rejection-risk mitigations

Risk: Patient safety objection (CMIO/patient safety officer)

Position all Fractal analytics and AMP spatial alerts as **decision-support tools for clinicians**, not autonomous decision-makers. Pursue **Class I or Class II SaMD pathway** under FDA guidance early in the engagement. Produce a software as a medical device classification summary before any CMIO technical demo. Do not wait for the objection — preempt it in the discovery conversation.

Risk: HIPAA fear (CCO/Privacy Officer default-reject)

Before any technical demo: (1) deliver a written BAA scope document identifying every PHI touch point; (2) produce de-identification methodology documentation (Safe Harbor or Expert Determination per 45 CFR §164.514); (3) map the complete subprocessor chain including Escherbridge (Fractal), OmniOne (NEOS governance), OAS Web4 (OAS connectors), and cloud infrastructure providers. Privacy officers who receive this proactively become advocates; those who discover gaps in discovery become blockers.

Risk: "Rip and replace" resistance

The standard hospital objection: "We spent $300M on Epic. We're not changing it." Response: "We wrap Epic, we don't replace it. AMP adds the OR command center intelligence layer Epic doesn't offer. OAS compresses your integration spend by 35–60% vs Redox while remaining Epic FHIR R4 certified and App Orchard listed. Fractal adds population health analytics that Epic Cosmos does not provide for your MA payer arm." Every message must reinforce the augmentation framing.

Risk: Epic/Cerner integration lock-in

Epic's proprietary API surface (MyChart, Hyperdrive, App Orchard) creates real friction. **Epic FHIR R4 certification + App Orchard listing for OAS is a prerequisite for any large IDN sale** — not a differentiator, a requirement. The 6–18 month Epic technical review must begin immediately as a central PrimusNeo investment, not a per-engagement cost. Until App Orchard listing is achieved, large IDN sales require a lighthouse exception path (direct CIO sponsor, waived App Orchard requirement during pilot).

Risk: Procurement cycle mismatch (startup credibility gap)

12–18 month RFP cycles disadvantage early-stage vendors. Lack of healthcare segment references is a hard rejection criterion. The **lighthouse program** — securing 2–3 segment references (1 community hospital, 1 regional system, 1 pharma or payer) — is the prerequisite for substantive RFP participation. Tie Town serves as the reference architecture proof point but does not substitute for healthcare-specific references. Prioritize lighthouse deals above contract size optimization in the first 18 months.

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6. Regulatory + compliance considerations

HIPAA Security Rule (effective May 2026)

The overhaul mandates: MFA for all PHI access, universal PHI encryption at rest and in transit, 24-hour breach notification to HHS (down from 60 days for major breaches), and annual penetration testing. This is not optional and affects every PrimusNeo layer that touches PHI. **OAS connectors carrying PHI must comply by May 2026; AMP scenes with patient location data must comply; Fractal models trained on EHR data must comply.** Document the compliance posture per layer in every BAA and customer-facing technical spec.

HITRUST CSF

81% of hospitals and 83% of payers require HITRUST certification from technology vendors before RFP submission. **HITRUST i1 (~$70K) is the non-negotiable minimum** for any substantive healthcare RFP participation. HITRUST r2 ($100K–$500K, 12–15 month timeline; $400K–$800K per 3-year cycle) is

expected for core platforms at large IDNs and payers. **PrimusNeo must fund HITRUST i1 as a central investment — not a per-engagement line item.** Until i1 is achieved, the healthcare pipeline is restricted to pilot-stage conversations that do not require RFP submission.

ONC TEFCA / QHIN

Common Agreement v2.1 went into effect January 2025; 21,086 organizations are live. QHIN designation takes approximately 12 months. OAS as TEFCA-native FHIR fabric is a **direct differentiator vs Redox** (which is not QHIN-native) and vs Health Gorilla (QHIN participant but not a full interop fabric). Pursue QHIN designation for OAS as a network-level investment.

EU EHDS (Reg EU 2025/327)

In force March 2025; general application March 2027. EHR vendors must certify for interoperability and security by **January 2026**. Priority health data exchange (patient summaries, ePrescriptions, lab results, medical images, hospital discharge reports, prescriptions) mandatory EU-wide by **March 2029**. **NEOS consent and governance schema is directly revenue-generative for EU-operating health systems, pharma companies, and biotech** — DPIA workflow, cross-border data-sharing governance, and patient consent management are all addressable EHDS compliance requirements via NEOS.

EU MDR

If Fractal Engine outputs constitute clinical decision support (Class IIa/IIb SaMD under EU MDR), conformity assessment applies: **$20K–$80K per high-risk system**. Bundle into the engagement deliverable as a scoped item. Position Fractal as decision-support (not autonomous clinical action) where possible to maintain Class I pathway and minimize MDR overhead.

FDA SaMD (US)

Fractal analytics that inform clinical treatment decisions are subject to FDA digital health guidance. Class I and Class II SaMD (low-to-moderate risk decision support) require classification documentation and potentially a 510(k) or De Novo pathway depending on intended use. Prepare the SaMD classification summary early — before the first CMIO technical demo.

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7. Steward + service partner notes

Escherbridge — Fractal Engine steward (RWE and population health specialty)

Escherbridge is the primary delivery partner for all Fractal RWE and population health model development. Their public corpus provides the pretrained baseline for health system deployments; customer EHR + claims + SDoH data fine-tunes the models under the per-engagement IP schedule (customer owns blended weights; Escherbridge retains public-corpus-derived components). Escherbridge

earns **18% of Fractal Engine line-item license revenue** per the steward royalty schedule. Healthcare is one of their highest-density use cases — route all Fractal healthcare engineering and model-training SOWs through Escherbridge.

OmniOne — NEOS steward (consent and IRB governance design)

OmniOne leads all NEOS governance design using their ETHOS methodology. Healthcare consent management, IRB governance, and EHDS-compliance data-sharing governance are high-value applications of NEOS — OmniOne's governance design workshops ($35K–$60K per engagement) are the entry point for NEOS in healthcare. OmniOne earns **18% of NEOS line-item license revenue**. For EU pharma and health systems navigating EHDS, OmniOne's DPIA and consent architecture capability is a standalone sellable service that pulls Fractal and OAS into the engagement.

OAS Web4 — OAS steward (Epic/Cerner/HIE connector engineering)

OAS Web4 leads connector development for Epic FHIR, Cerner FHIR, Meditech, Athenahealth, and state HIE endpoints. Their connector engineering is the delivery capacity for OAS healthcare integrations ($120K–$400K per EHR connector). OAS Web4 earns **15% of OAS line-item license revenue** and captures services revenue directly when they deliver connector buildouts on subcontract. The Epic App Orchard certification work must be led by OAS Web4 with PrimusNeo investment backing.

PLAN Systems — AMP steward (OR command center and ED spatial)

PLAN Systems delivers the AMP spatial environment for hospital operations. The OR command center and ED throughput spatial builds are their specialty; sensor/OT integration for patient flow and equipment tracking maps to their existing capability. PLAN earns **15% of AMP line-item license revenue**. Tie Town serves as the reference architecture for offline-first spatial operations — relevant for rural critical access hospitals where connectivity is unreliable.

SI channel — delivery at scale

The healthcare SI channel commands premium day rates: Deloitte Health / Accenture Health at $4,000–$8,000/day (Principal/Director); EY Health / Optum Advisory at $3,000–$6,000/day; Chartis / Huron at $3,500–$7,000/day. Epic implementation at large IDNs runs $50M–$500M in services spend. PrimusNeo engages SI channel for: (1) large IDN deployments where SI project management is buyer-required; (2) pharma RWE programs where a Big-4 advisory relationship already exists; (3) change management at scale where Chartis/Huron clinical ops credibility is needed. **SI channel deals must be structured to preserve PrimusNeo's services margin — SI markup on PrimusNeo services is typically 40–80%; structure as a reseller or co-prime model with PrimusNeo retaining direct license relationship.**

Tie Town reference

Tie Town [TX] provides the reference architecture proof point for the full stack in an offline-first, privacy-preserving, TEFCA-capable deployment context. While not a hospital deployment, Tie Town demonstrates the stack's composability for prospects who require a live reference before advancing an RFP. Until

healthcare-specific lighthouse references exist, Tie Town is the fallback reference for proof-of-concept conversations.

HITRUST i1 — central investment prerequisite

HITRUST i1 certification (~$70K, recurring 3-year cycle) is a central PrimusNeo investment that unlocks the healthcare pipeline. It is not a per-engagement cost and should not be billed to any customer. It is the price of admission to substantive healthcare RFPs for 81% of hospital buyers and 83% of payer buyers. Begin immediately. Do not pursue large healthcare RFPs without it.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Healthcare engagement model

Logistics — Engagement Model

> ***Vertical:*** *Logistics (freight forwarders, 3PLs, shipping lines, port operators, last-mile, warehouse, supply chain control towers)* ***Research brief:*** `../research/logistics.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The Friction-Cost Anchor

Logistics technology costs are a useful fiction. Published list prices mask the total friction: failed consortium bets, integration debt, and compliance liability that compound into a per-buyer friction window large enough to make PrimusNeo's value capture defensible at every tier.

TMS ($400K–$3M/yr enterprise). Oracle OTM runs ~$450/user/month; enterprise TCV commonly $400K–$2M/yr. SAP TM embedded in S/4HANA estates reaches $500K–$3M/yr before implementation, which Blue Yonder alone prices at $1M–$5M. For freight forwarders, CargoWise has become the de facto operating system at **$6–$10/shipment** bundled — a per-shipment anchor that calibrates OAS conversations at every tier.

Visibility platforms ($100K–$1M+/yr). project44 platform fees run $150K–$600K/yr mid-market, $1M+ at enterprise; FourKites $100K–$500K/yr. Shippers paying $300K+/yr for a visibility point solution arrive at PrimusNeo conversations already aware they're paying for fragmented intelligence.

WMS ($500K–$3M Y1 large; $150K–$500K mid). Manhattan Active WMS at $2,000/user/yr quickly reaches $1.5M–$4M/yr for enterprise DC estates. Blue Yonder is comparable. Ongoing costs run $150K–$500K/yr after implementation.

Port/terminal OS ($300K–$1.5M/yr mid-size terminal). Navis N4-as-a-Service occupies this band; Octopi (lower-tier Navis) runs $50K–$200K/yr for simpler terminals. Awake.AI and Portchain target AI planning overlays with custom pricing.

TradeLens: the canonical failure. The November 2022 shutdown of the IBM/Maersk TradeLens consortium represents $1B+ of collectively lost value across four years. The failure was not technical — it was governance. Maersk's perceived ownership meant MSC, CMA CGM, and Evergreen never joined; Asian carriers cited governance concerns and never trusted the platform. This is the shadow every logistics network play must address head-on.

Per-buyer friction window (worked): A national 3PL paying $200K/yr to a TMS, $150K/yr to a visibility platform, $300K/yr to a WMS, and carrying $500K/yr in EU CBAM compliance costs (effective Jan 2026) faces a $1.15M/yr coordination and compliance cost baseline before any consortium ambition. A port operator running 8 terminals — already burned by TradeLens — faces $3M–$8M/yr in visibility-spend, customs-integration, and carrier-SLA-friction across its network. These windows establish the capture band.

PrimusNeo capture band: 10–20% of consolidated platform spend (per the pricing model). At $1.15M/yr friction, a mid-tier target is $115K–$230K/yr in defensible license capture — well inside the Mid-tier pricing anchor.

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2. PrimusNeo Positioning Per Layer in Logistics

The framing per layer must be precise because logistics buyers will compare against existing point solutions and will ask exactly where PrimusNeo fits or displaces.

AMP — Port/Terminal/Yard Spatial Twin + Control Tower

AMP is the **terminal-and-yard spatial twin that Navis N4 describes in static data but cannot render as a live operational scene**. For port operators, last-mile depot managers, and warehouse control towers, AMP provides the navigable spatial layer that ties vessel positions, crane allocations, yard slot state, inbound rail manifests, and truck gate queues into a single scene that ops teams can actually work from. AMP's offline-first architecture matters for port edge cases (spotty connectivity on the quay, terminal edge nodes, cold-store environments). The pitch to a CSCO at a port operator: "AMP is the terminal-and-yard spatial twin — Navis N4-adjacent but vendor-neutral, and it runs on the dock."

NEOS — Cross-Org Consortium Governance + the TradeLens Answer

NEOS is **the explicit, deployable answer to the question every logistics consortium must now answer before signing**: "Who owns the network?" The answer is nobody, or rather, the council does — and the council is governed by a NEOS charter that no single carrier, 3PL, or port operator controls. NEOS enforces SLA agreements, dispute resolution workflows, data-sharing policies, and contribution accounting across the consortium, with every governance event signed, anchored, and replayable. DCSA, GS1, and OpenAPI alignment is built in from day one, not retrofitted after the governance crisis.

The NEOS pitch in the carrier-shipper-3PL triangle: "No single party owns the network. You don't trust Maersk to run TradeLens. You shouldn't. NEOS governance means the charter owns the network — and the charter is yours, collectively, on day one."

OAS — Carrier-Shipper-3PL Integration Fabric

OAS is the **governance-neutral integration layer the entire carrier-shipper-3PL triangle runs across**. Its critical differentiator in logistics is coverage depth: OAS uniquely spans Web2 TMS/WMS/ERP connectors AND Web3 chain interop AND identity/SSI credentials AND IoT/device ingest under one API surface. No incumbent iPaaS (MuleSoft, Boomi, Workato) covers the Web3 + identity + IoT surface. project44 and FourKites cover the carrier-integration surface but not governance or cross-org data fabric.

Pricing mechanic: per-connector + base platform fee (default) or per-API-call/per-integration-event metered overage ($0.001–$0.05/call at volume). The CargoWise $6–$10/shipment anchor calibrates per-shipment equivalence for OAS conversations with freight forwarders.

For the EU Maritime Single Window (effective 2025, per IMO FAL and EU regulation), OAS is the port-operator and shipping-line integration play: pre-arrival data, port clearance, customs Single Window — all flowing through OAS connectors rather than bespoke EDI integrations.

Fractal Engine — Predictive ETA + Emissions Optimization + Demand Sensing

Fractal is the **operating intelligence layer** that converts raw AMP spatial events and OAS integration events into predictions that inform actual decisions. Three use cases anchor the logistics pitch:

1. **IMO CII compliance and voyage optimization.** Vessels rated D or E for three consecutive years face mandatory corrective action. CII tightens ~2%/yr through 2026. Fractal's voyage optimization model — fed by AMP spatial data and OAS carrier/weather/port-call feeds — produces slow-steaming recommendations and route alternatives that close the CII gap before the next annual reporting window.
2. **EU ETS emissions cost reduction.** At €60–70/tonne CO_2, large carriers face tens of millions of euros per year in ETS compliance from 2024. AMP-informed route and speed optimization, quantified by Fractal, converts into direct cost avoidance — the kind of CFO-legible ROI that unlocks budget in Q3–Q4 planning cycles.
3. **Demand sensing and ETA prediction across the global network.** Fractal aggregates multi-carrier, multi-3PL, multi-port data (via OAS ingest) into network-wide ETA predictions and demand signals that individual participants cannot produce from their own data alone. The data-value-exchange mechanic — carriers contribute raw feeds, receive aggregate benchmarking in return — is how Fractal addresses the carrier-data-hoarding rejection risk.

OAS + NEOS is the consortium-savior duo. Neither alone solves the TradeLens problem. OAS provides the integration fabric so data flows; NEOS provides the governance layer so the terms of that flow are trusted. Together, they are what TradeLens needed and didn't have.

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3. Engagement Tiers — Small / Mid / Large

Small: Regional 3PL / Freight Forwarder (<$50M revenue)

Profile: Regional freight forwarder or 3PL running CargoWise or a legacy TMS, handling 5K–50K shipments/yr, small compliance and IT team, 1–3 warehouse or depot sites.

Current 3-yr TCO: $120K–$400K (CargoWise subscription, EDI middleware, manual compliance, spreadsheet-based reporting).

PrimusNeo bundle: OAS Small (TMS/customs connectors, <10 integrations) + Fractal Small (shipment ETA model, emissions reporting for CBAM, <5 data sources).

Worked quote: - OAS Small: $40K/yr (8 integrations: CargoWise + customs authority + 3 carrier APIs + 2 shipper portals + 1 IoT tracker feed) - Fractal Small: $45K/yr (4 data sources: shipment feed, carrier API, customs data, emissions reference data) - Duet bundle: -10% → $76K/yr license - Y1 services (~2.3x OAS-

weighted blend): $175K (connector buildout: TMS $80K, customs $60K, ontology mapping $35K) - Y1 upkeep (Silver SLA, 22%): starts Y2 at $17K - Infra: $1,500/mo

Y1 total: ~$270K — 3-yr TCV: ~$480K

Friction anchor: CargoWise at $8/shipment × 20K shipments/yr = $160K/yr + manual compliance FTE at $80K/yr = $240K/yr direct. Capture: $76K ÷ $240K = **32% value share** on direct displacement — aggressive, defended by the combined displacement of manual compliance + carrier EDI middleware. If the buyer retains CargoWise, frame OAS as the enrichment layer (+ EU CBAM readiness, + ETA prediction) at 32% of the cost being displaced.

PrimusNeo target: $60K–$180K/yr; **3-yr TCV $180K–$540K** (per research brief).

Mid: National 3PL ($50–500M) / Port Operator

Profile: National 3PL with regional DC network (5–10 sites), or a port operator running 3–8 terminals with cross-carrier, rail, and customs coordination needs. Existing stack: TMS, WMS, point visibility platform, legacy EDI, manual customs. EU CBAM exposure growing.

Current 3-yr TCO: $1.5M–$6M (TMS $600K–$2M, WMS $500K–$1.5M, visibility $300K–$900K, customs middleware $200K–$600K, IT/integration overhead).

PrimusNeo bundle: OAS Mid + NEOS Mid + Fractal Mid (Triad). Port operator variant replaces Fractal with AMP Mid for terminal spatial twin.

Worked quote (national 3PL variant): - OAS Mid: $160K/yr (20 integrations: TMS + WMS + 8 carrier APIs + 3 shipper portals + CBAM data + customs + 3 internal) - NEOS Mid: $55K/yr (3 councils: carrier SLA council, data-sharing council, compliance council) - Fractal Mid: $180K/yr (12 data sources, network-wide ETA model + CBAM emissions reporting) - Triad bundle: -18% → $323K/yr license - Y1 services (~2.0x blend): $646K (TMS connector $120K, WMS $100K, carrier connectors $200K, ontology $80K, governance design $35K, dashboards $60K + others) - Y1 upkeep (Silver SLA, 22%): starts Y2 at $71K - Infra: $8K/mo

Y1 total: ~$1.06M — 3-yr TCV: ~$2.1M

Friction anchor: TMS $900K + WMS $750K + visibility $400K + CBAM compliance cost $400K–$1.5M (definitive Jan 2026) = $2.45M–$3.55M/yr mid-range. Capture: $323K ÷ $3M = **~11% value share** — inside the 10–20% logistics band.

Port operator variant (OAS+NEOS canonical example, per `layers/oas.md §7.2`): OAS Large $720K/yr + NEOS Large $220K/yr, Duet -10% → $846K/yr license. Y1 services ~2.2x: $1.86M. 3-yr TCO $6.5M–$8M. Friction anchor: TradeLens collective loss precedent + $3M–$8M/yr standing visibility + compliance friction. Capture: 17% value share.

PrimusNeo target: $300K–$900K/yr; **3-yr TCV $900K–$2.7M** (per research brief).

Large: Top-20 Ocean Carrier / Top-5 3PL / Global Integrator

Profile: Tier-1 ocean carrier (Maersk, MSC, CMA CGM, Evergreen, Hapag-Lloyd) or top-5 3PL (DHL Supply Chain, Kuehne+Nagel, DB Schenker, XPO, CEVA) or global integrator (FedEx, UPS). Existing stack: proprietary TMS, WMS, visibility, and often internally built platform capabilities. EU ETS exposure: tens of millions of euros/yr. CSCO procurement cycle 9–18 months.

Current 3-yr TCO: $15M–$80M+ globally (TMS, WMS, visibility, customs, IT integration, internal-build programs).

PrimusNeo bundle: Full Stack (AMP Large + NEOS Large + OAS Large + Fractal Large). Paid pilot ($250K–$500K) precedes full rollout.

Worked quote (top-20 carrier — Full Stack): - AMP Large: $600K/yr (port/terminal spatial twins across 6 primary hub terminals + control tower) - NEOS Large: $260K/yr (cross-carrier consortium governance: SLA, dispute resolution, data-sharing, eBL attestation councils) - OAS Large: $800K/yr (60+ integrations: Navis N4, customs single-window, 15 carrier APIs, rail carriers, IoT vessel/container tracking, CBAM declarant flows) - Fractal Large: $900K/yr (20+ data sources: vessel telemetry, port-call data, weather, ETS pricing, customs, carrier network; CII optimization + EU ETS cost modeling + demand sensing) - Full Stack bundle: -25% → $1.92M/yr license - Paid pilot (Y0): $350K (Atom + connector audit, 90-day, credits to Y1 services) - Y1 services (~2.0x blend, minus pilot credit): $3.49M - Y1 upkeep (Gold SLA, 25%): $480K (starts Y2) - Infra: $400K–$900K/yr

Y1 total: ~$5.8M–$6.3M (including pilot) — 3-yr TCV: ~$10M–$14M

Friction anchor: EU ETS compliance at €65/tonne for a 100-vessel fleet running 2M TEUs/yr → $30M–$80M/yr ETS liability. CII corrective action plans at D/E-rated vessels: $500K–$5M per vessel-class remediation. TradeLens-pattern internal build programs: $50M–$200M sunk in failed or stalled digital platforms. Combined friction window: $50M–$150M/yr. Capture: $1.92M ÷ $100M = **~2% value share** — conservative by design at this scale; the pilot ROI story (3-month CII optimization + EU ETS avoidance demo) does the commercial work before the full commitment.

Co-sell/embed motion for internally built platforms (Maersk Captain, CMA CGM CEVA, MSC One, DHL myDHLi, UPS): white-label Fractal Engine for network intelligence, or OAS as the cross-platform integration fabric behind the carrier's own consumer-facing product. This is not displacement; it's a revenue-share embed.

PrimusNeo target: $1.5M–$5M+/yr; **3-yr TCV $4.5M–$15M** (per research brief).

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4. Industry-Specific Procurement Signals

Buyers and cycles:

Buyer	Cycle	Agenda
CSCO / CPO	9–18 months (large)	2024/2025 agenda: resilience + tech ROI + sustainability. TradeLens hangover: governance neutrality is table stakes before any cross-org network conversation. EU ETS + CBAM as hard financial drivers.
COO / Head of Network Ops	6–12 months (mid)	Operational pain: visibility gaps, carrier SLA disputes, customs delays. Responds to demonstrated friction-cost math and peer-operator reference.
CTO / VP Engineering	Integration veto	OAS connector library depth and pre-built TMS/WMS integrations are the technical proof-point. Publish connector library for top-5 TMS + top-5 WMS before enterprise sales scale.
Head of Trade Compliance / VP Customs	3–9 months (specific tooling)	NEOS + OAS for 21CCF-compliant data flows. EU CBAM declarant workflows (definitive Jan 2026). EU Maritime Single Window (2025 mandate). Responds to concrete compliance-gap analysis.

Carrier-shipper-3PL triangle dynamic: No single party controls the network. Carriers own vessel and container position data; 3PLs own last-mile and warehouse data; shippers own cargo and purchase-order data. TradeLens failed because Maersk tried to be the hub in a network where its competitors were the spokes. PrimusNeo's answer is structural: NEOS governance is charter-owned from day one; OAS is the neutral integration fabric; no party sees competitor raw data — only what the charter permits.

Entry motion: The NEOS + OAS consortium-savior framing wins the first conversation with a CSCO burned by TradeLens. The CII/ETS Fractal use case wins the CFO's attention. AMP terminal spatial twin wins the COO's curiosity. Run them in parallel during the discovery sprint.

RFP cycles: Top-20 carriers and top-5 3PLs run formal tech RFPs annually or biannually, typically Q1 for budget alignment. Target Q3–Q4 budget season conversations to be on the shortlist before the Q1 RFP issues.

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5. Rejection-Risk Mitigations

Consortium fatigue (TradeLens precedent): This is the dominant rejection risk in any cross-org logistics play. Mitigation is structural, not rhetorical: NEOS governance neutrality must be demonstrable — no single carrier may access competitor data; charter terms are cryptographically enforced, not contractually promised; DCSA/GS1/OpenAPI alignment is documented and published before the first enterprise sales conversation, not promised as a roadmap item. The correct objection-handler is not a slide — it is a signed reference charter with a neutral third-party port operator or freight association as the founding council steward.

Legacy TMS/WMS integration complexity: Oracle OTM, SAP TM, Manhattan, Blue Yonder implementations at large accounts are deeply customized. The only credible response is a published OAS connector library for the top-5 TMS (OTM, SAP TM, Blue Yonder, MercuryGate, JDA/BluJay) and top-5 WMS (Manhattan, Blue Yonder, Körber, SAP EWM, Infor WMS) — certified, maintained, and demonstrable in a 30-day pilot window. This must exist before enterprise sales scale, not be built during each enterprise deal.

Carrier data hoarding: Ocean carriers treat real-time vessel and container position as competitive. project44 and FourKites spent years building carrier network agreements with persistent gaps. PrimusNeo's route is a data-value-exchange model: carriers contribute raw feeds via OAS connectors; Fractal aggregates across the network and returns benchmarking intelligence (CII performance vs peer fleet, ETA accuracy vs port benchmarks, EU ETS cost-avoidance modeling) that no individual carrier can produce from its own data. This is the model that converts a competitive instinct into a network participation incentive.

Overlap with existing visibility spend: A shipper or 3PL paying $400K/yr to project44 or FourKites is not going to add another layer on top. The conversation must position OAS + Fractal as a superset (spatial context from AMP + governance from NEOS + intelligence from Fractal) that either displaces the point solution or absorbs it via an OAS connector, making the total cost comparable while expanding the value surface. Displacement is the preferred outcome; augmentation is the fallback.

"We built it ourselves" at top carriers: Maersk (Captain Peter), CMA CGM (CEVA), MSC (MSC One), DHL (myDHLi), UPS — all have internal platforms. The motion here is co-sell or embed, not displacement. White-label Fractal Engine network intelligence behind the carrier's own product; offer OAS as the cross-platform fabric for carrier-to-shipper or carrier-to-3PL integrations that the carrier's internal team doesn't want to build. Revenue-share embed agreements are more likely to succeed than head-on displacement conversations at these names.

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6. Regulatory + Compliance Considerations

IMO CII / EEXI (in force January 2023). Vessels of 5,000 GT and above must calculate and report their Carbon Intensity Indicator annually. A D rating for three consecutive years or an E rating triggers a mandatory corrective action plan. CII tightens approximately 2% per year through 2026. This is a direct Fractal Engine voyage-optimization use case: model slow-steaming alternatives, route options, and port-call sequences against CII targets, and surface the optimal decision in real time via the AMP control tower.

EU ETS — Maritime (2024 onward). The EU Emissions Trading System now covers 100% of intra-EU voyages and 50% of extra-EU voyages calling EU ports. At approximately €60–70 per tonne of CO_2, compliance costs reach tens of millions of euros per year for large carriers. This is the hardest financial pressure in maritime logistics today and the most compelling entry point for a Fractal + AMP ROI conversation with a CFO or sustainability officer.

EU CBAM (definitive phase January 2026). The Carbon Border Adjustment Mechanism in its definitive phase requires authorized declarants for covered goods (steel, aluminium, cement, fertilizers, electricity, hydrogen) to purchase CBAM certificates tied to the EU ETS price. This creates new documentation workflows, customs classification requirements, and supplier carbon-data collection needs — all addressable by NEOS (governance of declarant workflows) and OAS (customs and supplier data integration). National 3PLs and freight forwarders handling covered commodities must be CBAM-ready; this is a procurement trigger for the mid tier from mid-2025 onward.

US 21st Century Customs Framework (21CCF). Mandatory e-record-keeping, expanded supply chain data sharing requirements, and ACE 2.0 are in progress. Customs brokers face breach reporting obligations. OAS is positioned for 21CCF-compliant data flows: structured, auditable, machine-readable customs data pipelines that satisfy expanded reporting requirements without bespoke broker-by-broker integration.

Single Window Mandates — IMO FAL + EU Maritime Single Window (2025). Both frameworks require pre-arrival and port clearance data submission via national single-window systems. OAS port-operator and shipping-line integration is the direct play: standardized pre-arrival data, port state control submissions, and departure manifests flowing through OAS connectors into the national single-window, displacing the current patchwork of EDI, email, and portal submissions.

DCSA / BIMCO eBL target (100% by 2030). The digital bill of lading sector is in pre-consolidation. WaveBL, CargoX, essDOCS, and Bolero operate at varying scales; TradeLens' eBL function died with the platform. NEOS governance event anchoring provides a path to DCSA-aligned eBL attestation without requiring a new platform — the eBL event is a NEOS-signed, OAS-transported document that any party in the charter network can verify.

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7. Steward + Service Partner Notes

PLAN Systems (AMP steward, 15% royalty on AMP license). PLAN delivers AMP terminal/yard spatial environment buildouts — the specialized 3D scene authoring, Navis N4 data ingest, crane/berth asset modeling, and IoT binding that logistics spatial twins require. PLAN is the primary subcontract channel for AMP deployment services in logistics; their domain expertise in operational spatial environments is not replicable by a generalist SI in a short engagement window.

OmniOne (NEOS steward, 18% royalty on NEOS license). OmniOne brings the ETHOS governance-design methodology — the specific intellectual work of structuring a carrier-shipper-3PL consortium charter that no single party controls. The TradeLens-failure answer is not a product feature; it is a governance-design capability. OmniOne's ETHOS method is the delivery vehicle for that capability. Every mid and large logistics consortium engagement should open with an OmniOne ETHOS workshop ($35K–$60K) as the first SOW line.

OAS Web4 (OAS steward, 15% royalty on OAS license). OAS Web4's integration depth is the commercial value. Their connector authors are the fastest path to certified TMS/WMS/customs integrations. The pre-built connector library for top-5 TMS and top-5 WMS — built and maintained by OAS Web4 — is the technical proof-point that displaces the "integration complexity" rejection. OAS Web4 integration engineers ($2,200/day) should be staffed on every mid and large OAS engagement in logistics.

Escherbridge (Fractal Engine steward, 18% royalty on Fractal license). Escherbridge delivers Fractal ETA prediction, CII optimization, and EU ETS cost-modeling — the three Fractal use cases that anchor the logistics pitch. Their ML engineers ($2,800/day) build the voyage optimization models; their public corpus provides the pre-training base for network-level benchmarking. Escherbridge is the primary delivery partner for any Fractal engagement where predictive logistics and emissions analytics are the primary use cases.

SI channel:

Firm	Relevant practice	Day rate (USD)	Note
Accenture SC&O	Supply chain digital transformation, TMS/WMS implementation	Analyst $1,500–$2,000; Manager $2,500–$3,500; Partner $5,000–$8,000	Preferred SI for large-carrier and top-5-3PL engagements; Accenture has carrier relationships PrimusNeo does not yet have
Deloitte SC	Strategy + digital transformation, sustainability	Manager $2,500–$3,500; Partner $5,500–$8,500	Strong on CSCO advisory and regulatory compliance framing (CBAM, ETS)
BCG / McKinsey Logistics	Pure strategy, network optimization	BCG Associate $2K–$3K, Partner $8K–$12K; McKinsey EM $6K–$9K, Senior Partner $10K–$15K+	Engage for the friction-cost math validation; their imprimatur on the ROI model unlocks CSCO sign-off
GEP	Procurement + SC managed services	$1,500–$4,000	Mid-market 3PL channel; strong CPO relationships
Maersk Consulting	Logistics-specific advisory	$2,000–$5,000	Ex-Maersk operators; carrier-negotiation credibility; paradoxically credible precisely because they know where TradeLens failed

Comprehensive SC transformation programs with these firms run $250K–$2M+ for 3–6 month engagements and typically precede or accompany a PrimusNeo platform decision at the large tier. Position PrimusNeo as the platform the SI's transformation strategy runs on.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Logistics engagement model

Manufacturing — Engagement Model

> ***Vertical:*** *Manufacturing & Industrial Production* ***Research brief:*** `../research/manufacturing.md`
> ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Manufacturing is the vertical where PrimusNeo's value proposition is most directly measurable, and most frequently stalled by pilots that never ship.

Macro anchors (sourced):

Metric	Figure	Source
Global manufacturing value	$16.8T/yr	UNIDO 2024
Global unplanned downtime cost	~$50B/yr	Siemens / McKinsey
Industry 4.0 pilots that never scale	70%	McKinsey
Average industry OEE	60–65%	Siemens benchmark
World-class OEE target	85%	Industry consensus
OEE gap × addressable sector	~20–25 OEE points	Derived

The pilot-purgatory number is the dominant friction mechanism in this vertical. A plant that runs 3–6 IIoT POCs before one scales typically sinks $500K–$3M in sunk integration cost per failed pilot, with no reusable infrastructure. That sunk cost — not the software license — is what makes the first successful deployment so valuable. McKinsey's 70% failure rate applied to the installed base of active pilots means **$35B+ in annual wasted pilot spend** is in play globally.

Per-customer friction windows by scale:

Scale	Definition	Estimated annual friction window	Math
Small	1 plant, 1 line, <50 machines	**$500K–$2M/yr**	~$1M/yr downtime (Siemens benchmark at this plant size) + $200K/yr in tools that don't talk to each other + 1–2 stalled POCs at $150K sunk each
Mid	3–12 plants, mixed OT estate	**$5M–$20M/yr**	$50B / ~10K plants global midcap ≈ $5M/plant/yr in downtime + $1–2M/yr integration debt + pilot-purgatory sunk cost across the fleet
Large	Tier-1 OEM, 25+ plants, global	**$50M–$200M+/yr**	AMP §7.2 reference: $4M/yr/plant downtime × 12 plants = $48M/yr, plus $12M+ sunk across 8 stalled POCs; scaled linearly to 25+ plants = $100M+ friction window

Capture rate target: 5–15% of the per-customer friction window per `01-pricing-model.md` §9. The math must hold at every scale.

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2. PrimusNeo positioning per layer in manufacturing

Manufacturing is the **full-stack default** vertical. Every layer has a clear and non-redundant job in the plant.

Layer	Role in manufacturing	Key substitutes displaced	Why this beats the point solution
AMP (Spatial OS)	Plant-floor spatial twin: CAD ingest, PLC telemetry binding, MES event overlay, worker location, asset-attached scene. The IT/OT bridge that makes "the plant" a navigable, offline-capable, federated data object.	PTC ThingWorx ($150K–$500K+/yr/plant), Siemens Insights Hub ($50K–$300K/yr/plant), AVEVA System Platform	AMP runs offline-first — no cloud dependency for the operational loop. Hexon file standard is open, so the buyer's spatial data is never locked. Premium PLAN 3D components deliver CAD-class viewport without forcing rip-and-replace of Rockwell or AVEVA.
NEOS (Governance)	Change control, management-of-change (MOC), supplier qualification attestations, CIP-013-equivalent supplier risk acceptance, signed approval logs for CAPA and FSMA traceability.	Intelex, Veeva Vault QMS ($100K–$400K/yr), custom SharePoint MOC workflows	NEOS produces a *cryptographically signed, replayable governance log* — not a PDF approval chain stored in a folder. This is what ISO 9001 / IATF 16949 auditors actually want and what the EU ESPR product-passport audit trail will require.
OAS (Interoperability Fabric)	SAP S/4HANA and ERP connectors, Rockwell FactoryTalk MES, AVEVA PI System / OSIsoft historian, Siemens MindSphere, GE Proficy, OPC-UA broker, MQTT ingest for PLC and sensor fleets. The layer that eliminates the "3 Python ETL scripts and a MuleSoft instance we hate" architecture.	MuleSoft ($80K–$2M+/yr), Boomi ($6.6K–six-figure), in-house Python ETL teams (2 FTE = $300K/yr loaded)	OAS carries the full IT/OT connector library from OAS Web4 — the connector authors who know Rockwell, Siemens, and AVEVA from the inside. Connector reuse is built into the business model, so each successive plant rollout costs less than the first.
Fractal Engine (Intelligence)	Predictive OEE modeling per line and per asset, downtime root-cause classification, scrap and yield prediction, Scope 3 emissions estimation per SKU for DPP compliance, real-time signal generation from the IT/OT data fabric OAS feeds.	AspenTech Mtell ($300K–$2M/yr), C3 AI APM ($1M–$5M/yr), Cognite ($500K–$3M/yr), Sphera ESG ($150K–$600K/yr)	Fractal Engine treats AMP spatial events and NEOS governance events as first-class signal types. OEE predictions are anchored to scene positions, not abstract asset IDs. Scope 3 MRV feeds directly from the production telemetry, not from spreadsheet estimates.

Full-stack is the default recommendation in manufacturing — not because we want the bigger deal, but because a twin without intelligence is a viewer, and intelligence without integration has no data. The four layers compose into a single operational fabric; selling fewer than three creates an incomplete

solution that the 70% pilot-purgatory statistic will eventually claim.

Tie Town [TX] serves as the proof-of-capability reference deployment. The same AMP + OAS + Fractal architecture used in Tie Town's operational coordination is the template for plant-floor deployments. Buyers skeptical of a new vendor can tour the reference deployment and inspect the live Hexon spatial model before signing.

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3. Engagement tiers — Small / Mid / Large

3.1 Small: 1 plant, 1 line

Definition: Single plant, 1–2 production lines, <50 machines, <100 OT-connected assets. Typical buyer is a $20M–$150M revenue manufacturer or a corporate innovation team running a line-level pilot with a defined productionization path.

Recommended bundle: AMP + OAS + Fractal (Triad). NEOS is optional at this scale unless there is a regulatory driver (FSMA, ISO 9001 MOC, DPP pilot). If NEOS is added, it becomes a Full Stack engagement.

Friction-cost anchor for this scale: $500K–$2M/yr. Targeted capture: 8–15% = $40K–$300K/yr, justifying the license spend at payback <24 months.

Worked quote — Small Triad (AMP + OAS + Fractal):

Line item	Amount	Notes
AMP Small license	$28K/yr	1 site × $28K/site/yr (mid of $18K–$45K range)
OAS Small license	$36K/yr	<10 integrations (SAP connector + Rockwell MES + AVEVA historian + 2 OPC-UA endpoints)
Fractal Small license	$50K/yr	<5 data sources (PLC telemetry, MES, historian, ERP quality module, energy meter); <1TB managed
Subtotal pre-bundle	$114K/yr	
Triad bundle discount (18%)	–$20.5K	
Licensed ARR	**$93.5K/yr**	
Y1 services (Molecule SKU deployment + connectors)	$200K	Molecule SKU: $180K (1 site, full-stack, ≤100 users, 8–14 weeks) + OAS connector bundle: SAP $60K + Rockwell $80K + historian $50K = $190K × 15% connector-bundle discount = $162K; blended ~2.1x license
Sustained upkeep (Silver SLA, 22%)	$20.6K/yr	Starts Y2
Infra (pass-through + 12% mgmt fee)	~$24K/yr	Cloud reference small: ~$1,200/mo + sensor ingest $600/mo
Y1 total	**~$318K**	License + services + Y1 infra
3-yr TCO	**~$550K–$620K**	Y2–Y3: license + upkeep + infra; services taper to ~0.4x in Y2

Capture rate check: $93.5K license ÷ $1.25M friction midpoint = **7.5% value share** — inside the 5–15% manufacturing band.

Cross-reference: This is the line-level analog to the AMP Small tier in `layers/amp.md` §3, composed with the OAS Mid-entry and Fractal Small anchors in `layers/oas.md` §3 and `layers/fractal-engine.md` §3.

3.2 Mid: 3–12 plants

Definition: 3–12 plants, mixed legacy and modern OT, typically $200M–$3B revenue manufacturer pursuing a digital transformation wave. Often a phased program: Wave 1 (2–3 pilot plants) → Wave 2 (remaining fleet). Average site rollout cadence: 1–2 plants per quarter after the template is established.

Recommended bundle: Full Stack (AMP + NEOS + OAS + Fractal). At 3+ plants, MOC and supplier qualification governance (NEOS) generate enough audit surface area to justify the layer. The NEOS investment pays for itself in CAPA cycle time reduction and auditor satisfaction alone.

Friction-cost anchor for this scale: $5M–$20M/yr across the fleet. Targeted capture: 6–10% = $300K–$2M/yr.

Worked quote — Mid Full Stack (6-plant wave):

Line item	Amount	Notes
AMP Mid license	$150K/yr	6 sites × $25K/site/yr (within $80K–$250K range)
NEOS Mid license	$55K/yr	3 councils (plant-ops, quality/MOC, supplier-risk) × $18K/council + governance design overhead
OAS Mid license	$160K/yr	20–30 integrations: SAP × 6 plants + Rockwell MES × 6 + historian × 6 + 3 cross-plant OPC-UA bridges + procurement connector
Fractal Mid license	$220K/yr	10–15 data sources across 6 plants: PLC telemetry, MES, historian, ERP, quality, energy, workforce; 2–5TB managed; predictive OEE + downtime + scrap + Scope 3 models
Subtotal pre-bundle	$585K/yr	
Full Stack bundle discount (25%)	–$146K	
Licensed ARR	**$439K/yr**	
Y1 services (~2.2x license blend)	$966K	Hex SKU deployment (2–5 sites, 14–24 weeks): $600K–$700K + OAS connectors (SAP $120K + Rockwell $160K + historian $70K, 15% bundle discount) = $296K; blended services ~2.2×
Sustained upkeep (Silver SLA, 22%)	$97K/yr	Starts Y2
Infra (pass-through + 12% mgmt fee)	~$72K/yr	Cloud reference mid per site: $6K/mo ÷ 6 sites in Y1; scales with fleet
Y1 total	**~$1.5M–$1.6M**	License + services + Y1 infra
3-yr TCO	**~$2.8M–$3.4M**	Y2–Y3 license + upkeep + infra; services taper; Wave 2 plants add per-site rollout cost of $80K–$200K/plant

Capture rate check: $439K license ÷ $10M friction midpoint = **4.4% value share.** Low end of the band — defensible because the full-stack friction-cost displacement (downtime + pilot-purgatory + MuleSoft/integration-debt + ESG compliance) is better quantified at the discovery sprint before quoting. If downtime at 6 plants runs $3M/yr and OEE uplift from Fractal Engine is validated at 3 OEE points, capture rate on downtime alone is $439K ÷ $3M = 14.6% — top of the band.

Note on new-site rollout economics: Per the research brief, template-based additional plant deployments run $50K–$250K per plant once the reference architecture is established. The Hex SKU deployment covering the first 2–5 plants absorbs most of the architectural cost; subsequent plants use the Atom ($35K–$90K) or a scaled Molecule SKU.

Cross-reference: The 6-plant full-stack example aligns with the `01-pricing-model.md` §4 worked mid-tier example ($443K license ARR / $1.5M–$1.7M Y1) and with `layers/amp.md` §7.2 (the 12-plant Tier-1 OEM example which uses this template scaled to 12 sites).

3.3 Large: Tier-1 OEM, 25+ plants

Definition: Tier-1 OEM or large industrial conglomerate, 25+ plants globally, typically $3B+ revenue. Full IT/OT transformation program with SAP/Oracle ERP, multi-vendor MES (Rockwell + Siemens + AVEVA in the same fleet), OSIsoft PI historian estates, and active ESG/Scope 3/DPP obligations. COO and CFO sponsorship required. SI partner (Accenture Industry X or Capgemini) typically embedded.

Recommended bundle: Full Stack, with Gold or Platinum SLA, and likely an outcome-based addendum on OEE uplift. Tessellation deployment SKU when the program spans >6 simultaneous plant waves.

Friction-cost anchor for this scale: $50M–$200M/yr. Targeted capture: 5–8% = $2.5M–$16M/yr. The friction anchor is documented and defensible: per the `layers/amp.md` §7.2 reference deployment, a 12-plant discrete manufacturer runs ~$4M/yr/plant in unplanned downtime ($48M total), with $12M+ in sunk pilot POC costs and 8 stalled initiatives. At 25+ plants, the total friction window exceeds $100M/yr before ESG compliance costs are added.

Worked quote — Large Full Stack (25-plant global program):

Line item	Amount	Notes
AMP Large license	$700K/yr	25 sites × ~$28K/site/yr (within $350K–$1.2M range; volume discount applies at 20+ sites)
NEOS Large license	$220K/yr	Multi-council federated: plant-ops councils × 25, quality/MOC council, supplier risk council, ESG/DPP governance council; 500+ members across the enterprise
OAS Large license	$750K/yr	50+ integrations: SAP global + Oracle Fusion variants + Rockwell FactoryTalk fleet + Siemens MindSphere + AVEVA PI × multiple historian estates + OPC-UA broker per plant cluster + supply-chain connector to Tier-2 suppliers
Fractal Large license	$900K/yr	20+ data sources per plant cluster, >20TB managed; predictive OEE per line, downtime root-cause, scrap, yield, Scope 3 per SKU for ESPR DPP; 6–10 custom models in Y1
Subtotal pre-bundle	$2.57M/yr	
Full Stack bundle discount (25%)	–$643K	
Licensed ARR	**$1.93M/yr**	
Y1 services (Tessellation SKU + connectors)	$4.2M–$5.0M	Tessellation SKU: $4.5M midpoint (multi-wave, 9–18 month horizon, Accenture Industry X co-delivery) + OAS connector program: 15+ connectors at bundle rates = $800K–$1.2M; blended ~2.3×
Sustained upkeep (Gold SLA, 25%)	$483K/yr	Starts Y2; Gold is default for regulated/enterprise
Infra (pass-through + 12% mgmt fee)	$400K–$700K/yr	Multi-region cloud for EU + NA + APAC; edge nodes per plant cluster; Tie Town reference colocation for primary federation layer
Y1 total	**~$6.5M–$7.5M**	
3-yr TCO	**~$14M–$18M**	Y2–Y3: license + upkeep + infra; services taper to ~0.5× in Y3; additional plant waves billed per Atom/Molecule SKU

Capture rate check: $1.93M license ÷ $100M friction midpoint = **1.9% value share** at nominal. With outcome-based addendum: if Fractal Engine documents 3 OEE points across 25 plants (conservative for a world-class implementation), at $10M revenue/plant/yr and 3% OEE-to-revenue correlation, documented savings ≈ $7.5M/yr. Performance fee at 20% of savings = $1.5M/yr on top of the base license. Combined effective value share: ($1.93M + $1.5M) ÷ $100M = **3.4%** — well within the defensive 5–15% band and below the 2× performance-fee cap.

Cross-reference: This is the full-stack extension of `layers/amp.md` §7.2 (the 12-plant Tier-1 OEM example) and `layers/fractal-engine.md` §7.2 (the large full-stack manufacturing program). The AMP + OAS + Fractal triad quote from `amp.md` §7.2 ($1.27M/yr license at 12 plants, –18% triad discount) scales here to 25 plants with NEOS added and the 25% full-stack discount applied.

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4. Industry-specific procurement signals

Decision makers by deal size:

Threshold	Decision maker	Priorities	Notes
<$250K	Plant Manager / VP Operations	Payback <18 months, minimal IT dependency, OT-safe rollout	Approves the pilot; does not approve the fleet program
$250K–$2M	CIO / VP Digital / CDO	Architecture fit, OT cyber posture, cloud strategy, vendor roadmap, OAS connector coverage for the existing stack	Often co-signs with Plant Manager above $500K
>$2M	COO / CFO (co-sign required)	Program ROI, supply-chain resilience, ESG/Scope 3 obligations, CapEx vs OpEx classification	CFO increasingly focused on DPP compliance timeline post-2025
ESG / DPP scope	CSO / Head of Sustainability	Scope 3 accuracy, ESPR audit trail, CBAM disclosure format	Enters the buying committee when Fractal Engine MRV is in scope
>$500K (formal)	Procurement	RFP/RFQ, MSA terms, SLA benchmarking, TCO disclosure, source-escrow availability	Require binding TCO disclosure; hidden-cost discovery is the top deal-kill trigger

RFP cadences:

- **Enterprise CapEx > $5M:** 6–18 month cycle. Typical flow: discovery sprint → business case → RFI → RFP → proof-of-value (paid, $50K–$150K) → vendor selection → MSA/PFEL → SOW. Accenture or Capgemini often runs the RFP process on behalf of the buyer.
- **Mid-market OpEx SaaS $500K–$2M:** 4–9 months. Discovery sprint → business case → shortlist → paid POC → PO. Procurement enters formally above $500K.
- **ESG/DPP compliance-deadline-driven:** 2–4 months when mandate is imminent (CBAM, DPP batteries 2025). This is the fastest procurement motion in the vertical right now.

CapEx vs OpEx bias:

Post-COVID, the dominant preference is **OpEx/subscription** (annual SaaS PrimusNeo ARR + services SOW) to preserve capital. OT infrastructure (edge hardware, OPC-UA gateways, historian server upgrades) remains CapEx-classified. The deal structure that wins: software OpEx (PrimusNeo ARR), connectivity CapEx (OT HW the SI or plant team procures separately), services as project CapEx (Y1 transformation budget) tapering to OpEx upkeep.

Discovery-to-PO timeline: 3–6 months for mid; 9–18 months for large. A paid discovery sprint ($40K–$80K, crediting 100% to the engagement) is the right first step at any scale. It produces a binding SOW and establishes the friction-cost baseline that makes the COO/CFO approval tractable.

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5. Rejection-risk mitigations

Top rejection trigger	Research brief figure	PrimusNeo response
Vendor lock-in	Single largest recurring rejection	Hexon v1.0.0 file standard is open Apache-2.0 — the buyer's spatial data is portable on day 1. OAS Connector SDK is open MIT/Apache-2.0. NEOS and Fractal Engine are CC-NC — the community version is always forkable. Data export is a non-negotiable contract term (PFEL §6). Source escrow via Iron Mountain for AMP/OAS closed-core above $500K ARR.
IT/OT integration complexity	42% cite OT integration as primary scaling barrier	OAS ships with first-class MES/OT connectors (Rockwell FactoryTalk, Siemens MindSphere, AVEVA PI, GE Proficy, OPC-UA broker) authored by OAS Web4 — the connector authors who know these systems from the inside. The Tie Town [TX] reference architecture demonstrates live IT/OT integration. Fixed-fee connector SOWs with 90-day warranty eliminate the "integration cost overrun" failure mode.
Unclear ROI / unquantified payback	48% report buyer's remorse from integration overruns; <24-month payback required at COO/CFO	Fractal Engine predictive ROI model is instrumented in the 60-day baseline phase before the contract goes hot (per `01-pricing-model.md` §8 outcome-based addendum). Downtime, OEE, and scrap baselines are established from live data before the buyer commits. Binding TCO disclosure is PrimusNeo standard in every RFP response.
Pilot-to-scale gap	70% of pilots never scale	The engagement model is designed around a defined productionization path from day 1 — the Atom/Molecule/Hex/Lattice/Tessellation SKU ladder gives the buyer a fixed-fee path from pilot to fleet with known per-plant unit economics ($50K–$250K/plant for template rollouts). NEOS governance councils for the digital transformation program ensure internal buy-in is treated as a deliverable, not an assumption.
Hidden cost discovery	Integration cost overruns cited by 38%; delays by 35%	All connector costs are fixed-fee with explicit scope. OAS per-connector overages are flagged in the connector-audit discovery output. Infra is always a separate pass-through line (never bundled into license), so OT hardware, historian per-tag overages, and cloud compute costs are visible before signature. Sustained upkeep at 22% Silver SLA covers connector maintenance against upstream API changes — the most common hidden-cost cliff.
Data sovereignty / cloud residency	European manufacturers reject US-only cloud deployments	BYO-cloud option is standard: the buyer owns the cloud account, PrimusNeo manages. AMP runs offline-first with optional EU-region cloud. Fractal Engine compute can be EU-only. NEOS coordination server is deployable on-premise or in an EU-residency cloud region. The Tie Town reference architecture supports hybrid deployments where production IP and quality data never leaves the buyer's cloud boundary.

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6. Regulatory + compliance considerations

Manufacturing is subject to the most acute near-term regulatory pressure of any PrimusNeo vertical. Three mandates are live or imminent:

EU ESPR / Digital Product Passport (DPP):

Category	Mandatory DPP deadline	What is required
Batteries (≥2 kWh)	February 2027 (phased from 2025 pilots)	Lifecycle data, Scope 3, recycled content, supply chain traceability
Textiles	2027	Material composition, origin, sustainability metrics
Electronics / ICT	2027–2029	Material, repairability score, recycled content
Industrial machinery	2029	Performance, energy, end-of-life data

PrimusNeo answer: NEOS generates cryptographically signed supply-chain attestations (supplier qualification, CIP-013-equivalent risk acceptance, incoming material traceability) that feed directly into the DPP audit trail. OAS connects to Tier-2 and Tier-3 supplier systems to aggregate the upstream data without manual data collection. Fractal Engine computes per-SKU Scope 3 emissions from live production telemetry and outputs them in the structured formats ESPR will require. The full-stack engagement produces DPP-ready data as a **by-product of normal operations** — not a separate compliance project.

EU Carbon Border Adjustment Mechanism (CBAM):

CBAM mandatory disclosure is in force for steel, cement, fertilizer, aluminum, electricity, and hydrogen as of January 2026, with expanded scope ongoing. The mechanism requires **verified Scope 3 embedded-carbon disclosure** per imported tonne. For EU-importing manufacturers, non-compliance is a cost line, not just a reputational risk.

PrimusNeo answer: Fractal Engine MRV (Measurement, Reporting, Verification) signal generation from the OAS-connected production data fabric produces CBAM-compatible embedded-carbon estimates with the audit trail CBAM verifiers require. The governance layer (NEOS) produces the attestation that the data was collected under a defined, auditable methodology — the difference between a number and a verified number.

Scope 3 mandatory disclosure (CSRD / SEC Climate Rule):

EU CSRD mandatory Scope 3 disclosure applies to ~50,000 large and listed companies in scope. Scope 3 Category 1 (purchased goods/services) and Category 11 (use of sold products) are the hardest to compute from manufacturing floor data.

PrimusNeo answer: The OAS + Fractal Engine combination produces Category 1 supplier data directly from the OAS-connected supply chain fabric (no spreadsheet collection), and Category 11 estimates from the AMP product lifecycle spatial model. This is the data flow that point-solution ESG platforms (Sphera, Watershed, Circularise) approximate from approximations; PrimusNeo generates it from live operational telemetry.

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7. Steward + service partner notes

PLAN Systems (AMP steward): Primary delivery partner for plant-floor spatial scene work — CAD ingest, PLC telemetry binding, Hexon scene authoring, and the premium PLAN 3D industrial interface components. PLAN Systems subcontracts at the AMP Spatial/3D Engineer day rate ($2,400/day per `03-`

`services-catalog.md` §2.1). For any large engagement with significant scene-authoring scope, PLAN Systems should be named in the SOW as the primary AMP delivery sub-partner. PLAN receives 15% steward royalty on AMP license revenue per `02-licensing-model.md` §4.

Escherbridge (Fractal Engine steward): Primary delivery partner for Fractal predictive maintenance model development, OEE classifier training, downtime root-cause model engineering, and Scope 3 / DPP MRV signal generation. Escherbridge ML Engineers ($2,800/day) are the right resource for the 4–8 custom model buildout that the large-tier engagement requires. Escherbridge receives 18% steward royalty on Fractal Engine license revenue.

OmniOne (NEOS steward): Primary delivery partner for NEOS change-control council design, MOC charter authoring, supplier risk governance council instantiation, and ETHOS methodology facilitation. OmniOne Governance Designers ($2,400/day) run the governance design workshop and charter authoring engagements. OmniOne receives 18% steward royalty on NEOS license revenue.

OAS Web4 (OAS steward): Primary delivery partner for OT/historian connector authoring (Rockwell FactoryTalk, AVEVA PI, GE Proficy, OPC-UA, MQTT), IT integration (SAP, Oracle), and the OAS federation engine deployment. OAS Integration Engineers ($2,200/day) carry the connector buildout work. OAS Web4 receives 15% steward royalty on OAS license revenue.

SI channel — Accenture Industry X / Capgemini Engineering: For large-scale Tessellation-tier programs ($5M+ TCV), co-delivery with a Tier-1 global SI is standard. Accenture Industry X and Capgemini Engineering are the recommended partners given their Industry 4.0 transformation practices and OT/MES certification pools. SI margin on implementation is 20–40%; PrimusNeo retains the license ARR and sustains upkeep revenue. SI co-delivery is the right structure when the buyer's procurement team insists on a named SI as the primary delivery party — PrimusNeo is the software vendor and the SI is the prime implementation contractor. Introduce SI partners at the RFP stage for any deal above $3M TCV.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Manufacturing engagement model

Robotics — Engagement Model

> ***Vertical:*** *Robotics (industrial, AMR, humanoid, drone, surgical, agricultural, fleet/RaaS)* ***Research brief:*** `../research/robotics.md` ***Anchors:*** `../01-pricing-model.md` , `../research/layer-comparables.md`

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1. The Friction-Cost Anchor

Robotics is the vertical where the infra layer has been commoditized by the largest player in the market — and that is the opportunity. NVIDIA Isaac ROS and ROS 2 are free. AWS RoboMaker deprecated for new customers in September 2025, pushing teams to AWS Batch or direct Isaac Sim on EC2 (GPU cost ~$3–$6/hr on G6e L40S instances). This commoditization of the robot OS and simulation layer creates an acute demand for a differentiated coordination and intelligence layer sitting above it.

Comparable anchors that set the price ceiling and floor:

Vendor / Reference	Model	Rate
InOrbit	Fleet management SaaS	~$500/robot/month (PAYG high-water; volume discounts annual)
Locus Robotics RaaS	Warehouse AMR all-in (hw + sw + support + optimization)	$2,000–$4,000/robot/month
Knightscope K5	Security patrol RaaS	~$60K–$70K/robot/year (~$11/hr)
Foxglove Studio	Robotics observability	$90+/user/month enterprise
Formant	Telemetry + fleet ops	Quote-based; data-egress cliff at scale
ROS 2 / NVIDIA Isaac ROS	Robot OS + middleware	Free (commoditized infra)
AWS RoboMaker	Cloud robotics runtime	Deprecated Sept 2025 for new customers
ISO 10218-1:2025 / -2:2025 cell certification	TUV / Intertek / UL; 7–12 mo timeline	$50K–$200K+ per certified cell
FDA 510(k) (surgical Class II)	FDA fee + consulting overhead	$26K–$54K + 142-day median review
EU AI Act conformity assessment (high-risk AI, Annex III)	Per high-risk system	$20K–$80K per system

Friction math: The direct cost of running an uncoordinated heterogeneous robot fleet is $50K–$200K+ in cert overhead per cell, 70% pilot-to-fleet failure rate (McKinsey), and $5K–$50K per unplanned downtime event in logistics contexts. A 100-robot fleet at InOrbit rates alone costs $600K/yr in fleet management software before any coordination intelligence. PrimusNeo positions the coordination/intelligence layer at **$200–$800/robot/month** — above the fleet-management floor ($50–$500/robot/month) and well below

the full RaaS ceiling ($2K–$4K/robot/month). For a 100-robot fleet at the midpoint ($500/robot/month), that is $600K/yr for the coordination layer — a defensible add-on when positioned against the alternative: a failed pilot ($150K sunk + 12 months lost) or a costly proprietary OEM solution.

PrimusNeo capture band: 10–20% of fleet coordination friction, per the pricing model's robotics row.

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2. PrimusNeo Positioning Per Layer in Robotics

The four-layer stack maps cleanly onto the robotics coordination problem. The key positioning principle: PrimusNeo operates **above the robot OS** (Isaac, ROS 2, proprietary OEM stacks) and **below the full RaaS operator bundle**. It is the coordination intelligence tier that neither NVIDIA nor Locus nor the OEM provides.

AMP — Fleet Spatial Coordination

AMP is the operational ground layer for robotics: the multi-site spatial model that attaches robot positions, mission states, safety zones, and anomaly events to a navigable scene. In robotics vocabulary it is the **fleet spatial OS** that sits above VDA5050-level fleet management and below the robot proprietary controller.

- Geofencing and safety-zone management that satisfies ISO 10218-1:2025 spatial constraints
- Multi-site dispatch: a single spatial scene spanning 3 DCs with 200 AMRs and 2 fixed-arm cells becomes navigable and governable from one pane
- Offline-first architecture makes AMP viable in low-connectivity environments (agricultural robots in the field, underground mining, edge-network industrial cells)
- Composable with NVIDIA Omniverse reference architectures (AMP is not Omniverse; it is positioned as the coordination-tier complement, not the simulation replacement)

NEOS — Fleet Governance and Safety Attestation

NEOS is the governance and accountability layer for a robot fleet: the council structure that turns SLA commitments, safety-event attestations, and change-control approvals into signed, auditable, replayable records.

- **Safety attestation traceability:** Every ISO 10218-compliant cell inspection, lockout/tagout approval, and HRC hazard sign-off becomes a NEOS governance event — a signed record that is cert-ready evidence, not a PDF in a SharePoint folder
- **Fleet SLA councils:** For RaaS deployments, NEOS structures the operator–vendor SLA governance as a council (uptime commitments, maintenance windows, fleet rebalancing approvals) with on-chain-anchored records that support EU AI Act human oversight requirements
- **EU AI Act Annex III compliance:** Fractal classified as a high-risk AI system under Annex III requires conformity assessment documentation, logging, and human oversight — NEOS provides the governance event structure that satisfies the human oversight requirement

OAS — Vendor-Neutral Robot API Fabric

OAS is the critical interoperability layer for robotics. Robot vendors ship proprietary controllers: FANUC KAREL, ABB RAPID, KUKA KRL, Universal Robots URScript, Boston Dynamics API, Agility Digit API. Each is a bespoke integration problem. OAS provides the connector library and federation engine that abstracts these into one API surface.

This is the connector-buildout work that defines the robotics engagement scope. Every robot vendor added to a fleet requires a custom OAS connector. The scope and pricing of that work is explicit (see §3 worked quotes) — it is not absorbed into the platform license.

- Runs alongside existing NVIDIA Isaac fleet management (VDA5050) without forking the ROS graph — additive, not replacement
- FAA Part 107 drone deployments: OAS includes UTM (Unmanned Traffic Management) API integration as a robotics-vertical connector class
- Post-AWS RoboMaker deprecation, OAS positions as the vendor-neutral integration fabric for teams migrating to AWS Batch + Isaac Sim on EC2

Fractal Engine — Predictive Maintenance and Fleet Analytics

Fractal is the intelligence layer that converts robot telemetry into operational signals. In robotics it is the **predictive maintenance and fleet performance analytics engine** — the layer that makes a robot fleet's operational data into a durable archive and real-time decision surface.

- Predictive maintenance ROI model: 1 avoided unplanned downtime event in a logistics AMR fleet = $5K–$50K. A 500-robot fleet avoiding 2 incidents/month across the fleet justifies the Fractal license tier on its own
- Custom model retraining per robot model class (AMR, fixed arm, drone) trained on customer's sensor data; blended weights remain customer IP per the PCL default
- Fractal audit logs are **cert-ready evidence** for ISO 10218 cell certification and EU AI Act conformity assessment — the Fractal telemetry archive is the paper trail TUV/Intertek/UL reviewers require
- Per-mission metering option: $0.002–$0.01/mission event for deployments preferring consumption-based pricing over flat annual license

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3. Engagement Tiers — Small / Mid / Large

Small: Startup Fleet or Single-Facility Integrator (<10 Robots)

Profile: Early-stage robotics startup deploying a pilot fleet, or a systems integrator managing a single-customer fixed installation of cobots or AMRs. Current fleet management at $0–$5K/month (free tier or entry InOrbit). Pain: no mission-level coordination, no governance, no cross-system telemetry, every ISO 10218 cert event is tracked in spreadsheets.

Bundle: OAS Small + Fractal Small. OAS provides the interop to ROS 2 / proprietary controller; Fractal captures telemetry and builds the predictive maintenance baseline. AMP added at renewal if multi-site; NEOS added when they need safety attestation traceability.

Worked quote (8-robot cobot cell, single facility, systems integrator):

Line item	Amount
OAS Small license (1 domain, <1M calls/mo; 2 connectors included)	$30K/yr
Fractal Small license (<5 data sources, <1TB)	$45K/yr
Duet bundle discount (–10%)	–$7.5K
License subtotal	**$67.5K/yr**
OAS connector buildout: FANUC KAREL + ROS 2 (2 connectors, fixed-fee)	$55K one-time
Fractal data pipeline buildout (2 sources: PLC + historian)	$40K one-time
Atom deployment SKU (single site, 2 layers)	$50K one-time
Y1 services total	**$145K**
Sustained upkeep Silver SLA (22% of license)	$14.8K/yr (starts Y2)
Infra pass-through + 12% mgmt fee	~$800/mo (~$9.6K/yr)

Y1 total: ~$222K. **3-yr TCO:** ~$290K–$320K (services taper Y2–Y3).

Friction anchor: ISO 10218 cert-prep spreadsheet labor ($15K–$40K/cell), InOrbit or equivalent fleet mgmt at $0–$3K/month, 1 avoided downtime incident/yr ($10K–$30K value). **Capture rate:** $67.5K license ÷ estimated $80K annual friction displacement = ~84% of direct friction displacement — high, because this buyer has almost nothing today and the connector buildout is real capability they could not otherwise buy.

Mid: Multi-Site Fleet Operator or Mid-Size OEM/Integrator (50–500 Robots)

Profile: Mid-size logistics operator running 200 AMRs across 3 DCs, or a Tier-2 OEM integrator delivering fleet-as-a-service to 5 customers with heterogeneous robot vendors. Current spend: InOrbit or Formant $25K–$250K/yr + custom SI $200K–$800K. Pain: 3 different robot vendors (KUKA, Universal Robots, a domestic AMR brand), multi-site dispatch without a unified spatial model, safety audit readiness before an ABB/FANUC partnership requires ISO 10218 attestation, no predictive maintenance.

Bundle: AMP Mid + NEOS Mid + OAS Mid + Fractal Mid (full stack). AMP handles multi-site spatial coordination. NEOS handles fleet SLA governance and safety attestation councils. OAS handles the heterogeneous vendor API surface. Fractal handles predictive maintenance and analytics.

Worked quote (200-robot fleet, 3 sites, 3 robot vendors):

Line item	Amount
AMP Mid license (4–15 sites, 50–500 concurrent spatial)	$140K/yr
NEOS Mid license (multi-council: SLA council × 3 sites + safety council)	$55K/yr
OAS Mid license (10–50 integrations, 1M–50M calls/mo)	$160K/yr
Fractal Mid license (5–20 sources, 1–20TB managed)	$200K/yr
Full stack bundle discount (–25%)	–$138.75K
License subtotal	**$416K/yr**
OAS connectors: KUKA KRL + UR URScript + domestic AMR API (3 connectors × ~$45K avg)	$135K one-time
AMP spatial environment buildout (3 sites)	$180K one-time
Fractal pipeline + 2 custom predictive maintenance models	$180K one-time
NEOS governance design + safety attestation council setup	$80K one-time
Hex deployment SKU (3 sites, full stack)	$600K one-time (midpoint)
Y1 services (blended ~1.8x license)	**~$750K**
Sustained upkeep Gold SLA (25%, robotics regulated default)	$104K/yr (starts Y2)
Infra pass-through + 12% mgmt fee	~$15K/mo (~$180K/yr)

Y1 total: ~$1.35M. **3-yr TCO:** ~$2.4M–$2.8M (within the $750K–$2.5M Mid research band; lands at the upper end because full-stack + 3 connector buildouts).

Friction anchor: Current InOrbit/Formant + SI spend: $250K–$800K/yr. 70% pilot failure rate means 1 failed pilot at this scale costs $100K–$300K in sunk SI + 9–12 months of delay. Predictive maintenance: 4 avoided AMR downtime events/month across 3 DCs × $15K/incident = $720K/yr in avoided logistics disruption. ISO 10218 cert-prep per cell: $75K avg per cell × 3 cells = $225K one-time displacement. **Capture rate:** $416K ÷ estimated $1.4M annual friction = ~30% value share — defensible inside the 10–20% capture band for robotics (the higher number reflects displacement of real line-items, not just advisory uplift).

RaaS opex framing: "You're already paying $2,500/robot/month to your AMR vendor all-in. We add $416K/yr ÷ 200 robots = $173/robot/month for coordination intelligence, safety attestation traceability, and predictive maintenance that cuts unplanned downtime 15%. That is a $208/robot/month net gain on your existing $2,500 RaaS invoice."

Large: Global OEM or National Logistics Fleet (1,000+ Robots)

Profile: Global robot OEM (ABB, FANUC, KUKA, Boston Dynamics class) seeking to offer a coordination intelligence platform as an embedded SKU in their fleet management software, OR a national logistics operator (Symbotic class) running 1,000+ AMRs across 20+ facilities. Current spend: proprietary OEM fleet software + SI ($1M–$5M+/yr); custom digital twins at $500K–$3M. Pain: cross-vendor fleet interop (EU

sites run KUKA; US sites run Fanuc; AMR layer is Symbotic; surgical arm in one facility is Intuitive Surgical); global compliance (ISO 10218:2025 across all cells, EU AI Act by 2026 enforcement, FDA SaMD layer for any clinical robotics); no unified fleet analytics.

Bundle: Full stack at Large tier. OAS federation engine on. NEOS federated governance councils per region. AMP multi-org spatial federation. Fractal large with custom model retraining per robot class.

Worked quote (1,200-robot global fleet, 20 sites, 5 robot vendors + OEM embedded licensing track):

Line item	Amount
AMP Large license (15+ sites, 500+ concurrent spatial, multi-org federation)	$700K/yr
NEOS Large license (federated governance councils per region + EU AI Act oversight councils)	$250K/yr
OAS Large license (50+ integrations, >50M calls/mo, cross-org federation)	$900K/yr
Fractal Large license (20+ sources, real-time + archive, >20TB, custom retraining)	$1.2M/yr
Full stack bundle discount (–25%)	–$762.5K
License subtotal	**$2.29M/yr**
OAS connectors: FANUC KAREL + ABB RAPID + KUKA KRL + BD API + Symbotic API (5 connectors × ~$60K avg)	$300K one-time
AMP federation enablement (cross-org, 20 sites)	$450K one-time
Fractal custom models per robot class (6 models) + regulatory validation for EU AI Act	$480K one-time
NEOS EU AI Act conformity assessment council architecture + audit-log buildout	$150K one-time
Tessellation deployment SKU (20+ sites, full stack, regulated)	$5M one-time (midpoint)
Y1 services (blended ~1.8x license)	**~$4.1M**
Sustained upkeep Platinum SLA (custom %, robotics regulated)	$600K/yr (starts Y2)
Infra pass-through + 12% mgmt fee	~$60K/mo (~$720K/yr)

Y1 total: ~$7.1M. **3-yr TCO:** ~$11M–$14M (within the $3M–$12M Large research band at the upper end; justified by the 5-vendor connector scope and EU AI Act regulatory depth).

OEM embedded licensing alternative path: Skip the services-heavy Y1 and embed OAS + Fractal as licensed modules in the OEM's existing fleet management software. **Revenue share: $50–$200/robot/year royalty** (depending on layer depth) billed by OEM to their end customers. At 1,200 robots × $150/robot/yr = $180K/yr royalty. At 10,000 robots across the OEM's installed base = $1.5M/yr royalty. 3–8% of operator RaaS revenue for deeper platform-partner arrangements.

Friction anchor: NVIDIA/ABB digital twin programs run $2M–$8M/yr in software + SI. ISO 10218 cert across 20 cells: $100K avg × 20 = $2M one-time. EU AI Act conformity assessment: $50K avg per high-risk system × 15 systems = $750K. Pilot failure avoidance: 70% failure rate applied to a 3-year $8M internal

build program = $5.6M risk. Estimated annual friction window: $8M–$15M/yr. **Capture rate:** $2.29M ÷ $11.5M (midpoint) = **~20% value share** — top of the 10–20% capture band, defensible for global OEM with EU AI Act exposure.

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4. Industry-Specific Procurement Signals

Buyer Coalition

Title	Authority	Entry Point
VP Robotics / Head of Automation	Technical champion; fleet roadmap owner	OAS interop demo across heterogeneous vendor fleet + Fractal predictive maintenance ROI model (avoided downtime = $X/incident × fleet size)
CTO	Architecture sign-off; multi-year platform decision	Full stack briefing + Tie Town reference architecture; position AMP + OAS as the coordination fabric above Isaac, not a replacement for Isaac
COO / VP Operations	Budget owner; RaaS opex framing is the unlock	"Add $X/robot/month to your existing Locus/$3K invoice and recover it in downtime avoidance within 6 months" — CFO-friendly math, not IT capex
Head of Safety / Compliance	Veto power in any regulated engagement; ISO 10218 + EU AI Act + FDA gating	Compliance posture document as first deliverable; Fractal audit logs as cert-ready evidence; NEOS safety attestation council design

Head of Safety is the non-negotiable stakeholder in regulated robotics. Any engagement touching industrial cobots (ISO 10218 HRC cells), surgical arms (FDA 510(k) SaMD exposure), or drone fleets (FAA Part 107 waiver contexts) that does not have the compliance posture doc in hand by end of discovery is a stalled deal.

Sales Motions

RaaS opex motion. The fastest path for warehouse AMR and security patrol contexts. Position PrimusNeo as an add-on line item to the existing RaaS operator invoice. "You're already paying $3,000/robot/month to Locus. We add $300 for cross-site spatial coordination, safety attestation traceability, and predictive maintenance — and we target cutting your unplanned downtime rate by 15%, which on a 200-robot fleet at $15K/incident is $540K/yr." Requires a partnership or referral relationship with the RaaS operator or a direct connection to the COO who signs the RaaS contract.

Pilot-to-fleet funnel. Three-to-six month fixed-fee pilot at $50K–$150K (Atom or single-site Molecule SKU) covering OAS connector buildout for 1–2 robot vendors + Fractal telemetry baseline for 1 facility. Pilot outputs: (a) connector library reusable across customer fleet, (b) Fractal downtime baseline that becomes the gainshare measurement period, (c) reference site for multi-site expansion pitch. McKinsey cites 70% pilot-to-fleet failure — Tie Town reference architecture is the counter-narrative that shortens the sales cycle from 9–18 months to 4–8 months.

OEM embedded licensing. Target ABB, FANUC, KUKA, Universal Robots, Boston Dynamics — embed OAS connector library + Fractal as licensed modules in their fleet management software or cloud portal. Revenue share or per-robot-per-year royalty at scale. Entry: technical partnership agreement + co-sell on OEM's existing deals. This is the highest-leverage distribution channel for reaching 10,000+ robot deployments without proportional services overhead.

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5. Rejection-Risk Mitigations

"You're not ISO 10218 / EU AI Act certified." Mitigation: Compliance posture document delivered as an engagement artifact, not a post-sale promise. Fractal audit logs are architected as cert-ready evidence from day one — the telemetry schema maps to the traceability requirements in ISO 10218-1:2025 Section 5.4 and EU AI Act Article 12 logging requirements. NEOS safety attestation councils produce signed governance events that TUV/Intertek reviewers can audit. The posture doc is a deal gate, not a nice-to-have.

"We're already on NVIDIA Isaac + ROS 2. We don't want to fork the ROS graph." Mitigation: OAS is additive. It runs alongside Isaac without touching the ROS 2 DDS layer. The OAS connector subscribes to ROS topics as a standard subscriber; it does not require modifying the robot's navigation stack or mission planner. AMP sits above the Isaac fleet manager — it consumes VDA5050 fleet state as a data source. The positioning is "coordination intelligence above Isaac, not instead of Isaac."

"Every robot vendor we run needs a custom integration. That's expensive and risky." Mitigation: Explicit per-connector fixed-fee scoping ($25K–$80K per connector) before any engagement begins. Connector scope, timeline, and acceptance criteria are defined in the SOW. PrimusNeo retains rights to harden non-bespoke connectors and redistribute through the OAS connector library — which means connector cost amortizes across multiple customers and future connector upgrades are covered under sustained upkeep. For a fleet with 3 vendors, the connector line is $75K–$180K one-time, not an open-ended integration tax.

"We can't calculate ROI per robot." Mitigation: Fractal's 60-day instrumentation phase (pre-contract measurement period, optionally part of the pilot) establishes the downtime baseline. The gainshare addendum structure (base license at 80% of standard anchor + 15–30% of documented annualized savings above baseline) converts the ROI question from a sales conversation into a contractual obligation. One avoided incident in a logistics AMR fleet at $15K–$50K converts the baseline math on its own.

"We don't have robotics engineers to manage another platform." Mitigation: Managed deployment service (all four services classes — customization, integration, deployment, sustained upkeep — delivered by PrimusNeo + steward partners). NVIDIA partner channel (Isaac/Omniverse certified, $200–$350/hr; projects $250K–$2M) and AWS partner network (post-RoboMaker migration projects, $100K–$500K) provide certified SI capacity that extends the engagement delivery capacity without requiring customer headcount.

"Humanoids are the future but we don't know when they'll be commercial." Mitigation: Do not over-index pricing or engagement design on humanoid robotics as of 2026. Figure, 1X, Agility Robotics, and Boston Dynamics Atlas are pre-commercial at enterprise fleet scale. Humanoid positioning is appropriate

for research and pilot agreements only; commercial fleet pricing models are premature. Track the space; activate humanoid-specific OAS connector development when a reference customer has a commercial fleet above 20 units.

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6. Regulatory and Compliance Considerations

Robotics sits under more regulatory overlay than any other PrimusNeo vertical except healthcare. Every engagement touching industrial cobots, surgical robotics, drones, or service robots in the EU must have a compliance posture document as a named engagement deliverable.

ISO 10218-1:2025 / -2:2025 (Industrial Robots and Robot Cells) Updated 2025 revision strengthens functional safety, HRC (Human-Robot Collaboration), and force/power limiting requirements for cobot cells. Certification via TUV Rheinland, Intertek, or UL; 7–12 month timeline; $50K–$200K+ per certified cell including engineering hours.

PrimusNeo contribution: AMP provides safety-zone geofencing that maps directly to ISO 10218-2:2025 collaborative workspace definitions. Fractal audit logs provide the traceability evidence that notified body reviewers require. NEOS safety attestation governance events provide the change-control record for cell modifications.

ISO 13482:2014 (Personal Care and Service Robots, including Humanoids) Applies to robots operating in proximity to humans in non-industrial settings: service robots, hospital logistics robots, and humanoids. Requires notified body engagement for EU market access.

PrimusNeo contribution: NEOS hazard accountability traceability — every human-proximity event, safety intervention, and operational exception becomes a signed governance event with replay capability for post-incident analysis.

EU AI Act (2024, enforcement 2026) Fractal Engine, when processing operational decision data for robot navigation, fleet dispatch, or safety intervention recommendations, is classified as a high-risk AI system under Annex III (safety component of products covered by Union harmonisation legislation, which includes robots under the Machinery Regulation).

Requirements: conformity assessment, Article 12 logging (full audit trail of all inputs and outputs), Article 14 human oversight mechanisms, technical documentation per Annex IV.

PrimusNeo contribution: Fractal audit log architecture satisfies Article 12. NEOS human oversight councils satisfy Article 14. Conformity assessment documentation: $20K–$80K per high-risk system, bundled into the engagement deliverable as a named SOW item, not post-sale.

FAA Part 107 (Commercial Drones <25kg, US) Knowledge test $175 + $5/drone registration. 90–120 day waiver for operations beyond visual line of sight (BVLOS). UTM (Unmanned Traffic Management) API integration is required for BVLOS commercial fleet operations.

PrimusNeo contribution: OAS UTM API integration as a drone-vertical connector class. Priced per-connector ($30K–$60K fixed-fee) within the standard OAS connector framework.

FDA 510(k) (Surgical Robotics, Class II Medical Devices) FDA submission fee: $6,084–$24,335 (2025 fee schedule) + $20K–$30K consulting. 142-day median review period. Fractal Engine, if processing clinical decision data (e.g., surgical performance analytics, real-time guidance signals), may qualify as Software as a Medical Device (SaMD) under FDA guidance and require 510(k) clearance or De Novo classification.

PrimusNeo contribution: Fractal's telemetry architecture for surgical robotics is scoped to operational performance analytics (robot arm telemetry, procedure time, maintenance prediction) rather than clinical decision support, to stay below the SaMD trigger. Any surgical customer requiring clinical decision support integration must have a regulatory counsel review prior to engagement scoping. This is a deal gate.

RIA TR R15.806 (US Human-Robot Collaboration Technical Report) Not a certification standard — an internal compliance reference. Fractal risk signals and AMP spatial zone mapping support HRC safety documentation under this technical report.

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7. Steward and Service Partner Notes

Layer Stewards in Robotics

PLAN Systems (AMP) leads AMP fleet spatial coordination buildout. PLAN's Hexon file standard is the portable spatial format for robot cell layouts, AMR path maps, and multi-site fleet spatial models. PLAN subcontracts site-walk, spatial environment buildout, and sensor/OT binding services within robotics engagements.

Escherbridge (Fractal Engine) leads Fractal predictive maintenance and custom robot model development. Escherbridge's analytics depth is the differentiator for the robotics vertical — their team has domain context in time-series anomaly detection on robot actuator telemetry that makes the custom model training materially faster than a generalist Fractal deployment. Escherbridge royalty: 18% of Fractal line-item license revenue.

OmniOne (NEOS) leads fleet governance and safety attestation council design. OmniOne's ETHOS methodology is adapted for robotics safety governance: ISO 10218 cell approval councils, EU AI Act human oversight councils, and RaaS SLA governance councils are all deliverable templates OmniOne can instantiate within a standard Governance Design Workshop ($35K–$60K).

OAS Web4 (OAS) leads robot vendor connector buildouts — the critical work in every robotics engagement. Every robot vendor added to an OAS deployment requires a custom connector authored and maintained by OAS Web4. The connector library (FANUC KAREL, ABB RAPID, KUKA KRL, UR URScript, Boston Dynamics API, Symbotic API, DJI drone API, Intuitive Surgical telemetry) is the strategic moat that makes OAS the right choice for heterogeneous fleets. OAS Web4 royalty: 15% of OAS line-item license revenue.

Service Partner Channel

Robotics SI firms (US boutique): $150–$275/hr specialist; $100–$175/hr mid-level. These firms (Bastian Solutions, Honeywell Intelligrated, Dematic, and boutique AMR integrators) are the warm channel into mid-size fleet operators who do not have internal robotics engineering capacity. SI partnership = co-sell arrangement; PrimusNeo provides the coordination layer, SI provides the robot hardware + OEM relationship.

NVIDIA partner network (Isaac/Omniverse certified): $200–$350/hr; fixed-scope projects $250K–$2M. NVIDIA's certified partner channel is the fastest path into large OEM and enterprise fleet operator deals where Isaac Sim and Omniverse are already in the stack. PrimusNeo's positioning as "coordination intelligence above Isaac" makes this a complementary co-sell, not a competitive displacement.

AWS partner network (post-RoboMaker migration): $150–$250/hr; projects $100K–$500K. The AWS RoboMaker deprecation (September 2025) created a migration cohort of robotics teams that need a new cloud robotics runtime strategy. OAS as the vendor-neutral integration layer for AWS Batch + Isaac Sim on EC2 G6e is a specific positioning play for this migration cohort.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Robotics engagement model

Technology — Engagement Model

> ***Vertical:*** *Technology (software vendors, platforms, AI/ML companies, infra providers, telcos, gaming/metaverse, devtool ecosystems)* ***Research brief:*** `../research/technology.md` ***Anchors:*** `../01-pricing-model.md`, `../research/layer-comparables.md`

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1. The friction-cost anchor

Technology buyers are the sharpest cost-aware purchasers in any PrimusNeo vertical — they build the tools that other industries buy. But their internal coordination stack carries real, quantifiable friction:

IDP / developer platform spend. Managed IDP pricing has converged at **$22–$78/developer/month** (Roadie $22, Port Enterprise $60–$78, Cortex $65–$69, OpsLevel $39). These look cheap per seat until headcount scales. A 300-engineer scale-up pays $79K–$281K/yr for portal seats alone — before platform engineering salaries. The counter-anchor is self-hosted Backstage, which appears free but carries **3–5 senior platform engineers at $250K fully loaded = $750K–$1.25M/yr in carrying cost plus $12K–$24K/yr infra**. Three-year Backstage TCO regularly exceeds **$2M at scale**, a figure Roadie, Port, and Humanitec all cite in competitive decks.

AI agent infrastructure. LLMOps tooling (LangSmith Plus at $39/seat/mo, Weights & Biases at $50/seat/mo) adds $30K–$80K/yr for a 50-engineer org running active AI programs, before compute. GPU inference via managed serverless (Modal A10G $1.10/hr, H100 $3.95/hr; Replicate H100 $5.49/hr) introduces high per-experiment cost and zero commitment visibility. NVIDIA Omniverse Enterprise benchmarks GPU-level infrastructure at **$4,500/GPU/year** — the spatial and metaverse/gaming sub-segment pays this directly.

Compliance as a selection criterion. Per BCG 2026: **73% of enterprise AI initiatives cite compliance posture as a top-three vendor selection criterion**, up from 41% in 2024. CAIO roles now exist in **61% of large enterprises**, making compliance a named buyer with budget. SOC 2 Type II runs $35K–$150K in year one; ISO 27001 initial certification $25K–$80K; FedRAMP Moderate/High $500K–$2M+. EU AI Act full application August 2026 (high-risk Annex III potentially deferred to December 2027 under the AI Omnibus) adds conformity-assessment cost ($20K–$80K per high-risk system) for any platform with AI decision-support outputs.

Compute-per-customer friction. Multi-tenant SaaS platforms and AI infra companies face a per-customer coordination problem: governance of compute allocation, model versioning, multi-tenant observability, and cross-org federation that neither Backstage nor MuleSoft nor LangSmith was designed to solve in combination. That coordination gap is the technology vertical's primary PrimusNeo friction window.

Friction-cost anchor summary (3-yr): $130K–$340K for a 50-engineer startup; $550K–$1.5M for a 50–500-engineer scale-up; $2.3M–$7M+ for 500+ engineer enterprises, telcos, and hyperscalers. Large-tier buyers frequently embed these costs inside cloud EDP/CUD committed-use agreements at **20–40% effective discount off list** — which means the cash friction is lower than the TCO number, but the procurement path is easier (existing cloud commitment absorbs new line items).

PrimusNeo capture band: 8–18% of platform engineering spend, per the cross-vertical pricing model.

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2. PrimusNeo positioning per layer in technology

The technology vertical is the sharpest build-vs-buy context PrimusNeo will face. Buyers are themselves platform vendors, infrastructure builders, and developer-tooling companies — they have high tolerance for self-hosting and high skepticism of vendor lock-in. The engagement motion leads with **PLG + OSS inbound**, not with an enterprise RFP response.

AMP — Spatial OS. Lighter use in pure-software companies; heavier in metaverse/gaming studios, spatial-product companies, AR/VR platform vendors, telcos running physical network operations, and infra providers with data center and edge site coordination problems. For a pure-software SaaS vendor, AMP is not the primary entry layer. For a gaming company building a metaverse product, a telco managing 50K cell towers, or an AR SDK vendor running multiplayer spatial sessions, AMP is the differentiated operational layer — positioned against NVIDIA Omniverse Enterprise ($4,500/GPU/yr) and Unity Industry ($4,950/seat/yr) on price and against both on offline-first + federated governance characteristics.

NEOS — Governance. Highly relevant for telcos with subsidiary structures and post-merger integration challenges, multi-org devtool platforms (plugin ecosystems, marketplace governance), and AI/ML companies managing federated multi-org model governance (regulatory audit trails, cross-team contribution accounting). The CC-NC dual-license makes NEOS free for non-commercial OSS projects and open-source ecosystems — which is also the inbound funnel. Commercial deployment triggers the PCL. Position as the governance protocol layer above Snapshot/Tally (which handle signaling and execution but not durable contribution accounting or holonic federation).

OAS — Interoperability Fabric. The primary entry layer for pure-software technology buyers. OAS covers what MuleSoft covers for Web2 enterprise integration plus what Chainlink CCIP, LayerZero, Okta, and IPFS cover individually — unified under one contract. For a fintech or AI platform vendor managing integrations across a partner ecosystem, OAS is the Web2+Web3+identity+reputation fabric that no incumbent iPaaS covers. For a devtool company managing GitHub Enterprise, Jira, PagerDuty, Datadog, and Slack integrations plus emerging chain-based attestations, OAS is the integration layer that replaces brittle custom glue code and cleans up the engineering debt before it compounds.

Fractal Engine — Intelligence. The LLMOps and agent observability layer — Palantir Foundry for federated orgs, without the services-heaviness. Positions below Palantir commercial entry ($250K–$500K/yr) and above commodity Databricks/Snowflake in the operational intelligence tier. For AI/ML companies, Fractal provides the multi-org agent observability, model performance tracking, and governance audit-trail intelligence that neither LangSmith nor W&B provides at the

organizational/federated layer. The CC-NC dual-license makes it free for research and academic use, seeding adoption inside the ML engineering community that eventually becomes the enterprise champion.

Layer interaction for technology. The dominant bundle for technology buyers is **OAS + Fractal** (integration fabric + intelligence), escalating to **OAS + NEOS + Fractal** for multi-org governance programs and telco/hyperscaler deals. AMP enters primarily for gaming/metaverse/telco/spatial-product sub-segments. Full-stack is the right shape for a telco managing physical network operations + subsidiary governance + partner ecosystem integration + AI operations.

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3. Engagement tiers — Small / Mid / Large

Small: <50-engineer single-product startup ($130K–$340K comparable 3-yr market TCO)

Profile: Early-stage SaaS or devtool company with a growing API/integration surface, early AI features, and no dedicated platform team. Spending $22–$39/seat/mo on IDP tooling and ad-hoc on AI observability. May have one Backstage installation abandoned after 60 days.

Bundle: OAS Small + Fractal Small (Duet).

Worked quote:

```
OAS Small:      $40K/yr  (8 integrations: GitHub, Jira, PagerDuty, Datadog, Slack, Okta, 2 custom
REST)
Fractal Small:  $45K/yr  (4 data sources: event stream, logs, model traces, deployment telemetry)
Duet bundle:    -10% → $76.5K/yr license
Y1 services     (~2x blend for OAS-heavy: $153K — connector buildout $80K, ontology mapping $50K,
deployment $23K)
Y1 upkeep       (Silver SLA, 22%, starts Y2): $17K/yr
Infra (pass-through + 12%): $18K–$30K/yr
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Y1 total:       ~$247K–$259K
3-yr TCO:       ~$390K–$430K
```

Friction anchor: $22–$39/dev/mo IDP seats ($13K–$23K/yr) + LangSmith Plus for 5 active AI engineers ($2.3K/yr) + ad-hoc integration debt (estimated 0.5 FTE at $200K loaded = $100K/yr in maintenance engineering time). Direct friction ~$115K–$125K/yr; PrimusNeo 3-yr cost lands inside comparable market band with meaningfully lower integration debt ongoing.

Capture rate: $76.5K/yr ÷ $120K/yr friction midpoint = ~64% of direct friction displaced — justified because OAS eliminates a category of ongoing maintenance engineering work, not just a seat license.

Cloud EDP note: At this tier, buyers rarely have cloud EDPs; pricing is straightforward.

Mid: 50–500-engineer scale-up ($550K–$1.5M comparable 3-yr market TCO)

Profile: Growth-stage SaaS platform, AI infrastructure company, or devtool scale-up. Has a platform engineering team (2–4 engineers), is running Port or Cortex at $40–$65/seat/mo, and is actively evaluating LangSmith Enterprise or W&B for LLMOps. May have a nascent CAIO function.

Bundle: OAS Mid + Fractal Mid + NEOS Mid (Triad) for multi-org coordination; or OAS Mid + Fractal Mid (Duet) for single-org.

Worked quote (Triad, 200-engineer scale-up):

```
OAS Mid:        $165K/yr  (25 integrations: GitHub Enterprise, Jira, PagerDuty, Datadog, Slack, Okta,
                           3 chain connectors for on-chain attestations, 10 partner API connectors, 8
internal)
NEOS Mid:       $55K/yr   (3 councils: engineering guild, partner ecosystem governance, AI model
governance)
Fractal Mid:    $200K/yr  (10 data sources: deployment pipeline, model telemetry, SLO/SLA signals,
                           partner API usage, governance events)
Triad bundle:   -18% → $344K/yr license
Y1 services     (~2.0x blend: $688K — OAS connectors $320K, Fractal ontology+pipeline $250K, NEOS
governance design $118K)
Y1 upkeep       (Silver SLA, 22%, starts Y2): $76K/yr
Infra (pass-through + 12%): $60K–$120K/yr
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Y1 total:       ~$1.09M–$1.15M
3-yr TCO:       ~$2.0M–$2.3M
```

Friction anchor: Port Enterprise at $65/seat/mo × 200 engineers = $156K/yr + 2 platform engineers at $250K loaded = $500K/yr + W&B at $50/seat/mo × 20 AI engineers = $12K/yr + LangSmith Enterprise custom ~$30K/yr. Estimated direct stack: $698K/yr before custom integration debt. Comparable market 3-yr TCO $550K–$1.5M — PrimusNeo Y1 lands at the high end of comparable 3-yr range, which is defensible only with a clear friction-displacement demonstration in the discovery sprint. The OAS connector audit should identify at least 2 FTE-equivalent of integration maintenance engineering being displaced (at $200K loaded each = $400K/yr), which repositions the capture to ~49% value share — inside the 8–18% band for pure platform spend but reasonable when engineering time is included.

Cloud EDP note: Mid-tier scale-ups often have AWS or GCP EDPs with 20–30% discounts. Frame PrimusNeo as eligible for cloud marketplace procurement to absorb into existing EDP at effective ~20–25% net discount off the above quote.

Large: 500+ engineer FAANG/MAANG/telco/hyperscaler ($2.3M–$7M+ comparable 3-yr market TCO)

Profile: MAANG or near-MAANG software company, Tier-1 telco (AT&T, Verizon, Deutsche Telekom, NTT), hyperscaler partner/platform, or major gaming/metaverse company (EA, Activision Blizzard, Roblox-adjacent). Has a full internal platform engineering organization (10–20 engineers), existing Humanitec or custom-built IDP, enterprise W&B + LangSmith or internal tooling, and a growing CAIO function.

Bundle: Full-stack (AMP + NEOS + OAS + Fractal) for telco and metaverse/gaming sub-segments; OAS + NEOS + Fractal (Triad) for pure-software large enterprise.

Worked quote (Full-Stack, Tier-1 telco, 2,000-engineer, 800 cell-tower sites):

```
AMP Large:      $900K/yr  (800 tower sites; physical network ops spatial coordination, per-site
~$1,125/yr)
NEOS Large:     $280K/yr  (federated subsidiary governance: 12 regional councils + partner ecosystem
governance)
OAS Large:      $750K/yr  (60+ integrations: OSS/BSS systems, partner APIs, chain-of-custody
attestations,
                           multi-cloud, regulatory reporting endpoints, identity federation across
subsidiaries)
Fractal Large:  $800K/yr  (25+ data sources: network telemetry, customer experience signals,
governance events,
                           spatial events, LLMOps telemetry, regulatory audit trails)
Full Stack:     -25% → $2.03M/yr license
Y1 services     (~1.8x blend: $3.65M — OAS connectors $1.2M, AMP site buildout $1.1M,
                 Fractal ontology+models $900K, NEOS governance design $450K)
Y1 upkeep       (Gold SLA, 25%, starts Y2): $508K/yr
Infra (pass-through + 12%): $600K–$1.2M/yr
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Y1 total:       ~$6.3M–$6.9M
3-yr TCO:       ~$11M–$14M
```

Friction anchor: A Tier-1 telco running a comparable market stack pays $2.3M–$7M+ per year across Humanitec (~$2M/yr IDP for 2,000 engineers at scale), enterprise W&B + LangSmith ($200K+/yr), Omniverse Enterprise for network digital twin ($450K/yr, 100 GPUs), ISO 27001 + SOC 2 compliance overhead ($200K+/yr), and integration engineering maintenance (estimated 8 FTE at $220K loaded = $1.76M/yr). Combined friction window $4.8M–$9.5M/yr; PrimusNeo $2.03M capture = **21–42% value share** — at the high end of defensible, appropriate given full-stack displacement depth.

Cloud EDP note: Tier-1 telcos and hyperscalers negotiate cloud EDPs at 30–40% discounts. PrimusNeo on cloud marketplace reduces effective cost to $1.2M–$1.5M/yr equivalent, which repositions the 3-yr TCO to $7M–$9M — firmly inside the $2.3M–$7M comparable band at the high end.

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4. Industry-specific procurement signals

CTO (build-vs-buy for IDP + agent infra; entry via conference talks + OSS presence). The CTO or VP Engineering is the primary build-vs-buy authority in technology companies. They respond to architecture credibility, standards alignment (OpenTelemetry, CNCF, OpenUSD, W3C DID/VC), and federated multi-org deployment proof. They will not respond to a traditional enterprise deck — they want a GitHub repo, a runnable demo, and a peer reference from a respected engineering org. Entry via PlatformCon, KubeCon, QCon, or an AI engineering conference; not via analyst relations at this stage. The Humanitec 2023 DevOps Benchmarking finding is the right counter to the NIH objection: top-performing engineering orgs adopt hybrid buy+extend models, not full custom builds.

VP Platform Engineering / Head of DevEx (IDP budget + DORA metrics; entry via community + peer referral). This is the operational buyer with the IDP line item. They own DORA metrics: deploy frequency, change failure rate, mean time to restore. They model ROI in engineering time saved, typically at **$150–$250K/engineer-year fully loaded**. A 5% reduction in integration-related incidents across a 200-engineer org at $200K/engineer = $2M/yr in recovered engineering productivity — a number that makes a $344K/yr PrimusNeo license defensible without further math. Entry is through community (PlatformCon, Internal Developer Platform Slack, platform.sh/DORA report influence), not cold outreach.

Chief AI Officer / Head of AI (agent infra + LLMOps governance + data residency; entry via analyst relations + AI governance working groups). CAIO roles exist in 61% of large enterprises. This buyer owns the LLMOps stack (LangSmith, W&B), the governance audit trail (EU AI Act compliance), and the data residency posture (EU telcos, GDPR-adjacent scale-ups). Fractal + NEOS is the relevant product pair for this buyer — Fractal for agent observability and model intelligence, NEOS for the governance fabric that produces the signed audit trails the EU AI Act rewards. Entry via AI governance working groups (Partnership on AI, EU AI Office stakeholder engagement), analyst briefings (Gartner CAIO summit), and Escherbridge thought leadership in the LLMOps community.

Head of Developer Experience / PLG champion (hands-on before budget; entry via free tier + community). The PLG motion in devtools is well-established: Cursor reached $2B ARR by February 2026 on PLG inbound. OSS inbound (Backstage plugin, OAS connector SDK, Fractal community tier under CC-NC) drives free team adoption → PQL-triggered outbound → VP Platform/CTO expansion. Baseline PLG free-to-paid conversion is ~9%; top quartile devtool companies reach 25–30% with PQL-triggered outbound (PrimusNeo target). The Head of DevEx is the internal PLG champion — they run the trial, build the proof of value, and bring the budget conversation upward.

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5. Rejection-risk mitigations

Build-vs-buy / NIH syndrome. Engineering-led organizations, especially FAANG-adjacent, default to "our requirements are unique." Counter with evidence: the Humanitec 2023 DevOps Benchmarking study found that top-performing engineering organizations use hybrid buy+extend models, not full custom builds. PrimusNeo's open SDK architecture (OAS connector SDK on MIT/Apache-2.0, AMP community runtime on Apache-2.0, NEOS + Fractal on CC-NC for non-commercial use) is explicitly designed for hybrid buy+extend — buyers can extend any layer without forking the commercial stack.

Vendor lock-in across four layers / four stewards. The federated architecture raises legitimate portability questions. Counter: open SDK across all layers (OAS connector SDK open, AMP Hexon file standard open Apache-2.0, NEOS + Fractal CC-NC for non-commercial), CC-NC dual-licensing on NEOS and Fractal (so the governance schema and intelligence ontology are portable under the open license even if the commercial engine changes), non-negotiable data export guarantees in OAS (machine-readable export at any time), and OpenUSD / CNCF / W3C DID standards alignment in AMP and OAS respectively. Source escrow via Iron Mountain is available above $500K ARR for AMP and OAS closed-core.

Open-source competition (Backstage, Crossplane, Flux, ArgoCD, OPA, MLflow, LangFuse). Every commercial PrimusNeo layer has credible OSS alternatives. Counter is not feature competition but TCO and time-to-value. Backstage self-hosted runs $2M+ over 3 years for large teams (3–5 senior platform engineers at $250K loaded). OPA is free but implementation engagements run $150–$300/hr. MLflow and LangFuse lack the federated multi-org governance and operational ontology layer that makes Fractal's agent observability distinct from pure logging. Enterprise support SLA (Gold 24×7, 1-hour P1) is the final differentiator that OSS cannot match at scale.

Integration with existing devtools. GitHub Enterprise, Jira, PagerDuty, Datadog, Slack, Okta, and LaunchDarkly are non-negotiable for any technology buyer. The first 90 days of any engagement must demonstrate connector coverage for the buyer's core toolchain. OAS connector library includes these as standard connectors (CRM at $30K–$80K, Identity/IdP at $25K–$90K, Custom REST/GraphQL at $20K–$120K per the services catalog). Discovery sprint output is a connector coverage matrix showing which integrations are available immediately and which require buildout. Gap commitments go into the SOW with 90-day warranty.

Governance / data residency (EU telcos, post-GDPR scale-ups, AI Act compliance). EU-based technology companies — telcos, fintech platforms, AI companies subject to GPAI provisions — require in-region data processing and demonstrable EU AI Act readiness. OAS supports multi-region deployment with data-residency routing baked into the federation engine. Fractal's model governance and Escherbridge's audit-trail architecture are designed to produce the conformity-assessment documentation that Annex III high-risk systems require. NEOS governance events are immutably signed — the audit trail is the product. Frame these not as compliance checkboxes but as the architecture choices that make EU AI Act conformity a 3-week exercise rather than a 6-month legal program.

Procurement cycle length. MAANG and Tier-1 telco enterprise procurement averages 6–18 months from first meeting to signed contract. PLG inbound shortens this: product usage data from the OAS community SDK or Fractal CC-NC deployment substitutes for an RFP at the team level, escalating to procurement only after internal champions have demonstrated adoption. Target is VP Platform / Head of DevEx as the internal champion who drives usage above the PQL threshold before procurement is formally engaged.

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6. Regulatory + compliance considerations

SOC 2 Type II is mandatory for any sale to US enterprise, govtech-adjacent, or financial-services technology company. Cost: $35K–$150K in year one with automation; $20K–$50K/yr ongoing. PrimusNeo SOC 2 Type II is funded as a central investment shared across verticals, not charged per-engagement.

ISO 27001 is expected by EU enterprise technology buyers and all telco customers. Initial certification $25K–$80K; $15K–$30K/yr surveillance audits. Required for any Capgemini- or Deutsche Telekom-adjacent deal in the EU market. Funded centrally.

FedRAMP Moderate/High is relevant for govtech-adjacent technology companies (defense-tech platforms, civic-tech SaaS, government-facing AI infrastructure). Cost: $500K–$2M+ to achieve; typically 12–18 months process. The post-2025 "20x acceleration" brings package completion to under 5 weeks

once the authorization package is complete, but the package itself still requires the full investment. Recommended: pursue FedRAMP Moderate by 2027 for the govtech TAM.

EU AI Act enters full application August 2026. High-risk systems under Annex III (AI in critical infrastructure, employment, education, essential services) face conformity-assessment requirements potentially deferred to December 2027 under the AI Omnibus proposal. GPAI models above compute thresholds (initially ~10^25 FLOPs) face transparency and systemic-risk documentation requirements effective February 2025. Technology companies building on PrimusNeo's Fractal Engine should document which AI outputs constitute high-risk decision support (Annex III) early — conformity assessment $20K–$80K per high-risk system is scoped as part of the engagement deliverable for any buyer where it applies.

EU Data Act / DSA / DMA apply to OAS-class interoperability fabrics, platform gateways, and data portability mechanisms. Any PrimusNeo OAS deployment for a platform company with EU operations above DSA/DMA thresholds should include legal review of data portability obligations and interoperability mandates. Initial posture advisory $50K–$200K per jurisdiction.

US AI EOs and dual-use foundation model disclosure apply to technology companies building foundation models above compute thresholds. Internal governance program cost $100K–$500K+. Fractal Engine's audit-trail and governance-event architecture is positioned as the technical substrate that makes AI governance reporting tractable.

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7. Steward + service partner notes

PLAN Systems (AMP steward, 15% royalty) leads AMP delivery for metaverse/gaming/telco/spatial-product company deployments. PLAN's contribution is lightest in pure-software SaaS deals and heaviest in any engagement involving physical infrastructure (cell towers, data centers, studio sites) or spatial-product development (AR SDK vendors, game studios building shared spatial environments).

OmniOne (NEOS steward, 18% royalty) leads NEOS governance design using the ETHOS methodology. Particularly relevant for technology companies managing multi-org governance: telco subsidiaries, plugin/marketplace ecosystems, OSS foundation governance, and AI model governance councils. OmniOne's ETHOS workshops ($35K–$60K for governance design, $25K–$80K for charter authoring) are typically the first named service that NEOS-entry buyers sign.

OAS Web4 (OAS steward, 15% royalty) leads connector development and federation engine deployment. For technology buyers, OAS Web4 is the primary delivery partner for any devtool integration program — GitHub Enterprise, Jira, PagerDuty, Datadog, Okta connectors are built and maintained by the OAS Web4 connector author community. The OAS connector library reuse model means each new connector built for one technology buyer reduces the marginal cost of the next.

Escherbridge (Fractal steward, 18% royalty) leads Fractal ML engineering, custom model development, and agent observability architecture. For AI/ML company buyers, Escherbridge is the primary delivery partner for LLMOps instrumentation, model performance tracking, and multi-org agent observability. Escherbridge's public corpus is the pretrained foundation that reduces custom model training cost by 40–60% relative to training from scratch — a material time-to-value advantage the CAIO buyer will quantify.

SI channel for technology large deals. Large technology accounts (telcos, hyperscaler-adjacent) enter through SI partners who frame PrimusNeo inside a broader transformation program:

Firm	Rate	Technology vertical fit
IBM Consulting	$200–$350/hr onshore; $80–$120/hr offshore	Strong in telco (Telco Cloud, network automation), govtech-adjacent
Accenture	$220–$400/hr; $3.6B AI bookings	Broadest technology reach; leads CAIO-adjacent AI governance programs
Capgemini	$180–$320/hr; strong EU + telco	EU telco and GPAI-focused programs; right fit for Deutsche Telekom/Orange-adjacent deals
Globant	$50–$99/hr; Agile Pods ~$20K/mo	Gaming/metaverse delivery; LatAm nearshore for rapid connector buildout
EPAM	$75–$200/hr	Engineering-heavy platform delivery; strong devtool and AI platform integration track record
ThoughtWorks	$200–$350/hr	Advisory-then-delivery for platform engineering programs; PlatformCon presence
Boutique platform-eng (Syntasso, Kosli, Configure8)	$150–$250/hr; workshops $30K–$80K	Backstage/IDP specialization; right entry for Mid-tier VP Platform Engineering deals

SI commercial model: SI partners take 20–40% implementation margin; PrimusNeo takes license/commit. The PLG-to-enterprise motion is the primary technology vertical path and does not require SI involvement at the team/startup tier.

PLG conversion targets: 9% free-to-paid baseline; 25–30% top quartile with PQL-triggered outbound once OAS community SDK or Fractal CC-NC tier installs exceed internal adoption thresholds.

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8. Revisions

Version	Date	Author	Change
0.1	2026-05-23	Founding team + research squad	Initial Technology engagement model

Layer Comparables — Cross-Cutting Market Research

> ***Purpose:*** *Pricing benchmarks for the four PrimusNeo product layers (AMP, NEOS, OAS, Fractal Engine) to anchor the value-based commercial pricing model.* ***Date:*** *2026-05-23*

Layer 1 (AMP) — Spatial OS / 3D Collaboration

Pricing Benchmarks

NVIDIA Omniverse Enterprise - **$4,500 per GPU per year** (unit is the GPU running the simulation server, not named seat) - Small studio: 2–4 GPUs = $9K–$18K/yr for compute licensing alone, before add-ons

Unity Industry / Industrial Collection - **$4,950/seat/yr** (post-2025 price revision; was ~$4,500) - Includes Pixyz, Unity Mars, Unity Cloud, DevOps tooling - Unity Enterprise (non-industry): ~$2,200/seat/yr after 8% increase effective Jan 2025

Unreal Engine Enterprise - **$1,850/seat/yr** for enterprise seat license — notably lower than Unity, positioned aggressively for industrial/simulation deals

Spatial.io - Business tier (up to 50 users/server, 20 servers, analytics): public listing but $ behind sales conversation - Enterprise tier (1,000+ concurrent, custom games/XR, dedicated support, advanced security): custom - **Mozilla Hubs shut down May 31, 2024** — no longer a comparable

Improbable / Hadean - Both pivot to defense simulation + game multiplayer infra - Public pricing not disclosed; enterprise engagements are **six-figure minimums**

Service Multiplier

- Typical **1.5–3x** license (heavy SI deployment)

Layer 2 (NEOS) — Governance / Councils / Holonic Orgs

Pricing Benchmarks

Aragon / DAOhaus / Snapshot / Tally — All free or near-free at baseline: - **Aragon**: no subscription; L2 deployment gas $50–$200 one-time; custom smart contract buildouts $20K–$100K+ if bespoke - **DAOhaus**: entirely free, OSS, forkable - **Snapshot**: free for all off-chain voting; powers 96% of DAOs + 35,000+ communities; gas-free by design - **Tally**: free self-serve Governor deployer; enterprise/white-glove DAO launch via biz@tally.xyz (no public pricing)

Diligent Boards / Nasdaq Boardvantage - **Diligent**: $12K–$60K+/yr depending on board size, features, org type. Per Vendr, 20–30% discounts routinely negotiated on multi-year - **Nasdaq Boardvantage**: starting ~$12K–$15K/yr for small boards - **OnBoard**: affordable alt; similarly quote-based, often 20–40% below Diligent list

Microsoft Viva - Employee Communications & Communities: **$2.00/user/mo** - Workplace Analytics & Employee Feedback: **$6.00/user/mo** - Viva Suite (full): **$12.00/user/mo** ($144/seat/yr)

Loomio - Per-group subscription model (not per-seat) - Nonprofit/cooperative discount; enterprise/custom domain custom-quoted - Indicative: **$99–$599/month** per organization depending on member count

Service Multiplier

- Typical **0.5–1x** license (lighter services, mostly config/training)

Layer 3 (OAS) — Interoperability Fabric

Pricing Benchmarks

MuleSoft Anypoint Platform - **$1,250/vCore/month** on-demand - Min production deployment ~1 vCore = $15K/yr entry, but real enterprise starts **$80K–$250K+/yr** - Large enterprise: commonly **$500K–$2M+/yr** under multi-year ELA - Services-to-license: **2–3x Year 1** (impl, DataWeave dev, architecture) - Total Y1 TCO mid-market: $350K–$600K+

Boomi - Professional ~$550/month ($6,600/yr) for lower volumes - Mid-market: $30K–$150K/yr - Enterprise: six figures custom - Services multiple: **1–2x** vs MuleSoft's 2–3x

Workato - Entry: ~$10K/yr (SMB automation) - Enterprise: 20–30% lower than MuleSoft for equivalent volumes ($60K–$1.5M+ depending on recipe volume) - Services: 1–2x license

Blockchain interoperability (Chainlink CCIP, LayerZero, Axelar, Wormhole) - None publish per-tx list prices; costs denominated in gas + relayer fees (network-condition-dependent) - Axelar processed $8.66B of transfers + 1.85M transactions by May 2024 across 64 chains - Enterprise SLA (e.g., Chainlink BUILD) negotiated directly - Cost advantage flows to LayerZero + Wormhole for high-frequency use cases; CCIP holds mid-cost position with stronger security guarantees

Decentralized storage (Pinata / Web3.Storage) - Pinata: **~$0.15/GB/month** on paid plans; enterprise custom - Web3.Storage: free tier wound down; now enterprise-focused custom SLAs - Ceramic (identity/data streams): no public pricing; developer-tier free on hosted node; self-hosted OSS

Service Multiplier

- **2–3x Year 1** (connector buildout, architecture); 0.5–1x maintenance from Y2

Layer 4 (Fractal Engine) — Intelligence / Analytics / Operational Telemetry

Pricing Benchmarks

Palantir Foundry - No public per-seat price. Enterprise ACV: **$1M–$10M+/yr** for gov; commercial mid-market entry $250K–$500K/yr - Dimensions: compute-seconds + storage (GB-months) + Ontology (GB-months) + named-user software licenses bundled into ELA - **Gross margin: 80.25% (FY2024)** on combined platform - Services-to-license: bundles O&M services as single performance obligation with software. Pre-2022 service-heavy model ran close to 1:1 software-to-services; early Palantir gov contracts ran **3–5x services vs license** in total deal value — a model they are deliberately shrinking

Snowflake - Compute: $2–$4/credit on-demand; $1.50–$2.50/credit under committed-use - Storage: $23/TB/month (AWS US regions) - Enterprise commits: $250K–$1M+ to unlock meaningful credit discounts - **Gross margin: ~75–78%** software

Databricks - DBUs: $0.15–$0.40/DBU (Enterprise tier includes Unity Catalog + advanced governance) - Cloud compute billed separately (dual-bill model) - Storage: ~$0.02/GB/month on underlying object storage - 700+ customers spending $1M+/yr; net dollar retention >140%; annualized revenue $5.4B (Jan 2026) - **Gross margin: ~74–80%** (declining slightly as AI workloads increase compute intensity) - Services largely delivered through SI partners; Databricks direct services multiple: **0.3–0.5x**

Splunk - On-premises: ~$150/GB/day/year list; enterprise security (ES) $250–$400/GB/day - Cloud premium: 1.2–1.5x on-premises rates - Multi-year discounts: 10–20% - Mid-size (200 hosts, 500 GB/day): **$30K–$60K/month** under enterprise agreement

Datadog - Logs ingest: **$0.10/GB ingest + $1.70/M indexed** - 200 GB/day at list exceeds $1.3M/yr for log mgmt alone - Hosts (infra monitoring): ~$15–$23/host/month - Mid-size (200 hosts + 500 GB logs + APM): **$15K–$25K/month**

C3.ai / Cognite Data Fusion / AspenTech - All enterprise custom-quote, no disclosed per-seat list - C3.ai reported $108.7M in FY2025 Q4 revenue with consumption-based model - Cognite Data Fusion on Azure Marketplace with custom quotes; heavy-industry OT/IT convergence target - AspenTech remains industrial simulation/optimization benchmark, similarly opaque

Service Multiplier

- **1–2x Year 1** (data pipeline buildout, model training, ontology mapping); 0.3–0.5x from Y2

OSS-to-Paid Conversion Playbook

Relevant precedents for NEOS (CC-NC) + Fractal (CC-NC) + OAS SDK:

- **MongoDB SSPL** (2018): Switched from AGPL to SSPL after AWS launched DocumentDB. SSPL forces SaaS providers offering MongoDB as a service to open their entire stack — effectively a commercial

license requirement for cloud deployments. MongoDB gross margin ~73–75% on software.

- **Elastic (ELv2 + SSPL then AGPL)**: Moved from Apache 2.0 to dual SSPL+ELv2 in 2021 after AWS forked OpenSearch; returned to AGPL Sep 2024. **Lesson: CC-NC/SSPL-style restrictions stop cloud hyperscaler free-riding but must have a clean commercial escape valve.**
- **HashiCorp BSL** (Aug 2023, now IBM/HashiCorp): Switched Terraform, Vault, Consul from MPL 2.0 to BSL 1.1. BSL allows production use except by direct competitors; converts to OSI-recognized license after a defined "Change Date." **Closest precedent to PrimusNeo's CC-NC**: non-competing entities get near-open access; commercial SaaS competitors must buy a license.
- **Conversion rate reality**: OSS-to-paid in infra software runs **1–5% of active community users**. At 10,000 active NEOS deployments → realistic paid conversion 100–500 orgs. Revenue-per-converted-customer must be high enough to sustain dev — typically **$10K–$100K+/yr** for enterprise governance tooling.

Outcome / Value-Based Pricing Precedents

- **Zendesk AI agents (2024)**: Switched to outcomes-based — charges only for successful AI resolutions, not seat licenses or consumption
- **Intercom Resolution Bot**: Pay per successful resolution, zero cost for failed interactions
- **Gartner forecast**: **30%+ of enterprise SaaS will incorporate outcome-based components by 2025** (up from ~15%)
- **McKinsey**: Value-based pricing companies achieve profit margins 15% above peers
- **Palantir**: Early gov contracts included value-share clauses (% fraud recovered, % logistics savings). Materially harder to audit and collect on; Palantir moved away post-2018
- **ServiceNow pilots**: Outcome pricing with guaranteed efficiency improvement SLAs — full rate only when targets met

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Service Multiplier Summary by Layer

Layer	Comparable	Services : License Ratio
AMP	NVIDIA Omniverse, Unity Industrial	**1.5–3x** (heavy SI deployment)
NEOS	Diligent, Microsoft Viva	**0.5–1x** (lighter services, mostly config/training)
OAS	MuleSoft, Boomi	**2–3x Year 1** (connector buildout, architecture)
Fractal	Palantir (historical), Snowflake/Databricks (current)	**0.5–1.5x** (shifting software-led; SI partners absorb much of services multiple)

Pricing Anchors for PrimusNeo

> *Internal anchor ranges — not published list prices. Quote inside these ranges, defend with friction-cost data per the formula in* `01-pricing-model.md`.

AMP (Spatial OS)

Scale	Definition	Annual License Anchor	Notes
Small	1–3 sites, <50 concurrent spatial users	**$18K–$45K/yr**	Between Spatial.io Business (~sub-$2K/mo signals) and Unity Industry ($4,950/seat at ~5–10 power users)
Mid	4–15 sites, 50–500 concurrent	**$80K–$250K/yr**	Mid-tier Omniverse ($4,500/GPU × 4–8 GPUs + site licenses) + workspace subscription
Large	15+ sites or multi-org federation, 500+ concurrent	**$350K–$1.2M/yr**	Omniverse Enterprise large deployment; positions below NVIDIA, above Unreal Enterprise

Services multiplier target: **1.5–2.5x license in Y1**; **0.3–0.5x from Y2 onward**.

NEOS (Governance)

Scale	Definition	Annual License Anchor	Notes
Small	Single org, 1 council, <50 members	**$6K–$18K/yr**	Above Loomio ($1,200–$7,200/yr), below OnBoard/Diligent entry ($12K/yr). Justified by on-chain auditability + contribution accounting
Mid	Multi-council, 50–500 members, charter management	**$30K–$90K/yr**	Comparable to Diligent mid-tier ($12K–$60K/yr) with governance-intelligence premium for holonic structures
Large	Federated governance, 500+ members, multiple orgs	**$120K–$400K/yr**	Positions against Nasdaq Boardvantage ($12K–$15K/yr entry but scales with org count) and Microsoft Viva Suite ($144/seat × large headcount)

Services multiplier target: **0.5–1x license** (lighter engagement; mostly configuration, template instantiation, training). **CC-NC dual-license**: non-commercial use free; commercial use requires PrimusNeo commercial license at these rates.

OAS (Interoperability Fabric)

Scale	Definition	Annual License Anchor	Notes
Small	<10 integrations, single domain, <1M API calls/month	**$24K–$60K/yr**	Aggressive undercut of Boomi Professional ($6,600/yr for low vol but scales fast). 30–50% of MuleSoft entry ($80K+)
Mid	10–50 integrations, multi-chain/multi-cloud, 1M–50M calls/month	**$80K–$250K/yr**	Parity with Boomi mid-enterprise; 30–40% below MuleSoft equivalent
Large	50+ integrations, full federation engine, >50M calls/month, cross-org	**$350K–$1.5M/yr**	ELA territory. Workato/MuleSoft large enterprise; positions OAS as **Web3+Web2 unified fabric neither incumbent covers**

Services multiplier target: **2–3x license Y1**; **0.5–1x maintenance Y2+**. **Gas/chain fees pass-through at cost + 12% management fee.**

Fractal Engine (Intelligence / Analytics)

Scale	Definition	Annual License Anchor	Notes
Small	<5 data sources, departmental analytics, <1TB managed	**$30K–$75K/yr**	Above Databricks/Snowflake raw compute for equivalent vol but includes PrimusNeo operational ontology + signal layer — comparable to Looker Standard entry
Mid	5–20 data sources, operational telemetry + historical archive, 1–20TB	**$100K–$350K/yr**	Below Palantir commercial entry ($250K–$500K/yr), above commodity Snowflake/Databricks. **"Palantir for federated orgs"** without the services-heaviness
Large	20+ sources, cross-org intelligence, real-time + archive, >20TB	**$500K–$2M+/yr**	Palantir Foundry territory; justified by the **operational ontology layer (NEOS governance events as first-class signals)** that neither Palantir, Databricks, nor Snowflake provides natively

Services multiplier target: **1–2x license Y1**; **0.3–0.5x from Y2**. **CC-NC dual-license**: research/nonprofit use free; commercial production requires license at these rates.

Gross Margin Targets

Comparable software gross margins: Palantir 80%, Snowflake 75–78%, Databricks 74–80%, MuleSoft (within Salesforce) ~73–78%. **PrimusNeo target blended gross margin on license/subscription revenue: 72–80%** — achievable given the federated steward payout structure (stewards receive a share of license revenue, which is the primary cost-of-revenue). Services gross margin target: **35–50%** (consistent with mid-tier SIs). **Blended target at scale: 65–72%**.

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Agriculture Vertical — Market Intelligence Brief

PrimusNeo Engagement Model Research *Prepared: May 2026 | Coverage: Row Crop, Specialty, Livestock, Cooperatives, Ag Inputs, Regenerative Ag, MRV*

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1. Comparable Vendor Pricing

Farm Management / FMIS

Vendor	Model	Price Range
Climate FieldView (Bayer)	Per-acre/yr	**$4–$10/acre/yr**
John Deere Operations Center	Bundled w/ JD equipment; analytics tiers	$0 base (JD hw required) → $3–$6/acre add-on analytics
Trimble Ag	Mixed-fleet guidance + SaaS	$5–$12/acre/yr software; hw additional
Granular / Traction Ag (Corteva divestiture)	Enterprise per-acre SaaS	$3–$25/acre/yr by module depth
CropX	Soil sensor hw + SaaS	Sensors $500–$1,000/unit; software $15–$25/acre/yr
AGCO Fuse	Equipment-bundled; TerraStar correction	Correction ~$800–$1,500/seat/yr; analytics dealer-quoted

FieldView's $4–$10 reflects mature mass-market product with vol discounts. CropX high end reflects hw + analytics. AGCO opaque outside dealer relationships.

Ag-Fintech / Supply Chain Traceability

- **Indigo Ag (Market+ / Scope 3):** Service fee to grain buyers + CPGs **2–5% of contracted grain value**. Farmers receive up to **10% premiums** on sustainably-attributed deliveries. Carbon layer guaranteed farmers **$15/tCO2e** at launch; peak prices reached $60–$80/tCO2e before market contraction.
- **Cargill / Provenance-style traceability:** Grain premiums carry $0.10–$0.30/bu traceability overhead. Enterprise food-co API integrations: **$50K–$250K/yr**.

MRV / Soil Carbon

Platform	Revenue Model	Effective Price
Indigo Carbon	Credit issuance	$60–$80/tCO2e peak; market contracted **57%** in 2024
Nori	Marketplace fee	Shut down September 2024
Boomitra	AI/satellite MRV; smallholder focus	Credits <$40/tCO2e; MRV costs **<$10/acre at continental scale**
Regrow Ag	B2B SaaS for supply chain owners	~$5–$15/acre/yr for Scope 3 monitoring
CIBO Technologies	Platform license + program admin	~$3–$8/acre for scenario modeling + reporting
Truterra (Land O'Lakes)	Revenue share program	~$19/tCO2e effective farmer payout ($9M paid for 462K tCO2e over 2 yrs)
Bayer Carbon	Bundled w/ FieldView enterprise	Subsidized via seed/chemical margin; not separately priced

Voluntary ag carbon credit market contracted from **$84.9M to $36.1M in 2024** — 57% decline driven by quality concerns + methodology disputes. Boomitra claims 90%+ MRV cost reduction vs traditional soil sampling at $50–$100/acre.

Cooperative ERP / Digital Platforms

- **Conservis / Traction Ag:** $5–$15/acre/yr for full FM suite; 50K+ acre operators quoted **$200K–$500K/yr**
- **AgVend:** Retail-facing omnichannel platform for co-ops; **0.3–0.8% of digital orders processed** (GMV-based)
- **CHS / GROWMARK in-house:** Both orgs formed **Cooperative Ventures** ($50M joint VC fund, 2022) to invest in ag-tech rather than build proprietary. Traction Ag $10M Series A (May 2024) Cooperative Ventures-led — signals which FMIS platforms are on co-op shortlist

Remote Sensing

- **Planet Labs:** Enterprise ag contracts **$30K–$500K/yr** for field-level daily imagery. Insights-as-a-Service emerging at $few thousand/mo for regional monitoring
- **Descartes Labs (Earth Daily):** AI-processed crop intelligence $50K–$300K/yr
- **EOS Crop Monitoring:** Lower-cost entry **$1–$3/acre/yr** for NDVI + crop health indices; field-level reporting tiers from $500/mo

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2. Engagement Scale Tiers — 3-Year TCO

Tier 1 — Small: Family Farm / Small Co-op (<5,000 acres, <$10M revenue)

- Typical annual: $4–$10/acre/yr → $20K–$50K/yr for 5K-acre operation
- **3-yr TCO: $60K–$150K** (incl 2–3 guidance/precision units @ $5K–$15K each + connectivity + light impl + annual SaaS)
- PrimusNeo: AMP for field boundary + agronomic data; Fractal for yield prediction + input optimization. NEOS/OAS lightweight unless co-op-connected.

Tier 2 — Mid: Regional Co-op / Mid Corporate Farm (5K–100K acres, $10–200M revenue)

- Typical annual: $8–$20/acre blended → **$400K–$2M/yr for 100K-acre entity**
- **3-yr TCO: $1.2M–$6M** incl impl, ERP integration (AgVend, SSI, Agris), training, data governance setup, MRV enrollment
- PrimusNeo: Full AMP + OAS for ERP interop across member systems; NEOS for member-data governance + sovereignty assurance; Fractal for SC carbon attribution + Scope 3 reporting

Tier 3 — Large: Ag Input Giant / Global Trader / National Co-op Federation ($200M+ revenue)

- Typical annual: $5M–$25M/yr across technology, data licensing, consulting, compliance
- **3-yr TCO: $15M–$75M** incl multi-region rollout, custom integrations, regulatory compliance infra (EUDR, FSMA, Article 6), ongoing PM
- PrimusNeo: All four pillars — AMP authoritative spatial layer, OAS multi-partner interop (grain traders, CPGs, regulators, MRV verifiers), NEOS cross-jurisdictional data sovereignty, Fractal cross-commodity SC intelligence

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3. Services Rates

- **McKinsey Agriculture Practice (ACRE):** Engagements $350K–$2M+ for digital ag strategy; senior partner ~$10K–$15K/day. ACRE licensed separately at undisclosed enterprise rates
- **Bayer Crop Science / Digital Farming:** Impl + agronomy advisory bundled into FieldView enterprise; dedicated agronomist $500–$1,500/day for custom
- **AGCO Digital Solutions:** Dealer-delivered Fuse impl $150–$300/hr; national account fixed-fee fleet-wide rollout
- **ADM Outcomes:** Multi-year regenerative ag transition + Scope 3 advisory; **$500K–$3M per engagement** depending on grower network size
- **Co-op tech installations** (precision planting, VR, soil health networks): $15K–$80K per site for hw + config; annual maintenance 15–20% of license

4. Pricing Mechanics

Per-acre dominant for crop software ($2–$15/acre/yr). Low ($2–$4): basic field mapping + record-keeping. High ($12–$15): soil health indexing + satellite NDVI + weather integration + financial reporting. Most enterprise co-op deals negotiated as total-AUM contracts, not per-acre rack rates.

Per-head livestock standard. Entry: $1–$3.65/head/yr (~$0.003–$0.01/head/day). Full precision livestock farming (wearables, AI health, feed optimization): **$5–$20/head/yr**.

$/tCO2e (MRV): Farmer revenue $15–$80/tCO2e by registry + buyer. MRV service costs $3–$15/acre/yr or $2–$10/tCO2e verified. **Verra revised Dec 2024 fee schedule charges $5,000 upfront per verification review** + per-credit issuance fees — compresses developer margins. Gold Standard imposes higher co-benefit verification overhead.

% of harvest value: Indigo-style programs offer **2–10% premiums** on sustainably attributed grain. SC traceability platforms charge buyers 0.3–2% of sourced commodity value.

Per-farm site: Flat-rate farm-site licensing ($500–$3,000/site/yr) attractive for co-ops serving many small members — avoids per-acre metering complexity across variable field sizes.

5. Regulatory Overlay

EU Deforestation Regulation (EUDR): Covers soy, cattle, cocoa, coffee, palm oil, rubber, wood. Large operators must comply by **December 30, 2026** (delayed twice); micro/small operators by **June 30, 2027**. Requires plot-level geolocation for all commodities entering EU market. EC estimates simplification measures reduce annual compliance costs by 75%, but upstream costs (physical segregation, traceability, geolocation infra) remain substantial and can exceed thin ag margins in high-risk origins.

CBAM: Nitrogen fertilizer production in scope → indirect input-cost pressure. Direct commodity crops not yet in CBAM but under ongoing review; inclusion would fundamentally reshape ag SC economics.

USDA NIFA / FSMA: NIFA Food Safety Outreach Program (FSOP) provides technical assistance funding to specialty crop producers. FSMA Produce Safety Rule creates record-keeping, traceability, food safety plan requirements driving FMIS + SC software adoption. **USDA estimated farms with $25K–$250K annual produce sales spend up to 60% of profits on FSMA compliance** — substantial SaaS buying trigger for documentation/audit-trail tools.

Article 6 (Paris Agreement): COP29 (Nov 2024) finalized remaining technical guidance. As of March 2025, 97 bilateral agreements across 59 countries active; Article 6.4 UN carbon market expected to launch 2025–2026. UN-endorsed credits expected premium pricing vs voluntary — could re-energize soil carbon MRV investment after 2024's contraction.

Verra VM0042 / Gold Standard: VM0042 v2.0 (active May 2023) primary compliance pathway for row-crop carbon. Dec 2024 fee revision ($5K upfront + per-credit issuance fees) squeezes developer economics at low credit prices. Gold Standard co-benefit doc requirements add cost but serve premium

buyer segments.

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6. Procurement Signals

Decision-maker map: Co-op boards (member-elected, risk-averse, annual budget cycles) hold ultimate authority for large cooperative tech investments. CTO/CIO at national co-ops + large input companies (Bayer, AGCO, CNH Industrial) govern platform-level decisions. At mid-sized corporate farms, buying committees include lead agronomist, ops manager, increasingly an ESG/sustainability lead. **The agronomist is the critical influencer — controls field data access + vendor trial permissions.**

Sales cycle: 12–24 months for co-op enterprise software; 18–36 months for MRV/carbon program enrollment at scale. **Planting-cycle awareness non-negotiable:** decisions made Oct–Feb for spring deployment. Pilots must complete within one full growing season to generate board-credible ROI evidence.

Buying triggers: EUDR SC compliance deadlines; Scope 3 reporting mandates from major retail buyers (Walmart, Unilever, Nestlé SC programs); USDA EQIP/CSP conservation program eligibility; state-level carbon incentive programs; co-op member demand for input cost transparency.

CHS/GROWMARK gateway: Both channel tech investment through Cooperative Ventures fund. Vendor relationships typically begin with Cooperative Ventures portfolio teams before reaching operational buyers — **fund relationship = prioritized channel entry point**.

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7. Rejection Reasons

Data ownership / sovereignty (primary barrier): GAO January 2024 precision agriculture report formally identified farm data sharing/ownership as top adoption challenge. Farmers fear precision data (field boundaries, yield maps, soil tests, application records) shared with large vendors becomes competitive intelligence for input cos or grain traders. Civil Eats (March 2026): farm data remains "a hidden crop for corporate tech." **Only 36% of smaller farms planned AI adoption vs 81% of large farms** — trust gap is structural.

Rural connectivity: High-speed broadband unreliable across large portions of US Corn Belt, Great Plains, international growing regions. **Cloud-first platforms with no offline mode fail at field edge — fundamental disqualifier in many geographies.**

ROI per acre too thin: At $200–$500/acre commodity crop margins, a $10–$15/acre software stack requires **demonstrable 2–5% yield lift or input cost reduction** just to break even. Most platforms have over-promised + under-delivered.

Broken farm-tech promises: John Deere data lock-in controversies, Sep 2024 collapse of Nori, multiple failed carbon programs, Bayer's years of financial losses on Climate Corporation → deep market skepticism. New entrants need **verifiable 3rd-party outcome evidence** — not vendor case studies.

Seasonal cash flow: Farm ops highly seasonal w/ cash concentrated post-harvest. Annual subscription billing misaligned to farm cash flow creates real friction; **installment billing or harvest-contingent payment terms materially improve conversion** with smaller operators.

Summary

Agriculture is a deeply fragmented but high-stakes opportunity. Per-acre pricing ($2–$15/yr) dominates crop software; MRV/carbon layers add $3–$15/acre for regenerative programs — with credit revenues under structural pressure after 2024's 57% voluntary market contraction. Enterprise co-op + global trader deals scale to $5M–$25M/yr in blended technology spend, with 3-yr TCOs reaching $75M for largest operators. Regulatory tailwind significant: **EUDR (effective Dec 2026), FSMA, Article 6 carbon market formalization** all mandate traceability + data infrastructure that PrimusNeo's OAS interop + NEOS governance directly address. Single largest adoption barrier: **farmer data sovereignty** — go-to-market must lead with transparent data ownership terms, genuine interoperability, one-season pilot designs producing board-credible ROI. Co-op procurement flows through 12–24 mo board approval cycles; **Cooperative Ventures (CHS + GROWMARK) = critical gateway** to cooperative segment. Motions that bypass the agronomist + grower buying committee reliably fail.

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PrimusNeo — Construction & AEC Market Research Brief

Vertical: Construction (AEC, GC/Sub, Real Estate Developers, Infrastructure, MEP, Smart Buildings / FM)
Products mapped: AMP (Spatial OS) · NEOS (Governance) · OAS (Interop Fabric) · Fractal Engine (Intelligence) **Prepared:** 2026-05-23

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1. Comparable Vendor Pricing

1.1 BIM / Digital Twin

Vendor	Model	Indicative Range
Autodesk Construction Cloud (ACC)	Per seat / annual	Autodesk Docs $500/seat/yr · BIM Collaborate $705/seat/yr · Autodesk Build **$1,625/seat/yr**; vol discount at 5+
Bentley iTwin	Credits + enterprise	Community free non-commercial; enterprise custom — typical large infra **$200K–$1M+/yr**
Trimble Connect	Per seat	Pro $12.41/user/mo (~$149/yr); Innovate $29.08/user/mo (~$349/yr)
Oracle Aconex	Per seat	~$49/user/mo (<10 users); $39/user/mo (100 users); $29/user/mo (1,000+); impl $50K–$100K
Hexagon SmartBuild	Custom enterprise	Not public; AEC twin contracts typically **$150K–$500K/yr**
Procore	**% of Annual Construction Volume (ACV)**	~0.1–0.3% of ACV; small GC $4.5K–$10K/yr; mid $25K–$80K/yr; ENR-100 **$150K–$500K+/yr**; unlimited user seats within license

1.2 Project Management

Vendor	Pricing
Procore	ACV model; 25-project regional GC all-in $40K–$120K/yr
PlanGrid (→ Autodesk Build)	Rolled into ACC; legacy ~$39/user/mo
Buildertrend	~$499/mo flat (residential / light commercial)
Fieldwire (Hilti)	Free / Pro $54/user/mo / Business $74 / Business Plus $94

1.3 Common Data Environment (CDE)

Vendor	Pricing
Asite Cognitive CDE	Enterprise custom; mid-to-large $50K–$300K/yr
Newforma	Custom; mid-market ~$20K–$80K/yr
Conject / Oracle EPPM	Per-seat or project; rolled into Oracle C&E cloud

1.4 Reality Capture

Vendor	Pricing
Matterport	Pro (20–150 spaces) ~$450–$900/yr per site; Business custom
DroneDeploy	Individual $349/mo · Advanced $649/mo (annual) · Enterprise custom
Pix4D	Cloud entry $49.20/mo; Advanced $249/mo; site license enterprise
Open Space	~$500–$2,000/project or enterprise site license

1.5 Smart Building / FM Digital Twin

Vendor	Model	Range
IBM TRIRIGA (Maximo RE&F)	Per seat + modules	~$35/user/mo entry; large portfolio SaaS **$200K–$800K/yr**
Planon	Per seat + impl	~$50/user/mo; full IWMS $200K–$1M+ deployment
Honeywell Forge	Custom enterprise	$100K–$500K/yr commercial bldg portfolio
Willow Twin	Per building / portfolio	$50K–$250K/yr per campus; full portfolio deals $1M+

1.6 ESG / Embodied Carbon

Vendor	Pricing
One Click LCA	Custom tiers (Designer → Power Pack); **$15K–$120K+/yr**; onboarding $3K–$25K
Tally (acq C.Scale, Apr 2026)	BIM plug-in; ~$500–$2,000/project or annual studio license
EC3 (Building Transparency)	Free, open-access EPD database

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2. Engagement Scale Tiers — 3-Year TCO

Small: GC/Sub <$50M Revenue, 1–2 Active Projects

Procore entry ($5K/yr) + ACC Build (5 seats, ~$8K/yr) + Fieldwire Pro (5 seats, ~$3K/yr) + Matterport Starter. Impl: self-serve or 1–2 SI days; $5K–$15K. **3-yr TCO: $40K–$90K.** PrimusNeo entry: OAS as Procore/ACC connector eliminating duplicate entry; AMP for as-built spatial record without displacing incumbents.

Mid: Regional Contractor/Developer $50–500M, 5–30 Projects

Procore ($25K–$80K/yr) + ACC BIM Collaborate 20–50 seats ($14K–$35K/yr) + CDE Asite/Newforma ($20K–$80K/yr) + DroneDeploy Advanced ($8K/yr) + One Click LCA Designer ($20K/yr). Impl: SI partner 60–120 days; $50K–$150K. **3-yr TCO: $350K–$800K.** PrimusNeo play: NEOS for cross-project governance/approvals; Fractal replacing PowerBI/Excel portfolio dashboards; OAS federating CDE + PM + BIM data.

Large: ENR Top-100, Mega-Projects ($1B+ capex), National Portfolio

Procore enterprise ($200K–$500K/yr) + Bentley iTwin or ACC enterprise ($200K–$1M/yr) + Willow Twin or Honeywell Forge ($250K–$1M/yr) + One Click LCA Power Pack ($80K–$120K/yr) + AECOM/Arup digital advisory ($500K–$2M/yr). Impl: 6–18 mo, $500K–$3M. **3-yr TCO: $3M–$12M.** PrimusNeo play: Full AMP+NEOS+OAS+Fractal as federated intelligence/governance layer across portfolio.

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3. Professional Services Rates

Firm / Role	Day Rate (USD)
McKinsey Capital Projects — Senior Partner	$10,000–$25,000
McKinsey — EM	$3,000–$6,000
Accenture Industry X (Built Env) — Principal/MD	$5,000–$15,000
Accenture — Senior Consultant	$1,500–$3,500
AECOM Digital — Digital Delivery Director	$1,200–$3,000
Arup Digital — Associate/Senior Associate	$1,500–$4,000
Autodesk / Trimble Impl Partner — BIM Consultant	$150–$250/hr ($1,200–$2,000/day)
Bentley Authorized Distributor — iTwin Impl	$175–$300/hr

Typical 6-mo AEC digital transformation advisory + impl (mid-tier firm): **$300K–$1.5M all-in**.

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4. Pricing Mechanics

- **Per-project:** Reality capture (DroneDeploy, Matterport, Open Space), LCA scans (Tally per model), CDE project modules. $500/project (residential) to $50K+ (mega-project)
- **Per-seat/per-user:** Dominant for PM + CDE. Fieldwire $54–$94/user/mo; Aconex $29–$49/user/mo; ACC $42–$135/user/mo by module. Vol discounts at 5–25 seats
- **Per-square-foot (FM):** IWMS + twin licensing **$0.10–$0.50/sf/yr** for portfolios >500K sf. TRIRIGA, Planon, Willow Twin
- **% of Annual Construction Volume:** Procore proprietary; **0.1–0.3% of ACV**. Healthy owner digital program budgets run 1–2% of total capex (McKinsey)

- **% of capex (advisory):** AECOM + Arup digital retainers on mega-projects **0.2–0.5% of capex**. McKinsey Capital Projects 0.5–1% for strategic programs

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5. Regulatory Overlay

EU EPBD (recast 2024): All new buildings must meet Zero-Emission Building criteria. Fossil-fuel boiler subsidies ended Jan 2025. **Digital Building Passports** (per-building records covering energy, materials, lifecycle) mandated as part of EPBD impl. Immediate demand for data fabrics ingesting heterogeneous building system data — **direct OAS + AMP use case**.

California Buy Clean Act (BCCA) / AB 2446: GWP limits on structural steel, concrete rebar, flat glass, insulation on CA public works enforceable Jan 2025. AB 2446 targets 40% reduction in material carbon intensity by 2035. Drives One Click LCA + EC3; Fractal opportunity for automated EPD ingestion + real-time compliance reporting.

UK Net Zero Carbon Buildings Standard (NZCBS): v1 launches early 2026, integrating with PAS 2080. Upfront carbon (A1–A5) limits per m^2. From 2025, publicly funded projects must be designed without fossil-fuel heating.

ISO 19650 BIM: Global info-management benchmark; mandatory EU + UK (since 2016), increasingly GCC + India. Part 6 (H&S integration) released 2025. Requires structured CDE — **AMP can serve as the ISO 19650-compliant spatial information layer**.

LEED / BREEAM / Green Star: Embodied carbon docs, energy modeling, operational data submissions tied to credit thresholds → ongoing data obligations through bldg ops, supporting persistent AMP/Fractal usage post-construction.

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6. Procurement Signals

Primary champions: - **VDC / BIM Manager** at GC or design-build: controls day-to-day platform selection for ACC, Procore, CDE. Pain: design-side BIM vs field-side PM interop gaps; duplicate model mgmt - **CIO / CDO** at ENR-100 contractor: owns enterprise arch + vendor consolidation. Budget authority $500K+. Receptive to consolidation reducing vendor sprawl - **Project Director / Owner's Rep** on capital programs: schedule + cost + risk visibility across 10–50 active projects. Values federated data without forcing subs onto single mandated tool — **OAS value prop** - **Director of Sustainability / ESG Officer**: owns embodied carbon reporting under EPBD/BCCA/NZCBS. Needs automated EPD ingestion + LCA workflow

RFP trigger events: - Owner-mandated ISO 19650 BIM/CDE reqs on public infra tenders (Highways England, GSA, state DOTs, Network Rail) - Post-merger portfolio consolidation - Mega-project mobilization: capital programs >$500M trigger digital delivery planning phase - ESG reporting deadline (annual sustainability report, EU SFDR for EU real estate funds, SEC climate disclosure for US-listed owners)

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7. Rejection Reasons

Paper/email-based culture: Significant portion of workforce (esp subs, trades) on paper, WhatsApp, email. Adoption requires change mgmt buyers consistently underestimate. Sub resistance to yet-another-login is a top-3 Procore complaint.

Subcontractor adoption friction: ENR-100 GCs routinely report sub networks won't adopt owner-mandated platforms. Solutions requiring zero friction on subs (read-only views, email-triggered approvals, no account creation) consistently outperform full-onboarding.

ROI attribution difficulty: Each construction project is unique → no A/B comparison. Decision-makers struggle to prove platform caused schedule improvement vs weather or skilled superintendent. Vendor-published ROI reports (Procore, Autodesk) discounted by CFOs.

Integration lock-in: Most ENR contractors have Procore + ACC + CDE locked in by multi-year contracts. Adding PrimusNeo requires clear OAS story: bidirectional Procore webhook, IFC/RVT ingestion from ACC, write-back without breaking workflows. **Integration tax = primary reason pilots stall at POC.**

Price sensitivity on thin margins: GC net margins 2–4%. Premium add-on without near-term measurable ROI (rework reduction, fewer RFIs, faster financial closeout) gets deferred. Pilot pricing with clear documented-ROI path is table stakes.

Summary

Construction/AEC is a $16T global market with fragmented multi-vendor software spend and accelerating regulatory pressure on data transparency. Dominant pricing: per-seat ($29–$135/user/mo) and ACV-% (0.1–0.3%) for PM, shifting to per-sf ($0.10–$0.50/sf/yr) for FM digital twins. 3-yr TCO: $40K–$90K (small GC) → $3M–$12M (ENR-100 portfolio). EU EPBD digital building passports + CA Buy Clean Act + UK NZCBS + ISO 19650 directly validate OAS interop + AMP spatial record propositions. Primary champions: VDC/BIM Managers + CDOs at large contractors; Owner's Reps drive top-down mandates on capital programs. Most durable rejection risks: subcontractor adoption friction, ROI attribution opacity, Procore/Autodesk integration lock-in. PrimusNeo addresses with zero-friction OAS connectors, documented project-level ROI instrumentation, and pilot structure that proves value in single project lifecycle before portfolio expansion.

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PrimusNeo Energy Vertical — Market Research Brief

Products in scope: AMP (Spatial OS), NEOS (Governance), OAS (Interop Fabric), Fractal Engine (Intelligence) **Vertical:** Energy — utilities, oil & gas, renewables, grid/ISO, DERs, microgrids, carbon markets **Date:** May 2026

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1. Comparable Vendor Pricing

Operational Digital Twin / ADMS / DERMS

Full-scale ADMS deployments: **$5M–$50M software-only** depending on utility scale and module count, exclusive of hardware and integration services (DOE ADMS Advisory).

- **GE Vernova GridOS (ADMS + DERMS):** Enterprise quote-only; ranked #1 Guidehouse Insights 2024 DERMS Leaderboard. IOU packages estimated **$8M–$20M over 5-year term**.
- **Schneider Electric EcoStruxure ADMS:** ~23% ADMS mindshare. Modular SCADA + OMS + DMS; comparable IOUs report **$6M–$15M**.
- **Siemens Spectrum Power ADMS:** ~20% mindshare; comparable. Both Schneider and Siemens price via competitive tender for IOU/ISO procurements.
- **AVEVA PI System (OSIsoft):** $162K example proposal itemises tags + connectors; enterprise deployments with 1M+ tags for large refineries / grid operators: **$500K–$5M/yr** subscription + support.
- **Cognite Data Fusion:** Closed 2024 at ~$100M SaaS ARR (+38% YoY); priced on usage/MAU tiers. Supermajor deployments: **$2M–$8M ARR**.
- **AspenTech APM suite / Mtell:** Enterprise custom; starts **$250K–$1M+/yr** for mid-size process industry.
- **Bently Nevada (Baker Hughes) System 1:** HW-bundled APM; per rotating machine; large plant deployments: **$500K–$3M** incl sensors.

DER Orchestration / VPP

- **AutoGrid Flex / Uplight DERMS:** Utility-facing SaaS converging around **$3–$8/kW-year** enrolled capacity. 50 MW portfolio: **$150K–$400K/yr**.
- **Voltus / Leap:** No-cost-to-customer, market-revenue-share. Voltus customers earn **$100K–$470K/MW-year** depending on ISO (PJM/ISO-NE highest; SPP $100K). Voltus retains ~20–30% of cleared revenue — platform value **$20K–$140K/MW-year** to aggregator.
- **Tesla VPP / Generac Grid Services (formerly Enbala):** Retail incentive ~$2/kWh dispatch (CA, PG&E). B2B utility DERMS SaaS: **$2–$6/kW-month** estimated.

Carbon Accounting / MRV

- **Watershed:** Packages from ~**$50K/yr**; large corporates with complex Scope 3 + energy portfolios pay **$200K–$500K/yr**.
- **Persefoni:** Free "Pro" tier; "Advanced" enterprise **$55K–$250K/yr**.
- **Sylvera / Pachama / Toucan / Flowcarbon:** Project-level rating; Sylvera **$20K–$100K/yr** for portfolio rating; tokenization platforms take **1–3% of credit face value**.

ETRM (Energy Trading & Risk Management)

- **Allegro (ION Group) / OpenLink:** Perpetual licenses **$250K–$5M+**; SaaS **$10K–$100K+/month**. Implementation: 1–2× license cost. Annual support 18–25% of license. Allegro renewables-sector CAGR 14.5% (Chartis Research).

REC / Energy Attribute Certificate Tracking

- **M-RETS / PJM-EIS GATS:** Registry transaction fees low (cents/MWh); value in data quality + hourly granularity. PJM Tier I RECs hit **~$40/MWh** in 2024 (long-run average $8.74/MWh). EnergyTag granular certificate pilots running in six jurisdictions. **Attestation infrastructure cost is the primary WTP signal for buyers seeking 24/7 CFE compliance.**

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2. Engagement Scale Tiers — 3-Year TCO

Tier	Profile	Software TCO (3yr)	Services TCO (3yr)	Total 3yr
Small	Single microgrid / community solar (<10 MW)	$150K–$500K	$100K–$300K	**$250K–$800K**
Mid	Municipal utility / mid-size IPP (100–500 MW)	$1.5M–$6M	$1M–$4M	**$2.5M–$10M**
Large	IOU / ISO-RTO / supermajor (1 GW+)	$10M–$50M	$8M–$30M	**$18M–$80M**

Small tier: microgrid SCADA, basic DERMS, carbon dashboard. Mid adds ADMS, ETRM integration, MRV. Large covers full ADMS/EMS/DERMS, ETRM, APM, regulatory compliance tooling.

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3. Services Rates

Firm	Role	Day Rate (USD)	Notes
McKinsey Energy Solutions	Senior Partner	$9,500–$12,000	Grid strategy, energy transition
McKinsey	EM	$5,500–$7,000	Program governance
Deloitte E&R	Director/Principal	$4,000–$6,000	ADMS PMO
Accenture Resources	Managing Director	$3,500–$5,500	OT/IT convergence, SAP IS-U
Accenture	Senior Consultant	$1,800–$2,800	Nearshore/offshore lower
Black & Veatch	ADMS/OT Principal	$2,500–$4,000	Utility eng + OT advisory
Worley	Process/Digital Lead	$2,000–$3,500	O&G, renewables EPC + digital
IBM Consulting	Energy Practice Lead	$2,500–$4,000	Maximo APM, grid data

ADMS rollout package cost (turnkey, mid-size IOU): Black & Veatch or Accenture programs for 500K-customer utility = **$8M–$25M in SI services** over 24–36 months.

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4. Pricing Mechanics — What Is Normal in Energy

Mechanism	Where Used	Typical Rate
Per-MW (enrolled capacity)	DERMS, VPP, DR aggregation	**$3–$12/kW-year**
Per-asset / per-machine	APM (Bently Nevada, AspenTech)	$500–$5,000/asset/yr
Per-tag (data historian)	AVEVA PI System	$0.10–$2/tag/month at scale
% of energy traded	ETRM / VPP revenue share	20–30% of cleared market revenue
$/REC or $/MWh	Certificate tracking, PPA settlement	$0.02–$0.15/MWh registry fee; $8–$40/REC market
$/tCO2e managed	Carbon MRV SaaS	Implicit in $55K–$500K/yr platform fees
Enterprise annual license	ADMS, EMS, ETRM	$500K–$10M/yr large-utility tier
% of project CAPEX	EPC / advisory	2–8% for engineering services

Dominant pattern: enterprise license + modules with 3–5% annual escalators. DERMS/VPP migrating to capacity-enrolled SaaS with performance incentives. Carbon software flat annual SaaS today, trending toward $/tCO2e-verified as MRV accuracy becomes auditable.

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5. Regulatory Overlay — Willingness-to-Pay Drivers

FERC Order 2222 (DER Market Participation): Requires all ISOs/RTOs to allow aggregated DER participation in wholesale markets (down to 100 kW min). Mandatory metering + telemetry + settlement-data → direct tech spend trigger for utilities + aggregators needing interoperable dispatch. **OAS**

architecturally positioned for this integration layer.

NERC CIP (CIP-002 through CIP-015-1): Annual aggregate burden for CIP-015-1 (Internal Network Security Monitoring) alone: **$13.2M/yr industry-wide** across ~1,636 registered entities (Fed Register Sep 2024). CIP-013 supply-chain mandates → structured vendor assessment + attestation → **direct WTP increase for auditable provenance (NEOS)**. CIP-003-9 (effective April 2026) extends supply-chain controls to low-impact BES.

EU CSRD / CSDDD: Large EU-listed energy firms + non-EU subs face mandatory double-materiality ESG disclosures from FY2024. Omnibus simplification (Sep 2025) reduced scope but preserved core emissions disclosure. Cross-border EU exporters face **CBAM** financial obligations from Jan 2026 — requires certified embedded-carbon data per shipment.

GHG Protocol + Article 6 (Paris Agreement): COP29 finalised Article 6.4 rules for ITMOs. Energy exporters + carbon market participants need cryptographically auditable provenance chains to prevent double-counting across NDCs. **NEOS + Fractal attestation + lineage capabilities map directly.**

State/PUC Rate Recovery: IOU technology investments (ADMS, DERMS, cyber) are recoverable through rate cases. CA CPUC has effectively mandated ADMS deployment for large IOUs; similar in NY (REV), IL (CEJA). PUC approval introduces 18–36 mo procurement cycle but de-risks vendor revenue through rate-payer funding.

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6. Procurement Signals

Buyer	Role	Decision Authority	Key Concern
Chief Grid Officer / VP Grid Ops	Grid modernisation sponsor	Op approval, PUC testimony	Reliability, NERC CIP, OT security
CIO / VP Technology	Platform + vendor governance	Architecture, vendor risk	IT/OT convergence, integration complexity
COO	Overall business case	Budget authority	TCO, risk, regulatory deadlines
Head of Trading / VP Commercial	ETRM, carbon, VPP commercial	Revenue-side tech	Settlement accuracy, market connectivity
Chief Sustainability Officer	Carbon MRV, ESG disclosure	Board-level mandate	CSRD/CBAM accuracy, auditability
Regulatory Affairs / GC	PUC filings	Rate-case approval	Regulatory defensibility, vendor attestation

Procurement pathway: Large ADMS / DERMS contracts ($5M+) require PUC pre-approval or inclusion in an Integrated Resource Plan (IRP) filing. RFPs via regulated competitive bidding; short-listed vendors pass 3–6 mo technical eval. **Sales cycle: 18–36 mo for IOUs/ISOs. Municipal utilities + IPPs faster (6–18 mo).**

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7. Rejection Reasons

NERC CIP / OT Cyber Risk: Software touching BES Cyber Systems must be classified, change-managed, assessed under CIP-007/010/013. No documented CIP compliance architecture = rejection at technical eval. CIP-005 requires jump-server + MFA for vendor remote access; cloud-native platforms need explicit network segmentation attestation. CIP-013 supply-chain assessments — no published security attestation pack = immediate disqualification.

Vendor Concentration Risk: IOUs + ISOs enforce multi-vendor policies post-SolarWinds + Volt Typhoon disclosures. Platform simultaneously touching SCADA, ADMS, DERMS, carbon accounting triggers vendor concentration review. **PrimusNeo must enter as interoperability + governance fabric above incumbents — not displacement — to reduce this.**

OT Integration Complexity: Legacy distribution SCADA + EMS run proprietary protocols (DNP3, IEC 61968/70, ICCP). Governance/interop platform must demonstrate certified connectors + reference architectures. Abstract "open APIs" claims don't survive utility architecture review boards.

Regulatory Approval Delays: Rate-recovery tech spend faces 18–36 mo PUC cycles; deals stall in evidentiary proceedings if cost-benefit testimony weak. **Third-party ROI validation before the rate case is prerequisite.**

Budget Cycle Mismatch: Utility capital plans set annually through IRP + budget processes. Missing capital plan window by one quarter delays deal by 12 months.

Summary

Energy presents PrimusNeo's highest regulatory tailwind and longest sales cycles simultaneously. ADMS, DERMS, APM markets dominated by GE Vernova, Schneider, Siemens, AVEVA at **$5M–$50M per IOU deployment**, with 18–36 mo procurement windows gated by PUC rate-case approval. FERC Order 2222, NERC CIP-013/015, CSRD, CBAM collectively force utilities + IPPs + supermajors to pay for auditable provenance, OT supply-chain attestation, cross-border carbon data — precisely NEOS + OAS capability set. **Highest-probability entry: interop + governance fabric layered above incumbent ADMS + PI System stacks — bypasses vendor-concentration objections, reduces OT cyber risk exposure, positions PrimusNeo in ETRM + carbon MRV adjacency ($50K–$500K/yr SaaS) while building toward full Spatial OS deployments at mid + large tier ($2.5M–$80M 3yr TCO).** Carbon markets + Article 6 attestation + REC hourly granularity = fastest near-term land opportunity.

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Finance Vertical — Market Research Brief

PrimusNeo: AMP (Spatial OS) · NEOS (Governance) · OAS (Interop Fabric) · Fractal Engine (Intelligence)
Date: 2026-05-23

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1. Comparable Vendor Pricing

Financial Data Fabrics / Middleware

Bloomberg Terminal: **$31,980/seat/year** (single seat); **$28,320/seat/year** (multi-seat); negotiated enterprise at **$18,000–$20,000/seat** (50+ seats). Hardware + add-ons push true Y1 cost to $34K–$35.5K/seat. Refinitiv Eikon (LSEG) ~**$22,000/seat/year** full feed; stripped variants from $3,600/year; exchange pass-through adds 20–50%. FactSet ~**$12,000/seat/year** mid-range. Snowflake / Databricks: **$25K–$40K/year** for 200 TB analytics; **$2M–$10M/year** for 500-analyst enterprise footprints.

Tokenization Platforms

Securitize: minimum engagement **$50K–$100K** (setup + issuance bundled); ongoing compliance/transfer-agent annual retainers. Tokeny (T-REX, EU-focused): SaaS licensing plus setup, comparable to Securitize. Polymath/Polymesh: transparent on-chain transaction fees. TaaS SMB SaaS: **$2K–$5K/month** (~$24K–$60K/year). White-label platforms: **$50K–$150K** upfront; custom development from $150K.

For live issuance: $3M offering = **$75K–$200K** all-in; $30M offering = **$200K–$1M+**. Platforms charge **0.5%–2.5%** of asset-under-tokenization for smart-contract deployment + compliance wrapping; success-fee models charge **6%–10%** of funds raised + 2% equity allocation. Onyx (now Kinexys) by JPM and Provenance are relationship-gated.

Custody / Wallet-as-a-Service

Provider	Model	Rate
BitGo	AUM bps	5 bps/month (60 bps/yr) on AUC >$100K
Fireblocks	API-usage + bespoke infra	Negotiated; ~$1M+ AUC minimum
Anchorage	Bilateral negotiation	$10M+ AUC minimum

Compliance / KYC / AML

Chainalysis Reactor: ~**$40K/year** single-seat; enterprise KYT: **$100K–$500K+/year**. Elliptic Forensics: comparable to Chainalysis at enterprise. TRM Labs: modular chain-coverage; negotiated. ComplyAdvantage: **$500–$5K/year** SMB API; enterprise negotiated.

Governance / GRC

Diligent, Nasdaq Boardvantage: custom enterprise; Vendr shows Fortune 500 negotiate **30–50% off** initial quotes. ServiceNow IRM/GRC: Fortune 500 negotiate **60–80% off list**; mandatory **5–10% annual uplift** since 2024.

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2. Engagement Scale Tiers — 3-Year TCO

Tier	Profile	3-Year TCO
Small	Boutique manager / family office <$1B AUM	**$350K–$900K**
Mid	Regional bank or asset manager $1–50B AUM	**$1.5M–$5M**
Large	G-SIB, top-20 asset manager, major exchange	**$8M–$25M+**

Small: TaaS SaaS ($72K–$180K), legal/structuring ($100K–$250K), custody ~60 bps/yr on $100M AUC, KYC/AML ($30K–$90K), governance software ($30K–$60K), integration/dev ($75K–$150K).

Mid: White-label tokenization ($150K–$500K), issuance legal ($200K–$600K), Fireblocks/BitGo custody on $500M–$2B notional, Chainalysis ($150K–$300K), GRC/IRM ($300K–$600K), data fabric ($150K–$400K), Big-4 SI ($500K–$1.5M).

Large: Custom tokenization + interop fabric ($2M–$5M build), multi-custodian, Snowflake/Databricks at scale ($2M–$10M/yr), full Chainalysis + TRM ($400K–$1M/yr), ServiceNow IRM ($600K–$1.5M/yr), Big-4 transformation ($3M–$10M).

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3. Services Rates

Deloitte FSI (GSA): **$2,000–$3,000/day** (senior consultant), **$3,000–$5,000/day** (principal/director), **$5,000–$10,000/day** (partner, digital-asset specialty). EY Blockchain, PwC Digital Assets, KPMG Capital Markets, Accenture FS: similar bands. McKinsey/BCG senior partners: **$6,700–$9,500/day**.

Engagement sizes: - Tokenization readiness assessment: **$150K–$400K** (8–16 wks) - Full tokenization program delivery: **$1M–$5M** (6–18 mo) - MiCA / regulatory compliance implementation: **$500K–$2M** - Enterprise data fabric + governance transformation: **$3M–$12M**

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4. Pricing Mechanics Norms

Mechanic	Range	Used by
Basis points on AUM (custody)	5–60 bps/yr	BitGo, Anchorage
% of issuance value	0.5%–2.5%	Securitize, Tokeny, TaaS
Success fee on raise	6%–10%	Republic, crowdfunding-adjacent
Per-seat/user annual	$12K–$32K/yr	Bloomberg, FactSet, Refinitiv
Monthly SaaS	$2K–$15K/mo	TaaS SMB/mid-market
Annual enterprise license	$50K–$500K+/yr	GRC, board software, AML
Per-transaction / on-chain	Protocol-variable	Polymesh, Provenance
Volume discount	20–60% off list at enterprise	All major vendors

Dominant pattern for mid-to-large: **tiered enterprise license + bps overlay on notional/AUC**, supplemented by services retainer.

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5. Regulatory Overlay

MiCA (EU, fully applicable Dec 2024): CASP licensing build-out €50K–€100K (small); capital reqs €50K (advisory), €125K (custody/exchange), €150K (trading platforms). Non-compliance fines: €15M or 3% of revenue. Creates direct demand for immutable governance logs, auditable issuance records, automated sanctions screening, cross-border identity portability.

OCC Interpretive Letter 1184 (May 2025): National banks may provide crypto custody (fiduciary + non-fiduciary), use sub-custodians, but must demonstrate controls. Banks will pay **$200K–$500K+/year** for platforms that produce attestations on-demand.

SEC SAB 122 (Jan 2025): Rescinds SAB 121 on-balance-sheet liability — removes primary accounting disincentive for bank custody.

FCA (UK): Implicit compliance tax **£100K–£500K/year** in incremental monitoring/reporting under crypto-asset registration and stablecoin frameworks.

Aggregate signal: Auditability + immutable governance logs + automated attestations are highest-value features FIs pay a premium for — they directly reduce regulatory capital risk and legal liability.

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6. Procurement Signals

Tier	Primary Buyer	Tech Sign-off	Compliance Veto	Cycle
Small	CIO / COO	CTO	—	2–4 mo
Mid	Head of Digital Assets	CDO / CTO	CRO	6–12 mo
Large	Head of Digital Assets + C-suite sponsor	CTO + CDO committee	CRO + Legal	12–24 mo

Mandatory mid/large requirements: SOC 2 Type II (US default), ISO 27001 (EU/APAC), annual 3rd-party pen test, BCP/DR docs, GDPR/data residency, reference clients in regulated FS. Missing SOC 2 or ISO 27001 disqualifies at mid/large.

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7. Rejection Reasons

1. **Regulatory ambiguity** — Legal/compliance veto until local regulator gives written guidance.
2. **Custody model mismatch** — Must map to self-custody MPC/HSM, qualified custodian, or bank-grade sub-custodian.
3. **Legacy core-banking integration gap** — Temenos/Finastra/Murex/Calypso require certified FIX/API connectors.
4. **Open-source / IP ownership risk** — Unaudited smart contracts + no formal vuln disclosure + no IP indemnity = reject.
5. **Vendor concentration / lock-in risk** — Multi-custodian, multi-chain is a requirement at G-SIB.
6. **Missing security certs** — SOC 2 + ISO 27001 non-negotiable at mid/large.

Summary

FIs deploying tokenization, governance, and interop infrastructure face: Bloomberg ($32K/seat) and Refinitiv ($22K/seat) anchor information-layer reference price; tokenization platforms (Securitize, Tokeny) charge $50K–$150K setup + 0.5%–2.5% of issuance value; custody (BitGo, Fireblocks, Anchorage) at 5–60 bps/year on AUC. 3-year TCO: $350K (boutique) to $25M+ (G-SIB). Big-4 day rates: $2K–$10K. MiCA, OCC Letter 1184, SAB 122 make auditability + immutable governance logs binary procurement requirements. Primary deal-killers: regulatory ambiguity, absence of SOC 2 Type II / ISO 27001, inability to integrate with legacy core-banking. Sales cycles: 6–12 mo at mid, 12–24 mo at G-SIB.

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- MiCA 2026 Guide · EU MiCA Stats
- OCC Digital Assets Banking · SEC SAB 122
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PrimusNeo — Governance Vertical: Market Research Brief

Verticals: Public-sector / Civic Government (cities, states, federal, intergovernmental) AND DAO / Web3 Governance / Cooperatives / Federations **Components:** AMP · NEOS · OAS · Fractal Engine **Date:** 2026-05-23

1. Comparable Vendor Pricing

Civic / GovTech

Vendor	Product	Pricing Signal
Granicus	Citizen Experience Cloud (govDelivery, EngagementHQ, GovAccess)	Custom; ~$25K–$150K/yr mid-size cities; bundled state-level $200K–$500K+. Via Carahsoft + cooperative vehicles
Tyler Technologies	ERP, Courts, Permitting (EnerGov), Public Safety	Impl fees rival/exceed license. EnerGov starts ~$50K (small), up to $2.7M for large county. SaaS courts $125–$370/user/month. Annual maintenance 18–22%. **Illinois statewide ballooned to $250M with delays**
OpenGov	Budgeting + Planning, Permitting, Reporting	Population-scaled SaaS: $20K–$40K/yr small municipalities; six-figure ARR for counties + state agencies
Esri / ArcGIS	Spatial / GIS	Desktop Advanced Concurrent Use: **$11,958/yr** (WA/ID state contract 2024). Enterprise Standard (gov): $3K–$12K per concurrent seat. **Statewide EAs $2M–$10M+**
Bloomberg Cities	Urban innovation initiatives	Grant-funded ($1M–$5M per initiative); not commercial SaaS

Participatory Democracy

Vendor	Model	Pricing
Decidim	OSS (Ruby on Rails)	Free to deploy. Certified partner impls typically: **3% of contract value (companies) / 1.5% (nonprofits) reinvested into the commons**. Used by Barcelona, Helsinki, EU Commission, NYC, French Senate
Consul Democracy	OSS	Free; self-hosted. Integrator support custom-priced
Pol.is	OSS (Computational Democracy Project)	Free SaaS; gov/enterprise support custom
CitizenLab / Go Vocal	SaaS (Brussels)	Custom; est **$30K–$80K/yr** mid-size cities. 400+ government customers across 18 countries
Bang the Table / EngagementHQ	SaaS (Granicus-acquired)	Bundled within Granicus; standalone est $25K–$75K/yr. 800+ gov orgs

DAO Governance Tools

Tool	Model	Pricing
Aragon (OSx)	Protocol + no-code launcher	Free to deploy on L2 (gas only; 67% cheaper post-ANTv2). Enterprise custom build services per engagement. **Manages $6B+ in active DAO treasuries**
Snapshot	Off-chain gasless voting	Free. No tx cost. Default signal layer for Uniswap, Aave, ENS, thousands of DAOs
Tally	On-chain governance (Governor Bravo / OZ)	Free for DAOs. Protocol fees from Tally Protocol (2024) shared among Tally, delegates, rebalancers. L2 execution: cents/vote; Eth mainnet $1–$50/tx
Boardroom	Governance aggregator + delegate dashboard	Free for participants; enterprise intelligence custom
Llama	DAO financial ops + analytics	Fixed retainer. **~$2.5M/yr for Aave** (treasury mgmt + analytics + growth, 12-mo); $300K–$800K for mid-size DAOs
Karpatkey	DAO treasury mgmt + DeFi yield	Fixed retainer + performance upside on yield. **Steakhouse Financial won ~$322K/yr for Lido**. Karpatkey raised $7M (2024) to scale services

Board / Council Software

Vendor	Pricing
Diligent Boards	$15K–$35K/yr for boards under 10 members; large multi-entity enterprises **$100K–$250K+/yr**
Nasdaq Boardvantage	All-inclusive custom. Est $20K–$40K/yr small public councils; $80K–$200K/yr large gov/corp entities
OnBoard	Three tiers (Essentials, Premium, Ultimate); est $15K–$60K/yr

Identity & SSI

Vendor	Pricing
Microsoft Entra Verified ID	Free up to **50K credential txs/mo** (issuance + verification + revocation). Above: Entra ID P1/P2 (~$6–$9/user/mo). Face Check: PAYG per verification or 8 checks/seat/mo via Entra Suite
Trinsic	Credit-based with min commitment. Per-tx varies by identity scheme + IDV source
Indicio	Enterprise SSI network; node participation ~$18K–$36K/yr
Dock.io	Pay-per-credential + SaaS tiers; ~$0.15–$0.50/credential issuance at scale
Civic	Identity verification SaaS; enterprise + PAYG in Web3/KYC contexts

Federation / Inter-Org Coordination

Vendor	Pricing
Discourse	Hosted $100–$300/mo (Standard–Business); self-hosted free; enterprise custom
Loomio	Community groups: low one-time fee. Business/nonprofit ~$1,200–$3,600/yr. Gov/enterprise custom. Worker-owned cooperative since 2012

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2. Engagement Scale Tiers — 3-Year TCO

Small: 1 Town / 1 Cooperative / 1 Small DAO (<$10M treasury)

- Software $20K–$80K/yr (civic engagement + basic board portal + identity)
- Impl $15K–$60K one-time
- Services $0–$50K (self-service or boutique)
- Compliance: StateRAMP-lite or none; basic SSI; no FedRAMP
- **3-Year TCO: $75K–$300K**

Mid: City 100K–500K / State Agency / Mid DAO ($10M–$500M treasury)

- Software $80K–$400K/yr (GovTech ERP + engagement + spatial + board portal + identity fabric)
- Impl $250K–$1.5M one-time
- Services $200K–$800K/yr (Deloitte GPS or regional SI)
- Compliance: FedRAMP Moderate or StateRAMP/GovRAMP; GDPR if EU
- **3-Year TCO: $1.2M–$5M**

Large: State-Level / Federal Agency / G20 Megacity / Top-50 DAO ($500M+ treasury)

- Software $500K–$5M+/yr (Esri statewide EA + Tyler ERP + Granicus state suite + identity fabric)
- Impl $2M–$20M+ (multi-year, multi-team program)

- Services $2M–$15M/yr (Big-4 or Booz Allen PM)
- Compliance: FedRAMP High or DoD IL4/IL5; MiCA CASP licensing for EU DAOs
- **3-Year TCO: $10M–$60M+**

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3. Services Rates

Systems Integrators / Management Consultants

Firm	Range	Notes
Deloitte GPS	$200–$450/hr GSA schedule; senior partners $500–$700/hr	Federal via OASIS+, agency IDIQs, GSA MAS
Accenture Federal Services	$190–$420/hr (GSA MAS GS-35F-540GA); senior $300–$500/hr	Loaded labor rates on GSA Advantage
Booz Allen Hamilton	$187–$376/hr (validated GSA 2024); principal/fellow $400–$600/hr	Defense, intel, civil agency
Tyler Technologies (impl)	Bundled into license; T&M ~$150–$225/hr custom dev; fixed-fee $75K–$500K	Often sole-sourced via cooperative purchasing

DAO / Web3 Treasury Services

Firm	Model	Typical Fee
Llama	Annual fixed retainer	**~$1.5M–$2.5M/yr** for large protocol DAOs (Aave, Compound); $300K–$800K for mid-size
Karpatkey	Fixed retainer + performance	$200K–$500K base/yr + performance upside on DeFi yield
Steakhouse Financial	Fixed retainer	~$322K/yr (Lido DAO governance proposal, public record)
Generic treasury advisor	% of AUM or fixed	**0.25–0.75% of treasury AUM/yr**, or $50K–$300K fixed annual

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4. Pricing Mechanics

Per-resident: Granicus, OpenGov, CitizenLab, Bang the Table use informal population-banded pricing. City of 50K residents ~$25K/yr; 500K ~$120K/yr for comparable modules. Not published; negotiated.

Per-member / per-user: Board portals (Diligent, Boardvantage) + enterprise SaaS charge per seat. SSI platforms per credential tx. Tyler SaaS courts $125–$370/user/mo.

Per-vote / per-tx: Snapshot free (off-chain). Tally + on-chain Governor: gas pass-through $0.01–$50/vote depending on chain + congestion. Entra Verified ID per-tx above 50K/mo threshold.

% of treasury: Karpatkey, Llama, comparable DAO treasury managers charge **0.25–0.75% AUM** or fixed retainer. $100M treasury → $250K–$750K/yr before performance component.

Enterprise license: Esri statewide EAs, Tyler ERP, Granicus state contracts = flat annual against population counts + user volumes — typically **$500K–$5M+/yr at state scale**.

Grant / public funding: Decidim, Consul, Pol.is — OSS with deployment costs borne by agency. EU Horizon funds + US federal innovation grants regularly fund participatory democracy deployments.

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5. Regulatory Overlay

Framework	Applies To	Cost Implication
FedRAMP Moderate	US federal SaaS	**$400K–$900K to authorize**; $100K–$300K/yr ongoing
FedRAMP High	Highly sensitive federal data	450+ controls; JAB or agency ATO; **$2M–$5M+ authorization**
StateRAMP / GovRAMP	US state + local (rebranded 2025); no FedRAMP reciprocity	$150K–$500K
DoD IL2	Non-sensitive DoD public data	~325 controls; FedRAMP Mod equivalent; $400K–$800K
DoD IL4	CUI + contingency ops	Add'l DoD-specific controls beyond IL2; $800K–$2M+
DoD IL5	NSS data + higher-sensitivity CUI	450+ controls; restricted DISA-authorized cloud only; **$2M–$5M+**
GDPR / EU NIS2	EU public sector	Data residency in EEA; DPA; DPIA; privacy by design; up to €20M or 4% global turnover in fines
MiCA (EU)	DAOs + CASPs operating in EU	CASP licensing required by end-2024 (stablecoins from June 2024); 18-mo grandfathering ending mid-2026; **legal + licensing €200K–€1M+**
SAB 121 / SAB 122	US DAO + crypto custodians	SAB 121 rescinded Jan 2025 via SAB 122; custodied crypto no longer on custodian balance sheet — reduces custody compliance overhead
CMMC / CMMI	Defense-adjacent contractors	Maturity Levels 1–3; Level 3 requires C3PAO assessment; **$500K–$3M+**

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6. Procurement Signals

Buyer Personas

Persona	Org Type	Trigger
CIO / CITO	City, county, state, federal	Modernization mandate; legacy EOL; federal efficiency pressure
Chief Innovation Officer	Large metro, state, federal	Pilot/innovation budget; often outside traditional IT procurement
City Manager / County Admin	Local gov	Budget cycle; council mandate; constituent service pressure
DAO Steward Council / Foundation	Protocol DAO	Token holder vote; treasury diversification; ops maturity milestone
Cooperative Board / Federation Steward	Worker co-op, credit union, housing co-op	Board vote; governance reform; federation membership requirement

Cooperative + Expedited Procurement Vehicles

- **Sourcewell:** Pre-competed cooperative; 50,000+ member agencies. **Ideal first cooperative vehicle for PrimusNeo targeting US local government**
- **NASPO ValuePoint:** Lead-state model; all 50 states + DC + territories eligible. Strong for cloud/SaaS. Separate enrollment from Sourcewell
- **GSA MAS Schedule (SIN 518210C for IT; OASIS+ for prof services):** Federal + state/local cooperative purchasing authority built into contract clauses
- **State-specific schedules:** TX DIR, CA DGS, NY OGS — non-negotiated access to pre-priced agreements within those states

Process Norms

- Government procurement cycles: **6–18 months RFP to award** for mid/large. Small/micro-purchase thresholds (under $250K federal) allow sole-source + streamlined
- **DAO governance procurement:** Token holder vote via Snapshot signal → Tally on-chain execution. Full proposal-to-execution cycle: **2–8 weeks** incl discussion period
- Budget cycles: Federal FY Oct 1. State/local vary; most decisions compress into Q3–Q4. Missing a budget cycle = 12-month delay

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7. Rejection Reasons

Risk	Driver
Procurement red tape	New vendor not enrolled on approved cooperative vehicle or agency-specific approved list. Incumbent (Tyler, Granicus, Esri) lock-in with switching costs $500K–$5M+
FedRAMP gap	No ATO = automatic disqualification for most US federal + many state agencies. **FedRAMP 20x initiative (launched 2025) aims to accelerate via agency-path in ~90 days** using pre-authorized infra boundaries
DoD security classification gap	IL4/IL5 block commercially-hosted SaaS without DISA provisional. CMMC Level 2–3 required for defense-adjacent
DAO treasury security	**$4B+ lost to governance attacks + treasury exploits since 2016**. Multi-sig thresholds commonly too low (2/3 cited as insufficient; experts recommend min 3/5 for <$10M, 4/7 for larger)
Governance attack vectors	Sybil attacks on voting identity, delegation abuse, flash-loan voting power acquisition. MiCA regulators cite these as institutional adoption barriers. **Governance attack = existential risk for DAO clients**
Data sovereignty / GDPR	EU public sector requires data residency in EEA. US-headquartered vendor without EU hosting = automatic disqualifier in most EU public tenders
Proof of scale / references	Governments require **3–5 comparable-agency references**. Early-stage vendors without marquee public-sector references routinely eliminated in technical evaluation
Budget cycle mismatch	Missing a gov's annual budget decision window = 12-month pipeline delay regardless of product fit

Summary

Governance spans two structurally different buyer archetypes — traditional public-sector agencies + emergent Web3 organizations — but both share core need for credible, auditable, multi-stakeholder decision infrastructure at scale. **GovTech incumbents (Tyler, Granicus, OpenGov, Esri) command $20K–$5M+ annual contracts** and benefit from cooperative purchasing lock-in but face criticism for cost overruns, slow delivery, poor interop. Participatory democracy tools (Decidim, CitizenLab) occupy lower-cost lane ($0–$80K/yr) but lack enterprise orchestration. **DAO governance infra (Snapshot, Tally, Aragon) free at protocol layer; economic value concentrates in treasury management services ($200K–$2.5M/yr)** where Llama + Karpatkey define pricing benchmark at 0.25–0.75% AUM or fixed retainer. PrimusNeo's combo (AMP + NEOS + OAS + Fractal) positions above point solutions, below full SI programs. Primary barriers: **FedRAMP/StateRAMP compliance ($400K–$900K, 12–24 mo), cooperative purchasing vehicle enrollment, proof-of-scale references**. DAO adoption barriers: governance security architecture + MiCA licensing overhead in EU. **Optimal entry: Sourcewell or NASPO enrollment + a single marquee city or mid-DAO reference deployment to unlock broader pipeline.**

Sources

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- GSA MAS Hourly Labor Ceiling Rates · NASPO ValuePoint · Sourcewell · Loomio Pricing

PrimusNeo Healthcare Vertical — Market Research Brief

> ***Scope:*** *Hospitals, health systems, payers, pharma/biotech, public health, digital health* ***Date:*** *May 2026*
> ***Products mapped:*** *AMP (Spatial OS), NEOS (Governance), OAS (Interop Fabric), Fractal Engine (Intelligence)*

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1. Comparable Vendor Pricing

EHR / HIE

Vendor	Model	Price Signal
Epic	Per-bed + impl	$500–$1,000/bed license; $1,200–$3,500/provider/mo SaaS; full IDN deployment **$50M–$1.2B** (Trinity $800M, Northwell $1.2B, AdventHealth $660M)
Oracle Cerner	Per-user SaaS + enterprise	~$25/user/mo base; large system contracts multi-million/yr
Meditech Expanse	Per-user + setup	$200–$800/user/mo; setup $7M–$20M for large systems; community hospitals ~$397/physician/mo
Athenahealth	PMPM / rev-share	Custom; 3–8% of collections for ambulatory

Market share: Epic 42.3% · Cerner 22.9% · Meditech 14.8%.

Interop / FHIR Fabric (OAS direct comps)

Vendor	Model	Price Signal
Redox	Sub + per-message	Monthly floor **~$35K**; $0.50–$2.00/message
1upHealth	Per-API	Fractions of a cent/call; enterprise custom
Health Gorilla (QHIN)	Enterprise SaaS	Custom; TEFCA QHIN live Q1 2024
Particle Health	Enterprise SaaS	Custom; 320M+ FHIR records; ~12-wk CTV
Datavant	Data collaboration	80K+ hospital/clinic connections; acquired Aetion (2025)

Clinical Data Platforms (Fractal + AMP comps)

Innovaccer, Health Catalyst, Arcadia, Komodo Health: custom-quoted/NDA. Benchmarks: **$200K–$2M+/yr** for mid-large. PMPM analytics: $0.50–$3 basic, $5+ full population health, $15–$20 high-risk disease, $150–$200 intensive care mgmt.

Digital Twin / Hospital Ops (AMP spatial)

GE HealthCare Command Center Digital Twin: ~500 hospitals / 55K+ beds globally; ~5-mo implementation; enterprise-quoted. Siemens Healthineers + Epic Cosmos: bundled add-ons.

Governance / Consent / IRB (NEOS comps)

OneTrust Healthcare: modular; min $10K/yr; Privacy Essentials ~$3,680/mo; HIPAA-module custom. Bundled into broader compliance/governance contracts typically.

Pharma Analytics (Fractal)

Vendor	Model	Price
Veeva Vault CRM	Per user/yr + enterprise	$1,500–$1,800/user/yr list; F500 pharma $5–10M+/yr
IQVIA	Data license + analytics	Multi-million-$ annual data licenses
Aetion/Datavant	RWE SaaS	Acquired by Datavant 2025; RWE market $1.3B (2024) → $3.7B by 2031
Komodo Health	Healthcare Map	330M+ patient journeys; custom enterprise

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2. Engagement Scale Tiers — 3-Year TCO

Small: Community Hospital (50–200 beds)

EHR: Meditech/Cerner $3M–$8M total 3-yr. Interop: $1.2M–$2.5M. Analytics: $200K–$600K/yr. Governance: $50K–$150K/yr. **PrimusNeo 3-yr TCO positioning: $1.5M–$4M** competing on TEFCA-native OAS vs costly Redox overhead.

Mid: Regional System (200–2,000 beds)

EHR: Epic/Cerner $30M–$150M total impl. Interop: $5M–$15M over 3 yrs. Data platform: $600K–$3M/yr. Governance/consent: $150K–$500K/yr. Consulting (Huron/Chartis): $5M–$20M. **PrimusNeo 3-yr TCO positioning: $8M–$25M** displacing/wrapping the interop+data stack with AMP+OAS.

Large: National IDN, AMC, Top-20 Pharma

EHR: Epic $200M–$1.2B all-in. SI services (Deloitte/Accenture): $50M–$500M. Interop/data: $20M–$80M over 3 yrs. Governance + compliance: $1M–$5M/yr (HITRUST r2 alone $400K–$800K per 3-yr cycle). **PrimusNeo 3-yr TCO positioning: $50M–$200M enterprise license**.

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3. Services Rates

Firm	Role	Day Rate (USD)
Deloitte/Accenture Health	Principal/Director	$4,000–$8,000/day
Deloitte/Accenture	Senior Manager	$2,800–$5,000/day
EY Health / Optum Advisory	Director	$3,000–$6,000/day
Chartis / Huron	Managing Director	$3,500–$7,000/day
Offshore/nearshore	Analyst–Consultant	$750–$1,500/day

Epic impl at large IDN: **$50M–$500M services spend** over 3–7 years. HL7/FHIR/custom-API integration at regional health systems: **$2M–$15M per major initiative**. OAS directly compresses this — clear ROI pitch to CIOs.

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4. Pricing Mechanics

Per-bed standard for inpatient EHR ($500–$1,000/bed upfront). **Per-provider/user** for ambulatory/analytics ($25–$3,500/provider/mo). **PMPM** dominant for payer analytics/population health/VBC ($0.50–$200 PMPM by intensity). **Per-API-call** for FHIR fabric/interop. **Enterprise license (ELA)** for broad deployments (Veeva, Datavant). **Per-encounter** emerging for ambient AI (Nuance DAX, Suki, Epic ambient). PrimusNeo positioning: AMP on per-bed/per-facility ELA · OAS on per-API/flat interop ELA · NEOS on per-covered-life/enterprise governance ELA · Fractal on PMPM/per-research-query.

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5. Regulatory Overlay

HIPAA / HITECH: Legal floor. BAAs non-negotiable. HIPAA Security Rule overhaul (effective May 2026) mandates MFA, universal PHI encryption, 24-hr breach notification, annual pen testing.

HITRUST CSF: De facto procurement gate. **81% of hospitals and 83% of payers** require HITRUST from tech vendors. **i1 (~$70K)** is now minimum for most health system RFPs; **r2 ($100K–$500K, 12–15 mo)** expected for core platforms. 3-yr enterprise cycles: $400K–$800K. PrimusNeo must achieve at least HITRUST i1 before substantive RFPs.

ONC TEFCA / QHIN: Common Agreement v2.1 effective Jan 2025; 21,086 orgs live. QHIN designation: ~12 months. OAS as TEFCA-native fabric is a direct differentiator vs Redox/Health Gorilla.

EU EHDS (Reg EU 2025/327): In force March 2025; general application March 2027. EHR vendors must certify for interop/security by Jan 2026. Priority health data exchange (patient summaries, ePrescriptions) mandatory EU-wide by March 2029.

EU MDR / GDPR: MDR applies if Fractal outputs constitute clinical decision support (Class IIa/IIb). GDPR special-category health rules → consent management + DPIA — direct NEOS revenue opportunity.

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6. Procurement Signals

Buyer personas: CIO (infra, vendor consolidation, integration), CMIO (workflow, patient safety, physician adoption), CDO (data strategy, AI governance), CCO (HIPAA, HITRUST, GDPR). Pharma: VP Data Science, Head of RWE, CTO.

Capital committee: IT capital >$1M requires board/capital committee. Decision: 6–18 months. EHR replacement: 5–10-yr cycles. Analytics/interop refresh: 3–5 yrs. Budget cycles Jul–Sep; Q4/Q1 peak signing.

RFP cadence: 68% of providers manage procurement manually (2024 HIMSS). RFPs: 6–12 mo issuance to award. Evaluation committees: 3–5 hrs/wk per member; technical fit weighted 30–35%; regulatory compliance gating (pass/fail).

Vendor risk mgmt: HITRUST cert attachment gating before submission. BAA + SOC 2 Type II + pen test standard. CISOs require AI governance docs for any ML/AI platform.

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7. Rejection Reasons

Patient safety risk: Any clinical-workflow touch faces CMIO/patient-safety officer scrutiny. Position as decision-support, not autonomous; pursue Class I/II SaMD pathway early.

HIPAA fear: Privacy officers default-reject when PHI flows unclear. Document de-identification methodology, BAA scope, subprocessor chains BEFORE any technical demo.

"Rip and replace" resistance: Hospitals that spent $300M–$1.2B on Epic are committed. Position AMP+OAS as layered augmentation — "We wrap Epic, we don't replace it."

Epic/Cerner integration lock-in: Epic proprietary APIs (MyChart, Hyperdrive, App Orchard) create real friction. 6–18 mo Epic technical review + App Orchard certification fees. OAS must achieve Epic FHIR R4 cert + App Orchard listing.

Procurement cycle mismatch: 12–18 mo RFPs disadvantage startups. Lack of segment references = hard rejection. Lighthouse program (2–3 references per community/regional/national) prerequisite.

Summary

Healthcare is PrimusNeo's highest-complexity, highest-value vertical: 3-yr deals from $1.5M (community hospital) to $200M+ (national IDN / top-20 pharma). Epic's gravitational dominance + multi-billion-$ integration-services tax + regulatory stack (HIPAA, HITRUST, TEFCA, EHDS) function as moats for certified vendors. Per-bed and per-provider dominate EHR; PMPM dominates payer/population health; ELAs close largest pharma/IDN deals. HITRUST i1 non-negotiable before substantive RFP. Winning motion: wrap, don't replace Epic/Cerner — AMP for spatial ops intelligence, OAS as TEFCA-native FHIR fabric, NEOS for consent/data governance, Fractal for RWE + population analytics. CIOs + CMIOs co-buy; patient safety + HIPAA anxiety are primary emotional rejection levers — neutralize in discovery convo #1.

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PrimusNeo — Logistics Vertical: Market Research Brief

Date: 2026-05-23 **Scope:** Freight forwarders, 3PLs, shipping lines, port operators, last-mile, warehouse, supply chain control towers **Products in scope:** AMP (Spatial OS), NEOS (Governance), OAS (Interop Fabric), Fractal Engine (Intelligence)

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1. Comparable Vendor Pricing

Transportation Management Systems (TMS)

Vendor	Model	Indicative Cost
Oracle OTM	SaaS subscription	~$450/user/mo; enterprise **$400K–$2M/yr**
SAP TM	S/4HANA embedded or standalone	**$500K–$3M/yr** depending on ECC estate
Blue Yonder TMS	Cloud SaaS	Undisclosed; impl alone **$1M–$5M**; TCO commonly exceeds license 2× Y1
MercuryGate (now Infios)	Per-user / per-tx SaaS	~$500/user/mo; 100-user ≈$40K/mo
project44	Per-shipment + platform	**$150K–$600K/yr** mid-market; enterprise $1M+
FourKites	Annual platform + per-shipment	**$100K–$500K/yr** platform; $0.50–$2/shipment

CargoWise (freight forwarder default) shifted to value-packs at **$6–$10/shipment** bundling forwarding + Customs + T&T. Relevant per-shipment anchor for OAS conversations.

Warehouse Management Systems (WMS)

Vendor	Annual Range
Manhattan Active WMS	$2,000/user/yr SaaS; enterprise DC estate **$1.5M–$4M/yr**
Blue Yonder WMS	Custom; comparable to Manhattan at enterprise
Körber WMS	Implementation from $100K; ongoing **$300K–$1M/yr**
SAP EWM	Bundled with S/4HANA; standalone $400K–$2M/yr
Microlistics	Mid-market $80K–$350K/yr

First-year WMS incl impl: **$500K–$3M** large; **$150K–$500K** mid. Annual ongoing **$150K–$500K**.

Port Community / Terminal OS / Digital Twin

- **Navis N4** (Cargotec): on-prem → N4-as-a-Service; **$300K–$1.5M/yr** mid-size terminal
- **Octopi** (Navis lower tier): simple subscription; **$50K–$200K/yr**
- **Awake.AI**: per-port-call optimization SaaS; project + subscription
- **Portchain**: AI terminal planning; custom (Capterra: no public price)
- **GreyOrange (GreyMatter Foundry)**: robotics orchestration; hw + SaaS; full robotic DC **$3M–$15M+**

Track & Trace / IoT (In-Transit Visibility)

Vendor	Model	Indicative
project44	Platform + carrier integration	$150K–$1M+/yr; $0.50–$3/shipment at vol
FourKites	Platform license	$100K–$500K/yr
Tive	HW + data sub	Solo Lite tracker $30–$60/device; platform $1.5K–$5K/mo
Roambee	Device + SaaS	~$10–$15/device/mo; enterprise platform $50K–$250K/yr

Cold-chain (pharma/perishables) commands **40–60% premium** over ambient.

Trade Finance / Digital Bill of Lading (eBL)

Vendor	Model
WaveBL	Pay-as-you-go + tiered subscription
CargoX	Per-document / enterprise; blockchain-attested eBL
essDOCS (CargoDocs)	Per-tx; enterprise $100K–$500K/yr
Bolero	Enterprise subscription; legacy; $200K–$800K/yr large banks/carriers
TradeLens	Shut down Nov 2022 — **canonical failure precedent**

DCSA/BIMCO target 100% eBL adoption by 2030 — sector pre-consolidation.

Customs / Trade Compliance

Vendor	Model	Indicative
Descartes	Modular SaaS	$80K–$500K/yr
Thomson Reuters ONESOURCE Global Trade	Enterprise (210+ country content)	$200K–$1M+/yr
Amber Road (now E2open)	Integrated GTC	$150K–$600K/yr

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2. Engagement Scale Tiers — 3-Year TCO

Small: Regional 3PL / Freight Forwarder (<$50M revenue)

Current 3-yr TCO landscape: $120K–$400K (CargoWise, EDI middleware, manual compliance). **PrimusNeo target: $60K–$180K/yr** (AMP lite + OAS connectors); 3-yr TCV $180K–$540K.

Mid: National 3PL ($50–500M) / Port Operator

Current 3-yr TCO landscape: $1.5M–$6M (TMS, WMS, visibility, customs, middleware). **PrimusNeo target: $300K–$900K/yr**; 3-yr TCV $900K–$2.7M. Displacement of 1–2 point solutions accelerates ROI.

Large: Top-20 Carrier / Top-5 3PL / Global Integrator

Exemplars: Maersk, MSC, CMA CGM; DHL Supply Chain, K+N, DB Schenker; FedEx, UPS. **Current 3-yr TCO landscape:** $15M–$80M+ globally. **PrimusNeo target: $1.5M–$5M+/yr**; 3-yr TCV $4.5M–$15M. Paid pilot ($250K–$500K) precedes full rollout.

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3. Services Rates

Firm	Role	Indicative Day Rate (USD)
Accenture SC&O	SI + strategy	Analyst $1,500–$2,000; Manager $2,500–$3,500; Partner $5,000–$8,000
Deloitte SC	Strategy + digital transformation	Manager $2,500–$3,500; Partner $5,500–$8,500
BCG SC	Pure strategy	Associate $2K–$3K; Project Lead $5K–$7K; Partner $8K–$12K
McKinsey Logistics	Pure strategy	EM $6K–$9K; Senior Partner $10K–$15K+
GEP	Procurement + SC managed services	$1,500–$4,000
Maersk Consulting	Logistics-specific advisory	$2,000–$5,000

Comprehensive SC transformation programs: $250K–$2M+ for 3–6 mo. Channel partners to target: Accenture, Deloitte, GEP.

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4. Pricing Mechanics — Industry Norms

- **Per-shipment:** $0.50–$10 (CargoWise $6–$10 anchor); TMS, T&T, eBL
- **Per-TEU/container:** Software-rare; carrier/terminal surcharges (THC). Twin/analytics experiments per-vessel-call $200–$2K

- **Per-vehicle/per-asset:** Fleet telematics $10–$30/vehicle/mo SaaS GPS; $15–$50/tracker/mo for IoT
- **Per-warehouse/per-DC:** WMS SaaS $3K–$10K/site/mo + per-user + per-tx tiers; multi-DC discounts standard
- **% of freight spend:** Managed transport (4PL/TMSaaS): 0.5%–2%
- **Per-API-call:** Emerging for carrier data + port community; $0.001–$0.05/call at vol; project44 carrier network access partly metered this way
- **Annual platform + overage:** Dominant mid-market model

PrimusNeo positioning: OAS priced per-API-call/per-integration-event (mirroring MuleSoft/Boomi at $0.01–$0.10/tx). NEOS governance priced per-entity (carrier, 3PL, shipper) — network-effect pricing rewarding ecosystem growth.

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5. Regulatory Overlay

IMO CII / EEXI (in force Jan 2023): Vessels ≥5K GT must calculate/report CII annually. D-rated 3 consecutive years or E → corrective action plan. CII tightens ~2%/yr through 2026. Drives demand for voyage optimization, slow-steaming analytics — direct Fractal Engine use case.

EU ETS — Maritime (2024 onward): 100% of intra-EU + 50% of extra-EU voyages calling EU ports. At ~€60–70/tonne CO_2, compliance costs reach tens of millions €/yr for large carrier — hard financial incentive for AMP-driven route/speed optimization.

EU CBAM (definitive phase Jan 2026): Authorized declarants only for steel, aluminium, cement, fertilizers, electricity, hydrogen. CBAM certificates tied to EU ETS price. Creates new doc workflows + customs classification + supplier carbon data — addressable by NEOS + OAS.

US 21st Century Customs Framework (21CCF): Mandatory e-record-keeping, expanded SC data sharing, ACE 2.0 (in progress). Customs brokers must report data breaches. OAS positioned for 21CCF-compliant data flows.

Single Window Mandates: IMO FAL + EU Maritime Single Window framework (effective 2025) require pre-arrival + port clearance data via national single-window — OAS integration play at port-operator + shipping-line tier.

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6. Procurement Signals

Buyer	Cycle
CSCO / CPO	Large enterprise; 9–18 mo. 2024 CSCO agenda: resilience + tech ROI + sustainability
COO / Head of Network Ops	Mid-tier 3PLs / regional carriers; 6–12 mo
CTO / VP Engineering	Technical gatekeeper for API platforms; critical for OAS/NEOS pilot approval
Head of Trade Compliance / VP Customs	NEOS + CBAM/21CCF tooling buyer

Carrier–shipper–3PL triangle: No single party controls the network. Carriers own vessel data; 3PLs own last-mile/warehouse data; shippers own cargo data. TradeLens failed precisely because Maersk tried to own a network requiring trust from its competitors. **PrimusNeo's governance-neutral NEOS positioning is a direct answer.**

RFP cycles: Top-20 carriers + top-5 3PLs run formal tech RFPs annually/biannually; typically Q1 for budget alignment. Target Q3–Q4 budget season convos.

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7. Rejection Reasons

Consortium / network-effect fatigue (TradeLens precedent): TradeLens shut down Nov 2022 after 4 yrs + $1B+ combined investment. Maersk ownership perceived competitively by rivals (MSC, CMA CGM, Evergreen never joined). Asian carriers cited governance concerns. **Lesson for PrimusNeo:** NEOS must demonstrate governance neutrality + open standards alignment (DCSA, GS1, OpenAPI) from day one. No single carrier may access competitor data.

Legacy TMS/WMS integration complexity: Oracle OTM, SAP TM, Manhattan have deeply customized large-account implementations. Pre-built connectors required. **Action:** OAS must publish connector libraries for top-5 TMS + top-5 WMS before enterprise sales scale.

Carrier data hoarding: Ocean carriers treat real-time vessel/container position as competitive. p44/FourKites spent years building carrier network agreements; gaps remain. Either partner with existing visibility networks or offer carriers a data-value-exchange model.

Perceived overlap with existing visibility spend: Shippers paying $300K+/yr to p44/FourKites won't add another layer. Must displace OR offer superset (spatial context + governance + intelligence).

"We built it ourselves" at top carriers: Maersk (Captain), CMA CGM (CEVA), MSC (MSC One), DHL (myDHLi), UPS. Motion here is co-sell / embed (white-label Fractal or OAS data fabric), not displacement.

Summary

Logistics tech segments: TMS ($400K–$3M/yr enterprise), WMS ($500K–$3M Y1), visibility ($100K–$1M/yr), port/terminal OS ($300K–$1.5M/yr terminal). Consulting via Accenture/Deloitte/GEP at $1.5K–$8.5K/day depending on seniority. Pricing favors per-shipment ($0.50–$10), per-site, or annual platform; API/event-metered emerging. Regulatory pressure from IMO CII, EU ETS, EU CBAM (now definitive), US 21CCF creates genuine urgency for data-layer + compliance tooling. Primary buyers: CSCOs + COOs at 3PLs/carriers; CTOs hold integration veto. Central challenge: the TradeLens shadow — any cross-party logistics network play must demonstrate ironclad governance neutrality (NEOS), pre-certified integrations (OAS), and carrier-data value exchange (Fractal) before procurement commits.

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Manufacturing & Industrial Production — Engagement Model Research Brief

> ***Context:*** *$16.8T global manufacturing value (UNIDO 2024), 70% of Industry 4.0 pilots never scale (McKinsey), ~$50B/yr unplanned downtime, average OEE 60–65% vs. 85% world-class target. Regulatory headwinds: EU Digital Product Passport (ESPR), CBAM, Scope 3 mandatory disclosure.*

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1. Comparable Vendor Pricing

Digital Twin / IIoT Platforms

PTC ThingWorx — No published list price. Per-core (server) plus named-user and device counts. Enterprise deployments: **$150K–$500K+/yr** for a meaningful plant-level rollout; seven-figure annual spend is common for multi-site engagements. PTC sold the ThingWorx/Kepware portfolio for up to $725M in late 2025.

Siemens Insights Hub (formerly MindSphere) — Tiered Basic / Standard / Premium, scaled by asset-attribute count and cloud bundles. **$50K–$300K/yr** for a mid-size plant; large OEMs in the Premium tier often exceed $1M/yr across a fleet.

GE Vernova Proficy MES / Predix Manufacturing Data Cloud — Proficy MES starts publicly at ~**$5,800/month** ($69.6K/yr); real deployments **$150K–$400K/yr** per plant. Predix-layer contracts are custom enterprise.

AVEVA System Platform / MES — Subscription; ~**$350/month** at entry (single-line monitoring), scaling to **$100K–$500K/yr** for full MES deployments.

Cognite Data Fusion — Custom enterprise. Industry benchmarks for asset-heavy industrials place contracts at **$500K–$3M/yr** for enterprise deployments. Independent study found $21.6M value over deployment lifecycle.

Tulip (Frontline Operations) — **$1,200/interface/yr** (Professional) to **$2,400/interface/yr** (Enterprise); minimum 10 interfaces. A 20-station plant runs **$24K–$48K/yr** in software alone.

Litmus Edge — **$5K–$15K/edge device/yr**; Edge Manager fleet licensing additive. Professional services at **$200–$400/hr**.

Enterprise IIoT (general) — Platform licensing typically **$50K–$300K/yr** at enterprise. Full single-plant digital twin builds: **$500K–$2M** initial + **$100K–$400K/yr** ongoing. Azure Digital Twins is consumption-based.

MES / MOM Platforms

SAP Digital Manufacturing (DMC, formerly ME) — Published floor ~**$105/month** (entry SaaS tier). Real enterprise plant deployments: **$200K–$1M+/yr** depending on module count (production, quality, OEE, genealogy).

Rockwell FactoryTalk MES — Per production line/asset + named users. Single-plant deployment: **$200K–$600K** all-in for software + integration. Annual maintenance: 15–20% of license.

MES Market General TCO (single plant, 20–30 machines, 3-year): - Cloud-native SaaS MES: **$29K–$72K/yr**; 3-yr TCO **$87K–$216K** - On-premise MES: 3-yr TCO **$250K–$500K** - Professional services typically exceed software license cost by 2–3×

APM / Predictive Maintenance

- **AspenTech Mtell** — Custom; asset-heavy process industry contracts **$300K–$2M/yr** at scale.
- **C3 AI APM** — Outcome-linked; large-scale heavy industry deployments **$1M–$5M/yr** total.
- **Uptake** — Discrete mfg / fleet APM contracts generally **$250K–$1M/yr**.

ESG / DPP / Scope 3

- **Sphera** — Enterprise mfg contracts with full EHS + Scope 3 + product stewardship: **$150K–$600K/yr**. Average disclosed transaction ~$16,874 (SME pilot).
- **Watershed** — Most enterprise contracts **$50K–$150K/yr**; complex Scope 3 + CSRD: **$150K–$300K/yr**.
- **Circularise (DPP / traceability)** — Mid-size mfr supply chain layer: **$50K–$200K/yr**. EU ESPR mandates DPPs for batteries (2025), textiles/electronics (2027–2029).
- **Sweep / Optera** — **$40K–$120K/yr** based on employee count and supplier network.

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2. Engagement Scale Tiers (3-Year All-In: License + Integration + Ongoing)

Tier	Scope	3-Year Cost Range	Notes
Small	1 plant, 1 production line, <50 machines	**$200K–$750K**	Cloud MES + basic IIoT; 2–6 month go-live
Mid	3–12 plants, mixed legacy/modern OT	**$1.5M–$8M**	Enterprise IIoT + MES; heavy integration; 12–24 months per wave
Large / Enterprise	Tier-1 OEM, 25+ plants, global	**$10M–$40M+**	Full twin + MES + APM + ESG/DPP; 30–60 month horizon

Professional services dominate — typically 2–3× software license for complex OT-connected deployments. New-site rollout on an established template: **$50K–$250K per additional plant**.

3. Services Rates — SI / Consulting for IT/OT Integration

Tier-1 Global SIs (Accenture Industry X, Capgemini, Deloitte Digital, IBM) - Day rate: **$1,600–$3,200/day** ($200–$400/hr) senior OT/mfg consultants (NA/Western Europe) - Principal/Director architects: **$3,000–$5,000/day** - Offshore blended: **$800–$1,400/day** (India/Poland/Mexico COE)

Specialist OT/MES SIs (Hitachi Digital Services, ATS, Rockwell SI partners, Inductive Automation SI partners) - **$1,200–$2,400/day** certified MES/SCADA engineers - Fixed-fee single-plant cloud-MES rollout (8–12 wks): **$80K–$200K** - OT network segmentation + IT/OT gateway: **$100K–$400K** per plant

Implementation rates (general) - MES implementation consultants: **$150–$350/hr**; 100–700+ consultant-hours typical for configuration - Annual S&M: **15–25% of license value/yr**

4. Pricing Mechanics Norms

Mechanic	Prevalence	Notes
Per-asset / per-device	Common (IIoT, APM, twin)	Siemens, Litmus, AspenTech
Per-tag (data point)	Common in SCADA/historian	OSIsoft/AVEVA PI ($0.50–$5/tag/yr at scale)
Per-seat / named user	Common (MES, ESG, analytics)	SAP DM, Sphera, Watershed
Per-interface / per-station	Tulip model	Charges workstation, not headcount
% of CapEx	Uncommon	OT integrators sometimes 10–20% of plant CapEx
% of OEE / gainshare	Emerging	Fixed base + 10–30% share of documented savings; attribution-hard
Outcome-based / XaaS	Growing in equipment sector	Guaranteed uptime SLAs; $/ton, $/vehicle
Subscription (annual SaaS)	Dominant trend 2023–2026	Cloud MES, ESG, IIoT moving here

Hybrid (base platform + consumption/asset overage) is now the most common enterprise structure. Pure gain-share remains rare in mfg due to attribution disputes.

5. Procurement Signals

Decision Makers by Deal Size: - **Plant Manager / VP Ops** — Signs <$250K. Prioritizes payback <18 months, minimal IT dependency. - **CIO / VP Digital / CDO** — Signs $250K–$2M. Architecture fit, cloud strategy, OT cyber, vendor roadmap. - **COO / CFO** — Required above $2M. Program ROI, supply chain

resilience, ESG compliance. CSO sign-off on ESG/DPP. - **Procurement** — Enters formally above $500K; RFP/RFQ, benchmarking, MSA/SLA terms.

RFP Cadences: - Enterprise CapEx >$5M: 6–18 month cycle - Mid-market OpEx SaaS $500K–$2M: 4–9 months; often paid pilot/POC $50K–$150K - ESG/compliance deadline-driven: 2–4 months when mandated (CBAM 2024, DPP 2025–2027)

CapEx vs. OpEx: - Strong post-COVID bias toward **OpEx/subscription** — reduces upfront approval, preserves capital - OT infrastructure (edge HW, OPC-UA gateways, historian servers) still CapEx-classified — split: software OpEx, connectivity CapEx

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6. What Buyers Will NOT Pay For

Vendor Lock-In — Single largest recurring rejection. Buyers reject proprietary data formats, non-standard protocols, platform-bundled AI that blocks open tooling, single-vendor historian/edge lock.

Integration Complexity / "Black Box" OT — 42% of manufacturers cite OT integration as primary scaling barrier. No credible connector story for legacy PLCs (Siemens S7, Allen-Bradley, Modbus) = rejected at pilot.

Unclear ROI / Unquantified Payback — 48% report buyer's remorse, most often from integration cost overruns (38%) and implementation delays (35%). No quantified OEE/downtime scenario with payback <24 months = stalls at COO/CFO.

Pilot-to-Scale Gap — 70% of pilots never scale. Buyers now reject POC-first proposals lacking a defined productionization path and budget envelope.

Underestimated Total Cost — Hidden costs that kill deals post-signature: OT segmentation, historian per-tag overage, change management, retraining, ML model maintenance. Binding TCO disclosure increasingly required in RFP responses.

Data Sovereignty / Cloud Residency — European manufacturers reject US-hyperscaler-only deployments without EU data residency or on-premise/hybrid for production IP and quality data.

Key Pricing Anchors (One-Paragraph Summary)

IIoT/digital-twin platform licenses run **$50K–$300K/yr per plant** for mid-tier vendors (Siemens Insights Hub, AVEVA) and **$150K–$500K+/yr** for premium platforms (PTC ThingWorx, Cognite); MES software adds **$100K–$500K/yr per plant** for enterprise vendors (SAP, Rockwell), while cloud-native MES (Tulip, Symestic) starts under $30K/yr for small lines. Three-year all-in (license + integration + ongoing) is **$200K–$750K** for a single-plant small engagement, **$1.5M–$8M** for a mid-market 3–12 plant rollout, and **$10M–$40M+** for a Tier-1 OEM global program. SI day rates for OT/MES integration run **$1,200–$3,200/day** onshore. ESG/Scope 3/DPP platforms (Sphera, Watershed, Circularise) run **$50K–$300K/yr** at enterprise scale, with EU regulatory deadlines (CBAM, ESPR/DPP by 2027) compressing procurement cycles. Dominant rejection triggers: vendor lock-in, unquantified ROI, underestimated integration cost.

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PrimusNeo — Robotics Vertical: Market Research Brief

> ***Status:*** *Draft — internal research anchor. Sourced ranges, not quotes.* ***Scope:*** *Industrial robotics, AMRs, humanoids, drones, surgical robotics, agricultural robotics, fleet/RaaS.*

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1. Comparable Vendor Pricing

Robot OS / Middleware

Vendor	Model	Price
ROS 2 / Open Robotics	OSS, community-supported	Free
NVIDIA Isaac ROS	OSS packages on NGC; fleet mgmt via VDA5050	Free (GPU cloud ~$3–$6/hr AWS G6e L40S)
Foxglove Studio	Free (3 users) / Starter $18/user/mo / Team $42/user/mo / Enterprise	**$90+/user/mo** enterprise
Formant	Free tier early-stage; data-vol at scale	Quote-based; expensive at high telemetry vol
Microsoft (ROS on Windows / Azure)	Azure cloud pass-through	Compute cost only

Isaac Sim free for dev; production GPU cloud cost on EC2 G6e applies. **AWS RoboMaker deprecated for new customers Sept 2025** — teams migrate to AWS Batch or direct Isaac Sim on EC2.

Fleet Management

Vendor	Pricing
InOrbit	**~$500/robot/month** (PAYG, high-water mark); volume discounts on annual commits; Enterprise custom
Freedom Robotics	SMB fleets (10–20 robots); quote-based
Formant	Free tier → quote-based at scale; data-egress cliff
Anyware Robotics	Quote-based; OEM embedded licensing

Industry range: **$50–$500/robot/month** depending on telemetry vol, modules, SLA.

Simulation / Digital Twin

- **NVIDIA Isaac Sim:** Free software; GPU cloud $3–$10/hr per instance on EC2 G6e
- **Gazebo:** OSS, free
- **AWS RoboMaker:** deprecated Sept 2025 for new customers
- **CoppeliaSim:** free educational; Pro ~$1,500/seat/yr
- **Microsoft Project Bonsai:** Azure pass-through

Robotics-as-a-Service (RaaS)

Vendor	Segment	Pricing
Knightscope K5	Security/patrol	~$11/hr; **$60K–$70K/robot/year** annual contract (vs $35–$85/hr human guard)
Locus Robotics	Warehouse AMR	**$2,000–$4,000/robot/month** all-in RaaS (hw, sw, support, optimization); 2–3x picks/hr uplift
Cobalt Robotics	Security/facility	Hourly RaaS; robot + remote operator bundle; quote-based

RaaS market: 25.52% CAGR through 2034; projected $125B.

Safety / Certification

- **ISO 10218-1:2025 / -2:2025** (industrial): Cert via TUV Rheinland / Intertek / UL; **7–12 mo timeline**; **$50K–$200K+** for full cell certification incl engineering hours
- **ISO 13482:2014** (personal care / service): notified body engagement required for EU market
- **RIA TR R15.806** (US HRC): Technical report, no cert body; internal compliance cost
- **Riskbench / safety SW tooling:** Boutique; $10K–$50K/yr SaaS

Computer Vision / Perception

Vendor	Pricing
Roboflow	Free (public) / Core $79/mo / Enterprise custom
Scale AI	Self-serve $0.02/image, $0.06/annotation; Enterprise **$93K–$400K+/yr** (~$93K avg contract)
Encord	Custom enterprise (regulated + multimodal); SOC2/HIPAA; raised $60M (2025)

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2. Engagement Scale Tiers — 3-Year TCO for Coordination/Intelligence Layer

Small: <10 Robots, Single Facility, Startup/Integrator

Current fleet mgmt: $0–$5K/mo (free tier or entry InOrbit). Pain: no mission-level coordination, no governance, no cross-system telemetry. **PrimusNeo 3-yr TCO: $120K–$300K** · Y1 $60K–$120K (integration + deployment + first license year), Y2–3 $30K–$90K/yr. Primary layers: OAS (interop to ROS/proprietary) + Fractal (telemetry/analytics).

Mid: 50–500 Robots, Multi-Site, Mid-Size Integrator/OEM

Current spend: InOrbit/Formant $25K–$250K/yr; custom SI $200K–$800K. Pain: heterogeneous vendors, multi-site dispatch, safety audit readiness, fleet analytics. **PrimusNeo 3-yr TCO: $750K–$2.5M** · AMP (spatial across sites) + NEOS (fleet governance, SLA councils) + OAS (vendor-neutral API) + Fractal (predictive maintenance). Services: $250K–$600K integration; $150K–$400K/yr upkeep. Displaces 2–4 point solutions ($50K–$500K/yr combined).

Large: 1,000+ Robots, Global OEM / National Logistics Fleet

Reference class: ABB, FANUC, KUKA, Boston Dynamics, Symbotic. Current: proprietary OEM fleet sw + SI ($1M–$5M+/yr); custom twin $500K–$3M. Pain: cross-vendor fleet interop, global compliance (EU AI Act, ISO 10218, FDA/FAA overlay). **PrimusNeo 3-yr TCO: $3M–$12M** · All four layers; multi-tenant spatial OS; governance councils per site/region; OAS federation; Fractal with custom model retraining. Services: $800K–$2M integration; $500K–$1.5M/yr upkeep + infra. Benchmark: NVIDIA/ABB twin programs run $2M–$8M/yr SW + SI.

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3. Services Rates

Service Type	Rate
Robotics SI firm (boutique US)	$150–$275/hr specialist; $100–$175/hr mid-level
NVIDIA partner (Isaac/Omniverse certified)	$200–$350/hr; fixed-scope projects $250K–$2M
AWS partner (RoboMaker→Batch/Isaac migration)	$150–$250/hr; projects $100K–$500K
PrimusNeo customization (Fractal model + twin schema)	$1,500–$2,500/day T&M or fixed-scope
PrimusNeo integration (OAS connector per robot vendor)	**$25K–$80K per connector** fixed-fee
PrimusNeo deployment (per site)	$40K–$150K depending on site complexity
Sustained upkeep	18–22% of annual license value

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4. Pricing Mechanics

Robotics responds to five mechanics:

1. **Per-robot/month SaaS** — dominant for fleet mgmt + RaaS overlays. Industry $50–$500/robot/mo software; $2K–$5K/robot/mo full RaaS. PrimusNeo positions **$200–$800/robot/mo** for coordination/intelligence layer (depending on Fractal depth + OAS connector count).
2. **Per-mission / per-task** — emerging for logistics AMRs (cost/pick, cost/delivery). PrimusNeo can meter Fractal inference + OAS API calls at $0.002–$0.01/mission event.
3. **Enterprise flat annual license** — preferred OEMs/Tier-1 integrators for predictability. PrimusNeo offers this for 500+ robot deployments with per-robot-per-yr floor.
4. **RaaS revenue share** — Knightscope/Cobalt model. PrimusNeo participates as platform partner (**3–8% of operator RaaS revenue**) where coordination layer is embedded in partner's offering.
5. **Per-site SLA** — construction, ag, surgical. NEOS governance councils map to per-facility accountability.

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5. Regulatory Overlay

Standard / Regulation	Scope	Implication
ISO 10218-1:2025 / -2:2025	Industrial robots + cells	AMP must support safety-zone geofencing; Fractal audit logs cert-ready
ISO 13482:2014	Personal care / service robots (humanoids)	NEOS supports hazard accountability traceability
EU AI Act (2024, enforcement 2026)	High-risk AI in robots near humans	Fractal classified as "AI system" under Annex III; requires conformity assessment, logging, human oversight
FAA Part 107	Commercial UAS <25kg (US)	$175 knowledge test + $5/drone registration; 90–120 day waiver. OAS must support UTM API integration
FDA 510(k)	Surgical robotics (Class II)	$6,084–$24,335 FDA fee + $20K–$30K consulting; 142-day median review. Fractal as data layer must meet SaMD if processing clinical decision data
RIA TR R15.806	Human-robot collaboration (US)	Internal compliance. Fractal risk signals support HRC safety documentation

Any deployment touching surgical, cobot HRC, or drone fleet requires a **regulatory posture document** as engagement deliverable. **Deal gate, not nice-to-have.**

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6. Procurement Signals

Title	Authority	Entry Point
VP Robotics / Head of Automation	Technical champion; fleet roadmap	OAS interop demo + Fractal predictive maintenance ROI
CTO	Architecture sign-off; multi-yr platform	Full stack briefing + Tie Town reference arch
COO / VP Ops	Budget owner; RaaS opex framing	Cost/robot/mo vs current fragmented spend; downtime cost model
Head of Safety / Compliance	Veto in regulated contexts	ISO 10218 + EU AI Act + FDA compliance posture doc

Sales motion: - **RaaS motion (opex framing):** Add line item on existing monthly RaaS invoice. "Already paying $3K/robot/mo to Locus — we add $300 for coordination intelligence + cut unplanned downtime 15%." Requires RaaS vendor partnership or OEM channel. - **Pilot-to-fleet funnel:** Single-facility pilot (3–6 mo, fixed $50K–$150K) → multi-site → enterprise MSA. McKinsey: 70% of pilots fail to scale — Tie Town reference is the counter-narrative. - **OEM embedded licensing:** Target ABB, FANUC, KUKA, UR, Boston Dynamics — embed OAS connector + Fractal as licensed module in their robot mgmt software. **Revenue share or $50–$200/robot/yr royalty at scale.**

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7. Rejection Reasons

1. **Safety/cert gap** — "Not ISO 10218 / EU AI Act certified." Mitigation: compliance posture doc; Fractal audit logs as cert-ready evidence.
2. **Vendor concentration risk** — "Already on NVIDIA Isaac + ROS 2." Mitigation: OAS is additive, not replacement; runs alongside Isaac without forking ROS graph.
3. **Integration with ROS / proprietary stacks** — ROS 2 DDS + proprietary controllers (FANUC KAREL, ABB RAPID, KUKA KRL) need custom connectors. Mitigation: OAS connector library; scope/price per connector explicitly.
4. **ROI per robot unclear** — Mitigation: Fractal predictive maintenance ROI model (1 avoided downtime incident = $5K–$50K in logistics).
5. **Talent shortage** — Mitigation: managed deployment service; NVIDIA/AWS SI partner channel.
6. **Humanoid uncertainty** — Figure, 1X, Agility, Boston Dynamics Atlas pre-commercial at scale as of 2026; enterprise in "watch and wait." Avoid over-indexing RaaS models on humanoids until commercial fleet sizes materialize.

Summary

Robotics is PrimusNeo's highest near-term AMP + Fractal opportunity. Fleet mgmt SaaS: $50–$500/robot/mo (InOrbit anchors at $500). RaaS bundles: $2K–$5K/robot/mo (Locus, Knightscope). Sim + OS tooling (NVIDIA Isaac, ROS 2) free — commoditizes infra layer, creating demand for differentiated coordination/intelligence above. PrimusNeo 3-yr TCO: $120K (small) → $12M (global OEM). Key deal gates: ISO 10218:2025 / EU AI Act compliance posture + clear per-robot ROI. Fastest entry: OEM embedded licensing with NVIDIA partners or major robot OEMs, then RaaS revenue-share with fleet operators. Buyer coalition: VP Robotics + COO + Head of Safety; COO's opex framing via RaaS = path of least procurement resistance.

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Technology Vertical — Market Research Brief

PrimusNeo: AMP (Spatial OS) · NEOS (Governance) · OAS (Interop Fabric) · Fractal Engine (Intelligence)
Date: 2026-05-23

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1. Comparable Vendor Pricing

Platform Engineering / IDP

Managed IDP pricing has converged at **$22–$78/developer/month**:

Vendor	Pricing
Roadie (managed Backstage)	~$22/developer/month — entry anchor for managed OSS portals
Port	Standard ~$40/seat/mo; Enterprise ~$60–$78/seat/mo
OpsLevel	~$39/user/mo base; Startup ~$30/dev/mo
Cortex	~$65–$69/user/mo at scale (community-reported)
Humanitec Platform Orchestrator	From ~$1,979/mo (5 users); orchestration layer above portals; enterprise custom
Backstage (self-hosted)	$0 license. Effective TCO: 3–5 senior platform engineers @ $250K loaded = **$750K–$1.25M/yr** carrying cost + $12K–$24K/yr infra. **3-yr TCO typically exceeds $2M** before productivity gains

Federation / Multi-Org Coordination

- **Open Policy Agent (OPA):** Free, CNCF-graduated. Implementation engagements $150–$300/hr
- **Styra (OPA commercial wrapper):** $100K–$500K ACV typical
- **GitOps (Flux/ArgoCD):** OSS; integration services at standard platform-eng rates

Data Fabrics / Data Mesh

- **Snowflake:** ~$2–$4/credit; mid-market monthly $10K–$80K; large enterprise **$100K–$500K+/month**
- **Databricks:** ~$0.22–$0.55/DBU-hour SQL/ETL + cloud VM costs; typically 20–40% cheaper than Snowflake at scale but requires FinOps maturity
- **Starburst (Trino federated):** Cost-offload from Snowflake; enterprise custom
- **Dremio:** ~$75K–$250K ACV estimated from community reports

AI Gateways / Agent Infrastructure

Vendor	Pricing
LangSmith (LangChain)	Free (1 seat, 5K traces/mo); Plus $39/user/mo; traces $2.50/1K (14-day), $5.00/1K (400-day); enterprise custom
Weights & Biases	~$50/seat/mo (CoreWeave-acquired March 2025); enterprise custom
Modal (serverless GPU)	A10G ~$1.10/hr; L40S ~$1.95/hr; H100 **~$3.95/hr**
Replicate	H100 ~$5.49/hr; overall 1.4–2.9x higher than Modal/Fal for custom models
LlamaIndex	OSS framework; commercial Cloud tier custom

Spatial / 3D Dev Platforms

- **NVIDIA Omniverse Enterprise: $4,500/GPU/year** subscription; enterprise contracts vary by GPU count
- **Niantic Lightship Spatial SDK:** $0.80–$1.00/MAU beyond 100 MAU; shared AR sessions billed separately per multiplayer session
- **Unity Industry / Pro:** Pro ~$2,040/seat/yr; Unity Enterprise custom; industrial twin modules add-on
- **Unreal Engine / MetaHuman:** Free for games <$1M revenue; enterprise licensing/support **$50K–$500K/yr** for industrial deployment

Web3 Infrastructure

Vendor	Pricing
Alchemy	30M CUs/mo free; pay-as-you-go $0.40–$0.45/1M CUs; **1 eth_call = 26 CUs** so effective per-call ~$10.40–$11.70/1M calls
QuickNode	Flat-rate RPS from $799/mo (75 RPS EVM chains); per-call ~$3.27–$12.25/1M calls
Infura	55M credits add-on $200/mo; **1 eth_call = 80 units** → higher effective per-call cost than Alchemy
Chainstack	Pro $199/mo (80M req, 400 RPS); Business $499/mo (200M req, 600 RPS, 20 nodes); enterprise custom

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2. Engagement Scale Tiers — 3-Year TCO

Tier	Profile	IDP (3yr)	Agent Infra (3yr)	Spatial Collab (3yr)	**Comparable Market TCO (3yr)**
Small	<50 engineers, single-product startup	$80K–$200K (IDP SaaS, no build)	$30K–$80K (LangSmith Plus + Modal spot GPU)	$20K–$60K (Unity Pro + basic Omniverse)	**$130K–$340K**
Mid	50–500 engineers, scale-up	$350K–$900K (Port/Cortex enterprise + 1–2 platform engineers)	$120K–$400K (W&B + LangSmith enterprise + Modal committed)	$80K–$200K (Omniverse 5–20 GPUs/yr)	**$550K–$1.5M**
Large	500+ engineers, FAANG/MAANG, telco, hyperscaler	$1.5M–$4M (Humanitec + self-hosted portals + 5–8 platform engineers)	$500K–$2M+ (enterprise W&B/LangSmith + dedicated GPU fleet)	$300K–$1M+ (Omniverse 100+ GPU/yr, Unity Enterprise)	**$2.3M–$7M+**

Large-tier telcos and hyperscalers often embed these costs inside cloud committed-use agreements (EDP/CUD) at **20–40% effective discount off list**.

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3. Services Rates — SI / Boutiques

Firm	Positioning	Typical Rate
IBM Consulting	End-to-end; proprietary AI/cloud (watsonx); regulated industries	$200–$350/hr onshore; $80–$120/hr offshore
Accenture	Global scale; $3.6B AI bookings; multi-vendor transformation	$220–$400/hr onshore; $70–$130/hr offshore blend
Capgemini	Strong in European enterprise, telco, mfg; GPAI-focused 2025	$180–$320/hr onshore; $60–$110/hr offshore
Globant	LatAm nearshore specialist; Agile Pods ~$20K/mo	$50–$99/hr (nearshore mix); pods from $20K/mo
EPAM	Engineering-heavy; Eastern Europe; platform/AI focus	$75–$150/hr nearshore; $150–$200/hr onshore leads
ThoughtWorks	Modern eng practices, Lean architecture, DevEx	$200–$350/hr onshore; advisory-then-delivery model
Boutique platform-eng (Syntasso, Kosli, Configure8)	Backstage/IDP specialization; faster ramp	$150–$250/hr; fixed-scope workshops $30K–$80K

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4. Pricing Mechanics Norms

- **Per-developer/seat:** $15–$100/dev/mo established IDP norm. $22 (Roadie) low end; $65–$78 (Cortex, Port Enterprise) high end. >500-engineer orgs prefer enterprise commits
- **Per-API-call / per-compute-unit:** AI inference (LangSmith $2.50–$5/1K traces) + Web3 RPC (Alchemy, QuickNode, Chainstack). Compute-unit obfuscation common — effective per-call costs 10–200x advertised unit rate
- **Per-compute-hour (GPU):** $1.10–$5.49/hr managed (Modal → Replicate) vs $2–$4/hr raw cloud. Serverless premiums 1.4–2.9x raw. Reserved/committed instances reduce 30–50%
- **Per-GB (storage/egress):** Snowflake, Databricks. Egress $0.08–$0.12/GB routinely surprises eng teams — common TCO underestimate
- **% of platform spend:** Less common in devtools; FinOps SaaS (Apptio, Harness Cloud Cost) typically **1–3% of monitored spend ACV**
- **Enterprise commit / MAP:** >$500K/year orgs get 12–36 month multi-year commits at 15–30% discounts + private pricing. Hyperscaler EDPs bundle IDP-adjacent tooling at 20–40% below public list
- **Usage-based + seat hybrid:** Emerging norm (LangSmith, Humanitec) — base seat for governance + usage overage for compute/traces

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5. Regulatory Overlay

Framework	Applicability	Cost
SOC 2 Type II	Mandatory for SaaS to US enterprise / gov-adjacent	$35K–$150K first year; $20K–$50K/yr ongoing (with automation)
ISO 27001	EU enterprise + telco + regulated-sector clients	$25K–$80K initial cert; $15K–$30K/yr surveillance
FedRAMP (Mod/High)	US federal agency sales; govtech-adjacent SaaS	**$500K–$2M+** to achieve; 12–18 months (post-2025 "20x" acceleration <5 weeks once package complete)
EU AI Act	Full application Aug 2026; high-risk (Annex III) potentially deferred to Dec 2027	Conformity assessment $20K–$80K/high-risk system; GPAI docs + transparency ongoing
EU Data Act / DSA / DMA	Interop fabrics (OAS), platform gateways, data portability	Legal/compliance advisory $50K–$200K/jurisdiction initial posture
US AI EOs	Dual-use foundation model disclosure (above compute thresholds); federal procurement AI risk	Internal governance program $100K–$500K+

73% of enterprise AI initiatives cite compliance posture as top-three vendor selection criterion (BCG 2026), up from 41% in 2024. CAIO roles exist in **61% of large enterprises** — compliance champion is now a named buyer.

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6. Procurement Signals

Buyer personas and triggers:

- **CTO:** Platform strategy; build-vs-buy for IDP + agent infra. Responds to architecture credibility, standards alignment (OpenUSD, OpenTelemetry, CNCF), federated multi-org deployment proof. Entry via conf talks, technical whitepapers, OSS presence
- **VP Platform Engineering / Head of DevEx:** Owns IDP budget + dev productivity KPIs (DORA metrics, deploy frequency, change failure rate). Responds to ROI on dev time saved — typically modeled at **$150–$250K/engineer-year fully loaded**. Entry via community (PlatformCon, KubeCon), peer referral
- **Chief AI Officer:** Owns agent infra + LLMOps stack. Evaluates observability (LangSmith, W&B), governance (audit trails, policy enforcement), data residency. Entry via analyst relations + AI governance working groups
- **Head of Developer Experience:** Onboarding speed, cognitive load, golden path adoption rates. Responds to PLG trials — hands-on before budget conversations

Motion patterns:

- **PLG to enterprise:** Dominant dev-tool pattern. Free OSS/SDK inbound (Backstage plugin, OAS connector, Fractal community tier) → team adoption → VP Platform/CTO expansion. Cursor reached $2B ARR by Feb 2026. PLG companies see **~9% free-to-paid conversion; top quartile reaches 25–30%** with PQL-triggered outbound
- **OSS to paid tier upsell:** Backstage (free) → Roadie/Port/Cortex ($22–$78/seat/mo). OPA (free) → Styra (enterprise). OAS open SDK → OAS federation engine (closed-core). Standard playbook for infrastructure OSS
- **Partner-led (SI channel):** For large/MAANG/telco, Accenture or IBM Consulting frames the engagement; PrimusNeo enters as tech layer inside broader transformation. **SI takes 20–40% impl margin; platform vendor takes license/commit**

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7. Rejection Reasons

- **Build-vs-buy / NIH syndrome:** Engineering-led orgs (esp FAANG) default to "our requirements are unique." Counter: top-performing orgs use **hybrid buy+extend models**, not full builds. Humanitec 2023 DevOps Benchmarking: top performers build IDP from OSS + vendor, not from scratch
- **Vendor lock-in:** Federating four layers across four steward orgs raises legitimate portability concerns. Counter: open SDK + CC licensing on NEOS/Fractal; data export guarantees in OAS; OpenUSD/open standards alignment in AMP
- **Open-source competition:** Backstage, Crossplane, Flux, ArgoCD, OPA, MLflow, LangFuse — every commercial layer has credible OSS alt. Counter is TCO (Backstage = $2M+ over 3 yrs for large teams), time-to-value, enterprise support SLA

- **Integration with existing devtools:** GitHub Enterprise, Jira, PagerDuty, Datadog, Slack/Teams — deep integrations required. First 90 days of any engagement must demonstrate connector coverage
- **Governance / data residency:** EU-based tech (telcos, post-GDPR scale-ups) require in-region data. AI Act readiness + multi-region OAS deployment must be demonstrable
- **Procurement cycle length:** MAANG/telco enterprise procurement averages 6–18 months from first meeting to signed contract. PLG inbound shortens — product usage data substitutes for RFP at team level, escalating to procurement only after internal champions exist

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Summary

Technology vertical has highest density of comparable vendor benchmarks + sharpest build-vs-buy instinct of any PrimusNeo target. IDP seat pricing clusters at $22–$78/dev/mo, with self-hosted Backstage carrying hidden $2M+ 3-yr TCO that commercial portals undercut on value. AI agent infra: $39–$50/seat/mo for observability + $1–$5/GPU-hour for inference. Spatial platforms (Omniverse, Unity Industrial): $4,500/GPU/yr and up. 3-yr stack TCO: $130K (50-engineer startup) → $7M+ (hyperscaler). SI rates: $50–$400/hr. Compliance (SOC 2, ISO 27001, EU AI Act) now top-3 vendor selection criterion for 73% of enterprises. Primary rejection vectors: NIH syndrome, open-source competition, data-residency. PrimusNeo's strongest entry motion: OSS/SDK inbound (PLG) escalating through VP Platform Engineering + CAO champions.

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